SMITHIAN FOLD THEORY - CHEMISTRY BRANCH PAPER 001

From Fold to Chemistry

An Exact, Parameter-Free and Machine-Closed Reconstruction of Chemical Science from Smithian Fold Theory

Ernos Labs
Open Source Science Platform and Knowledge Tree

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Version 1.1 - Chemistry inventory complete
86 required Chemistry claims and 7 explicit upstream claims
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Abstract

This paper reports the Chemistry branch of the third clean-room reconstruction of Smithian Fold Theory. It documents all 86 obligations in the frozen inventory and the seven newly required upstream physical prerequisite/validation claims. Every claim passed one fail-closed admission engine with a complete generated grammar, exactly one survivor, trace to the premise-free foundational theorem, zero axioms, zero free or fitted parameters, exact permitted arithmetic, four adverse controls and an implementation-distinct reconstruction. Empirical claims add capability-closed prediction, target custody, post-seal opening, complete authoritative rows and tampered controls. The required Chemistry support contains 22016 generated candidates and 86 survivors; the terminal-prediction prerequisite appendix contributes 4224 candidates and seven survivors. The formal walk forces orbit widths 2, 6, 10, 14 and 18; nuclear closures 2, 8, 20, 28, 50, 82, 126 and 184; Smithium at Z=126, N=184, A=310 with active 8s2 5g6 organization; g-block opening at 121; and a structural endpoint at 137. NIST/CODATA, IAEA and IUPAC checks occur only after derivation seals. Element 121, Smithium and endpoint 137 remain explicit unobserved predictions.

Results first: exact chemical structure and standing predictions

Headline result	Exact result	Empirical status and meaning
Atomic orbit capacities	`2, 6, 10, 14, 18`	Forced from three-space, its rank-two boundary, two held Fold labels and exclusion; the known periodic organization is checked only after sealing.
Nuclear closure sequence	`2, 8, 20, 28, 50, 82, 126, 184`	Exact match to the complete registered IAEA closure sequence. The `126/184` coordinate supplies the island-of-stability carrier used below.
g-block opening	`Z = 121`	Standing unobserved prediction: the complete fill walk first opens the 5g organization at element 121.
Smithium	`Z = 126`, `N = 184`, `A = 310`; active organization `8s2 5g6`	Standing unobserved prediction. Known IUPAC support through element 118 is retained separately and does not count as observation of Smithium.
Atomic structural endpoint	`Z = 137`	Standing unobserved prediction forced by the greatest-whole order beneath the exact inverse-alpha carrier.
Physical prerequisite used at the chemical boundary	`alpha^-1 = 503846395469/3676744786 = 137.035999177180855...`	Derived and empirically validated in the owning Physics branch against CODATA; Chemistry cites that admitted dependency and does not relocate or re-fit the constant.
Complete branch evidence	86 Chemistry laws, 22,016 candidates and 86 survivors; seven separately admitted prerequisites add 4,224 candidates and seven survivors	Every law has a root trace, complete decisions, one survivor, four adverse controls, independent reconstruction and post-seal comparison where an external chemical record exists.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorize. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigor.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Chemistry, a periodic-table entry, handbook definition, fitted Hamiltonian or opaque predictor may test or translate a sealed result but cannot select chemical identity, shell capacity, nuclear closure or an endpoint. Known-domain agreement, standing prediction and specimen-dependent measurement are kept as different evidence classes so that persuasive presentation never erases scientific custody.

Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through https://discord.gg/ucwGryVxGr, or inspect the public project at https://github.com/MettaMazza.

1. Publication, authorship and license boundary

LOCAL PREPUBLICATION MANUSCRIPT. Publication is not yet authorized. No push, upload, DOI action or publication follows from building this file.

Maria Smith, independent researcher and founder of Ernos Labs. Contact: Maria.Smith.Sftoe@gmail.com. Submissions and reproducibility reports: https://discord.gg/ucwGryVxGr. GitHub: https://github.com/MettaMazza.

2. Exact scope

The inventory contains 86 obligations in ten ordered subbranches and all are admitted. Closure means each registered Chemistry question has one machine-admitted law at its exact generated and empirical boundary. Materials bulk properties, biological function, clinical intervention, Earth-system fate and applications remain later branches. Observed absence beyond element 118 is not promoted into proof. The 121, 126 and 137 results are standing predictions.

3. Constitutional arithmetic

Structural absence is empty One, never numerical zero. Proof magnitudes are positive generated counts and exact positive rational parts or ratios. Opposition, orientation, gain/loss direction and charge sense are held labels, never negative proof quantities. Irrational, imaginary and floating proof values, completed infinity, imported equations, fitted coefficients, learned parameters and target-selected forms are prohibited. Host counters may enumerate artifacts but have no scientific value authority.

4. Single-engine admission

Every claim enters `SFTAdmissionEngine`. Registration verifies root identity, zero axioms, zero free parameters, admitted dependencies and exact source identity. The engine then requires complete candidate identity/cardinality, one decision per candidate, exactly one survivor, closure, minimality, named-shape uniqueness, four controls and an implementation-distinct reconstruction. Empirical claims additionally require an evaluator-verified derivation seal, capability isolation, target absence before seal, matching custody release, all-row retention and a falsification condition. Failure halts and is retained; it cannot be edited into admission.

5. External evidence and blind prediction

IUPAC terminology and periodic-table records test categorical consequences. NIST and IAEA records test sealed numerical or ordered-sequence consequences. External data never select the grammar or survivor. The capability-closed predictor cannot read the filesystem, network, clock, environment, subprocess, dynamic import or foreign function. A distinct target custodian releases only after a matching prediction seal, every registered row is compared, and a deliberately changed row must fail.

6. Preserved failure and lawful correction

The original prerequisite programs retained two capacity survivors, two nuclear-schedule survivors and no endpoint survivor. Their rejection receipts remain immutable. V3 then derived new discriminators: generator three, stable three-space, boundary rank two, exact orbit orientation, exact colour coupling, complete three-direction shell composition, the finite inverse-alpha promotion ladder and a greatest-whole order certificate. The later admissions do not rewrite the failed runs; they change the admitted dependency surface through separately proved claims.

7. V1/V2 reconciliation without proof import

V1/V2 identify mandatory questions, including magic numbers, Smithium, the g-block and endpoint. They supply no answer-bearing input to V3. `census/lineage_reconciliation.json` binds those sources by hash and now records the elements/nuclear/island-of-stability group as closed at the current V3 standard. Other lineage groups remain open for their own branches and the final TOE paper.

8. How to read the claim ledger

Every section below projects the complete machine evidence: statement, dependencies, grammar, boundary, candidate cardinality, sole survivor, axis eliminations, controls, independent reconstruction, external rows, exclusions and immutable receipt identities. The JSON claim packages remain authoritative and are included in the release archive.

9. Measurement Identity

Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability.

1. Chemical entity and retained identity

Claim identity: `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`

Question and theorem. A chemical entity is the least singular Fold carrier that remains separately distinguishable after constitution, isotopic identity and the observation context's required precision are retained.

> *singular-carrier__constitution-and-isotope-retained__separately-distinguishable__context-bounded-precision*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`

- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001

Generated grammar. Generate the literal product of the registered entity carrier, constitution, isotope, distinction, precision, record, provenance and extension choices.

Boundary: Every finite singular chemical carrier formed from admitted distinctions and any generated positive-finite extension of its constitution, isotope, state or contextual precision record.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: singular-carrier__constitution-retained__isotope-retained__separately-distinguishable__context-bounded-precision__held-observation-record__source-bound-proof-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	singular-carrier	ensemble-without-member	An ensemble alone does not identify one chemical entity.
constitution	constitution-retained	constitution-erased	Erasing constitution merges chemically distinct carriers.
isotope	isotope-retained	isotope-erased	Erasing isotopic identity merges distinguishable entities.
distinction	separately-distinguishable	indistinguishable-answer-label	An answer label without a retained distinction cannot identify an entity.
precision	context-bounded-precision	unbounded-or-fixed-precision	A fixed universal precision either loses a required distinction or adds an unforced one.
record	held-observation-record	result-without-observation-record	Without an observation record the identity boundary cannot be audited.
provenance	source-bound-proof-trace	unbound-provenance	Unbound identity cannot be traced to the Fold dependency spine.
extension	no-extra-rule	free-extra-identity-rule	An extra identity rule can arbitrarily split or merge entities.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:5f110d5c9e6a40120b2d1a5cb5a6dfa2738fc19b611acdb41f21092208f2a84e.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:564377b083b5fd48f74bcd6ffe5e88fbe00d970d9fb10253723fd39ff85ba41f.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:5f6268c44c846a4e91dea61554c4b01f5c85b882c9e894009c0d04e325d2a0a2.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:eec730468fa661f596b4ac9b005665b9817aa9b7a63bb0878ca3cf0ec33b255d.

Independent reconstruction. Implementation hash sha256:1e162cfb721018f6da4cb2c45eccff8324a1cba92e801169174696005e395938; independent certificate sha256:ab5e4ae45a37d0b2ac8f02f883990992b7c6492fcd59e809620be8a416d6fd86; external-validation hash sha256:1b952ccd2e4c3740a1bb5774c459db17faff80d5e6a4cb7daa93e4794f2a1e62. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03986-2026

Comparison records:

- chemical-entity-iupac-m03986: predicted singular-carrier__constitution-and-isotope-retained__separately-distinguishable__context-bounded-precision; source-derived singular-carrier__constitution-and-isotope-retained__separately-distinguishable__context-bounded-precision; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the current source-derived IUPAC entity record lacks singular distinguishability, constitution/isotope identity or contextual precision, or if any altered row is accepted.

Measurement receipt: `sha256:6a9b6ee1de808a674690a9a9a99fd5f30323d47069e59eefe8585b5947bdda7f`. Isolation certificate: `sha256:96a541e22fd08099bd11c4e7d615141525231b3347b80e2bd9c78caf14ccfa28`. Custody certificate: `sha256:5784d9539afd879ed716b07d55763fa6bbdb94bec65bfe0a47728f00beca35bd`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/M03986/json (sha256:6b0f6e183ef109d6f3483bcd1d70ca73a6d57e496f9ca32da23964bca652de8e)

Registered target identities:

- chemical-entity-iupac-m03986 from IUPAC-GOLD-BOOK-M03986-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every finite singular chemical carrier formed from admitted distinctions and any generated positive-finite extension of its constitution, isotope, state or contextual precision record.

Immutable identities. Source manifest `sha256:9aac86602d9aeafe5045ac6aa7c5d8f91e1e90faca9d244bb831aa65f3003d63`; derivation seal `sha256:9de339302a77f6504cf71743ec67ac5e3ec4abfadcc74da783418fd88e2a9f4d`; engine receipt `sha256:130c21788e898b4b6f0cb012bc72b4cb29d5aeda5b86e67257c6008c13318f66` at `receipts/engine/model_admitted/SFT-CHEM-MEAS-CHEMICAL-ENTITY-001-130c21788e898b4b.json`; empirical hash `sha256:141bf9f942bd9ffff44a8fe96e7a49f31931130413064ecaeba0e805491e64de`; measurement receipt `sha256:6a9b6ee1de808a674690a9a9a99fd5f30323d47069e59eefe8585b5947bdda7f`.

2. Chemical species and observation-equivalence class

Claim identity: `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`

Question and theorem. A chemical species is the least Fold quotient whose support is an ensemble of chemically identical entities sharing accessible state support at one declared experimental timescale.

> *ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`

Generated grammar. Generate the literal product of the registered species carrier, member identity, state support, timescale, solid support, record, provenance and extension choices.

Boundary: Every finite observation-equivalence class formed from admitted singular chemical entities under an exact declared experiment-timescale relation and any positive-finite successor extension.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary__molecular-or-solid-unit-support__held-member-and-class-record__source-bound-proof-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`ensemble-carrier`	`singular-only`	One member alone is not a chemical species class.
`member_identity`	`chemical-identity-equivalence`	`mixed-chemical-identities`	Mixed chemical identities do not form one species.
`support`	`shared-energy-support`	`energy-support-erased`	Erasing accessible-state support merges experimentally distinct species.
`timescale`	`experiment-timescale-boundary`	`timeless-equivalence`	Interconversion can change whether two forms are experimentally distinct.
`solid_support`	`molecular-or-solid-unit-support`	`isolated-molecule-only`	A species law limited to isolable molecules loses solid structural units.
`record`	`held-member-and-class-record`	`class-without-member-trace`	A quotient without member trace cannot be reconstructed.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	An unbound class can silently change its equivalence rule.
`extension`	`no-extra-rule`	`free-extra-equivalence`	A discretionary equivalence can force any desired species partition.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a7193b15135c4dbbd92429905bc94c36075dd22dcadc3a76f4198f85f360f0a4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e41ee4909562f76a1c4e66cce01a19dc0f86a6e9eb482555c81b63598cf9d228`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

`sha256:ecff63e0a1fe45c9d7427360c1ff52460290dafcb6e8b33ace20a61491aa215e`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4363a31941fcec233e36ff41c3cc5d6f7fdef7a860258c722f8ef96cae519289`.

Independent reconstruction. Implementation hash `sha256:0d4dfc16c46530663cc1eb405e4ca6b2ed0599e69792c046629a88fb4f98ff35`; independent certificate `sha256:cee56a524f22fcf251ad18689682ad1b6469ea97710e2683277a0350c05c517e`; external-validation hash `sha256:5b7fb1676de0d49c2e985ad3dc0f7c8361560ea6ce3bfa8586079c67bbf368b5`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CT01038-2026

Comparison records:

- chemical-species-iupac-ct01038: predicted ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary; source-derived ensemble-carrier__chemical-identity-equivalence__shared-energy-support__experiment-timescale-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC species record does not retain ensemble identity, common accessible support and experimental timescale, or if an altered row is accepted.

Measurement receipt: `sha256:cef190fad510ddb2239cffed977546900c68888b9357a3e6825289630d02dbed`. Isolation certificate: `sha256:6caa3ed29c98414d4e888a779c419cba0c579faa5725435c5bbb57bc1f89bc8d`. Custody certificate: `sha256:ed48c33c70767039f305f77ee378a8980046322871a4819c5dbd0c6e766b1620`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/CT01038/json (sha256:538dd1087bc057eec8b68fdb713987cc8e1f0ff7548b4d03192d67b3e328eafe)

Registered target identities:

- chemical-species-iupac-ct01038 from IUPAC-GOLD-BOOK-CT01038-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every finite observation-equivalence class formed from admitted singular chemical entities under an exact declared experiment-timescale relation and any positive-finite successor extension.

Immutable identities. Source manifest `sha256:1e419ad3d572f88eb13a42e738886b7e6dd56e382dfdfa5c52121021a9c087b0`; derivation seal `sha256:5ae3a8fef3137c61ff35cab5e3bb870ee623f3799971bf3c757fcb143efaf4c4`; engine receipt `sha256:5cde2482e85a9e7469dac34f4379a164979abac66a484aed5ebcb7b12639bd5b` at `receipts/engine/model_admitted/SFT-CHEM-MEAS-CHEMICAL-SPECIES-001-5cde2482e85a9e74.json`; empirical hash `sha256:be270d0c83872e53b8772bf007dbdc8b09790e24b3e44bfd3aceb575a740d43a`; measurement receipt `sha256:cef190fad510ddb2239cffed977546900c68888b9357a3e6825289630d02dbed`.

3. Chemical substance and composition identity

Claim identity: `SFT-CHEM-MEAS-SUBSTANCE-001`

Question and theorem. A chemical substance is the least source-bounded Fold carrier that retains constant composition, its constituent entity support and the complete registered property characterization.

> *constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`

Generated grammar. Generate the literal product of the registered substance composition, constituents, properties, source, phase, record, provenance and extension choices.

Boundary: Every finite chemical substance carrier formed from one generated constant-composition support of chemical entities plus any positive-finite extension of its source-bound property record.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity__phase-record-retained__complete-property-record__source-bound-proof-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`composition`	`constant-composition`	`variable-unrecorded-composition`	Unrecorded variable composition cannot identify one substance.
`constituents`	`constituent-entity-support`	`anonymous-matter`	Matter without retained constituent entities loses chemical identity.
`properties`	`property-characterization`	`composition-only-answer`	Composition alone can leave observation-equivalent candidates unresolved.
`source`	`source-bounded-substance-identity`	`source-free-substance-label`	A source-free label cannot distinguish sample and convention boundaries.
`phase`	`phase-record-retained`	`phase-silently-collapsed`	Silently collapsing phase can erase an observed substance distinction.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`record`	`complete-property-record`	`property-selection-only`	Selecting one favorable property is not a complete characterization record.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound composition and property claims cannot be audited.
`extension`	`no-extra-rule`	`free-extra-substance-rule`	A discretionary rule can split or merge substances after observation.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:ad2432d5ad2ab35e20e37628d4c7a34f5468b37f79470eecbe483f318bea4b2a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d3b75c1e2afca58bd1ace3b6c79c0db08b4790c8b5614a06f227e517436b2806`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b2afcb7ccd08f7de7686f06d0ee02426d69c0b76843fe38af1c13d393267e211`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:aa8c859776acb4b9507dadcdb36885aa443478164dce3f33ad26e21c9f415a45`.

Independent reconstruction. Implementation hash `sha256:c2a593300db78987c89c58b98d64c4f82f87cf79cbd340127f5f33ff93303b38`; independent certificate `sha256:bb690d7cc17309542fe5180ead6830f124ab9fa1db52c15739778d23046a5210`; external-validation hash `sha256:d069634773fcd9bfd04bf555fc350469bd45b504e44847c22f87bd6b61e21d5d`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01039-2026

Comparison records:

- chemical-substance-iupac-c01039: predicted constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity; source-derived constant-composition__constituent-entity-support__property-characterization__source-bounded-substance-identity; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC substance record lacks constant composition, constituent entities or property characterization, or if an altered row is accepted.

Measurement receipt: `sha256:c66662a1a574bf8b3358a0a4308abad3c8d232b612cb6d78ed2d4b1e944db4a6`. Isolation certificate: `sha256:21e6ccdcbe853b5176f0fdf5dbeeae2531bad16581a6135ab163ca1fc0659177`. Custody certificate: `sha256:4fa96f00cf7136b60fec8f4b6b73c71998dcdbf1e904c29bfb6fa42007f1e699`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01039/json (sha256:97af068278c31568a03402f8dcb3e5de2327833ab9ac8da78c1322e730997498)

Registered target identities:

- chemical-substance-iupac-c01039 from IUPAC-GOLD-BOOK-C01039-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every finite chemical substance carrier formed from one generated constant-composition support of chemical entities plus any positive-finite extension of its source-bound property record.

Immutable identities. Source manifest `sha256:a5d94367f08d367b2c628c0a7e9b4cbab4bfcace5aa5042433ef1e6255b6b87a`; derivation seal `sha256:e0134f8366c295cf62cd3a8cd00af0a215c3f9b4f0f2812017ea6b81006b128a`; engine receipt `sha256:264b52127a204f58e2bcce66e422f2f8b3c4c41d7bfd0f9d64bf6a2413454220` at `receipts/engine/model_admitted/SFT-CHEM-MEAS-SUBSTANCE-001-264b52127a204f58.json`; empirical hash `sha256:0b6623170beda97be63c42b4b4cbf6ee8d23e11811d8c13a8e61ce69c6acb73a`; measurement receipt `sha256:c66662a1a574bf8b3358a0a4308abad3c8d232b612cb6d78ed2d4b1e944db4a6`.

4. Amount-of-substance count and reference carrier

Claim identity: `SFT-CHEM-MEAS-AMOUNT-001`

Question and theorem. Amount of substance is the exact Fold measure carried by a positive-finite generated count of one specified elementary-entity kind, with the entity reference and count trace retained.

> *amount-carrier__count-measure__specified-entity-reference__generated-entity-support*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

Generated grammar. Generate the literal product of the registered amount carrier, entity reference, count, measure, support, record, unit and extension choices.

Boundary: Every positive-finite generated collection of one specified admitted chemical-entity kind, together with its exact member/reference trace and separately registered unit comparison.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `amount-carrier__specified-entity-reference__generated-cardinality__count-measure__generated-entity-support__member-and-reference-trace__separate-unit-comparison__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`amount-carrier`	`answer-only-amount`	An answer label does not retain the entities being counted.
`entity_reference`	`specified-entity-reference`	`unspecified-entities`	A count without a specified elementary-entity kind changes meaning under substitution.
`count`	`generated-cardinality`	`mass-proxy-count`	A mass proxy can vary with isotope and cannot establish entity cardinality.
`measure`	`count-measure`	`unrelated-count-and-quantity`	An unrelated quantity does not measure how many referenced entities occur.
`support`	`generated-entity-support`	`single-presupposed-entity-kind`	Presupposing one conventional entity kind excludes other generated elementary carriers.
`record`	`member-and-reference-trace`	`total-without-members`	A total without its entity reference cannot be independently reconstructed.
`unit`	`separate-unit-comparison`	`unit-scale-as-proof`	A conventional scale cannot select the amount law.
`extension`	`no-extra-rule`	`free-amount-factor`	An extra multiplier is a free answer-selecting parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:530a1ab514b800a0675a6f969439574a59117f367012e3ef8055b6cec3ebcc12`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:caafe4fbfd3413d2a4eeb2d4e1e893b744e6af78312e4a16209bccb295e91fef`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b4ba2f18586ba7dd2c311b1e9d1a1be3fdbd7f3980668604bd20becae6b4cd02`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:939ec2105ede1067163abcfe2685d946baa15236b910a9e6f925a5d31ef0d46f`.

Independent reconstruction. Implementation hash `sha256:7876fb1d2c9a700c5019ad446b918591401f0b872b74241cd017f0208c2f1a22`; independent certificate `sha256:64d8d926722782a15875c9eaf0a522aae7879683644af93d38b2082865374622`; external-validation hash `sha256:d2f7b45fd1f7add2b9d3206a58cca2999f9519aaecc41f2d3dafc8eddc94834a`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-A00297-2026

Comparison records:

- amount-of-substance-iupac-a00297: predicted amount-carrier__count-measure__specified-entity-reference__generated-entity-support; source-derived amount-carrier__count-measure__specified-entity-reference__generated-entity-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC amount record lacks number measure, specified entities or general elementary-entity support, or if an altered row is accepted.

Measurement receipt: `sha256:47cffd8c0f8e1ee747770cf58fb42a626bb62cba92c8b3431fdf548fd299db26`. Isolation certificate: `sha256:2a8dd4298c04a59a3f32910aefd475e794e2f6d440c16dee3ac4770d0aac7d31`. Custody certificate: `sha256:110d0ab7d20d5d6fb319b23299c4a88d5ca2f1eab0247796c4442bbe274af9e2`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/A00297/json (sha256:ddf59230caa12b510125795af6f624797e904a1724fe970b3c144af47167077e)

Registered target identities:

- amount-of-substance-iupac-a00297 from IUPAC-GOLD-BOOK-A00297-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite generated collection of one specified admitted chemical-entity kind, together with its exact member/reference trace and separately registered unit comparison.

Immutable identities. Source manifest `sha256:2583444b420c81a2c7b1d7ca3040e4d3801feafee2830b73ef1392be6cceece2`; derivation seal `sha256:e123e8f72a4f87a8188e392ea5f1e1526d778296998b7ba1cbec1429ffc0adc6`; engine receipt `sha256:c0dd23aedf3372bf9012540d9f80474ab7e34fa07b38664c9b0d3633f0cafa03` at `receipts/engine/model_admitted/SFT-CHEM-MEAS-AMOUNT-001-c0dd23aedf3372bf.json`; empirical hash `sha256:d4e5ab18313a931276ffb357cb2851acd039d3ccd496b054715835c6474e3b01`; measurement receipt `sha256:47cffd8c0f8e1ee747770cf58fb42a626bb62cba92c8b3431fdf548fd299db26`.

5. Chemical formula as exact composition encoding

Claim identity: `SFT-CHEM-MEAS-FORMULA-001`

Question and theorem. A molecular formula is the least decodable Fold word that retains a discrete molecule's constituent entity labels, exact positive multiplicities and any structure coordinate required by the declared identity boundary.

> *discrete-molecule-domain__formula-encoding__composition-mass-correspondence__structure-correspondence*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-MEAS-AMOUNT-001`

Generated grammar. Generate the literal product of the registered formula domain, carrier, constituents, multiplicity, mass, structure, decoding and extension choices.

Boundary: Every positive-finite discrete-molecule composition word generated from admitted entity labels, exact multiplicities and a held structural-order record.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `discrete-molecule-domain__formula-encoding__held-entity-symbols__exact-positive-multiplicity__composition-mass-correspondence__structure-correspondence__composition-decodable__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`domain`	`discrete-molecule-domain`	`unbounded-object-label`	A formula without a declared chemical carrier domain can encode unrelated objects.
`carrier`	`formula-encoding`	`name-only-string`	A name alone does not retain exact constituent multiplicity.
`constituents`	`held-entity-symbols`	`anonymous-symbols`	Anonymous symbols erase chemical-entity identity.
`multiplicity`	`exact-positive-multiplicity`	`unrecorded-repetition`	Unrecorded repetition cannot reconstruct composition or relative mass.
`mass`	`composition-mass-correspondence`	`mass-unrelated-to-composition`	A mass label not reconstructed from the constituent record is not a formula consequence.
`structure`	`structure-correspondence`	`order-erased`	Equal counts can represent distinct molecular structures.
`decode`	`composition-decodable`	`nondecodable-inscription`	An inscription that cannot return its constituent record is not an exact encoding.
`extension`	`no-extra-rule`	`free-formula-exception`	A discretionary exception can manufacture a desired formula.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3845fb5f1a0403ba69cc287efe987ff69f8c3e9f1acf7a480f9aea44f03fdadd`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2a15902eb8c0b79689cca8acb12b7c410d6e39222d9f24e4135db5e49740ba4f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6d635c7023bcc25b4967611c84b32596233cfd7d5cf35afa38b87999e78d31a0`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:aa7e7ab21abb28659b496319db0ef2d0e3706b69d707305a9813d1f5ef243140`.

Independent reconstruction. Implementation hash `sha256:16580eadc4660d4c0de70006350f24437e5dbafe427f71d0925f4cb62a62eb45`; independent certificate `sha256:cae7f79d06ad779f29123b54de65efd692475f4b1b10617a6ff41ce3049089a9`; external-validation hash `sha256:f5e253f8e69d5774f6dc6be627dbd002f626eabefc0d09c95519379d10b4246a`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03987-2026

Comparison records:

- molecular-formula-iupac-m03987: predicted discrete-molecule-domain__formula-encoding__composition-mass-correspondence__structure-correspondence; source-derived discrete-molecule-domain__formula-encoding__composition-mass-correspondence__structure-correspondence; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC molecular-formula record lacks its discrete-molecule domain, formula carrier, relative-mass or structure correspondence, or if an altered row is accepted.

Measurement receipt: `sha256:c002bc140bf7f658b9bd74f3cdccdb5940d9de7b0d41e8a7788042e3c9f5d49d`. Isolation certificate: `sha256:ec7b847ec7b441599c22fd7a69884695d8977e86eb2b7f7c1b53a531d74e402d`. Custody certificate: `sha256:28d7f9b6ad63d9a800288da13612553c37c13bb97e838098fd9ae08952a13086`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/M03987/json (sha256:b33959596338e0e2797247c379e8214ecd6a58abd3e929a4dfad36f42b081704)

Registered target identities:

- molecular-formula-iupac-m03987 from IUPAC-GOLD-BOOK-M03987-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite discrete-molecule composition word generated from admitted entity labels, exact multiplicities and a held structural-order record.

Immutable identities. Source manifest `sha256:a27c10a35cee4c3af142d2a480cb927c84c9cb972fe22356725e274ed4e439ae`; derivation seal `sha256:921e694baddb6f35e3da05f404a690f9dc9c5762d5a3ad062c76027127895889`; engine receipt `sha256:7cf0d7fa8f26712a49a5dab92f8019e46473a27e61cb1d94dc0ef8d98c861319` at `receipts/engine/model_admitted/SFT-CHEM-MEAS-FORMULA-001-7cf0d7fa8f26712a.json`; empirical hash `sha256:366a61563aaefc5fb17c55e4e15ecd7faa08268006f6e77b74c2d1410f3fff56`; measurement receipt `sha256:c002bc140bf7f658b9bd74f3cdccdb5940d9de7b0d41e8a7788042e3c9f5d49d`.

6. Reversible chemical nomenclature and identity boundary

Claim identity: `SFT-CHEM-MEAS-NOMENCLATURE-001`

Question and theorem. A systematic chemical name is a complete ordered Fold word assembled from registered name parts and optional count prefixes; exact reversibility is admitted only on the generated injective name-to-identity boundary.

> *complete-name-carrier__generated-name-parts__count-prefix-support*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-FORMULA-001`

Generated grammar. Generate the literal product of the registered name carrier, parts, prefix, order, formula binding, decoding, record and extension choices.

Boundary: Every positive-finite ordered chemical-name word assembled wholly from registered syllable labels and optional generated count prefixes, with an explicit injective-decoding boundary.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-name-carrier__generated-name-parts__count-prefix-support__held-name-order__formula-bound-name__injective-boundary-decoding__complete-name-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`complete-name-carrier`	`partial-name-fragment`	A fragment alone does not carry one complete registered chemical name.
`parts`	`generated-name-parts`	`unregistered-free-text`	Free text can import arbitrary answer-bearing distinctions.
`prefix`	`count-prefix-support`	`implicit-count`	An implicit multiplicity cannot be recovered from the name.
`order`	`held-name-order`	`unordered-syllable-bag`	Unordered parts can merge names with distinct composition.
`binding`	`formula-bound-name`	`name-detached-from-formula`	A detached name cannot preserve chemical identity.
`decode`	`injective-boundary-decoding`	`unconditional-reversibility`	Not every conventional name is uniquely decodable, so reversibility cannot be assumed.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`record`	`complete-name-trace`	`surface-name-only`	A surface name alone hides its assembly and decoding boundary.
`extension`	`no-extra-rule`	`free-naming-exception`	A discretionary naming exception can select any desired identity.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:363d36bc6f6549e4e0530b91da9cf081f8d8618f41d78e08b72d9307ca63e793`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5eed2098f5a9a626e7aa35bb5ca2da5d3c9ae4c9a5bf310a635b7bcfc898655e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:213e9b19aa90d67471f010fc8b8919b0c63247f7c71195d457576cf1b0da6e6e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:19185c053f8a90242fa96295f5bcd2329eb4dee3c69ece49d5d51993607c075a`.

Independent reconstruction. Implementation hash `sha256:35e60d416791137fa320e94b41a12eed0e18a1099c1b862537158c24d45f4fb9`; independent certificate `sha256:33534ecd55d79bd46e31fa8e85c98a78d1a0d48f961fa421774c1c17b1936a14`; external-validation hash `sha256:8735d81e6460be0dc718bd57b42f95898b7419725482134f4100650b95f5dc67`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S06236-2026

Comparison records:

- systematic-name-iupac-s06236: predicted complete-name-carrier__generated-name-parts__count-prefix-support; source-derived complete-name-carrier__generated-name-parts__count-prefix-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC systematic-name record lacks complete name composition, selected syllable parts or optional numerical prefixes, or if reversibility is claimed outside the registered injective boundary.

Measurement receipt: `sha256:d712d539a814c6a82ea1bdb5448a4f07f9bd1c481211d208234b6d83f68c07e4`. Isolation certificate: `sha256:20ebbcdf200ebc000e4026c0676d8fff1cfd1bfe8ccde37bda93fccfc5b884af`. Custody certificate: `sha256:c5a80bfffa574ea176c61f65735cbb6b9d06cbc34d56447c7423d725ed1d29d5`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/S06236/json (sha256:a3163fe9e9694283c7280d87679fb49286dd44631d908c19fcc164ce8f2984ac)

Registered target identities:

- systematic-name-iupac-s06236 from IUPAC-GOLD-BOOK-S06236-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite ordered chemical-name word assembled wholly from registered syllable labels and optional generated count prefixes, with an explicit injective-decoding boundary.

Immutable identities. Source manifest `sha256:80c5d76427e08fd47b0cf27d682c64de46a3874b5b1e7e16dab7ce6a3c5da173`; derivation seal `sha256:02607a2faf000f2fb512cdc4dc3e555a32eb5342469e3ac7cafcba1dc54d4c30`; engine receipt `sha256:33b780889dc8278c1dc64c63d16eb372d3222c6955f9b6c3f9077929411a4e8c` at `receipts/engine/model_admitted/SFT-CHEM-MEAS-NOMENCLATURE-001-33b780889dc8278c.json`; empirical hash `sha256:7ab983fba64884599030fec4cbdda2fd8069b531d4e84be88f32238500a8be98`; measurement receipt `sha256:d712d539a814c6a82ea1bdb5448a4f07f9bd1c481211d208234b6d83f68c07e4`.

7. Chemical measurement uncertainty and retained alternatives

Claim identity: `SFT-CHEM-MEAS-UNCERTAINTY-001`

Question and theorem. Chemical target uncertainty is a source-bound Fold record of unresolved measurement alternatives with a preregistered upper limit fixed by intended use and attached to the exact result identity.

> *uncertainty-carrier__upper-bound-support__use-bounded-resolution__result-record-boundary*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-AMOUNT-001`

Generated grammar. Generate the literal product of the registered uncertainty carrier, alternatives, bound, use, result, source, record and extension choices.

Boundary: Every source-bounded finite chemical measurement record whose unresolved alternatives, exact upper bound, intended use and result identity are retained without entering the derivation as proof values.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `uncertainty-carrier__retained-alternative-class__upper-bound-support__use-bounded-resolution__result-record-boundary__source-bound-uncertainty__complete-uncertainty-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`uncertainty-carrier`	`single-answer-without-resolution`	A lone answer erases the unresolved observation class.
`alternatives`	`retained-alternative-class`	`discarded-alternatives`	Discarding compatible alternatives falsely turns resolution into exact identity.
`bound`	`upper-bound-support`	`unbounded-vagueness`	Unbounded uncertainty cannot falsify a measured correspondence.
`use`	`use-bounded-resolution`	`context-free-resolution`	Required resolution depends on the declared purpose of the result.
`result`	`result-record-boundary`	`bound-detached-from-result`	A bound detached from its result cannot be applied or audited.
`source`	`source-bound-uncertainty`	`source-free-bound`	A source-free bound can be changed after comparison.
`record`	`complete-uncertainty-record`	`favourable-row-only`	Selected rows cannot test uncertainty transport.
`extension`	`no-extra-rule`	`free-uncertainty-margin`	An adjustable margin can force agreement with any target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f33978481bee18a785e521a82b83da05e66ad23051bf34af7e6ba8e6222893ac`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2f46e0952e59ff9947a89d6942dd04a2175a7182e033d5c137c0d7d097b50848`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:259fc17f95d9dab542b48fca5a0d0362fabb280b7315b7130540653701ef78d6`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:de5d0150aeecfffacf4ac229cb1552a22b70e6afa6c0f27ecb72cdd228a34927`.

Independent reconstruction. Implementation hash `sha256:8b0ba2478d537cb6f9383f894be69dab1485aecb826ef31b9205e556bdc6fbaf`; independent certificate `sha256:e9f1432b80fd5ed55c607e74ee387292fbb941e02ed52f76f923b53cca2f70dd`; external-validation hash `sha256:21608d02fd9a335c3693b3f297002da11da3af14faf5c0376041274c73179df5`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08133-2026

Comparison records:

- target-uncertainty-iupac-08133: predicted uncertainty-carrier__upper-bound-support__use-bounded-resolution__result-record-boundary; source-derived uncertainty-carrier__upper-bound-support__use-bounded-resolution__result-record-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC target-uncertainty record lacks an upper limit, intended-use boundary or measurement-result attachment, or if an altered row is accepted.

Measurement receipt: `sha256:146e60e995906ada26b57276843c2fe43560331cda2643c439a385911896cda3`.
Isolation certificate: `sha256:549188b9c54e2df8fb1c36f07e85fdfcba7c45e10061c391c5ed4e056282630a`.

Custody certificate: `sha256:5592e7a03248c5b1b239fead57d09a5ae99899b692cf9f79db6ebaf29555aa28`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/08133/json (sha256:014b6b034695f2b25c07cd4e5474b856aeecf8907e4022595cc93df2f6674dc1)

Registered target identities:

- target-uncertainty-iupac-08133 from IUPAC-GOLD-BOOK-08133-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every source-bounded finite chemical measurement record whose unresolved alternatives, exact upper bound, intended use and result identity are retained without entering the derivation as proof values.

Immutable identities. Source manifest `sha256:28c377ca52afd1ce1bf78e732ad37fa441f4dc6ad80d4c31eac50f86621b42b3`; derivation seal `sha256:158d9174013868a2a0218a64a4806c9891e067cecd1d40fc913c76878ff743c6`; engine receipt `sha256:32f136b934cb1114201bb7126bb4c83116018780973264d902def43937d23cbe` at `receipts/engine/model_admitted/SFT-CHEM-MEAS-UNCERTAINTY-001-32f136b934cb1114.json`; empirical hash `sha256:0637803ce25d0b17f78078fafe60eb873203497641208a42c08a71aabebdd6f1`; measurement receipt `sha256:146e60e995906ada26b57276843c2fe43560331cda2643c439a385911896cda3`.

8. Chemical reference traceability and complete record

Claim identity: `SFT-CHEM-MEAS-TRACEABILITY-001`

Question and theorem. Chemical traceability is the complete Fold predecessor chain binding a property value to an identified reference material while retaining uncertainty, material identity and its certificate record at every link.

> *reference-material-property__uncertainty-retained__traceability-retained__certificate-bound-record*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`

- SFT-PHYS-MEAS-UNCERTAINTY-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-MEAS-REFERENCE-REALIZATION-001
- SFT-PHYS-MEAS-CALIBRATION-001
- SFT-CHEM-MEAS-UNCERTAINTY-001

Generated grammar. Generate the literal product of the registered property, reference, uncertainty, trace, certificate, identity, record and extension choices.

Boundary: Every positive-finite certified chemical-property chain whose reference material, property identity, uncertainty, predecessor links and certificate record are retained to an admitted reference realization.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `reference-material-property__identified-reference-realization__uncertainty-retained__traceability-retained__certificate-bound-record__material-identity-held__complete-chain-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
property	reference-material-property	value-without-property	A value without a property identity cannot be traced.
reference	identified-reference-realization	anonymous-reference	An anonymous comparison object cannot anchor a trace chain.
uncertainty	uncertainty-retained	uncertainty-erased	Erasing uncertainty overstates what each comparison preserves.
trace	traceability-retained	broken-predecessor-chain	A missing predecessor prevents reconstruction to the reference.
certificate	certificate-bound-record	uncertified-result-label	A result label alone does not bind property, reference and trace.
identity	material-identity-held	substitutable-sample	Silent sample substitution breaks traceability while preserving a surface value.
record	complete-chain-record	terminal-value-only	A terminal value cannot reproduce the chain.
extension	no-extra-rule	free-trace-link	An unregistered link can hide a discontinuity.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:97e310c97ca4153e8085aa29d2b588832b6425e4220e59493ed9a11b6d0a9c48`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0a72df6d765e4516c4b2e09d9092b1bcef4c9665423eb32a35aa1eec79899488`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b1151e9de01fe193c6b7934ff2bca5000776b28dc01480a985bb08dc9de5e007`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6743deb0003e20c5af2dac5d032ad5d67023c98dd7446991ec75a06c01abee11`.

Independent reconstruction. Implementation hash `sha256:87aa49285efc58f739ef9b98990ae232ca5b846a3d052775904c353924df74bd`; independent certificate `sha256:1de8c820992757a3ec3499dda3159ad7f6e874af66e63b62d67c21a873a17a8c`; external-validation hash `sha256:5349788bb8f683ad56d0ec122803afd7568f82ded454f8dbc49a2605502ba541`. The independent

program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08013-2026

Comparison records:

- certified-property-iupac-08013: predicted reference-material-property__uncertainty-retained__traceability-retained__certificate-bound-record; source-derived reference-material-property__uncertainty-retained__traceability-retained__certificate-bound-record; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC certified-property record lacks reference material, associated uncertainty, traceability or certificate binding, or if an altered row is accepted.

Measurement receipt: `sha256:bbaf754c42c20b65ad35ff2c5245f8058ec184990f96e3ae024e462c39f8f1dc`. Isolation certificate: `sha256:be1c1fdcc2d815663fdaac62ebb81e2ccc704729c7b5406fafae95777dfaec0e`. Custody certificate: `sha256:d1fcabab015c47623821e98f2026ef93d20f4cd20d464b5939139e5a30b5fce6`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/08013/json (sha256:34656c42ec7d741c00a0006b462487115d51dcd3d918d2fa64982ce100e24868)

Registered target identities:

- certified-property-iupac-08013 from IUPAC-GOLD-BOOK-08013-2026

Meaning. Chemistry begins with exact identity and measurement boundaries: entity, species, substance, amount, formula, nomenclature, uncertainty and traceability. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, conventional chemistry model or target content may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- external target content is absent from this specification and opens only through post-seal custody
- Every positive-finite certified chemical-property chain whose reference material, property identity, uncertainty, predecessor links and certificate record are retained to an admitted reference realization.

Immutable identities. Source manifest `sha256:ad5f1df58a9beab76e6e3034a81a8e2e8fd65d318f6467bddc09d6f0ef21918b`; derivation seal `sha256:23fcc09debac7c710bfa5bbed6d301d1229db7fc994fc4202eb4755cc2ff8a86`; engine receipt `sha256:7482443dfeb4205a7bcd2a89f0ac89014486ab7f8d960a0e0821f917247108fe` at `receipts/engine/model_admitted/SFT-CHEM-MEAS-TRACEABILITY-001-7482443dfeb4205a.json`; empirical hash `sha256:1eff947400af8d1bc6293865d4476377c46cc6df48de302129f7e37dbca253c6`; measurement receipt `sha256:bbaf754c42c20b65ad35ff2c5245f8058ec184990f96e3ae024e462c39f8f1dc`.

10. Elements Periodicity

Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries.

9. Chemical element identity

Claim identity: `SFT-CHEM-ELEM-ELEMENT-001`

Question and theorem. A chemical element is the complete Fold class of generated atoms sharing one held nuclear proton count; isotope mass refinements preserve that identity, while pure-substance use remains a separate correspondence.

> *atom-species-carrier__common-proton-count__nuclear-identity__substance-carrier-correspondence*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001`
- `SFT-PHYS-NUCLEAR-LEVELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

Generated grammar. Generate the literal product of the registered element carrier, identity, location, membership, isotope, substance, record and extension choices.

Boundary: Every positive-finite atom-species class generated from one held nuclear proton-count identity, including all mass-number refinements and its separately recorded pure-substance correspondence.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `atom-species-carrier__common-proton-count__nuclear-identity__all-generated-atoms-with-identity__mass-refinement-permitted__substance-carrier-correspondence__complete-nuclear-membership-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`atom-species-carrier`	`substance-name-only`	A substance name does not define the atom class.
`identity`	`common-proton-count`	`mass-selected-identity`	Mass can vary between isotopes of one element.
`location`	`nuclear-identity`	`protons-without-nuclear-carrier`	An unlocated count does not preserve atomic constitution.
`membership`	`all-generated-atoms-with-identity`	`sampled-atom-list`	A sample cannot close the complete generated class.
`isotope`	`mass-refinement-permitted`	`one-mass-number-only`	Restricting an element to one mass number erases its isotopes.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`substance`	`substance-carrier-correspondence`	`species-substance-conflation`	An atom species and a pure substance are related but not identical carriers.
`record`	`complete-nuclear-membership-trace`	`element-label-only`	A label alone cannot reproduce class membership.
`extension`	`no-extra-rule`	`free-element-exception`	A discretionary exception can split or merge element classes.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:76dd7261cbd3b67042302e0d1db175f36c10d018ae297517613179d05638a9d5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:13b7765dcc0f65b3252315344aad5c8a70c3fc576d445f435ec090dcd81e8e61`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:09f6d83fe72cd5c693cec7b72ad03d414521081d8b0934b0289a7580b0df232f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f00796ad6f37a439619cea9a44469afcd69aa1d406af0949ab0b2172048d418a`.

Independent reconstruction. Implementation hash `sha256:099700048127608e9d9b7d95a855a9c6cf5f79d2e8a37ebc5198fadbba1ca6b5`; independent certificate `sha256:be152c894873bde8bc0c4fa13caed3619dcde05f4a97b6d1351f0fcbf5bc3027`; external-validation hash `sha256:a33fb7f0513f61005f561158de442bccb71f5d3a62ea43fe1bd3bf86ee76b87a`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01022-2026

Comparison records:

- chemical-element-iupac-c01022: predicted atom-species-carrier__common-proton-count__nuclear-identity__substance-carrier-correspondence; source-derived atom-species-carrier__common-proton-count__nuclear-identity__substance-carrier-correspondence; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC element record lacks an atom-species carrier, common proton count, nuclear identity or pure-substance correspondence, or if an altered row is accepted.

Measurement receipt: `sha256:f79af5351c60739735dc774cc20043ed452b76ee03ea6088cc064d01fac53829`. Isolation certificate: `sha256:26eb6acd9dbf27a6b226638bb7d6b485f8a05e4cb6c6337002c95c4338461d43`. Custody certificate: `sha256:f030e4eea55cdd5a6703a715589e085ad181ff0a9ea72af42c408e7df840fc90`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01022/json (sha256:6d75c5b2d5299402b7f35243e248cb60d4972553d3bb622490236e7b889d9575)

Registered target identities:

- chemical-element-iupac-c01022 from IUPAC-GOLD-BOOK-C01022-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every positive-finite atom-species class generated from one held nuclear proton-count identity, including all mass-number refinements and its separately recorded pure-substance correspondence.

Immutable identities. Source manifest `sha256:b9d75c92672c039d7f4a5deb483bcdfe8ee1a6f61a2c8a0fe13853de6f027e83`; derivation seal `sha256:1aa52affee6653414842ec851ef6a5e55473108f1638ef8939184c330e1bac21`; engine receipt `sha256:ea7aa6a543e4710c2901ad39a6d67445bebda012e735322cff802f05720434d7` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-ELEMENT-001-ea7aa6a543e4710c.json`; empirical hash `sha256:6977b7243cf2ec9f6c028016233c7ba0fbb3bba0bbb789a00450822f9a1c30af`; measurement receipt `sha256:f79af5351c60739735dc774cc20043ed452b76ee03ea6088cc064d01fac53829`.

10. Atomic-number ordering and element distinction

Claim identity: `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`

Question and theorem. Atomic number is the exact generated positive-finite count of protons in an atom's nucleus and therefore the unique order coordinate of its chemical-element class.

> *exact-proton-count__nuclear-count-carrier*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001`
- `SFT-PHYS-NUCLEAR-LEVELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-ELEM-ELEMENT-001`

Generated grammar. Generate the literal product of the registered atomic-number carrier, location, order, element map, isotope, charge, record and extension choices.

Boundary: Every generated positive-finite nuclear proton collection and its one-to-one order coordinate over admitted chemical-element classes.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `exact-proton-count__nuclear-count-carrier__positive-count-order__one-count-one-element-class__isotope-invariant__charge-state-invariant__count-nucleus-element-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`exact-proton-count`	`symbol-position-only`	A table position cannot define the nuclear quantity.
`location`	`nuclear-count-carrier`	`whole-atom-particle-count`	Counting every atomic particle changes across isotopes and charge states.
`order`	`positive-count-order`	`unordered-label`	An unordered label cannot supply periodic succession.
`element_map`	`one-count-one-element-class`	`many-elements-one-count`	Two element classes with one proton count violate their identity condition.
`isotope`	`isotope-invariant`	`mass-dependent-number`	Mass-number variation must not alter element order.
`charge`	`charge-state-invariant`	`electron-count-dependent`	Ion formation changes electrons without changing the element.
`record`	`count-nucleus-element-trace`	`number-without-nucleus`	A bare number loses the physical carrier and class mapping.
`extension`	`no-extra-rule`	`free-order-offset`	An offset can relabel the periodic order arbitrarily.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4ddd54cd7947aa0e3b2aa3d03493a8c34c888edcfc7dea36e9282471cc6c308c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ed80251591ba927846d8a1247ed87eb592801a08102e5dbde5cac482563e7ef7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:be2dd24a591be28884b941432fdafcdb712120c27fd0008b55795cad15fab594`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:db5a33c3d4c0529fb072f64aaf74b52b89281189de1ad94dc76b985ad0466b81`.

Independent reconstruction. Implementation hash `sha256:a562cb4295bee04704e6d6f298f9400a671a00306a4c3e99f78b3192e9dec8c7`; independent certificate `sha256:50a96aa5a9fc736f9b208bcbaa5ad98be89a75d160feb645174f208c0c28123d`; external-validation hash `sha256:6efd610e28967cfbb5efa2627358944c4d44a76332c5f1b1e98230c81ccf9abe`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-A00499-2026

Comparison records:

- atomic-number-iupac-a00499: predicted exact-proton-count__nuclear-count-carrier; source-derived exact-proton-count__nuclear-count-carrier; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC atomic-number record is not the proton count in the atomic nucleus, or if an altered row is accepted.

Measurement receipt: `sha256:f59895f2b153d17da12ab9b09f875868e0451fc789fb53d6a979dc8bf3ad9eaa`. Isolation certificate: `sha256:b9634abd89c1680c95f58fe81bd47e7fb90589acb8fdf9f0abed916127fa01fd`. Custody certificate: `sha256:dff8feac17003dea3a7af44fa7a350a6e945797a2d2d6a2dd92179d4953c3605`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/A00499/json (sha256:6ba53740329bcdde38d8a8734282fb9414a1c7e62fad09743d0890890ff3e127)

Registered target identities:

- atomic-number-iupac-a00499 from IUPAC-GOLD-BOOK-A00499-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every generated positive-finite nuclear proton collection and its one-to-one order coordinate over admitted chemical-element classes.

Immutable identities. Source manifest `sha256:94777a429f8e0c22d9c3e5647936c8340570eca54e9f73a2c1d4e45cb3ea77f4`; derivation seal `sha256:7cd1aa9a886c6f4665acd5de9967d9c13b46eb3079c9f2e78e7e16e251904b22`; engine receipt `sha256:d0fccb17f5ec07aefad94d4b1118dc69e8e90be2ed1b2c9f742215b0fc9e50c6` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-ATOMIC-NUMBER-001-d0fccb17f5ec07ae.json`; empirical hash `sha256:4a059aea0ff9e110133f3beafc0986860757123ac9a632aece87e9dea0a6b421`; measurement receipt `sha256:f59895f2b153d17da12ab9b09f875868e0451fc789fb53d6a979dc8bf3ad9eaa`.

11. Isotope identity within an element

Claim identity: `SFT-CHEM-ELEM-ISOTOPE-001`

Question and theorem. Isotopes are the generated Fold family of nuclide carriers sharing one atomic number while retaining different exact mass-number coordinates.

> *nuclide-carriers__element-identity-retained__mass-number-distinction*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`

- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001`
- `SFT-PHYS-NUCLEAR-LEVELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`

Generated grammar. Generate the literal product of the registered isotope carrier, element, mass, complementary nuclear count, chemistry, observation, record and extension choices.

Boundary: Every positive-finite family of admitted nuclides sharing one atomic-number identity while retaining each distinct generated mass-number coordinate and complete nuclear record.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `nuclide-carriers__element-identity-retained__mass-number-distinction__complementary-nuclear-count-held__common-element-chemistry-boundary__resolution-bounded-distinction__complete-nuclide-family-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`nuclide-carriers`	`atom-label-without-nucleus`	An atom label alone does not establish nuclide identity.
`element`	`element-identity-retained`	`mixed-atomic-numbers`	Different proton counts are different elements, not isotopes.
`mass`	`mass-number-distinction`	`mass-number-erased`	Erasing mass number merges distinct nuclides.
`neutron`	`complementary-nuclear-count-held`	`unrecorded-nuclear-difference`	The source of mass-number distinction cannot remain anonymous.
`chemistry`	`common-element-chemistry-boundary`	`different-element-chemistry`	Changing element identity does not define an isotope relation.
`observation`	`resolution-bounded-distinction`	`always-indistinguishable`	Isotopic distinctions are resolvable in registered measurements.
`record`	`complete-nuclide-family-trace`	`isotope-name-only`	A name cannot reconstruct proton and mass coordinates.
`extension`	`no-extra-rule`	`free-isotope-equivalence`	An arbitrary equivalence can merge different elements or erase mass distinctions.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8b484361d78650d9a8282675607175a29a3f07500429fe30cdbc8be83fb806e8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d12deb5f406b68526427218796918b712da246387d65d777d60f2500d0d4f4a0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3b803889d00f08f745238ff5c3ba28066c73e87c0d9722104dc3c1b7ef514893`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:aca608666ff8ba3aa7e88511159717815c73d806611043bb4da1be7d86e7cc4b`.

Independent reconstruction. Implementation hash `sha256:d5f0ab0003ff6099b3b93cd20d190c1ff1c2b231255d493a8e3275e5c580c8d8`; independent certificate `sha256:79130b42b41148a1bc0c2d74d50493a322b3eba6375c8a84a6f992c33a84db56`; external-validation hash `sha256:b9c97d6ebf639546d9cd99cc7a53152fec627e8b5c36f65a93654d4e6684133c`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03331-2026

Comparison records:

- isotopes-iupac-i03331: predicted nuclide-carriers__element-identity-retained__mass-number-distinction; source-derived nuclide-carriers__element-identity-retained__mass-number-distinction; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC isotope record does not retain nuclides, common atomic number and distinct mass numbers, or if an altered row is accepted.

Measurement receipt: `sha256:92be2b66a7789623a970c015af9c8534dd7e798570078bee0e560045632cd1b5`. Isolation certificate: `sha256:2dfcbbd8c82ae8a225a018b4947bcd82c6cbdc21ec302d88b7fa8a580f2fb222`. Custody certificate: `sha256:b9649c33739d67c8af62b4c078d4a74c7aa85d22a1d618b223988a5ed0d70f09`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/I03331/json (sha256:7b5ca7fe293902804ca76194a6ac35a57a35bfa489245c93979eb81370a6635c)

Registered target identities:

- isotopes-iupac-i03331 from IUPAC-GOLD-BOOK-I03331-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every positive-finite family of admitted nuclides sharing one atomic-number identity while retaining each distinct generated mass-number coordinate and complete nuclear record.

Immutable identities. Source manifest `sha256:3313ef2e5af4b187fd2ef43be40fb53c248fbfd67283100d91b1cf219f569470`; derivation seal `sha256:d6dd20174811be1754c06fe8d2ad4de01aec548b5656a840206850b8e3cb051b`; engine receipt `sha256:814344a440e051068f0e54e91276fddd45b2d388ea1fe438f630e3aae0a4b28f` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-ISOTOPE-001-814344a440e05106.json`; empirical hash `sha256:8a4eede1026a036dc1160429f93a3ad2c26572173e864dc6c9c83299ab551304`; measurement receipt `sha256:92be2b66a7789623a970c015af9c8534dd7e798570078bee0e560045632cd1b5`.

12. Atomic-weight record and isotopic composition boundary

Claim identity: `SFT-CHEM-ELEM-ATOMIC-WEIGHT-001`

Question and theorem. Relative atomic mass is the exact source-bounded Fold ratio of a finite atom population's retained average mass to a separately registered unified atomic-mass reference, with isotope support and measurement uncertainty preserved.

> *exact-ratio-carrier__average-atomic-mass__reference-unit-comparison*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-QUANTITY-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-MATTER-PARTICLE-SPECTRUM-001`
- `SFT-PHYS-NUCLEAR-LEVELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-MEAS-AMOUNT-001`
- `SFT-CHEM-MEAS-TRACEABILITY-001`
- `SFT-CHEM-ELEM-ISOTOPE-001`

Generated grammar. Generate the literal product of the registered atomic-weight carrier, population, isotope support, mass pairs, reference, source, uncertainty and extension choices.

Boundary: Every finite source-bounded atom population with retained isotope membership, exact positive rational abundance parts, atomic-mass records and a separate unified-reference-unit comparison.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `exact-ratio-carrier__average-atomic-mass__isotope-support-retained__isotope-mass-pairs__reference-unit-comparison__source-bounded-population__measurement-interval-retained__no-extra-rule`. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	exact-ratio-carrier	decimal-answer-only	A decimal answer hides population, isotope and reference provenance.
population	average-atomic-mass	single-atom-mass	Atomic weight concerns a population average rather than an arbitrary single atom.
isotopes	isotope-support-retained	isotope-mixture-erased	Erasing isotope support loses the source of population variation.
mass	isotope-mass-pairs	unbound-mass-labels	Unbound masses cannot be paired with their isotope parts.
reference	reference-unit-comparison	unit-as-derived-law	A conventional unit scale cannot select the structural ratio.
source	source-bounded-population	universal-fixed-composition	Natural isotope composition can vary by source.
uncertainty	measurement-interval-retained	exact-decimal-pretense	A reported decimal must not erase its measurement interval.
extension	no-extra-rule	free-weight-adjustment	An adjustable correction can force a measured target.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:90d954fc2cd84571376d947dca60f0e84ee9fb1325ede38bfd707aa41ef2ea0f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2ff64c465529ab6a5c91e2ea3023965c9e499826385218803b3983644b4974ce.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:c341944a48f79104fa347f17ecb18ceeeddad0388d4e720e6ad69745b6e70ee8.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:fc1f589db5a76a8aadd4fd38cd7cce85db80eced58a1aa027d11df0c552742eb.

Independent reconstruction. Implementation hash sha256:039223c857c830d8414ee0110d8b2216a5f2c4f22714d0c2242d71e34dbb29e2; independent certificate sha256:a0c1484aa9c368048506eda8a5164f4678cc44bcde54be62e38a94bbcb126bb9; external-validation hash sha256:4969b91031a41ce1744454b4f52e0287a9002aed4ddf700ec0a25694f1a7884a. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-R05258-2026

Comparison records:

- relative-atomic-mass-iupac-r05258: predicted exact-ratio-carrier__average-atomic-mass__reference-unit-comparison; source-derived exact-ratio-carrier__average-atomic-mass__reference-unit-comparison; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC relative-atomic-mass record lacks a ratio, average atomic mass or unified atomic-mass reference, or if an altered row is accepted.

Measurement receipt: sha256:21313a418ae7996bdf96d2506cc95bf578404e55d8ba7b5fd7e33972d4a37041. Isolation certificate: sha256:7ed3d2c02eb1da57572bb9cd3be50d176e74daf990e644f474862c07416d43fd.

Custody certificate: `sha256:4ddd880e74630e1a4f4488a4d90669b6fefff6fc9230481d81b8b22be693659e`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/R05258/json (sha256:9a08b1dfe8699f2be546b1a0fcb9b359d49c78419099d853603c6c750a04f17a)

Registered target identities:

- relative-atomic-mass-iupac-r05258 from IUPAC-GOLD-BOOK-R05258-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC definition, periodic-table arrangement, measured element value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite source-bounded atom population with retained isotope membership, exact positive rational abundance parts, atomic-mass records and a separate unified-reference-unit comparison.

Immutable identities. Source manifest `sha256:5a744926c46b43835db76bd0e8deabed8ca4561e5c72675601f4ace437897719`; derivation seal `sha256:114a1de5d3715ee6d73bc7400aa8cb90e6e928a6bb500dabf706da100c64d279`; engine receipt `sha256:611aeb63a9ed5877d30862065a427b279c52aa8cd1f3b94bc81920643951fe4a` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-ATOMIC-WEIGHT-001-611aeb63a9ed5877.json`; empirical hash `sha256:3b2b46429f7461bc13d9eff3a61aaac08e2eed4fc365830d70e8a5e8e7e601db`; measurement receipt `sha256:21313a418ae7996bdf96d2506cc95bf578404e55d8ba7b5fd7e33972d4a37041`.

13. Periodic ordering by retained nuclear identity

Claim identity: `SFT-CHEM-ELEM-PERIODIC-ORDER-001`

Question and theorem. Chemical elements possess one invariant exact total order: the generated positive proton-count order, with the complete element class retained at each coordinate.

> *complete-element-class-support__atomic-number-coordinate__positive-count-total-order__element-identity-retained*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`

- SFT-PHYS-MEAS-VALUE-RECORD-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001

Generated grammar. Generate the literal product of the registered periodic-order carrier, coordinate, order, identity, succession, observation, record and extension choices.

Boundary: Every positive-finite chemical-element class carrying its admitted atomic-number identity, ordered only by generated proton-count succession and retaining the complete element record at each coordinate.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-element-class-support__atomic-number-coordinate__positive-count-total-order__element-identity-retained__one-proton-successor__post-seal-known-table-correspondence__number-element-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	complete-element-class-support	names-without-element-records	Names alone do not preserve the generated element classes.
coordinate	atomic-number-coordinate	layout-selected-coordinate	A drawn location can be changed without changing nuclear identity.
order	positive-count-total-order	unordered-element-set	An unordered set cannot express successor structure.
identity	element-identity-retained	order-erases-isotopes	Ordering must not discard refinements of an element record.
succession	one-proton-successor	free-table-rearrangement	A discretionary rearrangement can select any pattern.
observation	post-seal-known-table-correspondence	known-table-as-proof-premise	The observed table cannot create the order law.
record	number-element-source-trace	atomic-number-only-list	A number list loses symbols, identities and source boundary.
extension	no-extra-rule	free-order-exception	An exception can silently permute successors.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fb79be00e6697a390e2401a3f35d4a36e790dd0e25f17ce2d82cdf109459fdc8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2468f6a821c5f46777ffec9979273168f4fb4a44c9a3b07a459d3885cd591fe4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fcd4a72f1f0a76bcc6ab20b90e22587f439fd13e32a9a32b98ab2c2a256b2cf2`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8f2e3cc1418f0d11a932467a232d2806e7e2ad150c76df6cbf655f22d58a3a91`.

Independent reconstruction. Implementation hash `sha256:1ab5b9c59e4bf934778ced8adb3532d3129d525e09be9ba7215d8b53249aee1d`; independent certificate `sha256:e3e8d00b305566a1ce3a029a1b2fe23d581ff06b4c904848b79cadf71121781e`; external-validation hash `sha256:f4f6cbb70c7fd3237d454f6a746fed076402174d152c4fea167c0db37d1888f3`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- periodic-order-iupac-2022: predicted complete-known-element-support__atomic-number-coordinate-order; source-derived complete-known-element-support__atomic-number-coordinate-order; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official source lacks the complete registered element support or atomic-number coordinate, or if a changed row is accepted.

Measurement receipt: `sha256:d6bf48b62ba038232ce2da4f0d9f6a776405f68b04b8301f248c1548ef4dc775`. Isolation certificate: `sha256:e98f2de32003bb5ea5bf83bdc912a7f748ac5a5ab36346c62098915ea1e67efa`. Custody certificate: `sha256:a8c2761729e8122a72c1aea1ee0e98da0f0a8d830a7aa76e0f89507c316fa22c`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- periodic-order-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every positive-finite chemical-element class carrying its admitted atomic-number identity, ordered only by generated proton-count succession and retaining the complete element record at each coordinate.

Immutable identities. Source manifest `sha256:9b2389237f9ad69dd83892791369de88baee13f1c90a5209de80dac6286d0332`; derivation seal `sha256:6885e8766a0e478de2a057c2c68db53263289d2a942ba9e293b0539c7796c37c`; engine receipt `sha256:2c3e161729975fa71a7baffe345e886c873fdec44e54e05547ba1ce105a10b4c` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-PERIODIC-ORDER-001-2c3e161729975fa7.json`; empirical hash `sha256:43e7886c804f8cd2df62f76d209094f5861f7ed55856611bc38f3b33a8da9dc3`; measurement receipt `sha256:d6bf48b62ba038232ce2da4f0d9f6a776405f68b04b8301f248c1548ef4dc775`.

14. Periodic recurrence of outer chemical organization

Claim identity: `SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001`

Question and theorem. Periodicity is the return, along atomic-number succession, of an earlier complete outer chemical observation class generated by finite bound-state recurrence and exclusion-preserving occupation.

> *boundary-closed-outer-support__atomic-number-successor-progression__complete-chemical-trace-equivalence__return-of-observation-class*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`

Generated grammar. Generate the literal product of the registered recurrence carrier, progression, equivalence, closure, occupation, periodicity, record and extension choices.

Boundary: Every generated positive-finite atomic-number successor family whose boundary-closed outer observation support is quotiented only by complete chemical transition equivalence.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `boundary-closed-outer-support__atomic-number-successor-progression__complete-chemical-trace-equivalence__finite-bound-state-recurrence__exclusion-preserving-occupation__return-of-observation-class__support-transition-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`boundary-closed-outer-support`	`property-name-series`	A list of property names contains no physical recurrence carrier.
`progression`	`atomic-number-successor-progression`	`mass-selected-progression`	Isotope mass does not determine element succession.
`equivalence`	`complete-chemical-trace-equivalence`	`visual-column-equivalence`	A drawing cannot establish common chemical behavior.
`closure`	`finite-bound-state-recurrence`	`unbounded-orbital-continuum`	An ungenerated continuum violates the finite exact domain.
`occupation`	`exclusion-preserving-occupation`	`duplicate-state-without-trace`	Unrecorded duplicate occupation violates exclusion.
`periodicity`	`return-of-observation-class`	`assumed-fixed-period`	A fixed imported period can manufacture recurrence.
`record`	`support-transition-recurrence-trace`	`recurrence-label-only`	A repeated label does not reproduce the equivalence test.
`extension`	`no-extra-rule`	`free-shell-rule`	A discretionary shell equation would import the desired organization.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:79f9f2d5c12e6b82db58e3e375f1eacc44ecb05fab2fa50516b603680f28412c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:28b430eac9c449e5d0a8d0fb4bba092ed7abb057120feb4adf7acffbdc2322d4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bbc1a7eb2265d08c410bfba1e8e59977b6d22242f4ecc31d573126547e3564e9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:beaad842b2bffca59a6ff2d5afadcabaa2e10ef7901d916f73c931394e1d43f4`.

Independent reconstruction. Implementation hash `sha256:f41c5f8015683765486787511a50e71e5470ce686a2d01b82e98842cc0aa0c64`; independent certificate `sha256:497e7cb1c9504707525c356e731610f93b86140f319380641e6854ed1e9e1d85`; external-validation hash `sha256:f1f1dc9ad812144cd280bd0e7ec8f882baf24b26f8d7d38e0fa49d7e51b48229`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- periodic-recurrence-iupac-2022: predicted recurrent-element-organization__ordered-periodic-table-support; source-derived recurrent-element-organization__ordered-periodic-table-support; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official periodic source lacks recurrent ordered element organization at the registered boundary, or if a changed row is accepted.

Measurement receipt: `sha256:5e525ae34da77af4a8ca56a8c5d781501603c089b924bdff38b98d7cb0622fce`. Isolation certificate: `sha256:b4ed887f73fbdd06b09c8c63fec1e2ac3c9451e2669f433cede0ded45da8430d`. Custody certificate: `sha256:3ba444f8def3a520da6c2db126b1bcad5e56af2490e26a447820074181e0a859`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- periodic-recurrence-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every generated positive-finite atomic-number successor family whose boundary-closed outer observation support is quotiented only by complete chemical transition equivalence.

Immutable identities. Source manifest `sha256:a495a7d16184cb6a720db529dcc86261f9e6149e3686d05c78e502d3a3c7b46e`; derivation seal `sha256:c1593f4d94b272c104efa540ff269c897523d2841b7630baf0880f96ea778779`; engine receipt `sha256:c14383f6326b6c3cb950e80d48e1864d60ac7a8e794272abd80664e35338ca83` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001-c14383f6326b6c3c.json`; empirical hash `sha256:46ace4f08e9f48c8e0f447369c6a46466de8403197d7accc83bf7aec4639c08e`; measurement receipt `sha256:5e525ae34da77af4a8ca56a8c5d781501603c089b924bdff38b98d7cb0622fce`.

15. Group and period compositional coordinates

Claim identity: `SFT-CHEM-ELEM-GROUP-PERIOD-001`

Question and theorem. A Fold group is an outer-chemical-trace equivalence class and a Fold period is its counted closure-to-reopening depth; their pair is independent of how a table is drawn.

> *periodic-element-carrier__outer-trace-equivalence-class__closure-reopening-depth__joint-group-period-coordinate*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001

Generated grammar. Generate the literal product of the registered coordinate carrier, group, period, pairing, ordering, equivalence, record and extension choices.

Boundary: Every finite periodic-order prefix partitioned by exact outer-trace recurrence, with each element assigned one recurrence-class coordinate and one counted closure-to-reopening depth.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `periodic-element-carrier__outer-trace-equivalence-class__closure-reopening-depth__joint-group-period-coordinate__atomic-number-order-retained__complete-trace-quotient__element-order-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`periodic-element-carrier`	`printed-cell-coordinate`	A page coordinate is representational rather than chemical.
`group`	`outer-trace-equivalence-class`	`column-number-as-premise`	A conventional column number cannot select equivalence.
`period`	`closure-reopening-depth`	`row-number-as-premise`	A conventional row number cannot select recurrence depth.
`pairing`	`joint-group-period-coordinate`	`group-only-or-period-only`	Either coordinate alone loses a required distinction.
`ordering`	`atomic-number-order-retained`	`layout-dependent-order`	Alternate drawings must not alter chemical coordinates.
`equivalence`	`complete-trace-quotient`	`namesake-family-only`	A family name cannot prove matching transition support.
`record`	`element-order-recurrence-trace`	`coordinate-without-witness`	A coordinate without a recurrence witness is not auditable.
`extension`	`no-extra-rule`	`free-layout-exception`	A layout exception can move an element without a chemical distinction.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c3f412be7e6caf1ae399956b649c37ddf393710eddf9fdb12016eea631e0b019`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:98da74bd924801244f5e8caa8728810ec0af692816c78ff6a8d630faccfb69f0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:60a6d8de4a10afab9cd43595398e1dead9b886863e69e3047bce476d9cfe5b9d`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:137fa1bbe50de52fb9ced8a59a7e7c9aaba62081905c588226eb4b55378df7bd`.

Independent reconstruction. Implementation hash `sha256:70bed5c2690f2e67d9c06dae6a5d91346efc2ffdd3a831a7be754f51740875b3`; independent certificate `sha256:f4762e4f73f2b81be48db4aee3269731dd14ed8cf2cd2ee5f6e57bb13db1c7f2`; external-validation hash `sha256:5825dc7247ed53812a05f89b525015475d9d597df304ef669aa1af067f466a4e`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- group-period-iupac-2022: predicted group-column-classes__period-row-progression__atomic-number-order-retained; source-derived group-column-classes__period-row-progression__atomic-number-order-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official table does not retain column classes, row progression and atomic-number records together, or if a changed row is accepted.

Measurement receipt: `sha256:308a30c9c2357b6e6e2a1f5b0378a1dcb16d5f3230abf0ccb108b961ce6a8d77`. Isolation certificate: `sha256:ee7c666ade956cf327e9ba9cca9ac4e4ffe4ee9eb946355a7afb830040fc789b`. Custody certificate: `sha256:8317dd343d712159cc95d6ce534d6dafc844faf4122cf3c88bd3066033bd4ae5`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- group-period-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite periodic-order prefix partitioned by exact outer-trace recurrence, with each element assigned one recurrence-class coordinate and one counted closure-to-reopening depth.

Immutable identities. Source manifest `sha256:c87091d13914ec3c6740b16efe6029eb769b82f253673ab88a8aa14409ee1e00`; derivation seal `sha256:181bc9d5fdf0cbb0dd587f7a8706a9cc4cfd0ea94d104f00073dd582099f5641`; engine receipt `sha256:9e7a0fd48fc23d0b8ec3bb2656b1ba78fe5d760aea358e51c897a1ae9bd630da` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-GROUP-PERIOD-001-9e7a0fd48fc23d0b.json`; empirical hash `sha256:67dd185229d14dc8365a73b060046f333914e105df3bc20bd4361037811afd89`; measurement receipt `sha256:308a30c9c2357b6e6e2a1f5b0378a1dcb16d5f3230abf0ccb108b961ce6a8d77`.

16. Valence availability and chemical combination boundary

Claim identity: `SFT-CHEM-ELEM-VALENCE-001`

Question and theorem. Valence is the exact greatest realized univalent joining or substitution count in the complete generated combination support of a retained atom or fragment and chemical context.

> *atom-or-fragment-carrier__complete-generated-combination-support__univalent-partner-count__maximum-realized-count*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`
- `SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001`
- `SFT-CHEM-ELEM-GROUP-PERIOD-001`

Generated grammar. Generate the literal product of the registered valence carrier, support, unit, selection, substitution, context, record and extension choices.

Boundary: Every finite atom or chemical fragment with complete generated combination and substitution traces, where the largest realized univalent joining count is retained with its chemical context.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `atom-or-fragment-carrier__complete-generated-combination-support__univalent-partner-count__maximum-realized-count__combination-or-substitution__chemical-context-retained__combination-maximum-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`atom-or-fragment-carrier`	`element-name-only`	An element name contains no combination witness.
`support`	`complete-generated-combination-support`	`assumed-valence-number`	An assumed number imports the answer.
`unit`	`univalent-partner-count`	`mixed-partner-count`	Partners of unrecorded multiplicity cannot define one comparison.
`selection`	`maximum-realized-count`	`typical-count`	A typical value can omit a lawful larger combination.
`substitution`	`combination-or-substitution`	`combination-only`	Valence also constrains lawful substitution for the carrier.
`context`	`chemical-context-retained`	`universal-context-free-number`	Chemical environment may alter available combinations.
`record`	`combination-maximum-trace`	`number-without-traces`	A bare number cannot reproduce its maximum witness.
`extension`	`no-extra-rule`	`free-valence-correction`	An adjustment can force a conventional value.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5806447da7a755ea31ac1d6fd82ead9fe731b639e24caddb330920603419c201`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c1fe82f9e1c25112465debbf7a0addc85fad42abe7789fe1bd67cb615eaf18d2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:091cb85be6fa237c253e1c4bc53ae5f474f1aba2031682b507ccab02ce6a5752`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:549fd50e5b28eee9a33c5d54cf5a6e7e30bd1ea06f63e8cbcb36d869dd1c2b78`.

Independent reconstruction. Implementation hash `sha256:65ca8bb5246cd20eb6c823f8ffd231729d9340a5330aca2a3c2b19751167f6d0`; independent certificate `sha256:40019dd01e7a18c83373b3ae05cfe65130dac33c549325a093ecdb169e7d389d`; external-validation hash `sha256:f9beaa50054128034e329f57e42c26db09c5cc90d10f50e1123d3039fa80310f`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-V06588-2026

Comparison records:

- valence-iupac-v06588: predicted maximum-univalent-combination-count__atom-or-fragment-carrier__substitution-boundary; source-derived maximum-univalent-combination-count__atom-or-fragment-carrier__substitution-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC record lacks a maximum univalent combination count, atom/fragment carrier or substitution boundary, or if a changed row is accepted.

Measurement receipt: `sha256:1f1d5e7911ec5bd684094de0543366afaa5146e4dfe1dde279d608bf1913a310`. Isolation certificate: `sha256:f0e86d8677ea4dea42bcda19d6aeeebe2dddd42cea345f8919cbba017f42eec2`. Custody certificate: `sha256:0dc036295c288a80524fc61eaaf11c6ab8be89ced27645f1423ae9b13a0601c2`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/V06588/json (sha256:99ae406a9ac89a8f0a89bf3c775ff9ba7c5d9d0c96f89f83a57a0b35c76c4077)

Registered target identities:

- valence-iupac-v06588 from IUPAC-GOLD-BOOK-V06588-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite atom or chemical fragment with complete generated combination and substitution traces, where the largest realized univalent joining count is retained with its chemical context.

Immutable identities. Source manifest `sha256:51369027d90ed6e989ad967b42c4796457065baaee933c83d46692ec406ce46c`; derivation seal `sha256:18a74251e5eca02417c3febb4357618634c5aa241a8e33bee56d281bd287f0b1`; engine receipt `sha256:2110f0a13f9081622880753f9a8c1eb5948486d92b8376d780b8e428eb7abd61` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-VALENCE-001-2110f0a13f908162.json`; empirical hash `sha256:305f7d43aa0e56ee708a9301489fa21ca684838a4fd117b07a946550fec5be90`; measurement receipt `sha256:1f1d5e7911ec5bd684094de0543366afaa5146e4dfe1dde279d608bf1913a310`.

17. Ion formation with retained elemental identity

Claim identity: `SFT-CHEM-ELEM-ION-001`

Question and theorem. An ion is an atomic or molecular Fold particle whose closed electric-label transfer leaves a net held charge orientation while retaining its nuclear and chemical identity.

> *atomic-or-molecular-particle__net-held-electric-charge__label-transfer-with-identity-retained__closed-charge-transfer*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

- SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001
- SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-ATOMIC-NUMBER-001
- SFT-CHEM-ELEM-VALENCE-001

Generated grammar. Generate the literal product of the registered ion carrier, charge, formation, conservation, element, orientation, record and extension choices.

Boundary: Every finite atomic or molecular chemical carrier whose complete conserved electric-label accounting leaves one non-neutral held orientation while retaining its elemental and molecular identity records.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `atomic-or-molecular-particle__net-held-electric-charge__label-transfer-with-identity-retained__closed-charge-transfer__atomic-number-retained__cation-anion-held-fibres__carrier-charge-transfer-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	atomic-or-molecular-particle	charge-label-without-particle	A charge label detached from matter is not a chemical ion.
charge	net-held-electric-charge	signed-proof-number	A signed proof magnitude violates the positive exact domain.
formation	label-transfer-with-identity-retained	identity-destroying-change	Destroying the carrier does not create a charge state of it.
conservation	closed-charge-transfer	unpaired-charge-creation	Unpaired creation violates closed label flow.
element	atomic-number-retained	proton-count-change	Changing proton count changes element identity.
orientation	cation-anion-held-fibres	positive-negative-proof-scalars	Signed scalars are not SFT proof objects.
record	carrier-charge-transfer-trace	ion-name-only	A name cannot reconstruct carrier, charge and transfer.
extension	no-extra-rule	free-charge-exception	An exception can create or erase charge without transfer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2ae6650a4e3754c24fb9a4f8f09536c51e7b81ac2b96455c63b56e597cb94fe0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3b2456e07702e7ce9cbe5e02573a4d787dcd52859d789ab8bbc3c418bc1633ae`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:df200a2b7556795d20a2d92057fdcb90f979904158d594e9d314cd2223d281f1`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:75999b23650345a5d5e4effbad9ba3bbf7ec85d09e0d64c8e675cc381cb72a3c`.

Independent reconstruction. Implementation hash `sha256:87f51143641461df3f93e8bc4ab0dc75357216b47b04bf5df604d469ee8f223f`; independent certificate `sha256:7fbe160af0765a688078df30a929390192b67361cb72598c39e46183ca9d4863`; external-validation hash `sha256:38523bcd1b0ff10c6c6cfe24dc3a64d839a02d31534d387d63e01c1bc1e15be2`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03158-2026

Comparison records:

- ion-iupac-i03158: predicted atomic-or-molecular-particle__net-electric-charge; source-derived atomic-or-molecular-particle__net-electric-charge; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the source-derived IUPAC ion record lacks an atomic-or-molecular particle and net electric charge, or if a changed row is accepted.

Measurement receipt: `sha256:124ef271dda53df2fd9cf81cc15dc01fdc9c2339d437f79cded6f0d634a8dec3`. Isolation certificate: `sha256:85d33896465594805f1d8c7ce9f57f543a261bf737ed8bc8462562b533374859`. Custody certificate: `sha256:780242878558167 55f0c6c958ba47ffc77b434abd705cc8217983ba785eb190f`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/I03158/json (sha256:6b54a0f6d9cf83f30251501b58b75764a39396e56d819b420b1556476f9f13f6)

Registered target identities:

- ion-iupac-i03158 from IUPAC-GOLD-BOOK-I03158-2026

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite atomic or molecular chemical carrier whose complete conserved electric-label accounting leaves one non-neutral held orientation while retaining its elemental and molecular identity records.

Immutable identities. Source manifest `sha256:664be51c66193e7f6227d15f9fa7dce3ae3d0c44235d204918cc3e9dac4696a2`; derivation seal `sha256:6560d04c7ffdaec794c13af0400a37d9632dadfe10d59b2144e84036b3057e89`; engine receipt `sha256:9015e9496cd2acac1ded1af87e474a69cf1d8cad0eb15f6f928464b761148c8f` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-ION-001-9015e9496cd2acac.json`; empirical hash `sha256:169141dd9bda0ebe1813a601cb3a190c7e2d13dbb50df93f9277731449b3e410`; measurement receipt `sha256:124ef271dda53df2fd9cf81cc15dc01fdc9c2339d437f79cded6f0d634a8dec3`.

18. Known periodic-table observation boundary

Claim identity: `SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001`

Question and theorem. The known periodic boundary is the complete source-dated IUPAC element census through its greatest listed atomic-number coordinate; it is an observation boundary and not a theorem forbidding successors.

> *complete-source-bounded-census__greatest-listed-coordinate__element-record-retained__unobserved-successor-open*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MEAS-OBSERVATION-CARRIER-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-QUANTUM-INDISTINGUISHABILITY-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-CHEM-ELEM-GROUP-PERIOD-001`
- `SFT-CHEM-ELEM-ION-001`

Generated grammar. Generate the literal product of the registered known-boundary carrier, extent, identity, time, completeness, extension, record and rule choices.

Boundary: Every official finite periodic-table release treated as a source-dated observation census, retaining all listed atomic-number/element records while leaving any unobserved successor scientifically open.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-source-bounded-census__greatest-listed-coordinate__element-record-retained__release-date-retained__all-listed-rows-retained__unobserved-successor-open__source-hash-census-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`complete-source-bounded-census`	`largest-number-only`	A largest number alone omits the observed element census.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`extent`	`greatest-listed-coordinate`	`timeless-terminal-claim`	A current release cannot prove that no successor can exist.
`identity`	`element-record-retained`	`count-without-element`	A number without its element record is not an observation boundary.
`time`	`release-date-retained`	`undated-table`	An undated table cannot define a reproducible empirical boundary.
`completeness`	`all-listed-rows-retained`	`selected-terminal-row`	Selecting only the last row cannot establish the registered census.
`extension`	`unobserved-successor-open`	`observed-means-impossible-beyond`	Absence from one release is not impossibility.
`record`	`source-hash-census-trace`	`boundary-without-source`	An unsourced endpoint cannot be reproduced.
`rule`	`no-extra-rule`	`free-terminal-rule`	A terminal rule chosen from the table would overfit observation.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:07957a12479b7fc21cb0f19bd56d222e157ebb86d5ecc6de767af082f09b4529`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9529beed9e6c984672c342003db5011e89be4e68d0e1591ab8de1c93c429482e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ef04b83aec104f822c81ac402cd320bdea3fd18aa95096286e409badaa6e4a4f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2eec45d3cb99a30f59046b3098e2f4659fb5d313949bb9dd9f1ac6d56ffd2cf6`.

Independent reconstruction. Implementation hash `sha256:cf9ea6d2271b92fe7b4dddb45245248dc45abe31f824d86bde302e418fe92c65`; independent certificate `sha256:f7c1c445a2435d2d0997ca0b49d94cc60bf174d5f973464f39c4f256593279be`; external-validation hash `sha256:9b2b3f3e4fe85ffed184cc29ae0ccf5fd1a7d1e9cdc5f4b9f1660e991fca420d`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- periodic-boundary-iupac-2022: predicted source-bounded-known-element-census__greatest-listed-atomic-number-118; source-derived source-bounded-known-element-census__greatest-listed-atomic-number-118; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the registered official release does not contain the complete stated current census through atomic number 118, or if a changed row is accepted.

Measurement receipt: `sha256:68205bb3cf64338c6cb4df3bbcbdcaaa2fd7200c5996771517de4670997755f6`. Isolation certificate: `sha256:626e0b70b2d68de0b1f30887d0bd047017c22600d889c933da04f3c3f88366c7`. Custody certificate: `sha256:11f824e4af628af363afff788480c72da16a213b2e09456e08eff4879bdf0822`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf

(sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- periodic-boundary-iupac-2022 from IUPAC-PERIODIC-TABLE-2022

Meaning. Element identity is a held proton-count class; isotopes, atomic-weight records, periodic order, recurrence, group, period, valence and ions retain exact provenance and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC table arrangement, term definition, measured value, electron-shell equation or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every official finite periodic-table release treated as a source-dated observation census, retaining all listed atomic-number/element records while leaving any unobserved successor scientifically open.

Immutable identities. Source manifest `sha256:c19bd53cdeb7132b39723a417b54955507500f1b2b905fb3bfe87c0b32bdbe51`; derivation seal `sha256:686eeab8f18d74ee44e4b60cb62514ce3fcad82000980177b5ea5671b4e4d74d`; engine receipt `sha256:8fbea0a9f06035f41313a38182243b9e7c9e5510957ce95c3069d30213e73fc8` at `receipts/engine/model_admitted/SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001-8fbea0a9f06035f4.json`; empirical hash `sha256:8fa21da784af33f3540672619aec09eed73b9e7ca318766f0209204bd473db39`; measurement receipt `sha256:68205bb3cf64338c6cb4df3bbcbdcaaa2fd7200c5996771517de4670997755f6`.

11. Composition Stoichiometry

Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization.

19. Exact chemical composition

Claim identity: `SFT-CHEM-STOICH-COMPOSITION-001`

Question and theorem. Chemical composition is the complete finite Fold carrier of distinct constituent identities paired with exact positive rational parts that exhaust the One.

> *complete-chemical-carrier__distinct-constituent-identities__exact-positive-rational-parts__parts-exhaust-the-One*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-MEAS-AMOUNT-001`
- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`

Generated grammar. Generate the literal product of the registered composition carrier, components, quantity, closure, absence, identity, record and extension choices.

Boundary: Every finite chemical carrier whose distinct constituent identities have exact positive rational parts that exhaust the One and preserve the measurement/source boundary.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-chemical-carrier__distinct-constituent-identities__exact-positive-rational-parts__parts-exhaust-the-One__absent-component-omitted__constituent-identity-retained__identity-part-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`complete-chemical-carrier`	`bulk-label-only`	A bulk label erases constituent support.
`components`	`distinct-constituent-identities`	`anonymous-parts`	Anonymous parts cannot preserve chemical identity.
`quantity`	`exact-positive-rational-parts`	`floating-percentages`	Floating percentages cannot certify exact closure.
`closure`	`parts-exhaust-the-One`	`partial-selected-composition`	Selected components need not exhaust the carrier.
`absence`	`absent-component-omitted`	`zero-valued-components`	Numerical zero is not an SFT composition object.
`identity`	`constituent-identity-retained`	`interchangeable-constituents`	Equal amounts do not make substances identical.
`record`	`identity-part-source-trace`	`composition-answer-only`	An answer loses part and source provenance.
`extension`	`no-extra-rule`	`free-normalization-correction`	A correction could force closure after the fact.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a72500ce5db6e8ec585702f2e1a8dbfed043e7e52c2ee8304e5647b7e3abf5e7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c75c9b3e1be8fe6f88b5afb87d59e106cfc119080312c51c41d0e99922b5167a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fcecb84871bf9e043bc807dccc5e3c148e05b26ac43b05af6806cfd0025fc304`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:6961ea1b574acf85e4ab8c4dc2c15a306a8853e50ddf3e7c30bb6e707858c87f`.

Independent reconstruction. Implementation hash `sha256:17727dd7b5815338c9679ab3cc2682e0e423ae50b27cf1305f044b57369c4959`; independent certificate `sha256:ad7e04bbe9d538d460664d08bba33e86cee6c30555ff31191f5fc4cbce190eb7`; external-validation hash `sha256:a57b4f689cfd58532a3ba9d3348415a52c28454b049eaba3dc8b4c90bc9ae336`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03722-2026

Comparison records:

- composition-iupac-m03722: predicted constituent-carrier__constituent-part-over-total-mixture; source-derived constituent-carrier__constituent-part-over-total-mixture; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks constituent identity, a constituent-to-total relation or mixture support, or if a changed row is accepted.

Measurement receipt: `sha256:d4a6cc0ba56b7f6b94481f21014a9be2deca14df0f4ba7cc26b95a7fef4374b2`. Isolation certificate: `sha256:2a8f74b85fac0e37d95b18ad205da144f96cf966913800109647a4cf545da143`. Custody certificate: `sha256:0a657320e42bc4eefc3aeedbb705dd197c259007ae0d2dbd23cde01147eeac8e`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/M03722/json (sha256:cc715cdd306779e5032685105dacbbc7d1ef79ea8241269496befb1dc37a7f2f)

Registered target identities:

- composition-iupac-m03722 from IUPAC-GOLD-BOOK-M03722-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite chemical carrier whose distinct constituent identities have exact positive rational parts that exhaust the One and preserve the measurement/source boundary.

Immutable identities. Source manifest `sha256:8f2e8d5cc8143850e17f1f542360dd2e0bfbbaa0f272eef399b72d1bd3a2b57b`; derivation seal `sha256:165b5200764d9d49f6881b554db180129167625b1e98c79e12037a7bc4dfd5a0`; engine receipt `sha256:4749de58750e305820c39678a3a2f0f6d8332e721a17921651eafca98a8a1dbc` at `receipts/engine/model_admitted/SFT-CHEM-STOICH-COMPOSITION-001-4749de58750e3058.json`; empirical hash `sha256:58069201b824519064d26333ef7a65bc5843bf825a54910f51a007904f4f5750`; measurement receipt `sha256:d4a6cc0ba56b7f6b94481f21014a9be2deca14df0f4ba7cc26b95a7fef4374b2`.

20. Conservation of elemental identity through reaction

Claim identity: `SFT-CHEM-STOICH-CONSERVATION-001`

Question and theorem. A chemical reaction is closed only when complete held reactant/product words have exactly equal positive multiplicity totals for every retained elemental identity, including named boundary carriers.

> *complete-reaction-word__held-reactant-product-sides__elemental-identities-retained__all-element-totals-paired*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-MEAS-AMOUNT-001`
- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`

Generated grammar. Generate the literal product of the registered conservation carrier, orientation, identity, count, closure, boundary, record and extension choices.

Boundary: Every finite chemical transition word whose reactant and product sides retain complete species formulae, positive multiplicities and equal per-element totals.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-reaction-word__held-reactant-product-sides__elemental-identities-retained__exact-positive-multiplicity__all-element-totals-paired__named-boundary-carriers__species-formula-total-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`complete-reaction-word`	`product-list-without-input`	Products alone cannot demonstrate conservation.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`orientation`	`held-reactant-product-sides`	`signed-stoichiometric-scalars`	Signed proof quantities violate the positive domain.
`identity`	`elemental-identities-retained`	`mass-only-balance`	Equal bulk mass can hide changed element labels.
`count`	`exact-positive-multiplicity`	`approximate-decimal-balance`	Approximation cannot prove equality of discrete carriers.
`closure`	`all-element-totals-paired`	`selected-elements-only`	Omitting an element can manufacture balance.
`boundary`	`named-boundary-carriers`	`unrecorded-open-system`	Unrecorded boundary transfer can mimic loss or creation.
`record`	`species-formula-total-trace`	`balanced-label-only`	A label cannot reconstruct the equality.
`extension`	`no-extra-rule`	`free-balancing-exception`	An exception could excuse an unpaired carrier.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5108a653085bf72b6824763bc3d1b16ed015738b65ff8dcd78f9367699bc2f27`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:079af8a29365c118d92b3732ad4284de9697833b1c3698b2e3da9b48679a85fc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:455afb7670db3dca84beea9f74cbfc874dd8b6a3bbaa348262b0aafd2c7282d2`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:bf89eecbc5d713f6278f738bfc83f579a0c871960d060961c54922ca77d71ce0`.

Independent reconstruction. Implementation hash `sha256:77092730858fcc8d021aca7483bbc7c25c64ef8b6cd071c34b0f409bb5e6901a`; independent certificate `sha256:85f9be325874c16051fc4b181a3689b3516089ad25438316ae14404254699402`; external-validation hash `sha256:25a3c688ffe9847880d57fcbc28559933ac4e2aaf0715a004e15b7277e365f75`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S06026-2026

Comparison records:

- stoichiometry-iupac-s06026: predicted reactant-amount-support__product-amount-support__reaction-relation; source-derived reactant-amount-support__product-amount-support__reaction-relation; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The formal conservation theorem fails on any unequal element map; the categorical correspondence fails if IUPAC lacks related reactant and product amount supports or if a changed row is accepted.

Measurement receipt: `sha256:cf0214f22de339efff956717d2cdc8acb0ff8e6fdef23090126f6d18be727b27`. Isolation certificate: `sha256:8cd441eb74abde90e29877ce9f3ad59fb77f7f6251baccdef4791383bf21c93a`. Custody certificate: `sha256:c7e3c8ea5cabfd382ce210a899b3f7e10136eecb58f4642c611adb86b991e755`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/S06026/json (sha256:0354d161fb1754d50dee2cfcd3bf051c6a005dbde765a112398f8e717e9952bc)

Registered target identities:

- stoichiometry-iupac-s06026 from IUPAC-GOLD-BOOK-S06026-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite chemical transition word whose reactant and product sides retain complete species formulae, positive multiplicities and equal per-element totals.

Immutable identities. Source manifest `sha256:9b6bf9c5cd0488c8496f56be542d82c20afd343c7a85219e42394867ce67079d`; derivation seal `sha256:f0b0c46ec94d704cc104131476b8ee493a63b542f3f7792ce173996a9a4ff714`; engine receipt `sha256:a510b7f8cba3f007052fd1bab433ad8e5085cb0beab3eb85f247f8614be23d81` at `receipts/engine/model_admitted/SFT-CHEM-STOICH-CONSERVATION-001-a510b7f8cba3f007.json`; empirical hash `sha256:e052d1b6079e4ea26822a82ccb4c1640bfd1c140570ff0c01b35f5f0ae660661`; measurement receipt `sha256:cf0214f22de339efff956717d2cdc8acb0ff8e6fdef23090126f6d18be727b27`.

21. Positive stoichiometric coefficient and reaction balance

Claim identity: `SFT-CHEM-STOICH-COEFFICIENT-001`

Question and theorem. Stoichiometric coefficients are the unique primitive vector of positive generated species multiplicities balancing every element map, with reactant/product direction held structurally and nonparticipants omitted.

> *formula-bound-species-support__held-side-positive-count__element-balance-preserved__primitive-positive-vector*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`

- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-MEAS-AMOUNT-001`
- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`

Generated grammar. Generate the literal product of the registered coefficient carrier, orientation, balance, scale, absence, uniqueness, record and extension choices.

Boundary: Every balanceable finite chemical reaction support with participating species only, represented by the least common-factor-free vector of positive generated multiplicities.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `formula-bound-species-support__held-side-positive-count__element-balance-preserved__primitive-positive-vector__nonparticipants-omitted__unique-up-to-common-multiple__formula-matrix-reduction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`formula-bound-species-support`	`coefficient-list-without-species`	Counts detached from formulae have no chemical meaning.
`orientation`	`held-side-positive-count`	`negative-reactant-count`	Negative proof counts are inadmissible.
`balance`	`element-balance-preserved`	`unbalanced-count-vector`	An unbalanced vector violates element conservation.
`scale`	`primitive-positive-vector`	`arbitrary-common-multiple`	Common multiples are equivalent but not minimal.
`absence`	`nonparticipants-omitted`	`zero-coefficient-species`	A zero coefficient imports numerical zero.
`uniqueness`	`unique-up-to-common-multiple`	`one-picked-balancing-vector`	Picking one vector does not eliminate rescalings.
`record`	`formula-matrix-reduction-trace`	`coefficient-answer-only`	A count list cannot prove its balance.
`extension`	`no-extra-rule`	`free-coefficient-adjustment`	An adjustment can force a desired equation.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c078d573e8a1e9e8a75c4bf44728a255e36bf2ce05c9bdb553302bc964024fb0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9a54b1e4fc558b7b721a27cfb8aeff9fd9fd2b1b8a7f2a46ecebd85bdba37fd4`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:32577dd9fbd901b3ba894763049cc1ed4f6a95e6b0741fb56d9befed0823016f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e7f45e303be1880d3bd54dde084ced16329f3cbfbcd507ceb19c40ddd1200de9`.

Independent reconstruction. Implementation hash `sha256:11bea52dca022dcd31511c76455602348cffc106f1c11d30a6f48454a49b6f41`; independent certificate `sha256:864a5d480e06f6a3c429374da8dc9e45ef4c20957a0a4713cbf3f9f00d91f0ca`; external-validation hash `sha256:a787dc6241c9679c6d5db09d5cd017b8d480ac33ea5285b60e8a88115e3a3017`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01034-2026

Comparison records:

- reaction-equation-iupac-c01034: predicted reactant-product-held-sides__absolute-stoichiometric-counts__formula-bound-coefficients; source-derived reactant-product-held-sides__absolute-stoichiometric-counts__formula-bound-coefficients; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC equation record lacks reactant/product sides, absolute stoichiometric counts or formula binding, or if a changed row is accepted.

Measurement receipt: `sha256:55d7db99d8d29620cccf07bd1ee2bea8b0b5de14c7f53be0b825c719ef320afb`. Isolation certificate: `sha256:f60f3161e2c07f23daf83f18c763c813fb51cf2acb05b743a0990e3825c7f6cf`. Custody certificate: `sha256:d82d11e8dd27c83edf8e3e7252904837dcd770d5679c76350c6a48e47da2de70`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01034/json (sha256:a582f71bc6c555d5513f365d18d5d46a6a5da61e0c1668d231ad860b9a32dea3)

Registered target identities:

- reaction-equation-iupac-c01034 from IUPAC-GOLD-BOOK-C01034-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every balanceable finite chemical reaction support with participating species only, represented by the least common-factor-free vector of positive generated multiplicities.

Immutable identities. Source manifest `sha256:e5eed4f4b0dae1eaf2e61c9a650168b57729440ac461ec4ab898e98740db3639`; derivation seal `sha256:efe5701f6eedd86927d41aebeb0b7a2549c9e8ea6690c674ef065d66b67beafd`; engine receipt `sha256:fa3142d8b967e420b605b436d4fe897322b2976dc1ab730dfa79bb5d29622947` at `receipts/engine/model_admitted/SFT-CHEM-STOICH-COEFFICIENT-001-fa3142d8b967e420.json`; empirical hash `sha256:bbc5100b4599ade423c2cd3b9282d2403595649ca8001f116e59bbe37136c5d3`; measurement receipt `sha256:55d7db99d8d29620cccf07bd1ee2bea8b0b5de14c7f53be0b825c719ef320afb`.

22. Limiting-component and complete-consumption boundary

Claim identity: `SFT-CHEM-STOICH-LIMITING-001`

Question and theorem. The limiting support is the complete tied least set of exact available-amount to primitive-required-count ratios; it bounds reaction progress before any unavailable carrier would be required.

> *available-reactant-support__primitive-coefficient-requirement__exact-available-required-ratios__complete-least-ratio-support*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-MEAS-AMOUNT-001`
- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-CHEM-STOICH-COEFFICIENT-001`

Generated grammar. Generate the literal product of the registered limiting carrier, requirement, comparison, selection, consumption, excess, record and extension choices.

Boundary: Every finite reaction input with exact positive available amounts and primitive required coefficients, where complete reaction progress is bounded by every available-to-required ratio.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `available-reactant-support__primitive-coefficient-requirement__exact-available-required-ratios__complete-least-ratio-support__complete-consumption-boundary__retained-excess-support__amount-coefficient-ratio-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`available-reactant-support`	`named-reagent-only`	A name without amount and coefficient cannot limit progress.
`requirement`	`primitive-coefficient-requirement`	`unbalanced-requirement`	Consumption requirements must come from the balanced reaction.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`comparison`	`exact-available-required-ratios`	`floating-ratio-ranking`	Floating comparison cannot certify ties or minima.
`selection`	`complete-least-ratio-support`	`first-considered-reactant`	Ordering of inspection cannot select the limiter.
`consumption`	`complete-consumption-boundary`	`negative-remainder`	Overrunning the least support would require unavailable matter.
`excess`	`retained-excess-support`	`excess-erased`	Nonlimiting remainder is part of the result.
`record`	`amount-coefficient-ratio-trace`	`limiter-name-only`	A name cannot reproduce the comparison.
`extension`	`no-extra-rule`	`free-limiter-rule`	A preference could choose a nonminimal reagent.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6ebd09dcebbc38118e45114f8cc922497e0587e6cdfe8cd4eabb024443338070`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:74667f499635c03544e37dbda2e53204e74a332e76fe799df64c6be0f44a9754`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:656b446ccde1d50e6e371ff7dc0cad490f962eef10fabddfd8a43fcbcc4bfd4f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c0ba9f1cd9197180791356d6e661445fe7460b90b6b9ce6446d2f8f18bcf07bc`.

Independent reconstruction. Implementation hash `sha256:8495ce255510f5059960137d6e1e9b3ee7214201e8d830661353e4c7d1274ecf`; independent certificate `sha256:1406ee0f53cea26f83d16e7cd23a3d49c5a694e77edd6020d3544febd31e8c34`; external-validation hash `sha256:519a8a00f02ccc7a3f7fc3da597c67113efc403ccc7c61a865fec4559dfc746e`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-D01771-2026

Comparison records:

- limiting-reagent-iupac-d01771: predicted limiting-reagent-carrier__amount-boundary; source-derived limiting-reagent-carrier__amount-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks a limiting reagent and its amount relation, an exact nonleast carrier is selected, or a changed row is accepted.

Measurement receipt: `sha256:af4b2a71f1cf3ff3049af90956d74c521b1fb6bd1f967619e433e1299d44382b`. Isolation certificate: `sha256:49ca936b63fb7bc678ad1191e82c2d8c6eaf963667978a997220527ede439d67`. Custody certificate: `sha256:c45284506b0557a4d8b86cdca4c78e0f4df93c446c56b96ae4ff409c8b27349e`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/D01771/json (sha256:352e753acf025b61ce54a5b00b61d9dd5b6d112d0d35133b48f2eba212c3b807)

Registered target identities:

- limiting-reagent-iupac-d01771 from IUPAC-GOLD-BOOK-D01771-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite reaction input with exact positive available amounts and primitive required coefficients, where complete reaction progress is bounded by every available-to-required ratio.

Immutable identities. Source manifest `sha256:9da9d441722db6967b140f8ff0849f7e757695f59bdc80ff8c23c853a26b08c3`; derivation seal `sha256:b784c5f3c18d8664ea883c1c9bbbba2321a6db47675ddb505a48367c1fb478f4`; engine receipt `sha256:9fa6f3b55c889c675a54c5e137b08f9357ecbe860c59b9ced2b506925a73468b` at `receipts/engine/model_admitted/SFT-CHEM-STOICH-LIMITING-001-9fa6f3b55c889c67.json`; empirical hash `sha256:bfe6a293e23cb72a148862a36162aa6c4d713fc8ed68d4e26df6b547ac31d69a`; measurement receipt `sha256:af4b2a71f1cf3ff3049af90956d74c521b1fb6bd1f967619e433e1299d44382b`.

23. Reaction yield as retained product share

Claim identity: `SFT-CHEM-STOICH-YIELD-001`

Question and theorem. Chemical yield is the exact positive Fold part of a retained element-or-compound product amount relative to its registered expected support after a specified reaction or separation.

> *element-or-compound-product__specified-reaction-or-separation__complete-observed-product-amount__exact-positive-amount-fraction*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`

- SFT-CHEM-MEAS-AMOUNT-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-ELEM-ELEMENT-001
- SFT-CHEM-ELEM-PERIODIC-ORDER-001
- SFT-CHEM-STOICH-LIMITING-001

Generated grammar. Generate the literal product of the registered yield carrier, process, numerator, reference, quantity, absence, record and extension choices.

Boundary: Every specified reaction or separation with a retained expected product support and a positive observed product amount represented as an exact part of that reference support.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: element-or-compound-product__specified-reaction-or-separation__complete-observed-product-amount__registered-expected-support__exact-positive-amount-fraction__empty-product-outcome-form__product-process-reference-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	element-or-compound-product	percentage-without-product	A percentage detached from a product is not chemical yield.
process	specified-reaction-or-separation	unspecified-process	Yield depends on the declared reaction or separation.
numerator	complete-observed-product-amount	selected-favorable-product	Selecting a favorable amount hides other outcomes.
reference	registered-expected-support	unbound-denominator	A fraction without its reference is uninterpretable.
quantity	exact-positive-amount-fraction	floating-percent	Floating output is not exact proof.
absence	empty-product-outcome-form	numerical-zero-yield	Zero is not an SFT proof value.
record	product-process-reference-trace	yield-answer-only	An answer loses product, process and reference provenance.
extension	no-extra-rule	free-yield-correction	A correction can inflate the observed result.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:d25409ab09b66f54e66658f55ef35bd4c7b994814725c3cecd49c5a18ffc6ede.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:9a81c40f9ff9716ea4d44351b4d379929c6a2e124154eaa6bb55b4e775388ee4.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:f7c821f8d7494f3c08d13fd43f4971cebcd6ea3d5f45d85f62b112202583abcb.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:2819fb08332303920b39a0fc659c93cea8485c85ea07e708d3be83a8055bf9d7.

Independent reconstruction. Implementation hash sha256:e84db5b8638298ffd8290fb69d355e310f64a6201fd700ce8d188b614972d04c; independent certificate sha256:27ce7f5b9a159971c4c513624866a8244cf854b3cd65b9bb3f465fd4e67f8dfb; external-validation hash `sha256:dcface3941413fec1009e46c661ae31364b0b44b93042c4ed596278d4e57bbf1`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01041-2026

Comparison records:

- chemical-yield-iupac-c01041: predicted amount-fraction__element-or-compound-product__specified-reaction-or-separation; source-derived amount-fraction__element-or-compound-product__specified-reaction-or-separation; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks an amount fraction, element/compound carrier or specified reaction/separation, or if a changed row is accepted.

Measurement receipt: `sha256:f2825991402f7f43d0d997ed2187e32820013df3201d200377060d94424c4335`. Isolation certificate: `sha256:fdb2d43fefde6ac4a5a114c3f1f1ad979158e4f8aa3fbe254ad7a57359bbad6e`. Custody certificate: `sha256:0fd75ee7e7fce4bc1badc51e44cbcdb9a740bef3bc071676c1d6a737bca09bee`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01041/json (sha256:47979dfe9e05be8a7c207779853ce25538a8753e05b648a8a43471ed040b732f)

Registered target identities:

- chemical-yield-iupac-c01041 from IUPAC-GOLD-BOOK-C01041-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every specified reaction or separation with a retained expected product support and a positive observed product amount represented as an exact part of that reference support.

Immutable identities. Source manifest `sha256:a0116438e6f83db3d0d99e824726e84c88b578f7437cfe799bc4ddc360d92adc`; derivation seal `sha256:127d3ab3ebb275a09332cd72b1af9c50849b46a51bf96aad1fb561c2d39f8886`; engine receipt `sha256:87383bf6cedbbc08dae3f91f6932bbd4b151d8288c32ac5908cd6f62fff649d6` at `receipts/engine/model_admitted/SFT-CHEM-STOICH-YIELD-001-87383bf6cedbbc08.json`; empirical hash `sha256:c9261af3ac67d4ec093db933e4af89fe002d4ac1693c29b5984b240eef512249`; measurement receipt `sha256:f2825991402f7f43d0d997ed2187e32820013df3201d200377060d94424c4335`.

24. Mixture components and composition support

Claim identity: `SFT-CHEM-STOICH-MIXTURE-001`

Question and theorem. A mixture is a finite portion-of-matter Fold carrier with multiple retained chemical-substance identities and exact positive composition parts, without assuming formation of a new substance.

> *portion-of-matter-carrier__multiple-constituent-support__chemical-substance-identities__exact-positive-composition-parts*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-MEAS-AMOUNT-001`
- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`
- `SFT-CHEM-STOICH-YIELD-001`

Generated grammar. Generate the literal product of the registered mixture carrier, multiplicity, identity, composition, bonding, phase, record and extension choices.

Boundary: Every finite portion of matter containing multiple retained chemical-substance identities and their exact positive composition parts without requiring a new bonded substance identity.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `portion-of-matter-carrier__multiple-constituent-support__chemical-substance-identities__exact-positive-composition-parts__constituent-identities-retained__phase-support-recorded-not-forced__component-part-phase-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`portion-of-matter-carrier`	`substance-name-only`	One substance name does not define a mixture.
`multiplicity`	`multiple-constituent-support`	`single-constituent`	A single constituent is not a mixture.
`identity`	`chemical-substance-identities`	`anonymous-components`	Anonymous components erase chemical distinction.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
composition	exact-positive-composition-parts	unmeasured-presence-list	Presence alone cannot reconstruct composition.
bonding	constituent-identities-retained	new-substance-conflation	Mixing does not by itself force a new chemical identity.
phase	phase-support-recorded-not-forced	single-phase-required	Mixtures may contain one or several phases.
record	component-part-phase-trace	mixture-label-only	A label cannot reconstruct components.
extension	no-extra-rule	free-mixture-equivalence	An arbitrary equivalence can merge distinct mixtures.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:acb80f57db5c57ab1df7b007f1543f85ba3748e02f7d0e3d95d125a1cfb31630`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:de36e78e252d653293e37ec113b0dc81ece173288e089f8d49fe23ac16133fdd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:56afba8866e13bcfdc397b0f3eee00ada26e353e74d5095ac4466e87ba6615f3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:97238131131cae2f7c598609c639c60476b18cb2e6290608d7b8f37dfbd1b0de`.

Independent reconstruction. Implementation hash `sha256:85d3ff3531cced11d325c2211b6ea9680e0d6626a89522b8cd3e8872a322d103`; independent certificate `sha256:96bb6f5ffb2813d6bd37c857ebcefbaea52fb4dbc0be4cf181c61e211b5ed55f`; external-validation hash `sha256:d52d2fb26eae22622359bc13b920674b4633d994cc2c8cb1b782b9f304c3b6ae`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M03949-2026

Comparison records:

- mixture-iupac-m03949: predicted matter-portion__two-or-more-substances__constituent-identities; source-derived matter-portion__two-or-more-substances__constituent-identities; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC record lacks a matter portion, multiple chemical substances or constituent identities, or if a changed row is accepted.

Measurement receipt: `sha256:3b5d8220c456ede703a29c70a4a194fdb42cca38d763f4ef18bf947d8750d067`. Isolation certificate: `sha256:a2712fae535f20dd5d7c7f128176cc00c95fa410975d4e5b33030389a30e47fc`. Custody certificate: `sha256:d976c23c2014b49392fc31020aa7d4ed08f8d4995e7dceee239129204bb9bb50`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/M03949/json (sha256:71ba494096d995967227b9ef9fb4e6145b98c9cbcdbdb46460bc63b94c06966f)

Registered target identities:

- mixture-iupac-m03949 from IUPAC-GOLD-BOOK-M03949-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite portion of matter containing multiple retained chemical-substance identities and their exact positive composition parts without requiring a new bonded substance identity.

Immutable identities. Source manifest `sha256:bdc9b1e23fa1e8fcda55ede4f1803e82363189aa77af123609c2ebf8c2c33695`; derivation seal `sha256:b3b68cc273a3c95181481c58276e713fb631d80cadb79f154682f68e9b266d02`; engine receipt `sha256:d77d8e9e7e5fafc9a4a986b30bbc99a80af48adafccc45031c9190df75b29167` at `receipts/engine/model_admitted/SFT-CHEM-STOICH-MIXTURE-001-d77d8e9e7e5fafc9.json`; empirical hash `sha256:f490b60c9eb1cd26612a22e17d370715a1fb767dc32eda7613731c58b1a5fd1a`; measurement receipt `sha256:3b5d8220c456ede703a29c70a4a194fdb42cca38d763f4ef18bf947d8750d067`.

25. Solution composition and solute-solvent distinction

Claim identity: `SFT-CHEM-STOICH-SOLUTION-001`

Question and theorem. A solution is a liquid or solid single-phase mixture containing multiple retained substances under an explicit contextual solvent/solute role partition.

> *liquid-or-solid-phase__single-phase-support__multiple-substance-support__solvent-solute-role-partition*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-ENCODING-DECODING-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-CONSERVATION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-CHEMICAL-SPECIES-001`
- `SFT-CHEM-MEAS-SUBSTANCE-001`
- `SFT-CHEM-MEAS-AMOUNT-001`

- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-CHEM-STOICH-MIXTURE-001`

Generated grammar. Generate the literal product of the registered solution carrier, phase, composition, roles, identity, context, record and extension choices.

Boundary: Every finite liquid or solid single-phase mixture with multiple retained substances and an explicit contextual partition between solvent carrier roles and solute carrier roles.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `liquid-or-solid-phase__single-phase-support__multiple-substance-support__solvent-solute-role-partition__substance-identities-retained__declared-context-role__phase-component-role-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`liquid-or-solid-phase`	`gas-or-unclassified-matter`	The registered solution boundary is a liquid or solid phase.
`phase`	`single-phase-support`	`multiple-unresolved-phases`	Unresolved phase multiplicity does not define one solution carrier.
`composition`	`multiple-substance-support`	`single-substance-phase`	A pure phase lacks solute/solvent distinction.
`roles`	`solvent-solute-role-partition`	`unpartitioned-components`	A solution record distinguishes solvent from solutes.
`identity`	`substance-identities-retained`	`roles-replace-identity`	A role label must not erase substance identity.
`context`	`declared-context-role`	`intrinsic-permanent-role`	Solvent/solute assignment can depend on the description context.
`record`	`phase-component-role-trace`	`solution-name-only`	A name cannot reproduce phase and roles.
`extension`	`no-extra-rule`	`free-solution-exception`	An exception can relabel a multiphase mixture as one solution.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4fe30a8a55d56b5ab42bf2b742148820ad68162cf9d5e70ee5a38004e0f457b9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4ea951e01a0a70099147774aea8b7a15b2ec3dc32a151b75faf81596cee969d7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:24316754e898abeac2d19748aaf72de6a2f28f7cd99dd250b68a28cc3fa74def`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e1aad5725bea83acba570f25c4e0438cbff9c9620e5629755c76672e3f2b015a`.

Independent reconstruction. Implementation hash `sha256:c0b00a98c9f8125cb8cf283d7fa1e8a33b328efeb815cb3e559cedc6740126bd`; independent certificate `sha256:105ee387d1fc186480013933b78a95f785fb02c759ac692c1cb36547a421997f`; external-validation hash `sha256:3e720b08c2d95c2a714a714fc6432fbc6ab362d7c67d1a62567725f2a7566dae`. The independent

program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S05746-2026

Comparison records:

- solution-iupac-s05746: predicted single-liquid-or-solid-phase__multiple-substance-support__solvent-solute-role-partition; source-derived single-liquid-or-solid-phase__multiple-substance-support__solvent-solute-role-partition; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the IUPAC solution record lacks liquid/solid phase, multiple substances or solvent/solute distinction, or if a changed row is accepted.

Measurement receipt: `sha256:715f072b9ecb06487c766125d3dddb7cfadad15cbf708a3a420b7effbdbceacc`. Isolation certificate: `sha256:29378a1c0c44592cbc595fe9461242a245f0d11fe67e586dad244badd67fda25`. Custody certificate: `sha256:13f2783c070b3017363e017a98bba399b8db9cbfdd35ca942a4ffcd5a4210087`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/S05746/json (sha256:39f8916fdced6993ceb67e374641b953755708 66a24123508ce26f9ec0049f28)

Registered target identities:

- solution-iupac-s05746 from IUPAC-GOLD-BOOK-S05746-2026

Meaning. Composition and stoichiometry preserve every elemental carrier across complete reactant/product maps, primitive positive coefficients, limiting support, yield, mixture and solution organization. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC wording, balanced equation, measured amount or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no absent species represented by a zero coefficient; absence is an empty generated form
- no application result or opaque predictor
- official target content opens only through post-seal custody
- Every finite liquid or solid single-phase mixture with multiple retained substances and an explicit contextual partition between solvent carrier roles and solute carrier roles.

Immutable identities. Source manifest `sha256:db699f4ed4734f107bf620736f458a9f033160b02451a20467a4619f0f81c4d2`; derivation seal `sha256:6e5542e15258a967197185e1f07e767e97459a180c9fe7347d0e46ae97f96199`; engine receipt `sha256:9775d542364d5db8cb2ce3a2491cc224abfde1709129db3d42f0e64ed5cac371` at `receipts/engine/model_admitted/SFT-CHEM-STOICH-SOLUTION-001-9775d542364d5db8.json`; empirical hash `sha256:02cfbf84dce854f2b64540c9bf6b036d8dc337ad39637bfc42e50dbb0179d3d8`; measurement receipt `sha256:715f072b9ecb06487c766125d3dddb7cfadad15cbf708a3a420b7effbdbceacc`.

12. Bonding Molecular

Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures.

26. Chemical bond as retained interaction closure

Claim identity: `SFT-CHEM-BOND-CHEMICAL-BOND-001`

Question and theorem. A chemical bond exists when complete source-bound interaction support between distinct atomic carrier occurrences or groups closes into stable joint recurrence distinguishable as an independent chemical carrier.

atoms-or-atomic-groups__source-bound-interaction-support__stable-joint-recurrence__independent-chemical-entity

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-CONDENSED-BAND-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-VALENCE-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`

Generated grammar. Generate the literal product of the registered bond carrier, interaction, closure, independence, energy, reversal, record and extension choices.

Boundary: Every finite pair or group of admitted atomic carriers and complete interaction support whose joint recurrence is stable as an independently distinguishable chemical carrier at the registered boundary.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `atoms-or-atomic-groups__source-bound-interaction-support__stable-joint-recurrence__independent-chemical-entity__measured-stability-record-separated__opening-closes-to-separated-support__carrier-interaction-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`atoms-or-atomic-groups`	`bond-name-without-atoms`	A bond name detached from carriers has no chemical support.
`interaction`	`source-bound-interaction-support`	`coincidental-proximity`	Proximity alone need not produce a stable entity.
`closure`	`stable-joint-recurrence`	`transient-contact-only`	A contact without recurrence is not an independent bonded carrier.
`independence`	`independent-chemical-entity`	`parts-only-observation`	If only separated parts persist, no joined entity is formed.
`energy`	`measured-stability-record-separated`	`imported-potential-function`	A consensus potential cannot select closure.
`reversal`	`opening-closes-to-separated-support`	`irreversible-by-definition`	Bond opening may be lawful and must retain carriers.
`record`	`carrier-interaction-recurrence-trace`	`bond-label-only`	A label cannot reproduce interaction and stability.
`extension`	`no-extra-rule`	`free-bond-exception`	An exception can call any contact a bond.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c79949b0bfe9498cf4d374c3041cad4bd3d2c76f13ee438fb68d1e4fed80a468`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:67d9bddfde9089152683c855abe345423e079bb9c69cbb4c2ec01eaea88c89ff`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:829ed6daf9d8afea57c415f332a212cf2dd68a7834c99709e75ed5ca411522b8`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8647853beaa9c2398420258b61194b99ef938fb4a29bf75625cae36762f09e7b`.

Independent reconstruction. Implementation hash `sha256:13efe64c7d43bb48869c408cbab8c44f2ccf1d7ed6e0e875e332e021c448ec6b`; independent certificate `sha256:13b94ba61ffe44d8d734add0368d83768a5e8e2384b92b4517df7bc2422418a6`; external-validation hash `sha256:fc484bd2b42299be96d1d68707cafab3022c9e9c7c2fb5524e92e97aadeed349`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CT07009-2026

Comparison records:

- chemical-bond-iupac-ct07009: predicted atoms-or-atomic-groups__stable-independent-molecular-entity__interaction-forces; source-derived atoms-or-atomic-groups__stable-independent-molecular-entity__interaction-forces; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks atomic/group carriers, stable independent molecular entity or interaction-force support, or if a changed row is accepted.

Measurement receipt: `sha256:5b96d8da59dcd2182001f4b15391758cf3342d766bd58d70bc73e794aecd082e`. Isolation certificate: `sha256:5460f966fa502eaab9a5a9b4fbc7149e6d85791abb800b101079296116777c1d`. Custody certificate: `sha256:9d7857250559bc276af0441c1abd3f5d9a9ee8e424d50efa68144e0a21b979bd`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/CT07009/json (sha256:691e8522684b2e4290c207ae988a4689d1ab255cdd5fca4ed66a1bd557fa4207)

Registered target identities:

- chemical-bond-iupac-ct07009 from IUPAC-GOLD-BOOK-CT07009-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite pair or group of admitted atomic carriers and complete interaction support whose joint recurrence is stable as an independently distinguishable chemical carrier at the registered boundary.

Immutable identities. Source manifest `sha256:ad51f358991f2d60e1c5d5cfa71c4c37f8f7fbe1652cc5a5b3968d8058ba194d`; derivation seal `sha256:523dfb58508e28f1d3bbafdcf7d0b0b224e2a4522504ce4ab6a5b10646f3c786`; engine receipt `sha256:6f5a76351dd92e06a0982b6a611e3bf04d2d69dae0635e8dd6b4a3b4e92586eb` at `receipts/engine/model_admitted/SFT-CHEM-BOND-CHEMICAL-BOND-001-6f5a76351dd92e06.json`; empirical hash `sha256:3241f15a85770f40e37f28ca05b10426a7e3cd63b037a0a3f5bedc80a78c9e86`; measurement receipt `sha256:5b96d8da59dcd2182001f4b15391758cf3342d766bd58d70bc73e794aecd082e`.

27. Covalent shared-support bond

Claim identity: `SFT-CHEM-BOND-COVALENT-001`

Question and theorem. A covalent bond is stable joint recurrence in which one or more exclusion-preserving electron-labelled channels are jointly accessible between identified nuclear carriers without duplicating the conserved carrier.

> *identified-nuclear-pair__joint-electron-support__one-support-shared-access__internuclear-accessible-region*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-CONDENSED-BAND-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-VALENCE-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-CHEMICAL-BOND-001`

Generated grammar. Generate the literal product of the registered covalent carrier, electron, sharing, density, attraction, occupation, record and extension choices.

Boundary: Every finite bonded atomic support with one or more electron-labelled recurrence channels jointly accessible to both nuclear carriers and exclusion-preserving occupation.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `identified-nuclear-pair__joint-electron-support__one-support-shared-access__internuclear-accessible-region__stable-contractive-response__exclusion-preserving-channels__nuclei-channel-sharing-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`identified-nuclear-pair`	`bond-without-nuclei`	Covalent support relates identified atomic centres.
`electron`	`joint-electron-support`	`electron-assigned-to-one-centre`	One-centre support alone is not shared bonding support.
`sharing`	`one-support-shared-access`	`copied-electron-label`	Copying one conserved carrier violates exclusion and conservation.
`density`	`internuclear-accessible-region`	`empty-internuclear-path`	No connecting support cannot join the nuclei.
`attraction`	`stable-contractive-response`	`unbound-force-label`	A force name without recurrence does not establish a bond.
`occupation`	`exclusion-preserving-channels`	`duplicate-identical-cell`	Duplicated identical occupation violates exclusion.
`record`	`nuclei-channel-sharing-trace`	`covalent-label-only`	A label cannot reconstruct sharing.
`extension`	`no-extra-rule`	`free-orbital-rule`	An imported orbital choice can select a desired bond.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f59fe4de319dd05bd3b6b6d48f1de3d15a39d2deee4bc65cbf5214affe909a3c`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:92ed58420e5d68434c41c74d2cb5eb78fb3782d79080388c4cf7c4efdbed92d0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:ee7b74ae479c8c65c71393fac1927bf6dc7eb709d54e5f3a9f14167debde57f7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a6c12d5e1a6339f3a4c4318629455f975ea16a517e11d0bb1cf857097839ea88`.

Independent reconstruction. Implementation hash `sha256:4b68316cb8fb10965e70db523fc6335b89af5a692c612d7eaa9634157368aa0f`; independent certificate `sha256:3544a1af89d8576edf7c999b7ab8c70fada8cf2ad63bc3b0bef03541d0c4992c`; external-validation hash `sha256:65299913bf8ed63d2e7b7ecb64db0cad8cd0ab3e9fc9e155451e9c248442ebed`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-C01384-2026

Comparison records:

- covalent-bond-iupac-c01384: predicted internuclear-electron-density__electron-sharing__attractive-characteristic-distance; source-derived internuclear-electron-density__electron-sharing__attractive-characteristic-distance; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks internuclear electron density, sharing or attractive characteristic distance, or if a changed row is accepted.

Measurement receipt: `sha256:4389207d46667075d3136e495b54799342ddc50516f726bb0bb1caf5d15e65d8`. Isolation certificate: `sha256:7660d942fb6500602f9f53a9e426844d631d14a08448c79c63029be8d6ad69e7`. Custody certificate: `sha256:e1fde5da0ecc8b811a1d6525d4e18503f6be7fe623ef7f168ff148198764fe1c`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01384/json (sha256:5106a40f5b1c372a570fc42f3cb1641b63a270454e4a7138699c584754294622)

Registered target identities:

- covalent-bond-iupac-c01384 from IUPAC-GOLD-BOOK-C01384-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite bonded atomic support with one or more electron-labelled recurrence channels jointly accessible to both nuclear carriers and exclusion-preserving occupation.

Immutable identities. Source manifest `sha256:ac32a8598b14c48ec075df48f45bb5ddd1d5b331e64209e8e28ceb477190e775`; derivation seal `sha256:6c8f4e6687ca434ec1572a13444e6c53ce0bc6bc359b802d3b77e511cdff152e`; engine receipt `sha256:b1efa88f0ebfe5c562e52a5a39673b1242da71b91d7d3c24b66356bfd462399b` at `receipts/engine/model_admitted/SFT-CHEM-BOND-COVALENT-001-b1efa88f0ebfe5c5.json`; empirical hash `sha256:dd9f285006ea268f1df83b9432b26b249b9ca4ebaf0551f66b8407058f35b160`; measurement receipt `sha256:4389207d46667075d3136e495b54799342ddc50516f726bb0bb1caf5d15e65d8`.

28. Ionic transferred-label bond

Claim identity: `SFT-CHEM-BOND-IONIC-001`

Question and theorem. An ionic bond is stable joint recurrence of opposed cation/anion held charge fibres produced by closed charge transfer, with electrostatic attraction and both chemical identities retained.

> *cation-anion-carriers__closed-charge-transfer__opposed-held-charge-fibres__electrostatic-attraction-support*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-CONDENSED-BAND-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-VALENCE-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-CHEMICAL-BOND-001`
- `SFT-CHEM-BOND-COVALENT-001`

Generated grammar. Generate the literal product of the registered ionic carrier, formation, orientation, interaction, identity, character, record and extension choices.

Boundary: Every finite charge-transfer product containing opposed cation/anion held fibres whose closed electric interaction yields stable joint recurrence while both chemical identities remain retained.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `cation-anion-carriers__closed-charge-transfer__opposed-held-charge-fibres__electrostatic-attraction-support__chemical-identities-retained__exact-support-composition__ion-transfer-interaction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`cation-anion-carriers`	`neutral-atom-pair-only`	The ionic class requires retained ion carriers.
`formation`	`closed-charge-transfer`	`unpaired-charge-creation`	Charge cannot appear without conserved transfer.
`orientation`	`opposed-held-charge-fibres`	`signed-charge-magnitudes`	Negative proof magnitude is inadmissible.
`interaction`	`electrostatic-attraction-support`	`generic-proximity`	Proximity does not explain ionic stability.
`identity`	`chemical-identities-retained`	`element-identity-erased`	Ion formation preserves nuclear element identity.
`character`	`exact-support-composition`	`pure-binary-dogma`	Observed bonds may retain both ionic and shared support.
`record`	`ion-transfer-interaction-trace`	`ionic-label-only`	A label cannot reconstruct transfer and attraction.
`extension`	`no-extra-rule`	`free-electronegativity-cutoff`	A fitted cutoff would import the classification.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9cbd2a42338ef908d2f97d639e4e163a85f7c01e8f842a1027ea1bd7d82156f7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:e01b7d4d4f8d957a884aca8560cad6aa53ce0add2f825323d75db608900ab55d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2259224cc1c5e1e8fe927ed597b35a75ef8ace609b8f28dab7e2a23e6537117b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:92462b6af0cedacb2f75734bf5743e5454e360fc260e71256956609f5be56c39`.

Independent reconstruction. Implementation hash `sha256:d343bcfb4e46576e97186ef06aba3aa0ae1e3d63bb00cafe938cc6172a3320bb`; independent certificate `sha256:3fb249ae42a0f7cfef14ee71adccafd0dc851d4d7cc205a8b438663482a103ce`; external-validation hash `sha256:49d1839c410e0bcc7ebd5cce71525f7368d8e9dfb6c89edc7c27674d1e3e8ba1`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-IT07058-2026

Comparison records:

- ionic-bond-iupac-it07058: predicted cation-anion-carriers__electrostatic-attraction__ionic-character-boundary; source-derived cation-anion-carriers__electrostatic-attraction__ionic-character-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks cation/anion carriers, electrostatic attraction or the ionic-character boundary, or if a changed row is accepted.

Measurement receipt: `sha256:9bbd734ddfabcbc307e4f027b045a6d1f138551f43f77d142dc696342197b561`. Isolation certificate: `sha256:dc95344f2d54eab75cd6be7cc5c73ff8ba72e09117a2fedb144fc96a1e4b7ca2`. Custody certificate: `sha256:267fd55f24526b1cb9fed32621e9f52bb9ae246a1f02a27bf45b0b404149ea6e`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/IT07058/json (sha256:85d726bab39480e944dfd95e31889931459f4436480d751d73df058c68ad232e)

Registered target identities:

- ionic-bond-iupac-it07058 from IUPAC-GOLD-BOOK-IT07058-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite charge-transfer product containing opposed cation/anion held fibres whose closed electric interaction yields stable joint recurrence while both chemical identities remain retained.

Immutable identities. Source manifest `sha256:27f04cbd8c0b331ccc2f8e03a5124b7017dcf58c649dfa8b3c45670b658a5c45`; derivation seal `sha256:c24c59d4b1e5603d40c6787bad3b65dcd122ccce662d17071066ba2961539f16`; engine receipt `sha256:f0effeef99b4caec396b299ab13e5d4dffe97ed7d7e85abb035e0068ad4468fc` at `receipts/engine/model_admitted/SFT-CHEM-BOND-IONIC-001-f0effeef99b4caec.json`; empirical hash `sha256:197102353edfda54d286fa8743c8ab004c3f854dcb96ca6fe8aac78159d8127b`; measurement receipt `sha256:9bbd734ddfabcbc307e4f027b045a6d1f138551f43f77d142dc696342197b561`.

29. Metallic collective-support bond

Claim identity: `SFT-CHEM-BOND-METALLIC-001`

Question and theorem. Metallic bonding is stable collective recurrence of a connected atomic network supported by electron-labelled paths delocalized across multiple cells of the complete finite network.

> *connected-atomic-network__delocalized-electron-support__finite-whole-network-extent__collective-shared-support*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`

- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-CONDENSED-BAND-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-VALENCE-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-CHEMICAL-BOND-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`

Generated grammar. Generate the literal product of the registered metallic carrier, electron, extent, joining, recurrence, identity, record and extension choices.

Boundary: Every finite connected atomic network whose electron-labelled accessible support extends across multiple atomic cells and whose collective recurrence stabilizes the whole carrier.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `connected-atomic-network__delocalized-electron-support__finite-whole-network-extent__collective-shared-support__stable-collective-recurrence__atomic-and-network-identities__network-electron-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`connected-atomic-network`	`isolated-atom-pair`	Metallic support is collective rather than one fixed pair.
`electron`	`delocalized-electron-support`	`localized-pair-only`	A fixed pair cannot mediate the complete network.
`extent`	`finite-whole-network-extent`	`unrecorded-infinite-lattice`	A completed infinite lattice is not generated.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`joining`	`collective-shared-support`	`independent-local-bonds-only`	Independent local pairs omit collective redistribution.
`recurrence`	`stable-collective-recurrence`	`transient-conduction-only`	Transport alone does not establish stable bonding.
`identity`	`atomic-and-network-identities`	`anonymous-metal-label`	A metal label cannot create the law.
`record`	`network-electron-recurrence-trace`	`metallic-label-only`	A label cannot reproduce topology and support.
`extension`	`no-extra-rule`	`free-electron-sea-premise`	An imported metaphor cannot select the structure.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f274ec7ff0584a9b1aa9ba3b9c7ecc7e0bc0a0c9e2a02f0280cc33fb7cf4b0d5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:9e188caae50ec6514eba65a8d667abb1076a7ded34f9e0a45f6ec4ae466c1f27`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:6fe108c3748d6af8f85ab8790147389a9628a359676b39b909121c43441035ff`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1801270185f1d5190300b4a9bd1f172f07b407b7f606e915414dc3da23914cd1`.

Independent reconstruction. Implementation hash `sha256:f2495ddd58f17bc3d5859581fa2a310c528cd32e38b9434b51221ada9d9b22db`; independent certificate `sha256:32d15d7feb44f6a46c04372c2c1dd7eb026023f09f119126adbf729397408be7`; external-validation hash `sha256:83e8ceb16a401fd1ff76690a24d5de32bf14c372ee131916bc2a49af9ffd3f7a`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-08789-2026

Comparison records:

- metallic-delocalization-iupac-08789: predicted not-one-atom-or-bond__extended-whole-lattice-support__typical-of-metals; source-derived not-one-atom-or-bond__extended-whole-lattice-support__typical-of-metals; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks nonlocalized electrons, extended whole-lattice support or explicit metal correspondence, or if a changed row is accepted.

Measurement receipt: `sha256:27487aabfa2003834b1c4f418af9a100be24bf33823fc49e8f7a13cc38ad35ae`.
Isolation certificate: `sha256:7c7859fa1b0eef37bfd3a2b78bded47555ca2c7c7152352f6a4892a5b23e3db0`.
Custody certificate: `sha256:8ca4fb05f88c7606938f7920cba6007b185604ba7d434da49753d1494ec25c17`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/08789/json (sha256:570755940f01bfa32741b03b6b2f22b02742101605a2263e57369966ea433abd)

Registered target identities:

- metallic-delocalization-iupac-08789 from IUPAC-GOLD-BOOK-08789-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite connected atomic network whose electron-labelled accessible support extends across multiple atomic cells and whose collective recurrence stabilizes the whole carrier.

Immutable identities. Source manifest `sha256:fbe017c1cc8fe61261c1d93e072d0bd206d106665fc8061c65bf50af62984c6a`; derivation seal `sha256:4d1014a3eb98cd2461b3adfae4fa284c1b3de5d4b34ca484380f448f31e7f595`; engine receipt `sha256:2f6e0fa6b523ce81d7c4ab48b0413da06863c074cdeae580e39ef2f507771498` at `receipts/engine/model_admitted/SFT-CHEM-BOND-METALLIC-001-2f6e0fa6b523ce81.json`; empirical hash `sha256:9eb7097b10ed3df8d25bce50b38739628d4ad8f55b5339f4db3bb51d976c768e`; measurement receipt `sha256:27487aabfa2003834b1c4f418af9a100be24bf33823fc49e8f7a13cc38ad35ae`.

30. Bond order as retained joining multiplicity

Claim identity: `SFT-CHEM-BOND-ORDER-001`

Question and theorem. Bond order is the exact positive rational degree of complete joining-channel support for one identified bond relative to one localized electron-pair joining unit, with the observation partition retained.

> *identified-bond-carrier__complete-joining-channel-support__single-localized-pair-reference__exact-positive-rational-degree*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`

- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-CONDENSED-LATTICE-001
- SFT-PHYS-CONDENSED-BAND-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-VALENCE-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001

Generated grammar. Generate the literal product of the registered bond-order carrier, support, reference, quantity, delocalization, method, record and extension choices.

Boundary: Every finite identified bond with complete joining-channel support, measured relative to one localized electron-pair joining unit using exact positive counts or rational support parts.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `identified-bond-carrier__complete-joining-channel-support__single-localized-pair-reference__exact-positive-rational-degree__rational-support-sharing-retained__partition-bound-index__bond-channel-reference-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`identified-bond-carrier`	`index-without-bond`	A number detached from a bond has no meaning.
`support`	`complete-joining-channel-support`	`selected-channel-count`	Omitting channels can force the degree.
`reference`	`single-localized-pair-reference`	`arbitrary-scale`	A free scale can select any order.
`quantity`	`exact-positive-rational-degree`	`floating-index`	Floating arithmetic cannot certify the exact support ratio.
`delocalization`	`rational-support-sharing-retained`	`integer-only-order`	Delocalized support can distribute a joining unit across links.
`method`	`partition-bound-index`	`one-partition-as-absolute`	Different observational partitions can report different indices.
`record`	`bond-channel-reference-trace`	`order-answer-only`	An index alone cannot reproduce support.
`extension`	`no-extra-rule`	`free-order-correction`	A correction can force a conventional integer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:e2db93d7acdd9cf1733eff52d8fd837ae587de58815c0f9e206eff130e6bfc39`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:0730cf4b586be0e8feb19dc34f293a1d2078cf398ce3ca253c97db2a0c352073`.

- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:60a2093f539375e58bac443e95b50dbf98cd187090e8d91adb8100ffbc1a0d87`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1f104e993203597e06f02f2080cf9f28583704674402ade6ceb7e601696867a7`.

Independent reconstruction. Implementation hash `sha256:c65f2049ad12f2d618472090042a4f0a56dda8e31fabea284b65ecece6942c09`; independent certificate `sha256:cc744913cdce2a2bb18a73bddf76ea9c4e37f2ca0e0069b43d07769c5174fd49`; external-validation hash `sha256:159aa70869792c4cf42278a5f8d708222b95204bb90bec05f8a8502cf8097e2c`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-B00707-2026

Comparison records:

- bond-order-iupac-b00707: predicted bonding-degree-index__single-bond-reference__localized-electron-pair; source-derived bonding-degree-index__single-bond-reference__localized-electron-pair; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks a bonding-degree index, single-bond reference or localized electron-pair reference, or if a changed row is accepted.

Measurement receipt: `sha256:a120f4b897afe2d75d9c56645abd96804f50c3006662c01685120d99d0dd0d05`. Isolation certificate: `sha256:f2e6d4e1b7cb627ef32b40549779b2c5a63e70d3f4477c60c16f1133333bff48`. Custody certificate: `sha256:9ad5ecdc91597061cdb61a59b8148a89e9bbb210dee0fcc3bd8817f61aa4c5e9`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/B00707/json (sha256:f8056279c1ae14cc144d184f01797de9aa2d8fb69ed868b9753ba2e63255dccf)

Registered target identities:

- bond-order-iupac-b00707 from IUPAC-GOLD-BOOK-B00707-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every finite identified bond with complete joining-channel support, measured relative to one localized electron-pair joining unit using exact positive counts or rational support parts.

Immutable identities. Source manifest `sha256:d871f24eb935e7f08a3195bec4a6482219ee569a0333b2f00807c71660bff421`; derivation seal `sha256:7c78f302c8b6954108b1ad8c1255e7799e7ada125732370731e21f77f16ef140`; engine receipt `sha256:94f413630420c4d031cc217386f564020c765b8af20b9e107e42a15248ee076c` at `receipts/engine/model_admitted/SFT-CHEM-BOND-ORDER-001-94f413630420c4d0.json`; empirical hash `sha256:5eb9dba0e7fc2af3173b31cacbb28ef297d39be81679fa51200553ca76b34c10`; measurement receipt `sha256:a120f4b897afe2d75d9c56645abd96804f50c3006662c01685120d99d0dd0d05`.

31. Bond length, energy and identity correspondence

Claim identity: `SFT-CHEM-BOND-LENGTH-STRENGTH-001`

Question and theorem. Bond length and strength are separate source-bounded records paired by one specific bond identity: inter-centre distance and positive dissociation transfer under retained methods and conditions; no universal formula is imported.

> *specific-identified-bond__method-bounded-centre-distance__positive-dissociation-transfer__common-bond-identity-pairing*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-CONDENSED-LATTICE-001`
- `SFT-PHYS-CONDENSED-BAND-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-VALENCE-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-CHEMICAL-BOND-001`
- `SFT-CHEM-BOND-ORDER-001`

Generated grammar. Generate the literal product of the registered length/strength carrier, length, strength, conditions, correspondence, measurement, record and extension choices.

Boundary: Every identified bond with one source/method-bounded inter-centre distance record and one separately source/condition-bounded positive dissociation-transfer record, joined only by common bond identity.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `specific-identified-bond__method-bounded-centre-distance__positive-dissociation-transfer__measurement-conditions-retained__common-bond-identity-pairing__separate-source-bound-records__bond-length-strength-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`specific-identified-bond`	`generic-bond-type-only`	A type average can erase the specific molecular carrier.
`length`	`method-bounded-centre-distance`	`context-free-distance`	Measured bond length depends on method and state.
`strength`	`positive-dissociation-transfer`	`qualitative-strong-weak-label`	A qualitative label cannot reproduce dissociation.
`conditions`	`measurement-conditions-retained`	`conditions-erased`	Phase, state and cleavage path affect measured records.
`correspondence`	`common-bond-identity-pairing`	`universal-inverse-equation`	A categorical relation does not force one universal formula.
`measurement`	`separate-source-bound-records`	`decimal-as-proof-scalar`	Measured decimals cannot enter derivation.
`record`	`bond-length-strength-source-trace`	`paired-values-without-provenance`	A pair without methods cannot be compared.
`extension`	`no-extra-rule`	`free-length-strength-fit`	A fitted curve can force measured examples.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7e1b3cf5cdff81e62b593729ae380c6b6151fbb768b02e8bf683fd10d67c2c8f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:6d7e7989df91ca411877aaff4507ef5c4000a87e17468950721c429d0230f768`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fd7a974bad4e601cb6c1b60bb8f0389b4d00b597b8529173ea28367aedbff8f9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1026b70bc7c6d6d1fcaac20815b08bbbd0f66f656c3ee48a5cb01d4a3d22f86b`.

Independent reconstruction. Implementation hash `sha256:1ab7ee39098a0f17e5dc2489b48970f1f57d1444593a8701101beed99fc8fd04`; independent certificate `sha256:93ff0d4b30b1dc3cadf664a6d1271083634771cc896a074b5af4ad60dc8dc0b2`; external-validation hash `sha256:2322cd378ddc66143720e6084d3f91d582d9d9801cbff53a85e6fb1a0ae48b75`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-B00702-2026

Comparison records:

- bond-length-strength-iupac-b00702: predicted bond-length__bond-dissociation-energy__bond-order-empirical-relationship; source-derived bond-length__bond-dissociation-energy__bond-order-empirical-relationship; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if IUPAC lacks the registered empirical relationship among bond length, dissociation energy and order, or if a changed row is accepted; it does not fail merely because no universal monotone formula exists.

Measurement receipt: `sha256:bd6ee84cfb1fb9de9052f1412a4dd673ea49b85210da8a90d81a346348716a5f`. Isolation certificate: `sha256:8e095fef1f0d2c5f638d7cb5830d7a85081e2dca40418e5022d8f83adf882cea`. Custody certificate: `sha256:970841320705bb9ec75dc76317352a76cb270336e008f70cd9cf03ea8ec416b2`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/B00702/json (sha256:f53dadb6357406a52d41d9d4be1f7698d4db3b8954b04f798ba823b18c40c821)

Registered target identities:

- bond-length-strength-iupac-b00702 from IUPAC-GOLD-BOOK-B00702-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no IUPAC bond definition, orbital equation, electronegativity scale, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application result or opaque predictor
- no universal bond-length/strength formula inferred from a categorical correspondence
- official target content opens only through post-seal custody
- Every identified bond with one source/method-bounded inter-centre distance record and one separately source/condition-bounded positive dissociation-transfer record, joined only by common bond identity.

Immutable identities. Source manifest `sha256:2fb0972cd501ccaab9ccf8eb29080835cb64a954472f1f57fc6057a55dda2863`; derivation seal `sha256:600aa8b89c38a43f5c825c0f4550ff6465e06160764917147abe3a813823ad49`; engine receipt `sha256:83ea09203b02f1e1f9872073bbb65b2e2609785c15d465de4d006276ee54aada` at `receipts/engine/model_admitted/SFT-CHEM-BOND-LENGTH-STRENGTH-001-83ea09203b02f1e1.json`; empirical hash `sha256:20fe93c8b49082a6682be04e1c9c0e2034c5846046d7133e1d05dda320a16586`; measurement receipt `sha256:bd6ee84cfb1fb9de9052f1412a4dd673ea49b85210da8a90d81a346348716a5f`.

32. Molecular identity and complete bonded carrier

Claim identity: `SFT-CHEM-MOL-MOLECULE-001`

Question and theorem. A molecule is one finite neutral multi-atomic carrier whose complete bonded graph is connected and whose constituent occurrence and element identities remain retained.

> *multiple-atomic-occurrences__complete-bond-support__one-connected-component__closed-charge-boundary__constituent-identities-retained*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`

- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001

Generated grammar. Generate the literal product of the registered molecule cardinality, bonding, connectivity, charge, composition, identity, record and extension choices.

Boundary: Every finite generated multi-atomic carrier with complete connected bond support, retained constituent identity and no open net-charge fibre at the registered observation boundary.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `multiple-atomic-occurrences__complete-bond-support__one-connected-component__closed-charge-boundary__constituent-identities-retained__complete-structure-identity__complete-carrier-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`cardinality`	`multiple-atomic-occurrences`	`single-atomic-occurrence`	One occurrence has no internal molecular bond graph.
`bonding`	`complete-bond-support`	`unbonded-collection`	A collection without joining support is not one molecule.
`connectivity`	`one-connected-component`	`disconnected-components`	Disconnected components are separate carriers.
`charge`	`closed-charge-boundary`	`open-net-charge-fibre`	An open net fibre describes a molecular ion rather than the neutral molecular class.
`composition`	`constituent-identities-retained`	`anonymous-atom-count`	Counts without element identities cannot reproduce the carrier.
`identity`	`complete-structure-identity`	`formula-alone`	Equal composition need not imply one molecular structure.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`record`	`complete-carrier-trace`	`molecule-label-only`	A label cannot reconstruct atoms and bonds.
`extension`	`no-extra-rule`	`free-molecule-exception`	An exception can admit arbitrary aggregates.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bbed887164a606acb5cca576ab0d7da79e9254ab958d14e368f2d1b686224b7c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d21385a0ac8db65b285a1e5cf9b46eb58fb7d306b196d5c08fd900001d7916b1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:811a93ff2e246131afd67a2e53ecba9ba2d2c083c686188c278f8e537ee69785`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:809480cc84ffd38484c8ef07fe549a9a9e44f9c25a0943073cc937513b52886d`.

Independent reconstruction. Implementation hash `sha256:22db6e9a5a35d9e670f83677a119a13cb1eb94927c93ed2e9e10636410408ce1`; independent certificate `sha256:94b03b38ed013a00a089d3cb57cc45f51fba0e3cacc8ccc0b6697a98eb70c366`; external-validation hash `sha256:82e27bfe0c7c5647a8c165bb3d152c7d144675b2faad6e0d9b5f99b0cdf71104`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-M04002-2026

Comparison records:

- molecule-iupac-m04002: predicted multi-atomic-carrier__connected-complete-bond-support__closed-charge-boundary__constituent-identities-retained; source-derived multi-atomic-carrier__connected-complete-bond-support__closed-charge-boundary__constituent-identities-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official molecular record lacks multiple atomic constituents, one bonded chemical entity or retained constituent identity, or if a changed row is accepted.

Measurement receipt: `sha256:0a83e619481da1819ecc2cbf502875fd3e8bfdd67c1206492aebbed89d05e64a`. Isolation certificate: `sha256:c1841e6b99452e3af115174b89db9fe0f70ba921bdb7f1797d4a8462b303f047`. Custody certificate: `sha256:873154cb8a5e98c359916e137641c6ca06699e1fb54d05c94f001d0deca9a5ed`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/M04002/json (sha256:a5058dad716a0233d1a49fb8869d30b581cff6f483317ffdaaf10a3293c52e36)

Registered target identities:

- molecule-iupac-m04002 from IUPAC-GOLD-BOOK-M04002-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite generated multi-atomic carrier with complete connected bond support, retained constituent identity and no open net-charge fibre at the registered observation boundary.

Immutable identities. Source manifest `sha256:8c55560f345efdaf63aac199a5026cf893b0402107d6bf9325f71ea39711da44`; derivation seal `sha256:53751882f0e9e131f435d27edd8e68a879348afcc6e857d5acf8372250f84f66`; engine receipt `sha256:5aa379d0b14c341e04c6a041483415f94a04f45833ccb11af5277e53257ee7ce` at `receipts/engine/model_admitted/SFT-CHEM-MOL-MOLECULE-001-5aa379d0b14c341e.json`; empirical hash `sha256:34affa520b3d0bf26b1d70594cb1a9060663e08cebcdb01f57e08ae5c40d7b89`; measurement receipt `sha256:0a83e619481da1819ecc2cbf502875fd3e8bfdd67c1206492aebbed89d05e64a`.

33. Molecular geometry from held adjacency and orientation

Claim identity: `SFT-CHEM-MOL-GEOMETRY-001`

Question and theorem. Molecular geometry is the complete relative adjacency-and-orientation form of one identified finite molecular carrier, invariant under a global rigid host-frame relabelling and separate from measured coordinate values.

> *identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__complete-pair-support__rigid-relabel-equivalence*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`

- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001

Generated grammar. Generate the literal product of the registered geometry carrier, adjacency, orientation, completeness, equivalence, measurement, record and extension choices.

Boundary: Every finite admitted molecular carrier with a complete pairwise relative-orientation record, interpreted up to one global rigid relabelling and retained with its source and observation conditions.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__complete-pair-support__rigid-relabel-equivalence__source-bounded-geometry-record__adjacency-orientation-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	identified-molecular-carrier	geometry-without-molecule	A shape detached from an identified carrier is not molecular geometry.
adjacency	complete-atom-adjacency	atom-list-only	An atom list omits which occurrences are joined.
orientation	held-relative-orientation	distance-only-record	Distances alone can omit handed or ordered orientation.
completeness	complete-pair-support	selected-coordinate-subset	A favorable subset cannot establish the whole geometry.
equivalence	rigid-relabel-equivalence	absolute-host-frame	A host coordinate frame cannot define chemical identity.
measurement	source-bounded-geometry-record	coordinate-as-proof-premise	A measured coordinate cannot select the law.
record	adjacency-orientation-source-trace	shape-name-only	A shape name cannot reproduce the relations.
extension	no-extra-rule	free-angle-fit	A fitted angle can force a target geometry.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:37ade72544361890f8b377d0be9fb8246aad5cccc01e85702f531dfea43806cf`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2eef597bf64b1918b1dfd70e164ce86c8560f1d6cefe455e285fb5744b445b4a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:42d88d1105e6572412e6725c928b48cf33f934003dcae249480f642068779791`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:b52da0b02b3272f33b9bbfb53e964d411909027aac6b11fac7589e1149d05c24.

Independent reconstruction. Implementation hash sha256:4f780754facfb7667f6db5ee9eedb8ae99aaf3ea8043383e94de88c4647caafc; independent certificate sha256:8031cf31bbfbebc30b574d6ea6d4361d68c9fdb6cda3ce73f851af5f6bb1c512; external-validation hash sha256:ebe9f30414581e4ed31c537b8d8140ccb10e1d98b7bf74c37a6a04bac972a36f. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-S06061-2026

Comparison records:

- molecular-geometry-iupac-s06061: predicted identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__source-bounded-geometry-record; source-derived identified-molecular-carrier__complete-atom-adjacency__held-relative-orientation__source-bounded-geometry-record; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative geometry record lacks identified atoms, relational spatial organization or source-bounded coordinates, or if a changed row is accepted.

Measurement receipt: sha256:754bd429fa783348692940e3a757fbadb588fc1cd3d825a5db48053eacf78528. Isolation certificate: sha256:84176b1e992d227e441c27274b7af000342dc0ca13f2a2ac572f0d4bbabf7277. Custody certificate: sha256:c08889b7bb14bf7a4c0193bb6f2461f21670d180a912b7f699ebfa877d4aad2a.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/S06061/json (sha256:65cc9feaea028f9ab7bd7ed349276522555ebcc66d2f079dc94d55fe9dc7230b)

Registered target identities:

- molecular-geometry-iupac-s06061 from IUPAC-GOLD-BOOK-S06061-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite admitted molecular carrier with a complete pairwise relative-orientation record, interpreted up to one global rigid relabelling and retained with its source and observation conditions.

Immutable identities. Source manifest sha256:1f5e5f023b5b2dc5d4a9e4125a3642632913cdd52ce054a352ddc9b02c38c380; derivation seal sha256:7c7c69d3817c8e880c5621d6516d1773cf534ef508972d15409ca78c271f8cc2; engine receipt sha256:32416047fc77a3215966466b4a829e4658bcfb50c14652b3425b9aa1e8603a89 at receipts/engine/model_admitted/SFT-CHEM-MOL-GEOMETRY-001-32416047fc77a321.json; empirical hash sha256:e57949262e42f976fe560cebebf3e96e91920b87aaaba582d0688c8fd102d961; measurement receipt sha256:754bd429fa783348692940e3a757fbadb588fc1cd3d825a5db48053eacf78528.

34. Isomer distinction under equal composition

Claim identity: `SFT-CHEM-MOL-ISOMER-001`

Question and theorem. Isomers have exactly equal retained composition but inequivalent complete connectivity or orientation signatures after every identity-preserving finite relabelling.

> *two-identified-carriers__equal-exact-composition__complete-structure-signatures__all-identity-preserving-relabelings __connectivity-or-orientation-difference*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-CHEMICAL-BOND-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-BOND-ORDER-001`
- `SFT-CHEM-BOND-LENGTH-STRENGTH-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-GEOMETRY-001`

Generated grammar. Generate the literal product of the registered isomer pair, composition, structure, equivalence, distinction, identity, record and extension choices.

Boundary: Every finite pair of admitted chemical carriers with exactly equal composition signatures but unequal complete connectivity or held-orientation signatures after every identity-preserving relabelling.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `two-identified-carriers__equal-exact-composition__complete-structure-signatures__all-identity-preserving-relabelings__connectivity-or-orientation-difference__separate-chemical-identities__composition-and-structure-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`pair`	`two-identified-carriers`	`single-carrier-description`	Isomerism compares at least two chemical carriers.
`composition`	`equal-exact-composition`	`different-compositions`	Different composition is not isomerism.
`structure`	`complete-structure-signatures`	`formula-equated-identity`	A formula does not exhaust bonded organization.
`equivalence`	`all-identity-preserving-relabelings`	`name-string-inequality`	Different names can denote the same structure.
`distinction`	`connectivity-or-orientation-difference`	`equivalent-signatures`	Equivalent complete signatures denote the same isomeric form.
`identity`	`separate-chemical-identities`	`distinction-erased`	Erasing the difference destroys the chemical question.
`record`	`composition-and-structure-trace`	`isomer-label-only`	A label cannot reproduce equality and difference.
`extension`	`no-extra-rule`	`free-isomer-class`	A free class label can manufacture a distinction.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:cff91e3ddb362ea4243a3a751487636d8cbe5db830e215e91e1e8691584f0dcf`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c2dbbbc0bbb52a72785bc5da5d04267b55b28aa26d80002fe05bd5ed7543bbcc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:dc161f389ebcdec14659c79d57ce7f512f7f193d1ed845be8b7ae8da6361f5e3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ab3dfb81b50681603de1f62d122810cad4e7ad1bfa16ac3d1abc9d1abe830290`.

Independent reconstruction. Implementation hash `sha256:7aa0f2f5ddb4f83e66d9ad7ac46a812691bd7707808f34962af813d170388c67`; independent certificate `sha256:c465beb4071d7c4bbdf65b73e04f4e05d9f999142100dbbc350ac265c948d679`; external-validation hash `sha256:2a1fb4a835a698f1634a646f736dc6c10fb8fc4818de777f3920c7f3f9eab559`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03289-2026

Comparison records:

- isomer-iupac-i03289: predicted equal-composition__distinct-structural-signature__connectivity-or-orientation-distinction__separate-chemical-identity; source-derived equal-composition__distinct-structural-signature__connectivity-or-orientation-distinction__separate-chemical-identity; exact match True

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official isomer record does not require equal composition with distinct structure or spatial arrangement, or if a changed row is accepted.

Measurement receipt: `sha256:98b7ffae634ea86206b13f92518e34e5f2d1b65a4f5e2ace3b00f8cdf2895d62`. Isolation certificate: `sha256:fcdd26096b1379c7435d33ba9361489fb6708319418afb7f7a4915d4833ca0e7`. Custody certificate: `sha256:b9b37e84c0cb3f62e42776f51416789146cca89a56f2da537e7507f90175b448`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/I03289/json (sha256:64fa2c9cc58b18b399d22481ee7f37692c7f9c904e4d5fff27c65dd3325393e7)

Registered target identities:

- isomer-iupac-i03289 from IUPAC-GOLD-BOOK-I03289-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite pair of admitted chemical carriers with exactly equal composition signatures but unequal complete connectivity or held-orientation signatures after every identity-preserving relabelling.

Immutable identities. Source manifest `sha256:daf9b16232a55caa127373480a5fa56e69c990c5c78385a0ce31c76ca3f61746`; derivation seal `sha256:3b90b3058db11528d99a744d04912eddc94fdb979eb6305c822300143f77e849`; engine receipt `sha256:9a9604de7b5433fa0f6630c3840f91143bed30accc853ff8f4ad3495544536f3` at `receipts/engine/model_admitted/SFT-CHEM-MOL-ISOMER-001-9a9604de7b5433fa.json`; empirical hash `sha256:ab05cee07b44a2b1a227414f5f17a4c3639320518f0c1d84ee4507a78eb22ce7`; measurement receipt `sha256:98b7ffae634ea86206b13f92518e34e5f2d1b65a4f5e2ace3b00f8cdf2895d62`.

35. Intermolecular interaction and residual joining

Claim identity: `SFT-CHEM-MOL-INTERMOLECULAR-001`

Question and theorem. An intermolecular interaction is a registered response channel between already complete molecular carriers that retains each constituent molecular identity and supports collective recurrence.

> *distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence__explicit-molecular-boundary*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-GEOMETRY-TOPOLOGY-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-QUANTUM-ENTANGLEMENT-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-FORMULA-001
- SFT-CHEM-STOICH-COMPOSITION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-CHEMICAL-BOND-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-BOND-METALLIC-001
- SFT-CHEM-BOND-ORDER-001
- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001

Generated grammar. Generate the literal product of the registered intermolecular carrier, support, identity, recurrence, boundary, strength, record and extension choices.

Boundary: Every finite interaction support joining two or more already complete molecular carriers while retaining each molecular identity and the boundary between inter- and intramolecular support.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence__explicit-molecular-boundary__source-bounded-interaction-record__carrier-channel-recurrence-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carriers`	`distinct-molecular-carriers`	`one-molecule-internal-edge`	An internal edge is intramolecular support.
`support`	`between-whole-interaction-support`	`proximity-without-channel`	Proximity alone supplies no interaction trace.
`identity`	`constituent-identities-retained`	`constituents-merged-erased`	Erasing each molecule changes the class to a new bonded carrier.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`recurrence`	`collective-recurrence`	`transient-coincidence`	A coincidence without response cannot organize an assembly.
`boundary`	`explicit-molecular-boundary`	`inter-intra-conflation`	Conflation makes every covalent edge intermolecular.
`strength`	`source-bounded-interaction-record`	`universal-weaker-law`	Not every context forces one strength ordering.
`record`	`carrier-channel-recurrence-trace`	`force-name-only`	A force name cannot reconstruct endpoints and response.
`extension`	`no-extra-rule`	`free-force-family`	An imported family can select desired interactions.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8c200aa3e761e0e8e583df86cea1e401ef6ac0666629e7e42bf01790e147cfaa`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:66170d1985dd0b0b805f940cb21d00d932c3bc1d354a0ee4387e3f3b857262c0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d5722e6b4ee1020dd3d52b8d3223fe482d257809ca683e78114888c2684038a1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fa996e1ad0ab9e877ebe04f8ca07a8deb28146632f4b5b5e8a2ee67cc8a7cb34`.

Independent reconstruction. Implementation hash `sha256:d215c873502192ac1d7219b514b59406b547ce9b6be65c68c8902105a89cde0f`; independent certificate `sha256:1eb83e9304bfe476b0b2cd840e20e5867dd505fb5a069a8f8f95d35db08aeb49`; external-validation hash `sha256:0cd075555765f7f509d1845af1df851a092d3f265ba0815ec0d761fa9716f186`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03098-2026

Comparison records:

- intermolecular-iupac-i03098: predicted distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence; source-derived distinct-molecular-carriers__between-whole-interaction-support__constituent-identities-retained__collective-recurrence; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official record lacks interaction between distinct molecules or preserved molecular endpoints, or if a changed row is accepted.

Measurement receipt: `sha256:65d7ac7e6aaed6f7c27647ba3d5d7dfca452b9e02cefabea2145fe4bc57d0126`. Isolation certificate: `sha256:0e5f0f13353f306255f7a322f5a2d28d5208caa83bbdc70c629a7b3375a6a760`. Custody certificate: `sha256:d12f4da2ed33d511ff213173300012d603acba468e9ab374f7b8fbb17a41d447`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/I03098/json (sha256:ba7264c36447a463893ffdfe66e0bce7478e764db59b928911a8f82b59af4b3e)

Registered target identities:

- intermolecular-iupac-i03098 from IUPAC-GOLD-BOOK-I03098-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite interaction support joining two or more already complete molecular carriers while retaining each molecular identity and the boundary between inter- and intramolecular support.

Immutable identities. Source manifest `sha256:09a428c01beb574236e6d875f85e3f30281d2afa9c24e3ca3ca41b8d10365f77`; derivation seal `sha256:b3845294f802bd5cbb48345120d9351b6f86e33dc0fc7a3e5abf10bd4588749c`; engine receipt `sha256:ba1eabd232574b380b85aff50567257eb0b4c5318ba79e238410a737c38ea7ef` at `receipts/engine/model_admitted/SFT-CHEM-MOL-INTERMOLECULAR-001-ba1eabd232574b38.json`; empirical hash `sha256:de6ce0f2148ee584439dbc21a4562e84e201f9fb05b120a83ee9ed54b4435a3d`; measurement receipt `sha256:65d7ac7e6aaed6f7c27647ba3d5d7dfca452b9e02cefabea2145fe4bc57d0126`.

36. Supramolecular organization by reversible recognition

Claim identity: `SFT-CHEM-MOL-SUPRAMOLECULAR-001`

Question and theorem. A supramolecular whole is a finite assembly of multiple intact molecular carriers joined by reversible complementary recognition support, with all components recoverable.

> *multiple-molecular-components__noncovalent-interaction-support__complementary-recognition__reversible-component-release__assembled-whole-identity*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`

- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-CHEMICAL-BOND-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-BOND-ORDER-001`
- `SFT-CHEM-BOND-LENGTH-STRENGTH-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`

Generated grammar. Generate the literal product of the registered supramolecular components, joining, recognition, reversal, closure, scope, record and extension choices.

Boundary: Every finite assembly of multiple complete molecular carriers joined through reversible recognition support, with all components recoverable and the collective whole separately distinguishable.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `multiple-molecular-components__noncovalent-interaction-support__complementary-recognition__reversible-component-release__assembled-whole-identity__finite-declared-assembly__component-recognition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`components`	`multiple-molecular-components`	`single-molecule-only`	A supramolecular assembly contains multiple molecular components.
`joining`	`noncovalent-interaction-support`	`new-internal-covalent-graph`	Replacing components by one covalent graph erases the supramolecular boundary.
`recognition`	`complementary-recognition`	`arbitrary-collision`	A collision has no complementary selection trace.
`reversal`	`reversible-component-release`	`components-unrecoverable`	If components cannot be recovered, their identities were not retained.
`closure`	`assembled-whole-identity`	`components-only`	Separate components do not form an assembled whole.
`scope`	`finite-declared-assembly`	`completed-infinite-assembly`	A completed infinity is not generated.
`record`	`component-recognition-trace`	`assembly-name-only`	A name cannot reconstruct recognition and release.
`extension`	`no-extra-rule`	`free-host-guest-rule`	An exception can force any desired assembly.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4624527ef8a225f350b93ce389704a0ca9f1c3f27a4fea4781d818ed9fd8b1d5`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:39aa04f9f35d247da547a24dd03d6d7994e5812a23cd1563bb5fd6a73ee2f846`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:af9101491f206ffe81a9ab14550635b603e45d1c4ee776f501cb4326b4556fdb`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:d14f7bef34eb9b65bd346b19ba3fc422b78107c054db310315f56e20fd7e11ec`.

Independent reconstruction. Implementation hash `sha256:0a59a72d662c2a0a8f04c59801c5cf4384a68ee42c609bc133443b8c3589b0e4`; independent certificate `sha256:1d0d5cadf95fd4ea413ff2a3d3fcb215d96a64f1e82e125012659bd464ffa85c`; external-validation hash `sha256:f7abcdbd1385de8ccb0fc0c71462f91c8a4f9fa6839e5181c2e26d080baa871b`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-13867-2026

Comparison records:

- supramolecular-association-iupac-13867: predicted multiple-molecular-components__reversible-noncovalent-recognition__component-identities-retained__assembled-whole; source-derived multiple-molecular-components__reversible-noncovalent-recognition__component-identities-retained__assembled-whole; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official supramolecular record lacks multiple molecular components, noncovalent organization or retained component identities, or if a changed row is accepted.

Measurement receipt: `sha256:fe72bd84f6bd8a08a2daf4fe24ebf8643641c4f623bfec35a75fe37fb77ef829`. Isolation certificate: `sha256:9e8fac94eeb9b3319898781caf37f407c79baf98ca43f6fd46b87451325bc9f4`. Custody certificate: `sha256:4c43539c1e8c358ca5ddc4f8c4b60c1bf035777d3d1936c3dcc87d58340ff926`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/13867/json (sha256:34dd0e2a436c762eefbb6ddf8ab84b16b716b252632fc0835ec083d4d4a2551a)

Registered target identities:

- supramolecular-association-iupac-13867 from IUPAC-GOLD-BOOK-13867-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed

- Every finite assembly of multiple complete molecular carriers joined through reversible recognition support, with all components recoverable and the collective whole separately distinguishable.

Immutable identities. Source manifest `sha256:ef01da7f61edfa7a0bb00febaa3923c6f9494745cb856e97390422195c613802`; derivation seal `sha256:d600cd2296f751be3bbbc930c434a149828066f7103a0cf968ce8a0eedc1aaef`; engine receipt `sha256:03bc9e8b4e9ac2b3ef0a87f81bafeba98a31955ed55b226936c51f4f6dea045b` at `receipts/engine/model_admitted/SFT-CHEM-MOL-SUPRAMOLECULAR-001-03bc9e8b4e9ac2b3.json`; empirical hash `sha256:24a3138fd3845b956857932d3beef77d0dd313d332bfb1a1ecec9413382ccf83`; measurement receipt `sha256:fe72bd84f6bd8a08a2daf4fe24ebf8643641c4f623bfec35a75fe37fb77ef829`.

37. Molecular network and connected chemical whole

Claim identity: `SFT-CHEM-MOL-NETWORK-001`

Question and theorem. A molecular network is one finite maximal connected component of named molecular nodes and registered interaction edges with complete reachability and provenance.

> *finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-reachability-census__finite-maximal-boundary*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-GEOMETRY-TOPOLOGY-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-MECH-LOCATION-DISPLACEMENT-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-QUANTUM-ENTANGLEMENT-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-FORMULA-001`
- `SFT-CHEM-STOICH-COMPOSITION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-CHEMICAL-BOND-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-BOND-METALLIC-001`
- `SFT-CHEM-BOND-ORDER-001`

- SFT-CHEM-BOND-LENGTH-STRENGTH-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-MOL-SUPRAMOLECULAR-001

Generated grammar. Generate the literal product of the registered molecular-network nodes, edges, connectivity, paths, boundary, composition, record and extension choices.

Boundary: Every finite generated set of molecular nodes and registered interaction edges whose maximal declared component is connected, with complete node, edge, path and boundary provenance.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-reachability-census__finite-maximal-boundary__node-identities-retained__complete-network-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
nodes	finite-molecular-node-set	anonymous-node-count	A count cannot recover molecular identities.
edges	registered-interaction-edges	implicit-relatedness	An implicit relation cannot be audited.
connectivity	connected-maximal-component	disconnected-union-called-one	Disconnected components are not one connected whole.
paths	complete-reachability-census	selected-path-only	One favorable path cannot establish network connectivity.
boundary	finite-maximal-boundary	unbounded-network-name	An unbounded name does not enumerate a finite object.
composition	node-identities-retained	network-erases-molecules	Nodes remain chemical carriers, not anonymous points.
record	complete-network-trace	network-answer-only	A connectivity answer cannot reproduce the network.
extension	no-extra-rule	free-connectivity-edge	Adding an ungenerated edge can force connectedness.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:aecb155adfee2fa5c2569fcce2754efa066481a4569263794de20d763d1eb430`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:88618a1b00901ee0c581709ea8f5ed231b4d9e37c51f0c9d5a4ec7d4bcb94730`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0d5086ed6a62b7e51f3ae5bef5470bf27fc01435c48ec2fd806f412699448e52`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:134c1b01b41fb77ae299a443f0fe0be5edc83e08dd803e6331c14bdef0cefc7e`.

Independent reconstruction. Implementation hash `sha256:4803b87a071efe0f05f28748a53f932a80b12b5025d108a6bceb53444e66f1cf`; independent certificate `sha256:1c37d683d2ac8ff6c01fb6d7e2afe60fa7ea9d0dc2d0ba54915c360000eaacdf`; external-validation hash `sha256:800745f1353e9c562fa7aaa3fcf1e36bea7c949cc9b19d1da62c601ed3d10222`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-N04112-2026

Comparison records:

- molecular-network-iupac-n04112: predicted finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-network-trace; source-derived finite-molecular-node-set__registered-interaction-edges__connected-maximal-component__complete-network-trace; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the official molecular-network record lacks molecular nodes, interaction links or connected organization, or if a changed row is accepted.

Measurement receipt: `sha256:712805a35f631a8453bbd1d28c9949b1f18f3cd8bb0701420342432ef5e6367c`. Isolation certificate: `sha256:f2ed1f3080498cff0b42b999c2dc09f5abdeae96f3b6ce9644412ef5d07f541b`. Custody certificate: `sha256:e28b6e1bb189bf2708b659fe46f141792eea9ea12591fb2bbed521a5d0e33cc1`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/N04112/json (sha256:a146d63f8340f0cf5043e1f9fe2f1cfa8e5fa7c3bf27b8a409e665f40ae2c046)

Registered target identities:

- molecular-network-iupac-n04112 from IUPAC-GOLD-BOOK-N04112-2026

Meaning. Bonding and molecular structure are exact joint recurrences over held atomic, electron, adjacency and orientation support rather than imported molecular pictures. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no chemical dictionary, coordinate archive, fitted force field, measured geometry or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no application or opaque prediction result
- no claim that topology alone predicts every measured coordinate or energy
- external content remains inaccessible until after the consequence is sealed
- Every finite generated set of molecular nodes and registered interaction edges whose maximal declared component is connected, with complete node, edge, path and boundary provenance.

Immutable identities. Source manifest `sha256:3643fb6d040ae6d9e29e381e91403a69074a6e3d83482714d345f4e1f54e1bff`; derivation seal `sha256:7485b6cd62d05d1076ddd8a17d89baef6ded9be36adf7a1022549417fddc76be`; engine receipt `sha256:b8aca3860e24266b5d87c32e1ed0dff589097626ee821b8472a8849132ca0041` at `receipts/engine/model_admitted/SFT-CHEM-MOL-NETWORK-001-b8aca3860e24266b.json`; empirical hash `sha256:ce8ed0c73af4cc4a3e2a77988b055e034813888ba52cbaec158fcc452573363c`; measurement receipt `sha256:712805a35f631a8453bbd1d28c9949b1f18f3cd8bb0701420342432ef5e6367c`.

13. Acid Base Redox

Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier.

38. Conjugate acid-base partition

Claim identity: `SFT-CHEM-AB-ACID-BASE-001`

Question and theorem. A conjugate acid/base relation is a pair of retained chemical identities differing by exactly one named proton carrier: the protonated member donates it and the deprotonated member accepts it.

> *two-conjugate-identities__one-held-proton-occurrence__proton-donor-role__proton-acceptor-role__differ-by-one-proton*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`

Generated grammar. Generate the literal product of the registered conjugate pair, carrier, acid role, base role, difference, orientation, record and extension choices.

Boundary: Every finite generated pair of chemical carriers related by exactly one retained proton occurrence, with the protonated member able to donate that occurrence and the deprotonated member able to accept it.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `two-conjugate-identities__one-held-proton-occurrence__proton-donor-role__proton-acceptor-role__differ-by-one-proton__held-donor-acceptor-orientation__pair-and-proton-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`pair`	`two-conjugate-identities`	`one-species-label`	A conjugate relation compares two carrier identities.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`one-held-proton-occurrence`	`anonymous-hydrogen-count`	A count cannot trace the transferred occurrence.
`acid-role`	`proton-donor-role`	`acid-by-name`	A name cannot establish a transition.
`base-role`	`proton-acceptor-role`	`base-by-name`	A name cannot establish a transition.
`difference`	`differ-by-one-proton`	`unbounded-composition-change`	Additional changes destroy conjugacy.
`orientation`	`held-donor-acceptor-orientation`	`signed-acidity-scalar`	A signed scalar is not a carrier trace.
`record`	`pair-and-proton-trace`	`acid-base-answer-only`	An answer cannot reconstruct conjugacy.
`extension`	`no-extra-rule`	`free-acid-base-exception`	An exception can label arbitrary species.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c636c0c317cb207c4839f718a739e5a683b6040155125536f9280ca737ac0e4c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:05b2c6729d885da3646df12d9ee9a76888e057a0da289866cd518789a4558085`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f316b770b291518f418e84e5b494397d05ba8c8ec8525bd5a01ff8d30b72f537`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:408b29d1a7a54536d715a8da4f7f9947a771419db6481093b17a405f5ae716f4`.

Independent reconstruction. Implementation hash `sha256:69b4921f1868f68bf465f91a281a6c5f67e8da2de9f06b518a49325116c84b52`; independent certificate `sha256:88152d90d6de4349d732338892b209665ac0f6dff91ab5a7099a95bf1bfd25c4`; external-validation hash `sha256:0a138d9393293cbf97515a96c0640e5006a047eae2c093cae2fcce0201ff7775`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ACID-BASE-COMPOSITE-2026

Comparison records:

- conjugate-acid-base-iupac-composite: predicted proton-donor-acid__proton-acceptor-base__conjugates-differ-by-one-proton__paired-relation; source-derived proton-donor-acid__proton-acceptor-base__conjugates-differ-by-one-proton__paired-relation; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative conjugate acid/base record lacks proton donation, proton acceptance or a one-proton conjugate relation, or if a changed row is accepted.

Measurement receipt: `sha256:bfd39f473b2a824e2dc08fb73a6b5953e41112556bf93ecee09cd63dc43c3583`. Isolation certificate: `sha256:06b0a1e27a4cee94ced3a9a3a2c26e6f30c32a4b580c7822559f02a105ca5dae`. Custody certificate: `sha256:4108990961c157bd1e0b13333bc3ad0a9cbf8374f9e750c1da09abc435da09af`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01266 (sha256:1ac0d6138e316d6230d10b2e7f99d4837175fcb640e824a8cb2249100221a504)

Registered target identities:

- conjugate-acid-base-iupac-composite from IUPAC-GOLD-BOOK-ACID-BASE-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite generated pair of chemical carriers related by exactly one retained proton occurrence, with the protonated member able to donate that occurrence and the deprotonated member able to accept it.

Immutable identities. Source manifest `sha256:08459031c560cd3795677b3e9f2daac98746438dfef7bdc06ff3f87853fdac63`; derivation seal `sha256:76cfcdb1dedf829c2eb2b856cf3b614fe25284574d69d330651d2913b426dd53`; engine receipt `sha256:2f52c0900674b54387bc0168fc5c57ecc9ced3819c0423e31aa7bb0f39b9fdf7` at `receipts/engine/model_admitted/SFT-CHEM-AB-ACID-BASE-001-2f52c0900674b543.json`; empirical hash `sha256:66b231bbb4bc7172986aafbe91d31f8caecafa83ee356ca5b9ffae7c5378e1b5`; measurement receipt `sha256:bfd39f473b2a824e2dc08fb73a6b5953e41112556bf93ecee09cd63dc43c3583`.

39. Proton-transfer acid-base relation

Claim identity: `SFT-CHEM-AB-PROTON-TRANSFER-001`

Question and theorem. Proton transfer moves one named proton carrier from one registered donor occurrence to one distinct acceptor occurrence while conserving carrier identity and both endpoint records.

> *one-named-proton-carrier__registered-donor__registered-acceptor__donor-to-acceptor-transfer__same-carrier-before-after*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`

- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001

Generated grammar. Generate the literal product of the registered proton carrier, source, destination, transition, identity, closure, record and extension choices.

Boundary: Every finite chemical transition in which one named proton carrier leaves one registered donor occurrence and enters one distinct registered acceptor occurrence with identical carrier identity before and after.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `one-named-proton-carrier__registered-donor__registered-acceptor__donor-to-acceptor-transfer__same-carrier-before-after__complete-endpoint-closure__carrier-endpoint-transition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
carrier	one-named-proton-carrier	proton-count-only	A count cannot establish conservation of an occurrence.
source	registered-donor	source-erased	Without a donor, the carrier could be created freely.
destination	registered-acceptor	destination-erased	Without an acceptor, the carrier could be lost.
transition	donor-to-acceptor-transfer	copy-or-delete	Copying or deleting violates carrier conservation.
identity	same-carrier-before-after	before-after-carrier-change	A changed carrier is not transfer conservation.
closure	complete-endpoint-closure	unpaired-composition-change	Unpaired change leaves the reaction record open.
record	carrier-endpoint-transition-trace	product-label-only	A product label cannot reconstruct the path.
extension	no-extra-rule	free-proton-source	A free source can force any reaction.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:69777e7c94f864618f23bb8c316cb4755aa69932bbd5024824f7d651970eb1ed`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c4d8f6608680b8ea69ff9a3d1aff3134aeb226b21fc7553992373580dbbb6b56`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9e07bea854076486d00d939b92c32227e48d5576783837be7dc0119c913f1f5f`.

- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7183352ddde61f7554681ce6e9790f2a1b8e9c929bdf68cb50845f22b1c10ca3`.

Independent reconstruction. Implementation hash `sha256:4256a0b14274123ec7c79959d029bf6359f031402bce163da90a11f367076c4d`; independent certificate `sha256:59456ef8b2f6dbb57ed62c61f0c0384384fd8aa61d956eb93ad1981bf8a4102a`; external-validation hash `sha256:9ffb40fdfac3a2ca4d200b00134403d075d06a6967abe711854e93ae994710af`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-P04915-2026

Comparison records:

- proton-transfer-iupac-p04915: predicted one-proton-carrier__donor-to-acceptor-transfer__carrier-conserved__endpoint-identities-retained; source-derived one-proton-carrier__donor-to-acceptor-transfer__carrier-conserved__endpoint-identities-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative proton-transfer record lacks a proton donor, proton acceptor or conserved transfer relation, or if a changed row is accepted.

Measurement receipt: `sha256:27a7bad3dc157b9bbbb078e311faa01b52b090e49db53b82e626cb7f4ed513d1`. Isolation certificate: `sha256:da496511cabd9493a6c94231218cf200d4cc1da32cc5c24a6b92ff8ed59269e2`. Custody certificate: `sha256:a0c0faa1c58b19f307ca52e35599f0e7c27616c591c14e8e095ea14c744f2b3b`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/P04915 (sha256:72d9b1431f3daa6b0190094d8400bc73b31e786b7d69b4052632297b4b709d33)

Registered target identities:

- proton-transfer-iupac-p04915 from IUPAC-GOLD-BOOK-P04915-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical transition in which one named proton carrier leaves one registered donor occurrence and enters one distinct registered acceptor occurrence with identical carrier identity before and after.

Immutable identities. Source manifest `sha256:98afb03693336d7bab7a816aad595a967794e469dd77fb106019d9a07b0125c5`; derivation seal `sha256:97df51a71af6e82a16c67b9cb05ebce3b37ab689e84cbaf2676b164f69a65fb6`; engine receipt `sha256:bc5be8027f456ae522262e597279b16717f0831b2edf5008d8c268bc58d9814d` at `receipts/engine/model_admitted/SFT-CHEM-AB-PROTON-TRANSFER-001-bc5be8027f456ae5.json`; empirical hash `sha256:b376664238e671b494333ebeeb4d4e794574a78dce1eaa40f5b7b7b50cde4670`;

measurement receipt `sha256:27a7bad3dc157b9bbbb078e311faa01b52b090e49db53b82e626cb7f4ed513d1`.

40. Electron-pair donor-acceptor relation

Claim identity: `SFT-CHEM-AB-LEWIS-001`

Question and theorem. A Lewis acid/base relation joins a distinct electron-pair donor and acceptor through the same two exclusion-preserving electron occurrences, with both identities retained.

> *distinct-donor-acceptor__two-electron-occurrences__pair-donor-support__pair-acceptor-support__shared-or-transferred-pair*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`

Generated grammar. Generate the literal product of the registered Lewis endpoints, support, donor, acceptor, joining, conservation, record and extension choices.

Boundary: Every finite donor/acceptor pair joined by one exclusion-preserving pair of named electron occurrences, with both electron identities conserved and the donor/acceptor orientation retained.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `distinct-donor-acceptor__two-electron-occurrences__pair-donor-support__pair-acceptor-support__shared-or-transferred-pair__both-electron-identities-retained__endpoint-pair-joining-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`endpoints`	`distinct-donor-acceptor`	`one-unoriented-species`	The relation requires donor and acceptor roles.
`support`	`two-electron-occurrences`	`single-electron-only`	The registered relation concerns paired support.
`donor`	`pair-donor-support`	`electron-pair-name-only`	A name cannot demonstrate available support.
`acceptor`	`pair-acceptor-support`	`unregistered-vacancy`	An unregistered destination cannot close the interaction.
`joining`	`shared-or-transferred-pair`	`copied-electron-pair`	Copying violates exclusion and conservation.
`conservation`	`both-electron-identities-retained`	`electron-identity-erased`	Erasure cannot reverse the relation.
`record`	`endpoint-pair-joining-trace`	`Lewis-label-only`	A label cannot reconstruct pair support.
`extension`	`no-extra-rule`	`free-orbital-premise`	An imported orbital rule can select a desired classification.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2cc98aa40593481481ad39b0cb07b7ce428222ea723733f562c37b1394b2b6b6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:95cc34d98ca5a838b98c1074a1db31ed70571fbb0c387b70c5e910d134f18cab`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:c1fa64533d715cd9e3911fa10b34f9b093332ff954c844029a150f2f90ab4fcf`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:abc13b877a1bebe4f04b54c9618bdc2137ed324f325e4d0f3499a6153c8680f6`.

Independent reconstruction. Implementation hash `sha256:5e16152a66733442a6bde36d7de224f715f0f64f53c671e108eb36c57deef79f`; independent certificate `sha256:8bfe72a6239326e94c2ec7358fbe227e047702cb5ccf66ba4f41343be1deaa26`; external-validation hash `sha256:2107b13390cccd4a90480875a9e463b2c1f442858271c49024c3b68661f97025`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-LEWIS-COMPOSITE-2026

Comparison records:

- lewis-acid-base-iupac-composite: predicted electron-pair-donor__electron-pair-acceptor__shared-or-transferred-pair__pair-conserved; source-derived electron-pair-donor__electron-pair-acceptor__shared-or-transferred-pair__pair-conserved; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative Lewis record lacks an electron-pair donor, electron-pair acceptor or conserved pair relation, or if a changed row is accepted.

Measurement receipt: `sha256:7e1da91cd604bf1a3d0c5b196dfe2a0f0f7632670886763eeef0ab88649f93e7`. Isolation certificate: `sha256:d404805c02d76dbdba92d562da70a1c79b219d2e185760709ba33901f23d6332`. Custody certificate: `sha256:88b6adbf5b0ca93e411fde367c64a519f74c24348f4b5ef93993eb9223a8a223`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/L03508 (sha256:9fa013b12d6d0e3f305883a0db53dac7484e292e0288e2c3d0ba9625cfb2011c)

Registered target identities:

- lewis-acid-base-iupac-composite from IUPAC-GOLD-BOOK-LEWIS-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite donor/acceptor pair joined by one exclusion-preserving pair of named electron occurrences, with both electron identities conserved and the donor/acceptor orientation retained.

Immutable identities. Source manifest `sha256:e98cb4175058cfc641fd6d022f278c4f0098bd15d9d42e2e8c69982ecc03352c`; derivation seal `sha256:75683df9ae25bb776f74e5fb9971972ccd137cea37b5c50ab35c08ca4545cace`; engine receipt `sha256:05e8f96b150320e2809549b58d4e7f47862b5714153a8382309f958e1cc31959` at `receipts/engine/model_admitted/SFT-CHEM-AB-LEWIS-001-05e8f96b150320e2.json`; empirical hash `sha256:05d6954874a144897c0cdc4eb2e2bf0a304e0fb1f9a8f6a1dcb0840527ed22f7`; measurement receipt `sha256:7e1da91cd604bf1a3d0c5b196dfe2a0f0f7632670886763eeef0ab88649f93e7`.

41. Amphoteric dual-role relation

Claim identity: `SFT-CHEM-AB-AMPHOTERIC-001`

Question and theorem. An amphoteric species is the same retained chemical carrier appearing in two separately held contexts, once in a valid acid role and once in a valid base role.

same-species-identity__valid-donor-event__valid-acceptor-event__contexts-separately-held__both-roles-realized

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001

Generated grammar. Generate the literal product of the registered amphoteric species, acid context, base context, context separation, duality, identity, record and extension choices.

Boundary: Every finite chemical carrier for which two separately registered contexts contain a valid donor transition and a valid acceptor transition involving that same retained species identity.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `same-species-identity__valid-donor-event__valid-acceptor-event__contexts-separately-held__both-roles-realized__species-identity-retained__dual-event-context-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`species`	`same-species-identity`	`two-unrelated-species`	Dual behavior must belong to one retained carrier identity.
`acid-context`	`valid-donor-event`	`acid-label-without-event`	A label does not establish donation.
`base-context`	`valid-acceptor-event`	`base-label-without-event`	A label does not establish acceptance.
`context`	`contexts-separately-held`	`roles-conflated`	Conflation can assert incompatible events simultaneously.
`duality`	`both-roles-realized`	`one-role-only`	One role does not close amphoteric behavior.
`identity`	`species-identity-retained`	`species-changed-between-roles`	A changed carrier does not establish dual capacity.
`record`	`dual-event-context-trace`	`amphoteric-label-only`	A label cannot reproduce both contexts.
`extension`	`no-extra-rule`	`free-dual-role-exception`	An exception can admit any species.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5ade57913bbd3c0150acc9c9fb75df8d67ed170ae255e8b89209829a95841595`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f922ab1bf0af3b619c30a84ce5dabf9646cfe4aeedce383f3029f901c58cac3d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1ad519e6b1b5619447797c273c1f06084e45dfa0283bef8c23618549db219f5e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8ea2d3cfbcf557a1aa8f9c6a6400a0ebca9d6921702a109600abd4a47e87c374`.

Independent reconstruction. Implementation hash `sha256:ac8f026dbc2ed95ba69ffdfaac29bd5e7de7d5699e902f37d1b09665cf4528db`; independent certificate `sha256:48312bf23953149b2f7e9b86e4955f1fa4c4a2d0c1fc994bb1035905b0bdc8ac`; external-validation hash `sha256:a85ed53689c1d3bb8a41d06a081216e29135af2bc234686d8bceba3f0c6db119`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-A00306-2026

Comparison records:

- amphoteric-iupac-a00306: predicted same-species-acid-role__same-species-base-role__context-dependent-dual-response__both-traces-retained; source-derived same-species-acid-role__same-species-base-role__context-dependent-dual-response__both-traces-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative amphoteric record lacks acid and base behavior by one species under retained contexts, or if a changed row is accepted.

Measurement receipt: `sha256:ebe83d8610812de30385fc9a306ef23658a714603f04385dd7932e754b661651`.
Isolation certificate: `sha256:0e3e33ee179d3f0922e56c3784d3ab91b7760b0e2e0dd4c829daa39378872377`.
Custody certificate: `sha256:5c071a5b14f36df3201ef1f1255aad9d9738746d79e213533d4aa1936f0ab4a4`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/A00306 (sha256:70ab5021338400d9f02cbecc5c45b9f4271792e919d880aafff81684739022ab)

Registered target identities:

- amphoteric-iupac-a00306 from IUPAC-GOLD-BOOK-A00306-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical carrier for which two separately registered contexts contain a valid donor transition and a valid acceptor transition involving that same retained species identity.

Immutable identities. Source manifest `sha256:4d6bf3fa0ddc8b638be3957bf07a5aa1fd7235ea9fea906a160ee523e387db6a`; derivation seal `sha256:62fa5affb388260891509ffa90941ac314c52fad8f8ab8db3791c8525e6421f5`; engine receipt `sha256:9633c37ce51efb4c0d0109e6cecd161e45d21de52ce7c1d6b732235e7f9837dd` at `receipts/engine/model_admitted/SFT-CHEM-AB-AMPHOTERIC-001-9633c37ce51efb4c.json`; empirical hash `sha256:8e94f3de67b0214a8db493d5f07e34b30b581905d05489027544d930cba7c6d1`; measurement receipt `sha256:ebe83d8610812de30385fc9a306ef23658a714603f04385dd7932e754b661651`.

42. Buffer response and conjugate-pair retention

Claim identity: `SFT-CHEM-AB-BUFFER-001`

Question and theorem. A buffer is a finite positive reservoir of both members of a conjugate pair whose complementary paths consume added proton support or supply proton support to added base while retaining pair identity and finite capacity.

> *conjugate-pair-reservoir__positive-support-of-both__base-member-accepts-proton__acid-member-supplies-proton__conjugate-pair-retained*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`

Generated grammar. Generate the literal product of the registered buffer components, support, acid response, base response, retention, capacity, record and extension choices.

Boundary: Every finite reservoir containing positive support of both members of a conjugate pair and two complementary response paths that consume added proton support or supply proton support to an added base within capacity.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `conjugate-pair-reservoir__positive-support-of-both__base-member-accepts-proton__acid-member-supplies-proton__conjugate-pair-retained__finite-source-bounded-capacity__pair-count-response-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`components`	`conjugate-pair-reservoir`	`single-unpaired-species`	One species cannot supply both complementary responses.
`support`	`positive-support-of-both`	`one-component-absent`	An absent member leaves one response unavailable.
`acid-response`	`base-member-accepts-proton`	`added-proton-unanswered`	No response permits the proton distinction to remain open.
`base-response`	`acid-member-supplies-proton`	`added-base-unanswered`	No response permits proton removal without paired change.
`retention`	`conjugate-pair-retained`	`reservoir-identity-erased`	A response without conjugate tracking cannot be reversed.
`capacity`	`finite-source-bounded-capacity`	`unlimited-buffer-claim`	A finite reservoir cannot have unlimited capacity.
`record`	`pair-count-response-trace`	`pH-answer-only`	A pH value cannot reconstruct composition and response.
`extension`	`no-extra-rule`	`free-equilibrium-fit`	A fitted equation can force a measured response.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:15d9bacd7e2b1a04d502efb2b9edf19cc831eb2248f12f176669be2ed55b7727`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7e69a9d59c2e2894ca3ea5c30cf7467cf5128645d30c60e368cc800a0eb067ed`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1f4d2e4ef0767fc1a44344a407756603a283dad0ee6c5407c8604ac5338bc788`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:696c34f1a9d474e9875c4d03a58e6b5215831e6504ed23e52477dd7d20c84fe8`.

Independent reconstruction. Implementation hash `sha256:8646664e19c6946f77ce8eecb44b31ed57ac3ef2d67c745f9aa666f3a3246d09`; independent certificate `sha256:393f76310d7a38322171e44145f2cfcc0652668cdd4f8bdca9cd8eeeaeb112d6`; external-validation hash `sha256:13ee1a405b692d5e6debba2ed178b73f55528d90fd46e95bb980242be0927389`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-DIDAC-E15-2026

Comparison records:

- buffer-iupac-didac-e15: predicted weak-acid-conjugate-base-pair__responds-to-added-acid-or-base__composition-change-limited__source-bounded-capacity;

source-derived weak-acid-conjugate-base-pair__responds-to-added-acid-or-base__composition-change-limited__source-bounded-capacity; exact match True

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authoritative buffer record lacks a weak acid/conjugate base pair, response to added acid/base or finite composition-dependent capacity, or if a changed row is accepted.

Measurement receipt: `sha256:a7d7a39d4e3f869e1e0a7ab1fe909342503280e66423babd9d3452fb85a56b03`. Isolation certificate: `sha256:25359ccbb35e3a8a10125c2aa86ee066ac906e9e415ad3382e014cba6c4df7cc`. Custody certificate: `sha256:2c18aab35d14550dd2373ed5059fd1ff351d566c29a23a930a9587cd5a9b91ff`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://old.iupac.org/didac/Didac%20Eng/Didac02/Content/E15.htm (sha256:570a91d1a7de35b22a8469c117f10af9e4bed3012ae61f54362fe45534b3d01e)

Registered target identities:

- buffer-iupac-didac-e15 from IUPAC-DIDAC-E15-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no acid/base dictionary, pH scale, equilibrium equation, dissociation constant, measured value or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no unpaired creation, copying or loss of a proton or electron carrier
- no claim that structural closure alone fixes every solution-dependent strength or capacity value
- external target content remains inaccessible until the prediction is sealed
- Every finite reservoir containing positive support of both members of a conjugate pair and two complementary response paths that consume added proton support or supply proton support to an added base within capacity.

Immutable identities. Source manifest `sha256:dfd3b4141fab3d6c698fe25ad20f011d5eda39d7c3c557a6610576736d955c69`; derivation seal `sha256:d42c6cc2ed4c8a44fd95cd60a82c910ac9062dfd1eed14e9a1b426f59ecc0bc3`; engine receipt `sha256:c1503c569172a5cf32d266206aed2383ef3ca4998987c70907e138c93d930444` at `receipts/engine/model_admitted/SFT-CHEM-AB-BUFFER-001-c1503c569172a5cf.json`; empirical hash `sha256:b4ac25bc6410610189cbd06724947ec31532bd88a8b1b341b9a74a6d8c12633b`; measurement receipt `sha256:a7d7a39d4e3f869e1e0a7ab1fe909342503280e66423babd9d3452fb85a56b03`.

43. Electronegativity ordering

Claim identity: `SFT-CHEM-ELECTRONEGATIVITY-001`

Question and theorem. Electronegativity order is the transitive preorder of complete pairwise endpoint recurrence for identical shared electron support; equal traces remain an exact equivalence class rather than receiving an arbitrary rank.

> *two-named-atomic-occurrences__one-shared-electron-support__complete-endpoint-recurrence-comparison__held-preferred-endpoint__transitive-preorder-closure*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`

- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`

Generated grammar. Generate the literal product of carrier, support, comparison, orientation, tie, extension, record and extra-rule choices.

Boundary: Every finite pairwise comparison of two named atomic occurrences joined by one shared electron-labelled support, with the complete recurrence trace determining endpoint preference or an explicitly retained tie.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `two-named-atomic-occurrences__one-shared-electron-support__complete-endpoint-recurrence-comparison__held-preferred-endpoint__trace-equivalence-class__transitive-preorder-closure__complete-comparison-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carriers`	`two-named-atomic-occurrences`	`anonymous-element-values`	Values erase endpoint identity.
`support`	`one-shared-electron-support`	`free-affinity-scalar`	A scalar imports the ordering.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
comparison	complete-endpoint-recurrence-comparison	name-selected-order	Names cannot select preference.
orientation	held-preferred-endpoint	signed-difference	A signed magnitude violates the proof domain.
ties	trace-equivalence-class	forced-strict-order	Equal complete traces cannot be ordered uniquely.
extension	transitive-preorder-closure	pairwise-cycle-admitted	A cycle destroys lawful ordering.
record	complete-comparison-trace	rank-answer-only	A rank cannot reconstruct comparisons.
extra	no-extra-rule	free-scale-constant	A scale constant can tune any ordering.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:0b4cdcf1f4aeef52d241ee6c50286036754d52606fb34b3123efdf8788925569`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:75b4db050fe85f16506da2c636dce9c265cf2318fc57636085303aa2888a7b34`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e328e529b6c738adc4b931a31da3aa43513455bfb2797dff915c4b9c5f39bc88`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:c1c883deac6b9242a292b10d7a1e05a83ca81c2276fad9e8e34732e35f6f959f`.

Independent reconstruction. Implementation hash `sha256:a07b6ba2f87119929254b9ab82389f69c9e26a2659a799af6cfd6fdee3b42622`; independent certificate `sha256:f09179fb7686c4719baaceff6b28302171cbb14392ab3eb079be703c9f8acc70`; external-validation hash `sha256:6c345e0ee142a0f452732d235c9dc4fab7bcb10bb8959db65495d75368eacb24`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ELECTRONEGATIVITY-COMPOSITE-2026

Comparison records:

- electronegativity-iupac-composite: predicted atomic-attraction-for-shared-electrons__pairwise-affinity-order__ties-retained__scale-values-source-bounded; source-derived atomic-attraction-for-shared-electrons__pairwise-affinity-order__ties-retained__scale-values-source-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks atomic attraction for shared electrons or permits a scale value to replace the retained comparison relation, or if a changed row is accepted.

Measurement receipt: `sha256:bd86b140ac84feb20fcd1503c6284630317056d1b172d1063aab0832cf6f2deb`. Isolation certificate: `sha256:7728d84dacaf8a2d81d017ead378e76b352da93fc4b8106ad8b7845188118103`. Custody certificate: `sha256:6c14f366bf01e75f03c76d99e2b637c938cc5618785b6f4aae53054700cc92eb`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/E01990 (sha256:bc2888f84a2a3efc53f89738797c88528af75cd0564295e2a1d4138cd1c778e3)

Registered target identities:

- electronegativity-iupac-composite from IUPAC-GOLD-BOOK-ELECTRONEGATIVITY-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite pairwise comparison of two named atomic occurrences joined by one shared electron-labelled support, with the complete recurrence trace determining endpoint preference or an explicitly retained tie.

Immutable identities. Source manifest `sha256:ddb0747ac6471850417d28ff5ad8ad3decea2fead388f80b3fc664fde4a4dc5f`; derivation seal `sha256:9efe41b7f2f880257c5ad610b1b1cf5cc75636df410afe3c16d3fb518922611c`; engine receipt `sha256:eb149468f0bf1746f85bd081caebc00bf72c241c04ef4039e78a8a1d9fc735cb` at `receipts/engine/model_admitted/SFT-CHEM-ELECTRONEGATIVITY-001-eb149468f0bf1746.json`; empirical hash `sha256:665ff48f0d66dd7a54ef47601152879d5487449acdca892dc041056ad8eda8e9`; measurement receipt `sha256:bd86b140ac84feb20fcd1503c6284630317056d1b172d1063aab0832cf6f2deb`.

44. Bond polarity from unequal held affinity

Claim identity: `SFT-CHEM-BOND-POLARITY-001`

Question and theorem. Unequal endpoint affinity for the identical shared support forces an oriented but still shared bond partition, with opposed partial-charge fibres held as labels rather than signed proof numbers.

> *registered-covalent-bond__unequal-complete-affinity__same-shared-electron-support__unequal-shared-partition__held-partial-charge-fibres*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`

- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001

Generated grammar. Generate the literal product of bond, affinity, support, partition, orientation, closure, record and extra-rule choices.

Boundary: Every finite covalent bond whose two named endpoints have unequal complete affinity traces for the same shared support, producing one held orientation of the support partition while retaining the bond whole.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `registered-covalent-bond__unequal-complete-affinity__same-shared-electron-support__unequal-shared-partition__held-partial-charge-fibres__paired-endpoint-response__bond-affinity-partition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`bond`	`registered-covalent-bond`	`two-unjoined-atoms`	Polarity belongs to a shared bond.
`affinity`	`unequal-complete-affinity`	`equal-or-unknown-affinity`	No direction follows from equality or missing evidence.
`support`	`same-shared-electron-support`	`different-electrons-compared`	Different support cannot orient one bond.
`partition`	`unequal-shared-partition`	`full-transfer`	Full transfer changes the bonding class.
`orientation`	`held-partial-charge-fibres`	`negative-charge-number`	A negative scalar is unnecessary.
`closure`	`paired-endpoint-response`	`one-endpoint-only`	One endpoint leaves the bond record open.
`record`	`bond-affinity-partition-trace`	`polar-label-only`	A label cannot reproduce the comparison.
`extra`	`no-extra-rule`	`free-polarity-threshold`	A threshold imports a parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:bc9919833ca65448defb66482960c2ac35503f2445d3cf46e5a2a8d745c1a6d6`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:c24cafbc7496d10a8ec0cedbe90979502eb498af4dd5e8e8ee9dd9837b28933d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:81b2675f657711b9a4a484f076fe8dfb4d12779fae46fbd14f41a9f3aa753391`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:496e405f3e97843fe901246d4f831feb336c6e91ebb49dc12bf17fa6ca42503c`.

Independent reconstruction. Implementation hash `sha256:4c6ba5a086ff250256dee1c270231597286030abf637798a6138e036820db6fc`; independent certificate `sha256:97626f4d4ea6cc78c33c2a67460285297e027e25af4c7c7cec3f822cf909ff36`; external-validation hash `sha256:189db76256e2381a4b7dd2bba3182648b78e51a339d8507be86f53728cba2625`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-BOND-POLARITY-COMPOSITE-2026

Comparison records:

- bond-polarity-iupac-composite: predicted unequal-electron-attraction__polar-covalent-bond__partial-charge-separation__bond-support-remains-shared; source-derived unequal-electron-attraction__polar-covalent-bond__partial-charge-separation__bond-support-remains-shared; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks unequal attraction, polar covalent sharing or partial charge separation, or if a changed row is accepted.

Measurement receipt: `sha256:a88c1a4d6fc6f7e9271bf40141d884b3fec4148c2129033170b5e82b3dd03b05`. Isolation certificate: `sha256:2e00cdeb8f6e343c929edf4ade2b58c3664a6d05c05d6424d27f128cb8cbcba3`. Custody certificate: `sha256:1779fac6f4228e3daf6fa32d38d89148c85956f622b7adb121fa243b3939fceb`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/08195 (sha256:a907133200cc3296e7ecdb7c8970ae8ce35af7b6d3b44d6746f7d18ad5086d41)

Registered target identities:

- bond-polarity-iupac-composite from IUPAC-GOLD-BOOK-BOND-POLARITY-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite covalent bond whose two named endpoints have unequal complete affinity traces for the same shared support, producing one held orientation of the support partition while retaining the bond whole.

Immutable identities. Source manifest `sha256:e6017cc25cd31727ee3d2f5b4d0a4be83094d10b9b4ce160a9fbba2a3dc55a07`; derivation seal `sha256:bb37eaabf4cfe67424c181ad8adda039704a087602340374f4c07061edeaedfd`; engine receipt `sha256:b0a872705523d4edddb27b335602aa73685ead906bb0c466c1b0df5a1de6ab52` at `receipts/engine/model_admitted/SFT-CHEM-BOND-POLARITY-001-b0a872705523d4ed.json`; empirical hash `sha256:86ba63a8b53a6f70676c4d7ea2c6e5f1b84d92a1bdb52b59d156446a471343e5`; measurement receipt `sha256:a88c1a4d6fc6f7e9271bf40141d884b3fec4148c2129033170b5e82b3dd03b05`.

45. Oxidation-state accounting

Claim identity: `SFT-CHEM-REDOX-OXIDATION-STATE-001`

Question and theorem. Oxidation state is a complete source-bound ionic-approximation accounting: each bond-electron occurrence is assigned exactly once by endpoint affinity, with magnitude positive and orientation separately held.

> *complete-bonded-chemical-whole__declared-ionic-approximation__assign-to-preferred-endpoint__equal-endpoint-division__each-electron-assigned-once*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`

- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`

Generated grammar. Generate the literal product of whole, partition, unequal-bond, equal-bond, count, conservation, record and extra-rule choices.

Boundary: Every finite bonded chemical whole under one declared ionic-approximation partition, assigning each bond-electron occurrence exactly once by complete affinity order (or equal division for tied endpoints), retaining atom identity, orientation and total carrier conservation.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-bonded-chemical-whole__declared-ionic-approximation__assign-to-preferred-endpoint__equal-endpoint-division__positive-count-plus-held-orientation__each-electron-assigned-once__atom-bond-assignment-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`whole`	`complete-bonded-chemical-whole`	`isolated-atom-answer`	Oxidation accounting is composition-relative.
`partition`	`declared-ionic-approximation`	`unstated-electron-rule`	An unstated rule permits arbitrary answers.
`unequal-bond`	`assign-to-preferred-endpoint`	`split-despite-preference`	It contradicts complete affinity order.
`equal-bond`	`equal-endpoint-division`	`arbitrary-endpoint`	Equal traces cannot choose an endpoint.
`count`	`positive-count-plus-held-orientation`	`signed-integer-primitive`	A signed number hides orientation.
`conservation`	`each-electron-assigned-once`	`electron-created-or-lost`	Open assignment violates the whole.
`record`	`atom-bond-assignment-trace`	`oxidation-number-only`	A number cannot reproduce the assignment.
`extra`	`no-extra-rule`	`free-state-table`	A table can select desired values.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:eec9720b659074465f3bedf55d78324ab1d030d36810d96e0261e9ae158f95a9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b38971bae9d17c53c09af666594f47d248c37c82c5e5ac2f821cfea4eccf1a2b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b7ff129c257d37fbaeb72d84a6aca6cf5a240ecff39ca00dfee215ef9ba230ab`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f9451d77ea5492dbc9190e7320c55355c41a38dfc67711cf3c06479836059344`.

Independent reconstruction. Implementation hash `sha256:dde5adfefb754c437df1cfb00ac8e89e98023da3859ecc8e189227024cf5f072`; independent certificate `sha256:4c303af97dc7aa5fa8ab7803de872f79e74602902c10343f340eccf5f0457caa`; external-validation hash `sha256:87581483c009cc3ef228fb01d87001ad94075b6c92557fdb4514a93a8890b7c4`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-O04365-2026

Comparison records:

- oxidation-state-iupac-o04365: predicted ionic-approximation__bond-electrons-assigned-by-electronegativity__per-atom-charge-accounting__complete-assignment-conserved; source-derived ionic-approximation__bond-electrons-assigned-by-electronegativity__per-atom-charge-accounting__complete-assignment-conserved; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks ionic approximation, electronegativity-based assignment or complete per-atom accounting, or if a changed row is accepted.

Measurement receipt: `sha256:26213121965ff8f746f36669916d9591fdf2411cf348653da49258b56621fb3d`. Isolation certificate: `sha256:94464dc36dc6570e43deeac9a15d29c8bfca7a3bf58233efcdee56000a179f3c`. Custody certificate: `sha256:d9999eca3435ab84ed1a0ba8f1e32f6ebcab78a27bb191f09a52bbcea1561b00`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/O04365 (sha256:5413171bfaa9152ff9a6a2dee0c79a2ec3e31a4d1efbfff0f8c76804c7fdb06e)

Registered target identities:

- oxidation-state-iupac-o04365 from IUPAC-GOLD-BOOK-O04365-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite bonded chemical whole under one declared ionic-approximation partition, assigning each bond-electron occurrence exactly once by complete affinity order (or equal division for tied endpoints), retaining atom identity, orientation and total carrier conservation.

Immutable identities. Source manifest `sha256:8ba5ba8a14eb179b9dd7ca533ca5cdac2729479bca9c059c23b9ade08fb46088`; derivation seal `sha256:176aa2e998290d7f1ccffc12882274e0f19953b0ea05f527a1b0f7f875dcebda`; engine receipt `sha256:b08912e566745bea7d1a368aa201ca2f4d92c23e6a36de0c3c6e2ed0ad2482ed` at `receipts/engine/model_admitted/SFT-CHEM-REDOX-OXIDATION-STATE-001-b08912e566745bea.json`; empirical hash `sha256:4bc79cda405568128bebfd09f964d6f8c05a6db104ddb79254e0b676240ec141`; measurement receipt `sha256:26213121965ff8f746f36669916d9591fdf2411cf348653da49258b56621fb3d`.

46. Coupled oxidation and reduction

Claim identity: `SFT-CHEM-REDOX-COUPLING-001`

Question and theorem. Oxidation and reduction are the opposed endpoint records of one conserved electron transfer; neither half closes without the other.

> *named-electron-carrier__registered-oxidation-endpoint__registered-reduction-endpoint__one-paired-transfer__same-carrier-before-after*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`

Generated grammar. Generate the literal product of carrier, donor, acceptor, coupling, identity, accounting, record and extra-rule choices.

Boundary: Every finite closed electron-labelled transfer between distinct chemical endpoints, with identical carrier identity before and after and paired donor loss/acceptor gain or their complete oxidation-assignment equivalents.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `named-electron-carrier__registered-oxidation-endpoint__registered-reduction-endpoint__one-paired-transfer__same-carrier-before-after__opposed-assignment-updates__carrier-endpoint-coupling-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`named-electron-carrier`	`anonymous-charge-change`	It cannot establish electron conservation.
`donor`	`registered-oxidation-endpoint`	`donor-erased`	The carrier could be freely created.
`acceptor`	`registered-reduction-endpoint`	`acceptor-erased`	The carrier could be lost.
`coupling`	`one-paired-transfer`	`independent-half-changes`	One open half violates conservation.
`identity`	`same-carrier-before-after`	`different-carrier-after`	That is replacement, not transfer.
`accounting`	`opposed-assignment-updates`	`one-sided-state-change`	One side cannot close the assignment.
`record`	`carrier-endpoint-coupling-trace`	`redox-label-only`	A label cannot reverse the event.
`extra`	`no-extra-rule`	`free-electron-source`	A free source breaks closure.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:93dc0f0dca3a8d0637be0f8366a9c9f7efbabe274caf377dc0d68648b26162ca`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:135e22bd09a5e264309f9102ca7412c0debb155083e86f4b50edca884843fd63`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:f7c8cad06259b068ad6ad36db72aebfbdde432e2cb83719d9e7cf26d4c680e0b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fcdd87fce4203ac4f9b1284a0db8ea61f1f4ab2f08bbe6176dbaf3e6d2dcd12c`.

Independent reconstruction. Implementation hash `sha256:a4955d74f704c817aac676ffc3af783df9b6bb7f328576e68920563fed210eab`; independent certificate `sha256:76ec599dda0c5859c14d2c306a87f6205b3a6c018234b8512be63c2f3d9c3db8`; external-validation hash `sha256:ba80a84a3f87be5debcb6abaaf7d33cd4860b2326c5adacbdedbf2a41913d2f3`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REDOX-COMPOSITE-2026

Comparison records:

- redox-coupling-iupac-composite: predicted oxidation-electron-loss__reduction-electron-gain__paired-half-reactions__electron-transfer-conserved; source-derived oxidation-electron-loss__reduction-electron-gain__paired-half-reactions__electron-transfer-conserved; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks paired oxidation/reduction, electron loss/gain or conserved transfer, or if a changed row is accepted.

Measurement receipt: `sha256:56a9f13bed1324bb69831ef0046928a2c757d59f10f99d89c9a384b76a3844b1`.
Isolation certificate: `sha256:2c0211cda70ab9fe2c50674950761fe08cadb4808982525aae774882250d7edb`.
Custody certificate: `sha256:b0d147cff0011710434513be7176ed15d8eea1dd2523dc50fe6f1360e14061b1`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/O04362 (sha256:9e367d54ae7e7f9e5a40366f9298b1c8dfffbac4677847f60458b78e7e16fc68)

Registered target identities:

- redox-coupling-iupac-composite from IUPAC-GOLD-BOOK-REDOX-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints
- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite closed electron-labelled transfer between distinct chemical endpoints, with identical carrier identity before and after and paired donor loss/acceptor gain or their complete oxidation-assignment equivalents.

Immutable identities. Source manifest `sha256:55db1a263d7bd172615c685ba171b340f048cc2bda81ec82083dfa5393ce639a`; derivation seal `sha256:9a9262fd9a3f9e90c08a41eb5c2224ea19127901a4a9399f8addcf87454888c8`; engine receipt `sha256:69fc6428332eb2faeb8e2f40987067e351a39b0c0138e777a2438219723b657e` at `receipts/engine/model_admitted/SFT-CHEM-REDOX-COUPLING-001-69fc6428332eb2fa.json`; empirical hash `sha256:d4468eee8b0587e279b7126d893f7101d9e9faaaa47232ea88e086e0e7702633`; measurement receipt `sha256:56a9f13bed1324bb69831ef0046928a2c757d59f10f99d89c9a384b76a3844b1`.

47. Electrochemical cell and separated redox closure

Claim identity: `SFT-CHEM-ELECTROCHEM-CELL-001`

Question and theorem. An electrochemical cell is a separated coupled-redox process with complete external electron and compensating ionic paths, retaining site identities and source-bounded work transfer.

> *distinct-oxidation-reduction-sites__complete-external-electron-path__complete-compensating-ionic-path__one-redox-transfer-cycle__source-bounded-work-transfer*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001

Generated grammar. Generate the literal product of site, electron-path, ionic-path, coupling, work, orientation, record and extra-rule choices.

Boundary: Every finite separated pair of coupled redox sites with a complete electron path from oxidation to reduction, a complete compensating ionic path, retained chemical boundaries and source-bounded work transfer.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `distinct-oxidation-reduction-sites__complete-external-electron-path__complete-compensating-ionic-path__one-redox-transfer-cycle__source-bounded-work-transfer__held-cell-orientation__sites-paths-transfer-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`sites`	`distinct-oxidation-reduction-sites`	`one-conflated-site`	Separation is required for an external path.
`electron-path`	`complete-external-electron-path`	`electron-teleportation`	It omits the transfer trace.
`ionic-path`	`complete-compensating-ionic-path`	`charge-accumulation-unclosed`	Accumulation halts sustained transfer.
`coupling`	`one-redox-transfer-cycle`	`independent-half-reactions`	Unpaired halves violate conservation.
`work`	`source-bounded-work-transfer`	`free-energy-output`	Untraced work creates support.
`orientation`	`held-cell-orientation`	`negative-potential-primitive`	A negative scalar is not needed.
`record`	`sites-paths-transfer-trace`	`cell-label-only`	A label cannot reproduce either path.
`extra`	`no-extra-rule`	`free-electrode-potential`	A fitted potential can select behavior.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:63371e499dce6d50287fc484b0f0cbf70c3dfd532ed6b9703aa7bf01d4945db0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:8c3c9a1814cd2e0e60db3b4e95284e76c9a5eb2f3500e44588cc6f80b50c61cb`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:815ea97bdb3eea0fe85ee56fef6342a6e546c2c71f10bcfba66baee6eb6247ce`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:458f2a1f78edf5913a5538be6612662eb14b0c60edf88ce2f52524182f1a2383`.

Independent reconstruction. Implementation hash `sha256:4e3a0f2270afaa2bf36345a962acbb96bc9e41cdf8f268cc01a75f3ff5854c3f`; independent certificate `sha256:ce4e3f99a3026cab0531efad71c9733fc92a01eab2650628c4ed9bcce6636279`; external-validation hash `sha256:c0aaf6da4ea7ab9fabf07342aada8550cd487cf511d1a64b11957d6f2e8ecc58`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ELECTROCHEMICAL-CELL-COMPOSITE-2026

Comparison records:

- electrochemical-cell-iupac-composite: predicted separated-redox-half-cells__external-electron-path__ionic-circuit-closure__chemical-to-electrical-work-transfer; source-derived separated-redox-half-cells__external-electron-path__ionic-circuit-closure__chemical-to-electrical-work-transfer; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks separated redox sites, electron and ionic paths or chemical/electrical work conversion, or if a changed row is accepted.

Measurement receipt: `sha256:80b09725552c16e8044130c94315dde3c334ece64d749a2cca81d5375c5070e9`. Isolation certificate: `sha256:ecb151d1a3352b9c4412df686d5bdc3e89a457047d8b3294983704eea7f38a22`. Custody certificate: `sha256:9b84f7968931f9dc9b0ede41d8b1caf0328c45586668affee2b7869964848a78`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/09058 (sha256:0016d0976a51c44b0208458810ad455a459f988c7573d186293edff0aea33d68)

Registered target identities:

- electrochemical-cell-iupac-composite from IUPAC-GOLD-BOOK-ELECTROCHEMICAL-CELL-COMPOSITE-2026

Meaning. Acid/base and redox laws retain transfer orientation as held labels, not negative proof magnitudes, and preserve every donor, acceptor, charge and boundary carrier. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no electronegativity scale, oxidation-state table, electrode potential, external definition, measured value or V2 result may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no electron carrier may be created, copied, silently erased or assigned to two endpoints

- no qualitative structural law is represented as a parameter-free prediction of a measured decimal scale
- external target content remains inaccessible until the complete prediction is sealed
- Every finite separated pair of coupled redox sites with a complete electron path from oxidation to reduction, a complete compensating ionic path, retained chemical boundaries and source-bounded work transfer.

Immutable identities. Source manifest `sha256:c361a8df2ace000382f95e6b5d5001408aabf068043d398be65fde3845230252`; derivation seal `sha256:1de4b3b109b17025ece085f31ca01dd775867525a773cb277451718c2dcb247e`; engine receipt `sha256:05740565b573149c2c748a6c30271add9304a26fd1086aa6ffbd2c097d8374e5` at `receipts/engine/model_admitted/SFT-CHEM-ELECTROCHEM-CELL-001-05740565b573149c.json`; empirical hash `sha256:b2ef2d6f2c6542cea7535df4bdc260ab8bb5bdf08ad5a9d81300e9663ea624e3`; measurement receipt `sha256:80b09725552c16e8044130c94315dde3c334ece64d749a2cca81d5375c5070e9`.

14. Reaction Kinetics Thermodynamics

Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries.

48. Chemical reaction as source-bound identity transformation

Claim identity: `SFT-CHEM-RXN-IDENTITY-001`

Question and theorem. A chemical reaction is a complete source-bound reactant-to-product transformation conserving every elemental carrier while changing at least one chemical relation.

> *source-bound-transition__complete-reactant-organization__complete-product-organization__all-elemental-carriers-conserved__at-least-one-relation-changed*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`

- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`

Generated grammar. Generate the literal product of source, input, output, carrier, change, orientation, record and extension choices.

Boundary: Every finite source-bound chemical transition between complete named reactant and product organizations, conserving every elemental carrier while changing at least one registered chemical relation.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `source-bound-transition__complete-reactant-organization__complete-product-organization__all-elemental-carriers-conserved__at-least-one-relation-changed__held-reactant-product-orientation__complete-endpoint-carrier-relation-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`source`	`source-bound-transition`	`unregistered-transformation`	No chemical provenance is retained.
`input`	`complete-reactant-organization`	`reactants-erased`	A product alone cannot define change.
`output`	`complete-product-organization`	`products-erased`	An input alone cannot define change.
`carriers`	`all-elemental-carriers-conserved`	`element-created-or-lost`	It violates chemical closure.
`change`	`at-least-one-relation-changed`	`identity-transition`	No changed relation means no reaction event.
`orientation`	`held-reactant-product-orientation`	`unordered-endpoints`	Direction cannot be audited.
`record`	`complete-endpoint-carrier-relation-trace`	`reaction-name-only`	A name cannot reproduce the transition.
`extra`	`no-extra-rule`	`free-reaction-exception`	An exception can admit arbitrary change.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:fb754a6029d272793d88a9743f1a5ca1c2e28481c465d2910713413cb2b900f3`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:959bb4b2950fa5b8ca9f5ac5db294ffbcf0d889230a15c3bfffaf0026b4ccc46`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:767e576739cb6dcdc5eaa7f0d4ae141f26bf618676e521133b66f4407821b1e7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:15b247c519ea516c0b538768e82bebe17813c244712eb2495d2664222da5cd88`.

Independent reconstruction. Implementation hash `sha256:15c43d7930b426b88179e32fba7fbf6289d9c438b043a1e571c14243f6f908a2`; independent certificate `sha256:f250baeb1f01ff51426dcf57f3edc484ff4bfa408801c5e5aa6dd631239b1d9d`; external-validation hash `sha256:90e6ba56dc2f4993855f74d01bb735893ea0d60af9bb83a73576ac33728d789d`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026

Comparison records:

- chemical-reaction-iupac-composite: predicted reactants-transformed-to-products__chemical-identities-change__elemental-carriers-conserved__source-bound-reaction-record; source-derived reactants-transformed-to-products__chemical-identities-change__elemental-carriers-conserved__source-bound-reaction-record; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reactant/product transformation, chemical change or carrier conservation, or if a changed row is accepted.

Measurement receipt: `sha256:185372e20b90007da19bb1bb3d5007eccdd0313e05633693564d0cb92cd4f26e`. Isolation certificate: `sha256:9d31d049b62ad20873e79e82b014c747e29dff2f06eb52af633d7585b241200b`. Custody certificate: `sha256:1075f201a9d80dad816867b668546d316ea96c788cd9c55b5d4d7dc6688380df`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01033 (sha256:83d7891114ad65c012ce2e35f4021beaee6a37d4473c416170d6643db4dc7347)

Registered target identities:

- chemical-reaction-iupac-composite from IUPAC-GOLD-BOOK-CHEMICAL-REACTION-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed

- Every finite source-bound chemical transition between complete named reactant and product organizations, conserving every elemental carrier while changing at least one registered chemical relation.

Immutable identities. Source manifest `sha256:69c8d1b58730966ecc0addff5b0380c42fcec1a75f2140c33824d136f9fe1126`; derivation seal `sha256:8d997c8416d20ae98d74d99e1be40c6f8c64acd2c113caf785c19c4702bb9864`; engine receipt `sha256:554bb7fe234a60461eb8d475daff205fb91fa08c51da4e9fd598c1feefd0c212` at `receipts/engine/model_admitted/SFT-CHEM-RXN-IDENTITY-001-554bb7fe234a6046.json`; empirical hash `sha256:cb4ca23d7af4dee184fd5904c38247d456df7dba5e35a8bf43ed0dcf243e92dd`; measurement receipt `sha256:185372e20b90007da19bb1bb3d5007eccdd0313e05633693564d0cb92cd4f26e`.

49. Reaction mechanism as complete elementary-step trace

Claim identity: `SFT-CHEM-RXN-MECHANISM-001`

Question and theorem. A reaction mechanism is the complete ordered composition of elementary chemical transitions whose internal states join exactly and whose external endpoints equal the overall reaction.

> *ordered-elementary-step-word__adjacent-state-composition__declared-indivisible-steps__internal-states-held__same-overall-endpoints*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`

- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001

Generated grammar. Generate the literal product of step, adjacency, elementarity, internal-state, endpoint, conservation, record and extension choices.

Boundary: Every finite ordered sequence of elementary chemical transitions whose adjacent states compose exactly and whose endpoint composition equals one registered overall reaction after internal states cancel.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `ordered-elementary-step-word__adjacent-state-composition__declared-indivisible-steps__internal-states-held__same-overall-endpoints__carrier-conservation-each-step__complete-ordered-step-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
steps	ordered-elementary-step-word	unordered-step-list	Unordered changes do not form a path.
adjacency	adjacent-state-composition	unmatched-step-endpoints	The proposed path is discontinuous.
elementarity	declared-indivisible-steps	hidden-substeps	The trace is incomplete at its declared boundary.
internal	internal-states-held	internal-species-erased	Cancellation cannot be audited.
endpoints	same-overall-endpoints	different-overall-reaction	The path does not explain the claim.
conservation	carrier-conservation-each-step	carrier-loss-between-steps	Local openness breaks global closure.
record	complete-ordered-step-trace	mechanism-label-only	A label cannot reproduce the path.
extra	no-extra-rule	free-hidden-path	An unseen path can explain any endpoint.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2b08b15144e7448122e88a2a305aad7c8ac006ff06273c79a20de192a4534887`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:5a8ce953ed1770bff5b7b4cc8d8b8b44826a0eb87215bec2ebdbef36dc82ecdf`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:81abefbb5f378e8edd9b45f331410f29b0fc2b044732df627c2030f63501c9b9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:cca8d7ea96b3dd5513a244338663507233b8fa000c8323f450f93ae9113130ea`.

Independent reconstruction. Implementation hash `sha256:52cf78c584d78eea0045c48685d0bca81995b54d84f0a39dc59fa91be71cec1b`; independent certificate `sha256:e015218ecd7e1982bb5a476c05f79d02779995e38b0a668982c9638121c14058`; external-validation hash `sha256:3b7e30bc294fcac5ff8ed82c7d095f7cbc43a4570cf0bd89c3f8689d8edf2f2f`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-MECHANISM-COMPOSITE-2026

Comparison records:

- reaction-mechanism-iupac-composite: predicted overall-reaction-decomposed__ordered-elementary-steps__intermediate-states-retained__step-composition-reproduces-endpoints; source-derived overall-reaction-decomposed__ordered-elementary-steps__intermediate-states-retained__step-composition-reproduces-endpoints; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks ordered elementary steps, retained intermediates or composition to the overall reaction, or if a changed row is accepted.

Measurement receipt: `sha256:ce957ea02d3aa439297fa6c68e7585380101e9094fb5a5cd68ac8860bee79323`. Isolation certificate: `sha256:9cc3021837608f8cc64939c78ada877af842ad07d6af4edb555a19b168817182`. Custody certificate: `sha256:72738e2e206fb1adbf46a53f6aa2de242e2064d6c4e7bd0a34f576c081e65876`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/M03804 (sha256:b84ddab1f13dbc71482ed3f7653eeab3e2cfe5b7ffba64c30ef9d08513d6be33)

Registered target identities:

- reaction-mechanism-iupac-composite from IUPAC-GOLD-BOOK-REACTION-MECHANISM-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite ordered sequence of elementary chemical transitions whose adjacent states compose exactly and whose endpoint composition equals one registered overall reaction after internal states cancel.

Immutable identities. Source manifest `sha256:a99f963a1966c398b026249b9ff679cb2136238f538b2f8b574a8891c38305ff`; derivation seal `sha256:c020425e337f94fab0d38deec7533c1110d9f0f5ccd814bf1435c7c189c98cac`; engine receipt `sha256:0735442bcc4a807fe1c867dc97063a4fafea6963688fe9f02b3dff8f89897982` at `receipts/engine/model_admitted/SFT-CHEM-RXN-MECHANISM-001-0735442bcc4a807f.json`; empirical hash `sha256:71b3b85bbd94f28a469840d4fb578c4013df607b68a5b77d5987afde0e99add6`; measurement receipt `sha256:ce957ea02d3aa439297fa6c68e7585380101e9094fb5a5cd68ac8860bee79323`.

50. Reaction intermediate and retained path identity

Claim identity: `SFT-CHEM-RXN-INTERMEDIATE-001`

Question and theorem. A reaction intermediate is a named internal species produced by one mechanism step and consumed by a later step, canceling from net endpoints while remaining in the complete path trace.

> *named-internal-species__produced-by-one-step__consumed-by-later-step__production-precedes-consumption__cancels-from-net-endpoints*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`

- `SFT-CHEM-RXN-MECHANISM-001`

Generated grammar. Generate the literal product of identity, production, consumption, order, net, retention, record and extension choices.

Boundary: Every named chemical species produced by one internal elementary step and consumed by a later step of the same complete mechanism, absent from the net endpoint difference but retained in the path trace.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `named-internal-species__produced-by-one-step__consumed-by-later-step__production-precedes-consumption__cancels-from-net-endpoints__intermediate-path-identity-held__production-consumption-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`identity`	`named-internal-species`	`anonymous-transient`	Its recurrence cannot be checked.
`production`	`produced-by-one-step`	`not-produced`	It is not generated by the mechanism.
`consumption`	`consumed-by-later-step`	`not-consumed`	It remains a product, not an intermediate.
`order`	`production-precedes-consumption`	`consumed-before-produced`	The trace is causally open.
`net`	`cancels-from-net-endpoints`	`present-in-net-products`	It is then an endpoint product.
`retention`	`intermediate-path-identity-held`	`intermediate-erased`	Erasure destroys the mechanism proof.
`record`	`production-consumption-trace`	`intermediate-name-only`	A name lacks step witnesses.
`extra`	`no-extra-rule`	`free-short-lived-premise`	Lifetime alone cannot define path role.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a2752d09725cb5de86a3b147b38a4775c0d0becc9b26791e17fd354ce5226e48`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d64901b4b0f43026487180a4ab4442dca557b6ac66e399ebaa765e1d14e9028f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:2fdca90259d1e28cb742f7dc20ae1ba5f976a83c45c3f10a0c1e382fb943b0a7`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f2b5675a4b4d7b517d7b472aec706efc56e4d23acf3987a148244f3fbccde8f9`.

Independent reconstruction. Implementation hash `sha256:c903ed412ee204f58fc66bb48dd011c89b5c4b3b5fe31abbd90c28a67b8d4473`; independent certificate `sha256:068785565e96bdc6f82dc2adf032fa8837690bfab3da2ea0a7b9ccbddb92901d`; external-validation hash `sha256:4acb5fdf7c79667e0c65b328079b6a6937b3f641ca2f36b8e5e76d810375369a`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-I03096-2026

Comparison records:

- reaction-intermediate-iupac-i03096: predicted formed-in-one-elementary-step__consumed-in-later-step__absent-from-overall-reaction__retained-in-mechanism-trace; source-derived formed-in-one-elementary-step__consumed-in-later-step__absent-from-overall-reaction__retained-in-mechanism-trace; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks produced/consumed internal species behavior or net cancellation, or if a changed row is accepted.

Measurement receipt: `sha256:c92e03ebda1a8f51b0eb7c92c323ff29908c9edb55982ffa634603239131a2ca`. Isolation certificate: `sha256:20ee286750cb36b467a8df990444e0e08b4fcfcbd29bf513162d0fd3d06f9201`. Custody certificate: `sha256:c6f3cd23f5cdf1cf1ff8a689c62a2381d4acfd12aeca3ac90d51a589d4dbeda9`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/I03096 (sha256:9de39353e9af29cbe1b9dcb7f7972e2125378129e82924805f5d97696d93fd4a)

Registered target identities:

- reaction-intermediate-iupac-i03096 from IUPAC-GOLD-BOOK-I03096-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every named chemical species produced by one internal elementary step and consumed by a later step of the same complete mechanism, absent from the net endpoint difference but retained in the path trace.

Immutable identities. Source manifest `sha256:9c76d49c1d41800d93c513b98d52abca22f94cad8d7c03b733b6edb1d16f582e`; derivation seal `sha256:60419f37d0f2f2af2ab227df24f07cdf1492dea83d28ea32a32eaa52de479425`; engine receipt `sha256:d3d526c8adf59a9b3ec0a7c57e737f1d92396a72901748f64e0d35c426d28a4b` at `receipts/engine/model_admitted/SFT-CHEM-RXN-INTERMEDIATE-001-d3d526c8adf59a9b.json`; empirical hash `sha256:dd800138a2834ab9693b6852fab4a4d7951a54709ce78c59b2264deedc2f9219`; measurement receipt `sha256:c92e03ebda1a8f51b0eb7c92c323ff29908c9edb55982ffa634603239131a2ca`.

51. Activation barrier and transition boundary

Claim identity: `SFT-CHEM-KIN-ACTIVATION-001`

Question and theorem. Activation support is the least source-bound addition required for a generated reaction path to reach its highest transition boundary relative to the reactant boundary.

> *complete-generated-reaction-path__source-bound-support-account__highest-path-boundary__reactant-relative-support__least-crossing-support*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`

Generated grammar. Generate the literal product of path, source, transition, reference, minimality, orientation, record and extension choices.

Boundary: Every finite generated reaction path from a retained reactant organization to product organization, with the least additional source support needed to reach its unique highest transition boundary recorded relative to the reactant boundary.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: `complete-generated-reaction-path__source-bound-support-account__highest-path-boundary__reactant-relative-support__least-crossing-support__held-reactant-to-transition-orientation__path-source-boundary-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
path	complete-generated-reaction-path	endpoint-only	No crossing requirement is represented.
source	source-bound-support-account	free-energy-number	An imported number selects the barrier.
transition	highest-path-boundary	arbitrary-high-state	It need not lie on the path.
reference	reactant-relative-support	absolute-energy-origin	An arbitrary origin imports a parameter.
minimality	least-crossing-support	nonminimal-added-support	Extra support does not define activation.
orientation	held-reactant-to-transition-orientation	signed-barrier	A negative scalar is outside the proof domain.
record	path-source-boundary-trace	activation-value-only	A value cannot reproduce the path.
extra	no-extra-rule	free-temperature-fit	A fitted input can tune the result.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:660d0bba8291ba95de30f0439f20507f8a51eac2f38cbaf90287d61f61dd0645`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b027b4fdc5b38f0c0ef3c9a3232b0eabfa547393ee6c41f228abbb05e86328ad`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:94281f85f5d6d9d09d06e622c59d30d07f5246431edbba2ab67d03b2cc131be4`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4784092078887c2ffacaa9f8423c814871eb6d69a86186db2664831d2e22cac0`.

Independent reconstruction. Implementation hash `sha256:e383c27e7a4bb6acf54b280cfc8dbc77ee6df11b4dd68d4495893046a0b8c5d9`; independent certificate `sha256:2549feed7fb6748985ea7bfc80cd168b8d37152be45c05d441f5fc7a39972385`; external-validation hash `sha256:81b003fa87a790174213b42f0c5cdfc301b58cce5b0ab2e6bb5be25149bb3edb`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ACTIVATION-COMPOSITE-2026

Comparison records:

- activation-barrier-iupac-composite: predicted reactants-to-transition-state__minimum-required-energy__barrier-relative-to-reactants__reaction-path-source-bounded; source-derived reactants-to-transition-state__minimum-required-energy__barrier-relative-to-reactants__reaction-path-source-bounded; exact match True

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks a transition-state energy difference or activation barrier relative to reactants, or if a changed row is accepted.

Measurement receipt: `sha256:d322b1255c7d22f4a88b1ce7ef963ab0e3c0ed4f7c06fc3442f5474faa2980fc`. Isolation certificate: `sha256:f96a8486f9a212c48347a26e2794743b280f23151aef7dd473cb5f77a3c41fde`. Custody certificate: `sha256:97533f43d037a6e4fb53c50c27aea335f3312632f49bcc66a8f1dbd768179531`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/G02631 (sha256:c51e2c7167603bd633c54e11d238728c7313aa8584662ea4bfc3c3a4bf97a04c)

Registered target identities:

- activation-barrier-iupac-composite from IUPAC-GOLD-BOOK-ACTIVATION-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite generated reaction path from a retained reactant organization to product organization, with the least additional source support needed to reach its unique highest transition boundary recorded relative to the reactant boundary.

Immutable identities. Source manifest `sha256:ba9ef75073520a218b3d9b2733c17d66f85deb93f63bbbc4c6ff2d27a050bb4b`; derivation seal `sha256:5a6a90df0542817d3bb5ac872ebef9abcd3ce5a5a89126e41ba94d91f0ac73b1`; engine receipt `sha256:00815de40099c443e96ea38652bf6728bf86197c5ecbb5239f0886df392109a6` at `receipts/engine/model_admitted/SFT-CHEM-KIN-ACTIVATION-001-00815de40099c443.json`; empirical hash `sha256:923e4d7894816358f0e86e2de5bc28295e718de596a7b63ca28d3f6b0ca40591`; measurement receipt `sha256:d322b1255c7d22f4a88b1ce7ef963ab0e3c0ed4f7c06fc3442f5474faa2980fc`.

52. Reaction rate from counted transitions

Claim identity: `SFT-CHEM-KIN-RATE-001`

Question and theorem. Reaction rate is the exact positive count of completed instances of one registered reaction per positive reference recurrence and retained observation boundary, with conditions source-bound.

> *one-registered-reaction-event__positive-completed-event-count__positive-reference-recurrence-count__registered-species-observation-boundary__event-count-per-recurrence*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`

Generated grammar. Generate the literal product of event, count, duration, boundary, relation, condition, record and extension choices.

Boundary: Every finite observation boundary with a positive counted set of completed instances of one reaction transition and a positive counted set of registered recurrence intervals, retaining species amount and conditions as source records.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `one-registered-reaction-event__positive-completed-event-count__positive-reference-recurrence-count__registered-species-observation-boundary__event-count-per-recurrence__source-bound-condition-record__event-duration-boundary-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`event`	`one-registered-reaction-event`	`unidentified-change`	Events cannot be counted consistently.
`count`	`positive-completed-event-count`	`continuous-rate-primitive`	It imports a continuum quantity.
`duration`	`positive-reference-recurrence-count`	`clock-free-count`	No rate relation follows.
`boundary`	`registered-species-observation-boundary`	`unbounded-observation`	Amount change is undefined.
`relation`	`event-count-per-recurrence`	`fitted-rate-value`	A fitted value is target-derived.
`conditions`	`source-bound-condition-record`	`conditions-erased`	Rates cannot be compared reproducibly.
`record`	`event-duration-boundary-trace`	`rate-answer-only`	It cannot reconstruct events or time.
`extra`	`no-extra-rule`	`free-rate-constant`	A constant can tune any target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2ca5149a0037bdcb6c5af664a16fbe871bd044417c8461d3d1799bf414c4b85a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1ab78e2ba0ff39f6f97b554e3c8e84baeaf11592a3e925622243241c756f0513`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:a570e4838fecc768effd3446ebfe6dd2165ad3b10676081251be393901f4813f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:8c1067c6d6dd67c38da69b8631a81a9466230c9781a06506a5d3b2b95bcf7ab7`.

Independent reconstruction. Implementation hash `sha256:e6152457338984a6274e730a62ca2fe591bc2e4483bd7e7da9993b3f7eebef18`; independent certificate `sha256:6f370ecdb33bbca8628cb1164fc257a347847ae3548c54d3b7cb185ac3d21225`; external-validation hash `sha256:9610b18821a426fd436888599258a591f378fa1291f3ac3aaa9d85c96db2caf0`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-RATE-COMPOSITE-2026

Comparison records:

- reaction-rate-iupac-composite: predicted extent-of-reaction-change__per-time-interval__observation-boundary-retained__conditions-and-method-source-bounded; source-derived extent-of-reaction-change__per-time-interval__observation-boundary-retained__conditions-and-method-source-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reaction change per time or an observation/condition boundary, or if a changed row is accepted.

Measurement receipt: `sha256:926af1d645efd5da4142a2cb0c7271c25017b27a229092ad0c1d7c7920515683`. Isolation certificate: `sha256:c62c0366edd418c631e6b7a5c183aa562107e90bc15e9322b4d8e1a7a1f197b2`.

Custody certificate: `sha256:21609c1c1336495c948a13baac0c314b671dae43e321fd90a97bc28ee611e98c`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/R05147 (sha256:4f61cb4a9d72b0266dd464f84a2349aee57aec161521fb2466bcf8febed361fe)

Registered target identities:

- reaction-rate-iupac-composite from IUPAC-GOLD-BOOK-REACTION-RATE-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite observation boundary with a positive counted set of completed instances of one reaction transition and a positive counted set of registered recurrence intervals, retaining species amount and conditions as source records.

Immutable identities. Source manifest `sha256:5b37c081ee7916fb4c1590180a31e5c3ff8150c407f248c8ed34eb45df6b69a6`; derivation seal `sha256:a3bfaa4b92f014a8469b1efd9dd85aecc078e380a2b8843b4e54c1845405364b`; engine receipt `sha256:99c96fd6528df9b183cd7eaa7eefbb6ba0706cb3bc423712f861fcceae8cbc66` at `receipts/engine/model_admitted/SFT-CHEM-KIN-RATE-001-99c96fd6528df9b1.json`; empirical hash `sha256:35a59511424767f7bf92ceaa0c27aad4c89bcbe5c6839dd697703c255c3ae958`; measurement receipt `sha256:926af1d645efd5da4142a2cb0c7271c25017b27a229092ad0c1d7c7920515683`.

53. Kinetic order and dependency multiplicity

Claim identity: `SFT-CHEM-KIN-ORDER-001`

Question and theorem. Kinetic order is the exact condition-bound multiplicity of a named reactant coordinate in the complete intervention-to-rate dependency trace; independence is Empty rather than numerical zero.

> *named-reactant-coordinate__source-bound-coordinate-intervention__complete-rate-response-census__forced-dependency-occurrence-count__empty-dependency-form*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-KIN-RATE-001

Generated grammar. Generate the literal product of reactant, intervention, response, multiplicity, independence, scope, record and extension choices.

Boundary: Every finite source-bound intervention census for one reactant coordinate, retaining the exact multiplicity with which distinguishable occurrences of that coordinate participate in the rate dependency; absence is represented structurally.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `named-reactant-coordinate__source-bound-coordinate-intervention__complete-rate-response-census__forced-dependency-occurrence-count__empty-dependency-form__condition-and-mechanism-bound-order__reactant-intervention-response-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`reactant`	`named-reactant-coordinate`	`anonymous-concentration`	The dependency coordinate is lost.
`intervention`	`source-bound-coordinate-intervention`	`passive-correlation`	It cannot establish dependence.
`response`	`complete-rate-response-census`	`single-rate-answer`	Multiplicity cannot be recovered.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
multiplicity	forced-dependency-occurrence-count	stoichiometric-coefficient-assumed	Kinetic and endpoint counts need not coincide.
independence	empty-dependency-form	numerical-zero-order	Zero is prohibited as proof quantity.
scope	condition-and-mechanism-bound-order	universal-condition-free-order	Observed order can be condition-specific.
record	reactant-intervention-response-trace	order-number-only	It cannot reproduce the intervention.
extra	no-extra-rule	free-power-law-fit	A fit can select the exponent.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b2ec76b3e1f64ef6bbe60439d7c492f27614b93516d4c65092e90f2f94fc8985`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:295e33bd061e7b035038ac143c6ff58b9c0d6055338ea690a7eac1f7ef087903`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b1000a7635db0efabd0fbd160210eef648866a5cf90229cc0f5b0bdd55efeb31`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:dc3f77164ac93ae31817f766ae36505994210688eb8237c8a701a7f67084bc5e`.

Independent reconstruction. Implementation hash `sha256:ef4271771caeb278a9789ab2a7d72ab743de0a0cae928d1bec475ef3bdb934a2`; independent certificate `sha256:7d4b1f0011c6d5b0bce0edadeb270489f2a5924c3f7758d972ed5c640a79e26d`; external-validation hash `sha256:0230a4cc76598b16b09d36909165acbf8a4725fa7fb2e012d26b14b2eecd5284`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-O04322-2026

Comparison records:

- reaction-order-iupac-o04322: predicted rate-law-concentration-dependence__reactant-specific-exponent__experimentally-determined__not-forced-by-stoichiometry; source-derived rate-law-concentration-dependence__reactant-specific-exponent__experimentally-determined__not-forced-by-stoichiometry; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reactant-specific rate dependence, experimental determination or separation from stoichiometry, or if a changed row is accepted.

Measurement receipt: `sha256:b7890e640e17be8ec0a7adfabd3206761a25e02aaa84c609a674c4b107b1f4a4`. Isolation certificate: `sha256:81bbea60df23b9c7aa725798e54e899c18c5c86b16a7b1c954c251e14d855e4b`. Custody certificate: `sha256:54ed67554f342a9d0318a7bc538d3d62d96a410e35bb44df450049d24531bb7a`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/O04322 (sha256:acb0c1bcc39fca71b9fad77cfbc8813c6ffdc1728be3943d56dcd7631c90b9f4)

Registered target identities:

- reaction-order-iupac-o04322 from IUPAC-GOLD-BOOK-O04322-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite source-bound intervention census for one reactant coordinate, retaining the exact multiplicity with which distinguishable occurrences of that coordinate participate in the rate dependency; absence is represented structurally.

Immutable identities. Source manifest `sha256:069ae7a975d9f3116f2ee9a8203a90d4fda441c301daa5fbdab95d409558459b`; derivation seal `sha256:3c7cffc10bb4f18a24b259412017485981e62d7e11e5076b6053681e820b6d74`; engine receipt `sha256:e8260df43aa9bdf7fbdce385d508c2727ecc65a933199419d2a1dc2dce1e2807` at `receipts/engine/model_admitted/SFT-CHEM-KIN-ORDER-001-e8260df43aa9bdf7.json`; empirical hash `sha256:46440e2c4707141567a275ba3c3801f79be741177ed049dd4ea1ed5eaf3882c1`; measurement receipt `sha256:b7890e640e17be8ec0a7adfabd3206761a25e02aaa84c609a674c4b107b1f4a4`.

54. Chemical equilibrium as balanced reversible support

Claim identity: `SFT-CHEM-EQ-CHEMICAL-001`

Question and theorem. Chemical equilibrium is persistent forward and reverse reaction recurrence on a closed boundary whose paired carrier changes preserve the macroscopic composition class.

> *forward-and-reverse-transitions__persistent-opposed-events__paired-forward-reverse-support__macroscopic-composition-class-recurs__closed-observation-boundary*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`

- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-KIN-RATE-001

Generated grammar. Generate the literal product of reversibility, activity, balance, composition, boundary, condition, record and extension choices.

Boundary: Every finite reversible chemical transition on one closed observation boundary for which forward and reverse event recurrences persist while their complete carrier changes pair and the retained macroscopic composition class recurs.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `forward-and-reverse-transitions__persistent-opposed-events__paired-forward-reverse-support__macroscopic-composition-class-recurs__closed-observation-boundary__source-bound-condition-record__forward-reverse-composition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
reversibility	forward-and-reverse-transitions	one-way-reaction	No reverse support exists.
activity	persistent-opposed-events	static-no-events	Chemical equilibrium is dynamically supported.
balance	paired-forward-reverse-support	unpaired-event-counts	Composition would drift.
composition	macroscopic-composition-class-recurs	microscopic-state-fixed	Individual events need not stop.
boundary	closed-observation-boundary	open-material-flow	External flow can imitate balance.
conditions	source-bound-condition-record	condition-free-equilibrium	The class can change with conditions.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
record	forward-reverse-composition-trace	equilibrium-constant-only	A number cannot reconstruct dynamics.
extra	no-extra-rule	free-equilibrium-fit	A fitted constant can force a target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:efbdf446dafe76afa9d768e8ab86b366db99bb763a88014f1559f5279b7a246b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:101db4c738b823c2aaa66bf158c163a9c9f0b6dff0ebcaa24e4f669f0e8b77b6`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bef1a0371237a62a4e9f2ccebde1c97119c7797967f5081c7d6014b377fdd4fc`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:1d3e059d0e008a62401bd5313bb3466cef844adb001f118d0cdd58a821fa6dae`.

Independent reconstruction. Implementation hash `sha256:9795f2eafcfe4afe0c107b84963aeab091d2859dec0229c033bea11e3d13b0c6`; independent certificate `sha256:83220324729c01a4a2bdb47ec45c4b6bfc93606c5266a3cdca91c5e3165d6c21`; external-validation hash `sha256:d17f66994f5cd49c91591a208e5efdb6aa2ccfee607a6e268efda1c87908c4ad`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHEMICAL-EQUILIBRIUM-COMPOSITE-2026

Comparison records:

- chemical-equilibrium-iupac-composite: predicted forward-and-reverse-reactions-continue__rates-equal-at-equilibrium__macroscopic-composition-constant__condition-bound-dynamic-state; source-derived forward-and-reverse-reactions-continue__rates-equal-at-equilibrium__macroscopic-composition-constant__condition-bound-dynamic-state; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks continuing opposed reactions, equal rates or constant macroscopic composition under fixed conditions, or if a changed row is accepted.

Measurement receipt: `sha256:ff84b4286b37cf2810cbfe7e7df69ba44d3cd578188ab16641520059df284dcb`. Isolation certificate: `sha256:0259897be117fe27df8f7f19f3d31b3a2d62560d635f780fe0ef4e45f4f01197`. Custody certificate: `sha256:b553447a99205c4bd732e6551f68ff5a2c8f9c3ecffff9052885f96f68432eb2`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01023 (sha256:8720b60c684dd7869500da1577be0f0a91f2362e02420d755fbbd1abbad6582b)

Registered target identities:

- chemical-equilibrium-iupac-composite from IUPAC-GOLD-BOOK-CHEMICAL-EQUILIBRIUM-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no reaction dictionary, mechanism database, rate equation, Arrhenius law, equilibrium constant, measured target or V2 answer may select a candidate

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier may be created, copied or silently erased
- an absent dependence or event is an empty structural form, never numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite reversible chemical transition on one closed observation boundary for which forward and reverse event recurrences persist while their complete carrier changes pair and the retained macroscopic composition class recurs.

Immutable identities. Source manifest `sha256:d2a8ad0114607185b74f98b10700b4398001277e7b8f7e87b74d344375914eec`; derivation seal `sha256:018d95e3da5aa30562b76c4a00f079db3ef1eb3e0286a4d91808db3dbb450239`; engine receipt `sha256:978807f747d3d6af21f13d5b83308e9bde9ad276651aadffcdb721abd3edcfd8` at `receipts/engine/model_admitted/SFT-CHEM-EQ-CHEMICAL-001-978807f747d3d6af.json`; empirical hash `sha256:d2e8df7e234a5777b25e5724a4e5f69e063a6a4cf89409ab2fdb0436f09e05ef`; measurement receipt `sha256:ff84b4286b37cf2810cbfe7e7df69ba44d3cd578188ab16641520059df284dcb`.

55. Reaction thermochemical accounting

Claim identity: `SFT-CHEM-THERMO-REACTION-001`

Question and theorem. Reaction thermochemistry is the complete source-bound energy ledger across one conserved chemical transformation, with transfer orientation, phase, conditions and reference state retained.

> *registered-reaction-trace__complete-reactant-energy-support__complete-product-energy-support__energy-transfer-balanced__held-absorbed-or-released-orientation*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`

- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-RXN-INTERMEDIATE-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-KIN-ORDER-001`
- `SFT-CHEM-EQ-CHEMICAL-001`

Generated grammar. Generate the literal product of reaction, input, output, conservation, orientation, condition, record and extension choices.

Boundary: Every finite closed chemical reaction with complete source-bound energy carriers at reactant and product boundaries, held transfer orientation, phase, conditions and reference state.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `registered-reaction-trace__complete-reactant-energy-support__complete-product-energy-support__energy-transfer-balanced__held-absorbed-or-released-orientation__phase-condition-reference-held__reaction-energy-source-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`reaction`	`registered-reaction-trace`	`energy-value-without-reaction`	No chemical identity is retained.
`input`	`complete-reactant-energy-support`	`reactant-energy-erased`	The ledger cannot close.
`output`	`complete-product-energy-support`	`product-energy-erased`	The ledger cannot close.
`conservation`	`energy-transfer-balanced`	`free-energy-creation`	It violates closure.
`orientation`	`held-absorbed-or-released-orientation`	`signed-heat-number`	A signed scalar is unnecessary.
`conditions`	`phase-condition-reference-held`	`condition-free-value`	Thermochemical quantities are state-bound.
`record`	`reaction-energy-source-trace`	`enthalpy-answer-only`	It cannot reproduce accounting.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
extra	no-extra-rule	free-calibration-fit	A fit can select any value.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:f0d09549fc7aba9e3e959cdcd632a04ccd046cf9645b3a55925f0779021368d1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:57872b69512c1c9f80e2e7c241245dd8bd4bed71f6fdba2ff6b57b3ebc2f931a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:76d290aaa533e03e6abe301a16e9471b1e7120820c210f8187c6bef6d0a0a7d0`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:fdb60292a3bddf4eb85038e4d768a41a44dc18bdc79863a07913318ce58e8db3`.

Independent reconstruction. Implementation hash `sha256:f04a8631eb33adf1005011b19ed41ebf2b14be6ce7d9ebffcda79c8026cbd7fb`; independent certificate `sha256:3701008c6ea7b1b78f1f9909e75c5d5edac9a3877f7772c9607399a0232932d9`; external-validation hash `sha256:a880f1c747e82bd929353b86c60c814e7ec5e6be52ee739b47ed8b955a119dd6`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-THERMOCHEMISTRY-COMPOSITE-2026

Comparison records:

- reaction-thermochemistry-iupac-composite: predicted reaction-energy-change__products-minus-reactants__absorbed-or-released-transfer__state-and-condition-source-bounded; source-derived reaction-energy-change__products-minus-reactants__absorbed-or-released-transfer__state-and-condition-source-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks reaction energy difference, transfer direction or state/condition boundary, or if a changed row is accepted.

Measurement receipt: `sha256:ac1d842030c78560b84d78d2a9b4b9cad4644c9643f26fe814c8075d13ce1435`. Isolation certificate: `sha256:48b44c40284e3c5eafcbcde74672e91eb8116112abd94aba93e528cf9e4fe810`. Custody certificate: `sha256:b6b35c656d07aac110dda579955013faa44e5baccee8eb0863786d1e1a2475a2`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/S05923 (sha256:e976fc5090d3344af00af1a64fea3bc564f126f53d920a0387c1fbf56ba07bf6)

Registered target identities:

- reaction-thermochemistry-iupac-composite from IUPAC-GOLD-BOOK-REACTION-THERMOCHEMISTRY-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite closed chemical reaction with complete source-bound energy carriers at reactant and product boundaries, held transfer orientation, phase, conditions and reference state.

Immutable identities. Source manifest `sha256:69e956035fb491afd25f4dca581426cb39ea6281bb212b3208b43858ec22c968`; derivation seal `sha256:4ee0be5635e2212cfbf412062acaf3dfb7f7efb4b5eb4e72e007156bed858800`; engine receipt `sha256:840194dda329e635a6f566c8611fa14c264341b2dcb3df8c5023ed8d1b0a0ddc` at `receipts/engine/model_admitted/SFT-CHEM-THERMO-REACTION-001-840194dda329e635.json`; empirical hash `sha256:0925b392e7ec1d0556d5a245934271f2e02009aec55dc249873fcaa8a0e22d70`; measurement receipt `sha256:ac1d842030c78560b84d78d2a9b4b9cad4644c9643f26fe814c8075d13ce1435`.

56. Reaction direction under complete energy and distinction accounting

Claim identity: `SFT-CHEM-THERMO-DIRECTION-001`

Question and theorem. Reaction direction is the held orientation selected by strict complete support order between forward and reverse paths; equality yields no selected direction.

> *forward-and-reverse-paths__complete-energy-transfer-ledgers__complete-accessible-distinction-census__strict-support-order__held-favored-direction*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`

- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-RXN-INTERMEDIATE-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-KIN-ORDER-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`

Generated grammar. Generate the literal product of path, energy, distinction, comparison, orientation, equilibrium, record and extension choices.

Boundary: Every finite pair of forward and reverse reaction paths on one retained condition boundary, compared by complete source, energy-transfer and distinction-production support, selecting an orientation only when one path strictly closes the available transition support.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `forward-and-reverse-paths__complete-energy-transfer-ledgers__complete-accessible-distinction-census__strict-support-order__held-favored-direction__empty-direction-at-equality__path-energy-distinction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`paths`	`forward-and-reverse-paths`	`forward-path-only`	No directional comparison exists.
`energy`	`complete-energy-transfer-ledgers`	`energy-change-erased`	Available support is unknown.
`distinctions`	`complete-accessible-distinction-census`	`entropy-label-only`	A label cannot count support.
`comparison`	`strict-support-order`	`imported-spontaneity-sign`	A sign assumes the answer.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`orientation`	`held-favored-direction`	`negative-Gibbs-primitive`	A negative scalar is prohibited.
`equilibrium`	`empty-direction-at-equality`	`forced-direction-at-tie`	Equal support cannot select.
`record`	`path-energy-distinction-trace`	`direction-answer-only`	It cannot reproduce comparison.
`extra`	`no-extra-rule`	`free-temperature-threshold`	A parameter can force a target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:daf33949c5cd054da35a9146169943e1dda63e5891e02fb4492534265b797eb8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bf934a0d381cf363f0d6e66be8f507d530b60942e2322d5230310374c465f779`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:b87f78bc4cf618f78510d9a021bb6ce4de8289791c3af93cd66300979fbf8e65`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5343842832622e21645afc61d2c5a53702e7ee8f0ebee61619e03653bbbf8d1a`.

Independent reconstruction. Implementation hash `sha256:6576a2864e07da87d8d37adce128d68cdf2776c2fcfcc90f2b7e5deef9925c4e`; independent certificate `sha256:2189dca95b759411e57de2a862e881e2ded04255a51f0b74b44c3d7eb15631d0`; external-validation hash `sha256:5f99d50d30440c5391246981f586984022055369a8134aeee5ec6fc890c78d4f`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-DIRECTION-COMPOSITE-2026

Comparison records:

- reaction-direction-iupac-composite: predicted forward-or-reverse-spontaneous-tendency__Gibbs-energy-condition__equilibrium-at-no-driving-force__direction-state-bounded; source-derived forward-or-reverse-spontaneous-tendency__Gibbs-energy-condition__equilibrium-at-no-driving-force__direction-state-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks condition-bound forward/reverse tendency and equilibrium at no driving force, or if a changed row is accepted.

Measurement receipt: `sha256:abdeeefe3cbbffbba3e06d870eceeb641c90d50e615c6db77864a6c5e636e370`. Isolation certificate: `sha256:ca15d06d7c7be40abbf8aec72cd6d8abb2f312492c19bb46f036c50b83a2f4f5`. Custody certificate: `sha256:5921bd0bc5554b2c92ef339ace23e7c271de63e287eae04f4638d21837d051c3`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/A00178 (sha256:29e64f1b53aba905340496145e5393f853b118f32684689de8e79c573e2642ef)

Registered target identities:

- reaction-direction-iupac-composite from IUPAC-GOLD-BOOK-REACTION-DIRECTION-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite pair of forward and reverse reaction paths on one retained condition boundary, compared by complete source, energy-transfer and distinction-production support, selecting an orientation only when one path strictly closes the available transition support.

Immutable identities. Source manifest `sha256:73936f608dd0627a40b89cad3a1ed88af0d90ee17fd67bdc2406b020057ce4f6`; derivation seal `sha256:a5aa5ba02e12d483b2334877b1363d1c76e1146fda259c551bb7685d9aade456`; engine receipt `sha256:44288b966ad2be73f14444d40e340c5ae7b84124a568089710041c4e71fccf64` at `receipts/engine/model_admitted/SFT-CHEM-THERMO-DIRECTION-001-44288b966ad2be73.json`; empirical hash `sha256:f5dff69383067d491f8058244253651f77cf7658d9ff9468d604ebd952745865`; measurement receipt `sha256:abdeeefe3cbbffbba3e06d870eceeb641c90d50e615c6db77864a6c5e636e370`.

57. Chemical phase and coexistence identity

Claim identity: `SFT-CHEM-PHASE-CHEMICAL-001`

Question and theorem. A chemical phase is a maximal connected region of complete intensive-trace equivalence; coexistence retains two phase identities and their recurrent interface under fixed conditions.

> *maximal-connected-region__complete-intensive-trace-equivalence__extent-independent-class__retained-phase-interfac e__paired-interface-recurrence*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`

- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-RXN-INTERMEDIATE-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-KIN-ORDER-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`

Generated grammar. Generate the literal product of region, homogeneity, extent, boundary, coexistence, condition, record and extension choices.

Boundary: Every finite maximal connected chemical region whose complete intensive observation trace is homogeneous under the declared resolution, separated from a distinct region by a retained interface; coexistence is paired recurrence at that interface.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `maximal-connected-region__complete-intensive-trace-equivalence__extent-independent-class__retained-phase-interface__paired-interface-recurrence__temperature-pressure-composition-held__region-trace-interface-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`region`	`maximal-connected-region`	`arbitrary-sample-label`	It does not define phase identity.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`homogeneity`	`complete-intensive-trace-equivalence`	`single-point-observation`	It cannot establish uniformity.
`extent`	`extent-independent-class`	`size-dependent-phase`	Phase identity is intensive.
`boundary`	`retained-phase-interface`	`interface-erased`	Distinct phases cannot be separated.
`coexistence`	`paired-interface-recurrence`	`one-phase-only`	No coexistence relation exists.
`conditions`	`temperature-pressure-composition-held`	`condition-free-phase`	Phase depends on source conditions.
`record`	`region-trace-interface-record`	`phase-name-only`	It cannot reproduce homogeneity.
`extra`	`no-extra-rule`	`free-phase-diagram`	An imported diagram selects the class.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:eaad5bde6903bc5cb65e88d04f3ee3e8bd4a9421c6af7f9751047698beeb46b0`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:71c4e3eb21b71d48f16e94127cbf7c5b2bee89e466b76e2db5d2d8ee43e98db1`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bf9dcbde12718e36cc70d466631dfe841fa2855c168100de4cc9fc351f596967`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ae5241a8b94116a4a4db862b70868b989231718eb7d846d5376fdb5cacee158c`.

Independent reconstruction. Implementation hash `sha256:2b5773927b802554089e42a4a081ec5460410a5064e57aece41893d4a9408eab`; independent certificate `sha256:6c954b256c8c36582c6097aacb9c944371cfd17d01731f9fc6a336517e6cef23`; external-validation hash `sha256:62d3208c01b4e308cedebc0e3f314c5bfdaad58b61b2dab26edb9a7dbf7a9388`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHEMICAL-PHASE-COMPOSITE-2026

Comparison records:

- chemical-phase-iupac-composite: predicted homogeneous-material-region__uniform-intensive-properties__phase-boundary-retained__coexistence-condition-bounded; source-derived homogeneous-material-region__uniform-intensive-properties__phase-boundary-retained__coexistence-condition-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks homogeneous intensive properties, a phase boundary or condition-bound coexistence, or if a changed row is accepted.

Measurement receipt: `sha256:378e8c8a6055fa9ac51adbe2baedf501aea62e6a5b2ba476f2b22c20ea4666e8`. Isolation certificate: `sha256:6733cbc835038977074b56c9c85c585e3e5019620263c9251af4b8975350f6c7`. Custody certificate: `sha256:3256de62bbb1d0bdad6104e6e0db7b32ca48fa69be264bf83368a58738732c8f`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/P04528 (sha256:c903c27903b67f47a61403a4de408c23bb21c50100ee37e5d05e9a8aa168c4eb)

Registered target identities:

- chemical-phase-iupac-composite from IUPAC-GOLD-BOOK-CHEMICAL-PHASE-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite maximal connected chemical region whose complete intensive observation trace is homogeneous under the declared resolution, separated from a distinct region by a retained interface; coexistence is paired recurrence at that interface.

Immutable identities. Source manifest `sha256:d7047f0c37d679103c43e499b2a4ae61c751a601d38019c2c1d77d71d497cffd`; derivation seal `sha256:a966aa7be9fcab6078680631122c59e3596902f95a476543a43bd54e3a2a51d7`; engine receipt `sha256:5e82a4bf1e2a7cafa2e2016288ccc4389039c406cb713d9258817d01690cadb6` at `receipts/engine/model_admitted/SFT-CHEM-PHASE-CHEMICAL-001-5e82a4bf1e2a7caf.json`; empirical hash `sha256:d22585e67a829aaaf7d93acb4d3f5ee960f472e8d4986c3a93c47e704e35d8fa`; measurement receipt `sha256:378e8c8a6055fa9ac51adbe2baedf501aea62e6a5b2ba476f2b22c20ea4666e8`.

58. Solution and solubility equilibrium

Claim identity: `SFT-CHEM-SOLUTION-EQUILIBRIUM-001`

Question and theorem. Solubility equilibrium is paired dissolution and return recurrence on a retained solution boundary, preserving a condition-bound macroscopic dissolved-composition class at finite saturation.

> *retained-solute-solvent-organization__dissolution-and-return-transitions__paired-opposed-event-support__source-bounded-capacity-boundary__macroscopic-dissolved-class-recurs*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-RXN-INTERMEDIATE-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-KIN-ORDER-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`

Generated grammar. Generate the literal product of solution, dissolution, balance, saturation, composition, condition, record and extension choices.

Boundary: Every finite solution boundary containing a retained solvent/solute organization and opposed dissolution/return transitions whose complete event supports pair at saturation while the macroscopic dissolved-composition class recurs.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `retained-solute-solvent-organization__dissolution-and-return-transitions__paired-opposed-event-support__source-bounded-capacity-boundary__macroscopic-dissolved-class-recurs__phase-temperature-pressure-held__solution-event-composition-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
solution	retained-solute-solvent-organization	unidentified-mixture	Solute and solvent roles are lost.
dissolution	dissolution-and-return-transitions	one-way-dissolution-only	No equilibrium can close.
balance	paired-opposed-event-support	unpaired-event-support	Composition would drift.
saturation	source-bounded-capacity-boundary	free-solubility-number	A measured number imports the target.
composition	macroscopic-dissolved-class-recurs	microscopic-state-fixed	Individual events may continue.
conditions	phase-temperature-pressure-held	condition-free-solubility	Solubility changes with conditions.
record	solution-event-composition-trace	solubility-answer-only	It cannot reproduce the dynamics.
extra	no-extra-rule	free-equilibrium-fit	A fit can select any saturation.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:759effeb3a46056715c37ecb42ca70aab80ba89291af7b19bc00e39f9ff790ad`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1444804b3af5b2334fabc3e15a71486349ff605b48105b8b37a8e3f7916b0cbc`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:0794da732271434dc54f50e6cf7985ead0f54ea16e2550bd49c249b7947af18f`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:941cae2d85c4715f190d7e25bf31cc19ff04e0c64b3c5b8f0ec3d6218ced6651`.

Independent reconstruction. Implementation hash `sha256:04f14dc7cdbb5249ae6e469a5f9a7d95e1cdd028e8618ba000ecfe236c7bbda1`; independent certificate `sha256:7016e64075d4ce9a6ea43e2ca21334b72162eac1201bf3ca2634f086cff9ec6d`; external-validation hash `sha256:285f027c0ee4d166d12027761c3ac52d0f033fdf7f8e575f636324ec6ec138e3`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-SOLUTION-EQUILIBRIUM-COMPOSITE-2026

Comparison records:

- solution-equilibrium-iupac-composite: predicted dissolution-and-crystallization-continue__rates-balanced-at-saturation__dissolved-composition-constant__solubility-condition-bounded; source-derived dissolution-and-crystallization-continue__rates-balanced-at-saturation__dissolved-composition-constant__solubility-condition-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks opposed dissolution/crystallization, saturation balance or condition-bounded solubility, or if a changed row is accepted.

Measurement receipt: `sha256:e05637d5d7eb545895db3b52bea9e512eb68bd36a6a08dae932dd29b33c18ebf`. Isolation certificate: `sha256:582aa0e7333c4629b606aaa1764ab3315c6df9b6014f596fda80a3e9999d7b15`. Custody certificate: `sha256:f32ab21383a5df840a6c9c40887f1841e6ad05b8d5f6f9cc2f8fc51644097379`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/14656 (sha256:bd07661ff6120f150e0a53e67a72b1e212293bf96bfad41628c1e8f7ec9ffdcb)

Registered target identities:

- solution-equilibrium-iupac-composite from IUPAC-GOLD-BOOK-SOLUTION-EQUILIBRIUM-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite solution boundary containing a retained solvent/solute organization and opposed dissolution/return transitions whose complete event supports pair at saturation while the macroscopic dissolved-composition class recurs.

Immutable identities. Source manifest `sha256:9ce1a2df72de781c8a0e63ca571e8f76a12f4005a3897da83e2bf88ce50fd03b`; derivation seal `sha256:dbf569566576f95f51a09112cb46de35d86bc76ee5e6b0eefe1c41edd9b87a8d`; engine receipt `sha256:bf4b6867d9bdc697d9105e06e2774b8634ccc97bf843faf21b1ad37e6ddea9f6` at `receipts/engine/model_admitted/SFT-CHEM-SOLUTION-EQUILIBRIUM-001-bf4b6867d9bdc697.json`; empirical hash `sha256:37417706e059873afb2f0ce6b42c19164259c4dee15e5cfc4b416c3d9b44c09f`; measurement receipt `sha256:e05637d5d7eb545895db3b52bea9e512eb68bd36a6a08dae932dd29b33c18ebf`.

59. Photochemical excitation and reaction channel

Claim identity: `SFT-CHEM-PHOTOCHEM-001`

Question and theorem. A photochemical event absorbs one retained photon carrier into a named molecular carrier, produces distinct excited support and closes through a generated reaction or relaxation channel with complete energy accounting.

> *one-retained-photon-carrier__carrier-absorbed-once__distinct-excited-state-support__source-bound-reaction-or-relaxation-channel__complete-photon-molecule-energy-ledger*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-RXN-INTERMEDIATE-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-KIN-ORDER-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`

Generated grammar. Generate the literal product of radiation, absorption, state, channel, conservation, selection, record and extension choices.

Boundary: Every finite chemical path initiated when one retained radiation carrier is absorbed by a named molecular carrier, producing a distinct excited-state support and a source-bound reaction or relaxation channel with complete energy accounting.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `one-retained-photon-carrier__carrier-absorbed-once__distinct-excited-state-support__source-bound-reaction-or-relaxation-channel__complete-photon-molecule-energy-ledger__generated-channel-census__photon-state-channel-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
radiation	one-retained-photon-carrier	anonymous-energy-input	It cannot establish carrier identity.
absorption	carrier-absorbed-once	photon-copied	Copying violates conservation.
state	distinct-excited-state-support	ground-state-unchanged	No excitation occurs.
channel	source-bound-reaction-or-relaxation-channel	unregistered-outcome	No chemical path is retained.
conservation	complete-photon-molecule-energy-ledger	energy-erased	The path cannot close.
selection	generated-channel-census	free-quantum-yield-fit	A fit can force outcomes.
record	photon-state-channel-trace	photochemical-label-only	It cannot reproduce excitation.
extra	no-extra-rule	free-spectral-rule	An imported rule selects a target.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:604c53c9f236ae0f1bbe2349ede60a046b166a3e5310ea1874f6118f092477a4`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:db644f9f56463d159d682018a30e2c3e1b261e3080722afb120458ce503a11ee`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bbd4f835c874e51ff9de61a5de9b060ed22e3cc5400b77f7b6a1e19cda121d35`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:79ae19af5ff3de24c52ed6a09785a6d314a2b60b931535b8373fee3ff3743582`.

Independent reconstruction. Implementation hash `sha256:15de61fa6b9a5576d3eef1e3535ff6b4376d8041b93d8757fda8751a81ad1abe`; independent certificate `sha256:ff70f8241cc8f564e7d87df9ebdc5621ebd377a57b73bda7386a5ffdd1749376`; external-validation hash `sha256:7ddb6f6ef6b9a799d6f25bcb7c7522d96d304aa9d3857d408cc2158b28e2d858`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-PHOTOCHEMISTRY-COMPOSITE-2026

Comparison records:

- photochemistry-iupac-composite: predicted light-absorption-initiates-chemical-change__molecular-excited-state__reaction-or-relaxation-pathways__photon-energy-accounted; source-derived light-absorption-initiates-chemical-change__molecular-excited-state__reaction-or-relaxation-pathways__photon-energy-accounted; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if the authority record lacks light absorption, excited molecular state, reaction/relaxation pathways or photon-energy accounting, or if a changed row is accepted.

Measurement receipt: `sha256:8036e055d4e492ed9f5c5aa0c6ba35f582b0f505d4562be11c072be5343519d5`. Isolation certificate: `sha256:6b3830ac094fd71628d2cc484dc7beed556cf80b902333697ebcb5a88dab3c60`.

Custody certificate: `sha256:391fef0779e04135985b95dabecf4e4ad3fa00a0e8e244577d9c093fdbb7600b`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/P04585 (sha256:5e052ba4dda35e0152ec2de28c4d90a748933bd99a82481d5b0b31d9f0c0ed3d)

Registered target identities:

- photochemistry-iupac-composite from IUPAC-GOLD-BOOK-PHOTOCHEMISTRY-COMPOSITE-2026

Meaning. Reactions are complete transition maps; mechanisms, intermediates, activation, rate, equilibrium, direction, phase and photochemical response retain source, condition and observation boundaries. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no thermochemical table, spontaneity rule, phase diagram, solubility value, photochemical database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no energy, matter or photon carrier may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical path initiated when one retained radiation carrier is absorbed by a named molecular carrier, producing a distinct excited-state support and a source-bound reaction or relaxation channel with complete energy accounting.

Immutable identities. Source manifest `sha256:eeeef76938bbd5739b71a584660a924d9d29c5b3a23d4ea929825109a5e52de5`; derivation seal `sha256:e87d7d9d897f34a9de12c4f818d9b863c1e91a394eadbe8786df7aee2d48ccbb`; engine receipt `sha256:c3d62a2cb062977f327a9d9bf8db2ede9bd15e72a8b12b5cdbd690c8539f89af` at `receipts/engine/model_admitted/SFT-CHEM-PHOTOCHEM-001-c3d62a2cb062977f.json`; empirical hash `sha256:e20a75360518123d270b5a9eb3be2226df57fac0e9ffe44dd492e077d312c5a8`; measurement receipt `sha256:8036e055d4e492ed9f5c5aa0c6ba35f582b0f505d4562be11c072be5343519d5`.

15. Catalysis Networks Interfaces

Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates.

60. Catalyst identity conserved through a reaction cycle

Claim identity: `SFT-CHEM-CAT-CATALYST-001`

Question and theorem. A catalyst is a retained chemical carrier that opens an alternative closed reaction path and returns with exact identity while the net reaction endpoints are unchanged.

> *reaction-cycle-participant__exact-catalyst-identity-returned__alternative-step-path-retained__reaction-rate-increased__net-reaction-unchanged*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`

- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite source-bound reaction cycle whose catalyst entry, step occurrences, return identity, net endpoints and recurrence interval are completely recorded.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `reaction-cycle-participant__exact-catalyst-identity-returned__alternative-step-path-retained__reaction-rate-increased__net-reaction-unchanged__both-directions-share-path__complete-cycle-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`role`	`reaction-cycle-participant`	`ordinary-reactant-only`	The carrier is consumed.
`identity`	`exact-catalyst-identity-returned`	`carrier-identity-changed`	It is not recovered.
`path`	`alternative-step-path-retained`	`overall-reaction-only`	No catalytic path is distinguished.
`rate`	`reaction-rate-increased`	`rate-unchanged-by-cycle`	No catalytic action occurs.
`net`	`net-reaction-unchanged`	`catalyst-in-net-products`	The carrier was consumed or produced.
`direction`	`both-directions-share-path`	`one-direction-rule`	A reversible cycle is incomplete.
`record`	`complete-cycle-trace`	`catalyst-name-only`	Identity return cannot be checked.
`extra`	`no-extra-rule`	`free-catalytic-fit`	A fit can select the result.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a0ae5548ab64a5314989e01db584c23471703055d234de86e8ecf9285c346a9e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fc5ab89d3cfab7add282f8d230b65bbc434f7ef691e76e8ed714e1ca4f5f4c71`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9c78d99f45777e5234b96e68742b6f38d3807dddab51f0149c8aa228825b70df`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:0f4686e4369b29d1c58e4bad23b4d7954f59c0dbfe99122ffaddef061229f026`.

Independent reconstruction. Implementation hash `sha256:58ec12d7c67ad1fd917c9639d1ff8ced462e348d4c471e56192d4f5b8a7a314a`; independent certificate `sha256:f573cafbd6724ed8c57b766c376536ba7562110459be982c9209dae132e385f8`; external-validation hash `sha256:ef91777583f0538821a1e265b7a626a150d14043bb13afc9d614706e5205913a`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- CHEM-AUTHORITY-CATALYST-COMPOSITE-2026

Comparison records:

- catalyst-authority-composite: predicted reaction-rate-increased__alternative-reaction-pathway__catalyst-regenerated__net-reaction-unchanged; source-derived reaction-rate-increased__alternative-reaction-pathway__catalyst-regenerated__net-reaction-unchanged; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if an official authority does not retain changed rate, an alternative path, regenerated catalyst identity and unchanged net reaction, or if a tampered row is accepted.

Measurement receipt: `sha256:96d8e97195cd37027b33af2b192ef23d00925f57b66bff2ab20ba127440643ac`. Isolation certificate: `sha256:bf6cf0fee16d5c387c00fcba0ca8e6c284dcd727845c7addf275e9d825667d15`. Custody certificate: `sha256:712e07f3fc11a1f7a8494cf407b8f6630363cbe3a16a95cbb3de52dcd03a266d`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C00876 (sha256:a960e78123592da7dd8ad054b65a590961706c25e22a578f9088f91fb3c89e5e)

Registered target identities:

- catalyst-authority-composite from CHEM-AUTHORITY-CATALYST-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite source-bound reaction cycle whose catalyst entry, step occurrences, return identity, net endpoints and recurrence interval are completely recorded.

Immutable identities. Source manifest `sha256:f271c54c6bf485dbd04582f78a97baca117a1518aa8221b753dc87b84ffd1323`; derivation seal `sha256:dff7a7af9f63f2ce6f3acb8ebba8a8daef932c3556723ea419109b456507103a`; engine receipt `sha256:239ac25a2ef81dbb6f4a5a2664ee3310f3fb0e9c6388aa4a5fc6a80aaa214b35` at `receipts/engine/model_admitted/SFT-CHEM-CAT-CATALYST-001-239ac25a2ef81dbb.json`; empirical hash `sha256:5b97ac2db40346dd17a6fbd79f10d8f7ee129c5c3aac5cf36fce2fb428f3067b`; measurement receipt `sha256:96d8e97195cd37027b33af2b192ef23d00925f57b66bff2ab20ba127440643ac`.

61. Catalytic alternative-path relation

Claim identity: `SFT-CHEM-CAT-PATHWAY-001`

Question and theorem. A catalytic pathway is the condition-bound complete step trace sharing overall reaction endpoints with an uncatalysed path, ordered by activation support and closing the catalyst cycle.

> *same-overall-reaction__alternative-reaction-mechanism__lower-activation-support__catalyst-cycle-retained__all-intermediates-balanced*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-CATALYST-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite pair of balanced reaction mechanisms with common endpoints, complete intermediate and activation records, and one exact catalyst-return cycle.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `same-overall-reaction__alternative-reaction-mechanism__lower-activation-support__catalyst-cycle-retained__all-intermediates-balanced__conditions-held__pathwise-energy-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`endpoints`	`same-overall-reaction`	`different-overall-products`	The paths are not alternatives.
`mechanism`	`alternative-reaction-mechanism`	`step-trace-erased`	No pathway comparison exists.
`barrier`	`lower-activation-support`	`imported-energy-number`	A target value is not forced.
`carrier`	`catalyst-cycle-retained`	`catalyst-consumed`	The cycle does not close.
`conservation`	`all-intermediates-balanced`	`free-intermediate`	Matter is unaccounted.
`conditions`	`conditions-held`	`condition-free-path`	Path availability is unbounded.
`record`	`pathwise-energy-trace`	`endpoint-answer-only`	The alternative path is unreproducible.
`extra`	`no-extra-rule`	`free-mechanism-choice`	A choice can force any path.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d56c7c24dd710906db456ed73dff4648c5cbacb903334859dbc6eaedd87247d5`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:b1a3c0ce3bbe224fe6adfaca32428de47fcd2130bf09026d2d09617fb085506f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:62a01744198cbbe8cb2e61f83caf3adc1b591dcfe2bfd037895bec0be9f32c9e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a54f92031e4559f61065b6a954f4941b2298198daabf36d8b3930933d0279811`.

Independent reconstruction. Implementation hash `sha256:4d8c60ce565b3448c751cf5e52ccfefa1c9bcdd202ffd84f9e595d0d55365b4b`; independent certificate `sha256:f6bb5877f01fc2c5ebd935d2d55a1fe52134d695628d90f9755638631f692193`; external-validation hash `sha256:49310f8a512ebff0b4dd89b6eb4b823e25671e4a71707790cb830bce35d8158f`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- CHEM-AUTHORITY-CATALYTIC-PATHWAY-COMPOSITE-2026

Comparison records:

- catalytic-pathway-authority-composite: predicted alternative-reaction-mechanism__lower-activation-energy-path__same-overall-reaction__catalyst-cycle-retained; source-derived alternative-reaction-mechanism__lower-activation-energy-path__same-overall-reaction__catalyst-cycle-retained; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks an alternative lower-activation mechanism with unchanged overall reaction and retained catalyst cycle, or if a tampered row is accepted.

Measurement receipt: `sha256:61f3215190fac9070c00cea0bf3365a37660a3b7d2a45cb99b44562c31599b37`. Isolation certificate: `sha256:7d484dc4ef0c775af469c9efc9f70780f7b717f839452044e0211b6dbd42a8a8`. Custody certificate: `sha256:a4f0216152b5b357744336fe8e965ca0b06b612a97e474c3eb8fafeff4ebb423`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry and Royal Society of Chemistry: https://goldbook.iupac.org/terms/view/C00876 (sha256:a960e78123592da7dd8ad054b65a590961706c25e22a578f9088f91fb3c89e5e)

Registered target identities:

- catalytic-pathway-authority-composite from CHEM-AUTHORITY-CATALYTIC-PATHWAY-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite pair of balanced reaction mechanisms with common endpoints, complete intermediate and activation records, and one exact catalyst-return cycle.

Immutable identities. Source manifest `sha256:99a85f1de54f702ec08c3208b7e63ec4f447fa6b3daea3dc656d4091b72a2e37`; derivation seal `sha256:5e1ee14cf850b606ce8067cc51d25b66035f367c24c5cbb3cdfb1083aea494e9`; engine receipt `sha256:ba206769a2ff14720c048a7eafd71f61561086d11dcd6af2e98e6fc1731aadc8` at `receipts/engine/model_admitted/SFT-CHEM-CAT-PATHWAY-001-ba206769a2ff1472.json`; empirical hash `sha256:b7afd3ae80af2084bbdaf49b9af00adfb3728dac118da016658cb7468f933c6e`; measurement receipt `sha256:61f3215190fac9070c00cea0bf3365a37660a3b7d2a45cb99b44562c31599b37`.

62. Catalytic selectivity among generated products

Claim identity: `SFT-CHEM-CAT-SELECTIVITY-001`

Question and theorem. Catalytic selectivity is the exact maximal product-recurrence class over all generated competing products at one reactant, catalyst, condition and observation boundary; ties remain a cofavored class.

> *all-accessible-products-generated__complete-product-recurrence-order__one-product-favored__cofavored-class-at-tie__reactant-consumption-held*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`

- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-CAT-CATALYST-001`
- `SFT-CHEM-CAT-PATHWAY-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite complete product census for one retained catalytic input boundary with exact positive recurrence counts and all ties retained.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `all-accessible-products-generated__complete-product-recurrence-order__one-product-favored__cofavored-class-at-tie__reactant-consumption-held__condition-and-method-bounded__complete-competing-product-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`products`	`all-accessible-products-generated`	`one-product-presupposed`	Alternatives were not generated.
`comparison`	`complete-product-recurrence-order`	`yield-number-imported`	A measurement selects the answer.
`selection`	`one-product-favored`	`all-products-equated`	No preference is represented.
`ties`	`cofavored-class-at-tie`	`arbitrary-tie-break`	Equal support cannot force identity.
`reactant`	`reactant-consumption-held`	`consumption-erased`	Selectivity lacks a denominator.
`conditions`	`condition-and-method-bounded`	`universal-preference`	Selectivity can change with context.
`record`	`complete-competing-product-trace`	`selected-name-only`	Competing products are hidden.
`extra`	`no-extra-rule`	`free-selectivity-fit`	A fit can choose any product.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:5a1a0226f6360ac2b96c67319fff4be09265d2181106cd3396fc7cd908d5542e`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7a4e988ef2cfc8faa54e8492f557dc4e4d2b096938a9759bc4e90657f1ac86d0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:694038c24110a7ede1b6ebe7a63ee0b88c201fe288c773816c65416663b5d448`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:2788954e8e9dccabbf66dbe5c81d681fc83223ad6cd276ebe12ff20dc1c1e337`.

Independent reconstruction. Implementation hash `sha256:754cf0c2b2c7594c8784cc47def21254bbe89b6db3682727307ffad8ada6be12`; independent certificate `sha256:c45ba906fb4d3c4aa0f4d308389ebb355098d0766df03a451d237d5d88866196`; external-validation hash `sha256:714193c4b32267e9fc0f7f9d3ea403f7b0017e7b2ae8f1313b0d2b1ecb3a0552`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CATALYTIC-SELECTIVITY-COMPOSITE-2026

Comparison records:

- catalytic-selectivity-iupac-composite: predicted one-product-favored-among-alternatives__specified-reactant-consumption__condition-and-method-bounded__competing-products-retained; source-derived one-product-favored-among-alternatives__specified-reactant-consumption__condition-and-method-bounded__competing-products-retained; exact match True

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks product preference relative to alternatives, specified reactant consumption, a bounded method/condition or retained competitors, or if a tampered row is accepted.

Measurement receipt: `sha256:521c12218f929ef529d6f59320f28470cdddf57dd9eabd3c5199cb039673b03d`. Isolation certificate: `sha256:76ef47150ee7c3a9bb165a1de4560555e188915a78e73cf82cd221bb9dd17c93`. Custody certificate: `sha256:92f075412058de4db93764aa8125c147d44fb094629ec239e44ce347c4c17b10`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/S05563 (sha256:72489fb0beb2453befdf2628aa48d7fe5a18f8a8ce4e09573c09b3bb464d7506)

Registered target identities:

- catalytic-selectivity-iupac-composite from IUPAC-GOLD-BOOK-CATALYTIC-SELECTIVITY-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite complete product census for one retained catalytic input boundary with exact positive recurrence counts and all ties retained.

Immutable identities. Source manifest `sha256:45e34c1bd951fb10d5e97639ec24c57ea53d3b52ddb59ec2b4b0eb99c58d0fa2`; derivation seal `sha256:31e00c5a6c64c7da5f688349a84bf49340ed271f9dc47797a6440c0dbeba3c35`; engine receipt `sha256:a417163df1c29bfa35f2d677fbbc4453f6416bb9cac81d33e0a11ea22e9ea934` at `receipts/engine/model_admitted/SFT-CHEM-CAT-SELECTIVITY-001-a417163df1c29bfa.json`; empirical hash `sha256:5c4f47ecf21879ecfcee5a88bcca5a1de4a1cef30fae345f1cac1eaed0fa796d`; measurement receipt `sha256:521c12218f929ef529d6f59320f28470cdddf57dd9eabd3c5199cb039673b03d`.

63. Chemical reaction-network composition

Claim identity: `SFT-CHEM-NET-REACTION-001`

Question and theorem. A chemical reaction network is the complete finite directed incidence organization generated by balanced reaction steps sharing retained chemical-species identities, including branches, merges, cycles and external boundaries.

> *chemical-species-nodes__reaction-steps-connect-species__shared-species-compose-paths__source-product-boundary-held__all-generated-paths-retained*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-PATHWAY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite set of retained chemical-species carriers and balanced reaction-step incidences with complete external source/product boundaries.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `chemical-species-nodes__reaction-steps-connect-species__shared-species-compose-paths__source-product-boundary-held__all-generated-paths-retained__reaction-cycles-retained__complete-node-edge-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`species`	`chemical-species-nodes`	`anonymous-state-count`	Chemical identities are lost.
`reactions`	`reaction-steps-connect-species`	`unlabelled-adjacency`	Transitions are chemically ambiguous.
`composition`	`shared-species-compose-paths`	`isolated-reaction-list`	Shared carriers are not composed.
`boundary`	`source-product-boundary-held`	`open-unrecorded-flux`	The network cannot balance.
`branching`	`all-generated-paths-retained`	`single-path-only`	Network alternatives are erased.
`cycles`	`reaction-cycles-retained`	`cycle-collapsed`	Recurrence is lost.
`record`	`complete-node-edge-trace`	`network-picture-only`	The network cannot be replayed.
`extra`	`no-extra-rule`	`free-network-topology`	An imported graph selects the result.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c57d530691d21b8498f915c7441c1c1c725a6c7fe1e30d386eb9c97135bb20d9`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:204274c29c30cf943174df0510debdc011859a5541c2f550be0a193709e490e2`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:9b7d1d4cc915cbac4b24e9145301723b46be11253e0968d7d57b88349708238e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:5be5d2c477de356160384c12d2bafe67ce3b355ccce7adb22decd3b542764681`.

Independent reconstruction. Implementation hash `sha256:ffd05d6f83262ada8319d0cb30a8e8cdc9427acf976ee02b05d8020bb1e2584d`; independent certificate `sha256:f848abc22cd52e77c1a07e21dc4db6dc7f9fa6577f918a27f4c4c7b1dbb82768`; external-validation hash `sha256:f3ce744eb51ec16543e115842875f46f2e768a1832114eb6deb3e5e5a253bf05`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-REACTION-NETWORK-COMPOSITE-2026

Comparison records:

- reaction-network-iupac-composite: predicted chemical-species-nodes__reaction-steps-connect-species__combined-network-paths__complete-source-product-boundary;

source-derived chemical-species-nodes__reaction-steps-connect-species__combined-network-paths__complete-source-product-boundary; exact match True

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks species, reaction connections, composed network paths or a complete boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:a0e5902ebb2de9a2acd4fca5737e3434886bf2769fb7364158090f57226bb557`. Isolation certificate: `sha256:e6b8c7fd2342ad34dbcb9921216e09113f41ffefa009f17902dc5b963a77649d`. Custody certificate: `sha256:89356d5175bcf29197a8cef05b3d527da520202ae080feef574f711bba781db3`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01210 (sha256:f7949843026a8fe090e71f1d952633b8c63eb8277ad6b067b4bcaf5babf3598d)

Registered target identities:

- reaction-network-iupac-composite from IUPAC-GOLD-BOOK-REACTION-NETWORK-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite set of retained chemical-species carriers and balanced reaction-step incidences with complete external source/product boundaries.

Immutable identities. Source manifest `sha256:7e74fcc6796358d9b6922a7a9556f0037463586d4f95343216f78ec33aab630b`; derivation seal `sha256:58e90881b731b6595a516720f137da24f2032970bc515232fe3eeb8c5a52b831`; engine receipt `sha256:750b66a0cd8a1a3298b488e222fca2863ae173c09b92a75bde9515044d6b1259` at `receipts/engine/model_admitted/SFT-CHEM-NET-REACTION-001-750b66a0cd8a1a32.json`; empirical hash `sha256:c3a5a2cb271c5a164e83fe7f3f22489239e3a582478859528c166dcfee668741`; measurement receipt `sha256:a0e5902ebb2de9a2acd4fca5737e3434886bf2769fb7364158090f57226bb557`.

64. Autocatalytic closure and self-amplifying path

Claim identity: `SFT-CHEM-NET-AUTOCATALYSIS-001`

Question and theorem. Autocatalysis is a finite resource-bounded reaction-network cycle in which a retained product carrier re-enters as catalyst for another production cycle of the same identity.

> *reaction-product-is-catalyst__product-accelerates-own-formation__same-identity-feedback__resource-consuming-amplification__positive-feedback-cycle*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`

- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-CAT-CATALYST-001

- SFT-CHEM-NET-REACTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite reaction network whose product/catalyst identity equality, feedback edge, consumed inputs and stopping boundary are completely recorded.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: reaction-product-is-catalyst__product-accelerates-own-formation__same-identity-feedback__resource-consuming-amplification__positive-feedback-cycle__finite-resource-bounded__complete-feedback-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
product	reaction-product-is-catalyst	external-catalyst-only	Self-production is absent.
feedback	product-accelerates-own-formation	product-inert	Output cannot affect its formation.
identity	same-identity-feedback	product-role-erased	The feedback carrier is unknown.
growth	resource-consuming-amplification	unbounded-free-copy	Creation violates closure.
cycle	positive-feedback-cycle	linear-step-only	No feedback path closes.
limit	finite-resource-bounded	infinite-support-assumed	Resources are not counted.
record	complete-feedback-trace	growth-label-only	Carrier balance cannot be checked.
extra	no-extra-rule	free-growth-constant	A fitted rate can force behavior.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:0cee95067a1a5431ddc8780c8864ae551f96a26bb2c001743642747736cac51d.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:2b89cecb7863ccd95d6a1d95bf55533e2c1ad530a14eed634a88ca8534be5bac.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:b95e37ad7b16c81b5d7997fd345b2151209a5f626569efa1032891f153b64d4f.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:6f3686eed26f288c5bc85889d39921ff67795676ea2ba9075b06efdc5e039bfe.

Independent reconstruction. Implementation hash sha256:fdd0d83368dd716f7302b78183a3d93e151ae640703569d3fc20ae81a2299ebc; independent certificate sha256:dc9fca524512a204e305f1f2262f8ffc25587cb8c349b09c19b137519e8a8b88; external-validation hash sha256:2156c92b02cecc376144e9f781437a7d29137e858280b03cad69228218c99c0d. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-AUTOCATALYSIS-COMPOSITE-2026

Comparison records:

- autocatalysis-iupac-composite: predicted reaction-product-is-catalyst__product-accelerates-own-formation__positive-feedback-cycle__finite-resource-bounded; source-derived reaction-product-is-catalyst__product-accelerates-own-formation__positive-feedback-cycle__finite-resource-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks a product acting catalytically in its own formation, feedback closure or the declared finite boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:66d2251aab1445f74aa98b13c768c89e3cfe69b569e73821b7a0b9badb0400ba`. Isolation certificate: `sha256:f19eba414c114b3b34ec93cb821f88c4f04138d4ae921704e00edd80ecbd2948`. Custody certificate: `sha256:e77eb7d19e8706602456f42c2dcafbb7a442240a9a1af18a9acf79fdd9457868`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/A00525 (sha256:cd6e2b68b8eefaa71fd7212cb577bd9996810a0d64771508748f7d3599469b06)

Registered target identities:

- autocatalysis-iupac-composite from IUPAC-GOLD-BOOK-AUTOCATALYSIS-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite reaction network whose product/catalyst identity equality, feedback edge, consumed inputs and stopping boundary are completely recorded.

Immutable identities. Source manifest `sha256:3f7036a451de8e876c5ace7871e00b7d78cca3400001b7c8ec288b4d8c7d3489`; derivation seal `sha256:ae8a19e7cbe8513f19168ebdf9b53cb0d926dd9a5541f1c416569e853e96d472`; engine receipt `sha256:c84301ced7057f2e50d0535f4e7fa1b52543773aebdd8dfe367fd8dacbf03c63` at `receipts/engine/model_admitted/SFT-CHEM-NET-AUTOCATALYSIS-001-c84301ced7057f2e.json`; empirical hash `sha256:1ee148345c085815d08e6ee3bb71c9f7c033bf9c2c8dc3f0146cbf27139fd632`; measurement receipt `sha256:66d2251aab1445f74aa98b13c768c89e3cfe69b569e73821b7a0b9badb0400ba`.

65. Adsorption at a retained interface

Claim identity: `SFT-CHEM-SURFACE-ADSORPTION-001`

Question and theorem. Adsorption is reversible localization of retained adsorbate carriers at a retained adsorbent interface, producing surface support distinct from bulk support under declared conditions.

> *retained-interface__adsorbate-and-adsorbent-retained__accumulation-at-interface__surface-concentration-differs__reversible-desorption-boundary*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`

- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite two-region chemical boundary with named adsorbate and adsorbent, counted arrivals/residence/departures and source-bound conditions.

The engine regenerated 256 candidates and 256 one-for-one decisions. Exactly one survived: retained-interface__adsorbate-and-adsorbent-retained__accumulation-at-interface__surface-concentration-differs__reversible-desorption-boundary__condition-and-surface-bounded__complete-interface-event-trace__no-extra-rule. Closure is depth_independent; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
boundary	retained-interface	bulk-only	No interface is present.
roles	adsorbate-and-adsorbent-retained	carrier-identity-erased	Surface membership is ambiguous.
location	accumulation-at-interface	uniform-bulk-distribution	No surface accumulation occurs.
composition	surface-concentration-differs	surface-equals-bulk	The interface adds no distinction.
binding	reversible-desorption-boundary	irreversible-erasure	The adsorbate cannot be recovered.
conditions	condition-and-surface-bounded	condition-free-coverage	Surface support is context-bound.
record	complete-interface-event-trace	adsorption-name-only	Localization cannot be reproduced.
extra	no-extra-rule	free-isotherm-fit	A fit can select coverage.

Adverse controls. All mandatory controls passed:

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt sha256:1222829e939050aadd0f8822802fd259e38a34efc4956031923d6901d9376f77.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt sha256:5e79aced536f06e06417b9bd97561298a94bae3a6ed69a8edfc2238aa255bd21.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt sha256:4208793f4c9b2043c17f721fc4db84dacd13580da5eb86b7480bbabfd24e615b.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:59eae4ccdfd45006bede3138562ce34a170fd3cceaa0c58c81440c9b92f26b70.

Independent reconstruction. Implementation hash sha256:594768dbdeffe5e68cea02238fda64b8a1d762d3f6bf2af4706a08291bc6db7c; independent certificate sha256:083528ba959417a9c14e7a719aef5b1cc340b3bfbc296a7ef03ad2d78b7542f7; external-validation hash sha256:f11677f2823ddad32937c31ae3ade62fd9f54f6e158e169f378444f5d3238964. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ADSORPTION-COMPOSITE-2026

Comparison records:

- adsorption-iupac-composite: predicted accumulation-at-interface__adsorbate-and-adsorbent-retained__surface-concentration-differs-from-bulk__reversible-desorption-boundary; source-derived accumulation-at-interface__adsorbate-and-adsorbent-retained__surface-concentration-differs-from-bulk__reversible-desorption-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks interfacial accumulation, retained adsorbate/adsorbent roles, surface/bulk distinction or a desorption boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:d550367fcdee479c0559ff1fc1276a6a6a52308771fa6d22b3fd5528ecffdbd4`. Isolation certificate: `sha256:239b6b47fac5ed2eaa7fde95097f7fe600c9e051004544809afe5be723d1f1c6`. Custody certificate: `sha256:d2aae0aeb061606cb0227ab1ea72028d3fd79e6b4a6e513d4167c933a9ca27fb`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/A00155 (sha256:8d8d32d4fad6f36f8be4aeed8056bac1efb3ef5f9d03c3ec26fddaade7cce582)

Registered target identities:

- adsorption-iupac-composite from IUPAC-GOLD-BOOK-ADSORPTION-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite two-region chemical boundary with named adsorbate and adsorbent, counted arrivals/residence/departures and source-bound conditions.

Immutable identities. Source manifest `sha256:0be55d631dee35d8972bab503e9a130d958537fa6235088f72d7cc4722b693db`; derivation seal `sha256:1012d07cc5d979d4401c997892ff6aec336c1b6976dff62a08d214a7d3d35b27`; engine receipt `sha256:b89566d1d2c1059b72ea9b65da47b81b1c89d56c93678c6ebadc655bdde4ba34` at `receipts/engine/model_admitted/SFT-CHEM-SURFACE-ADSORPTION-001-b89566d1d2c1059b.json`; empirical hash `sha256:ad2d7c764bb84113a8b69aa87b15e8f0bfcef87a653fc39cc60418d7716cceb0`; measurement receipt `sha256:d550367fcdee479c0559ff1fc1276a6a6a52308771fa6d22b3fd5528ecffdbd4`.

66. Colloidal dispersion and phase-interface support

Claim identity: `SFT-CHEM-COLLOID-DISPERSION-001`

Question and theorem. A colloidal dispersion is a source-bounded population of finite dispersed-phase regions distributed through a retained continuous phase, each with a retained interface and observation-scale boundary.

> *dispersed-and-continuous-phases__finite-dispersed-particles__phase-interface-retained__particles-distributed-through-medium__particle-size-source-bounded*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`

- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite population of separated same-phase regions within a distinct continuous phase, with complete interfaces, sizes, positions, conditions and observation interval.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `dispersed-and-continuous-phases__finite-dispersed-particles__phase-interface-retained__particles-distributed-through-medium__particle-size-source-bounded__dispersion-condition-bounded__particle-interface-distribution-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`phases`	`dispersed-and-continuous-phases`	`single-uniform-phase`	No dispersion exists.
`extent`	`finite-dispersed-particles`	`molecular-solution-only`	No mesoscopic carrier is retained.
`interface`	`phase-interface-retained`	`phase-boundary-erased`	Particle identity is lost.
`distribution`	`particles-distributed-through-medium`	`one-settled-region`	Dispersion is absent.
`scale`	`particle-size-source-bounded`	`size-free-label`	Colloid identity is unbounded.
`stability`	`dispersion-condition-bounded`	`eternal-stability-assumed`	Persistence is condition-dependent.
`record`	`particle-interface-distribution-trace`	`colloid-name-only`	Phase support cannot be checked.
`extra`	`no-extra-rule`	`free-dispersion-fit`	A fit can force stability.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:edd8f0455d2f299a3ad9630f6622b49d804706b79d355ae6319912a942000e0c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:723ba74dce8f73823e5842be52b5b1dd8ecf3cde9c4f11da3b779a472b975234`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:083d54c83dfc2538aee457363b236556741bb91855fe1a35c5554550b3e837b5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f01f5eab860c62ac8679d4555e9eb9ce1c8f72eb35560ce05eff3a3858e0ae2a`.

Independent reconstruction. Implementation hash `sha256:1733d436397c2812555cf24e9c2225cf28f9d2e77c81560dbd60b58f74ae99aa`; independent certificate `sha256:455bdd9c0d4dc97eb1652ea2880c1a40fd3a2f25097e9e24371f13a601ed227a`; external-validation hash `sha256:a22012b42ae63a9b80400636e84a46866c7411fddd22ca8ade6d045a9f67fda2`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-COLLOIDAL-DISPERSION-COMPOSITE-2026

Comparison records:

- colloidal-dispersion-iupac-composite: predicted microscopic-phase-dispersed-in-continuous-phase__phase-interface-retained__particle-size-boundary__dispersion-condition-bounded; source-derived microscopic-phase-dispersed-in-continuous-phase__phase-interface-retained__particle-size-boundary__dispersion-condition-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks dispersed/continuous phases, retained interfaces, a particle-size boundary or condition-bounded dispersion, or if a tampered row is accepted.

Measurement receipt: `sha256:48e5f39d9e7a9213bfd4187c1cd0692968678126b908e4d8578b98c99b634dac`. Isolation certificate: `sha256:f2d6fdfdd14de0627cb79e6bc2c71f006c4d27bbd9a8fb4b86a1d330fced508a`. Custody certificate: `sha256:244dd247b747fc920addd450fdffc9e02fd293551a67738e69db705f7bc8fea5`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01174 (sha256:a8261e1400d58c847f37722d7c1cb39f782a97844d847648156bf01dfad7a636)

Registered target identities:

- colloidal-dispersion-iupac-composite from IUPAC-GOLD-BOOK-COLLOIDAL-DISPERSION-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite population of separated same-phase regions within a distinct continuous phase, with complete interfaces, sizes, positions, conditions and observation interval.

Immutable identities. Source manifest `sha256:1071e1711cfa513a50971c9cdbae95c642ed212f8f9e099613c69a42d6f3c0ca`; derivation seal `sha256:3f5b7909acd18202b17542290f3695277eebc733c7f4fe91354d45459b5c2d06`; engine receipt `sha256:b342599b09b46b62c530e7b466cc4a61a64a2727202f164053826c0058819a97` at `receipts/engine/model_admitted/SFT-CHEM-COLLOID-DISPERSION-001-b342599b09b46b62.json`; empirical hash `sha256:d8ae80a4d9f2c5742b5e4b9128f4506d4b0b567dcf1c7a4a9c9f0a3035ea053b`; measurement receipt `sha256:48e5f39d9e7a9213bfd4187c1cd0692968678126b908e4d8578b98c99b634dac`.

67. Chemical transfer across an interface

Claim identity: `SFT-CHEM-INTERFACE-TRANSFER-001`

Question and theorem. Interfacial chemical transfer is the counted, oriented crossing of a retained chemical-species carrier from a named donor phase to a distinct receiver phase, pairing departure and arrival under declared conditions.

> *two-phase-boundary__chemical-species-retained__held-donor-to-receiver-orientation__mass-balance-preserved__boundary-crossing-transition*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`

- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-SURFACE-ADSORPTION-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by carrier preservation, complete support, minimality and absence of an extra rule.

Boundary: Every finite two-phase boundary with named donor, receiver and chemical carrier, paired crossing events, complete balance and source-bound rate conditions.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `two-phase-boundary__chemical-species-retained__held-donor-to-receiver-orientation__mass-balance-preserved__boundary-crossing-transition__transfer-rate-condition-bounded__complete-donor-interface-receiver-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`interface`	`two-phase-boundary`	`one-phase-only`	No crossing boundary exists.
`carrier`	`chemical-species-retained`	`anonymous-flux`	Chemical identity is lost.
`orientation`	`held-donor-to-receiver-orientation`	`unsigned-transfer-number`	Direction is ambiguous.
`balance`	`mass-balance-preserved`	`carrier-created-at-boundary`	Conservation fails.
`process`	`boundary-crossing-transition`	`instantaneous-relabel`	No transfer event is represented.
`rate`	`transfer-rate-condition-bounded`	`universal-transfer-value`	Rate depends on context.
`record`	`complete-donor-interface-receiver-trace`	`flux-answer-only`	The crossing cannot be replayed.
`extra`	`no-extra-rule`	`free-transfer-coefficient`	A fitted coefficient can select any value.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c930e6944f1a09f686a08d3c96eb6287aba3fb827d5bad341ba8d8f4b7548113`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f24b244547dcfdb4e01c3b88b91ca4cb460bee0b24c241efe4f14e01f1406e96`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:72fe92a1dbf643723166962831da07bfc71712dda65fa11d6fd07ceb9cc88d78`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3d25aea62ec6d8302fefb5738b84969989268a5917e5d8a22d080a65215cd0b3`.

Independent reconstruction. Implementation hash `sha256:f9ce306e7d93983fb7c043ef150e796b853c53283ed531a9c34d2332848d44a9`; independent certificate `sha256:85cebf7d1bf236cc83718fb44d1f95b727ee51b66f36d86ba5e4608540032f44`; external-validation hash `sha256:bcc0c0c51ac48d5b57682cc8879e0ca7d4347b1f270d773d7d06296280a6ac77`. The independent

program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-INTERFACE-TRANSFER-COMPOSITE-2026

Comparison records:

- interface-transfer-iupac-composite: predicted chemical-species-crosses-phase-boundary__donor-and-receiver-phases-retained__transfer-rate-condition-bounded__mass-balance-preserved; source-derived chemical-species-crosses-phase-boundary__donor-and-receiver-phases-retained__transfer-rate-condition-bounded__mass-balance-preserved; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks a species crossing, two retained phases, a condition-bounded rate or mass balance, or if a tampered row is accepted.

Measurement receipt: `sha256:96e6ca08d7e05fe76facc2f4cc855b2564a0d7f597ccf589a7342bde233489f7`. Isolation certificate: `sha256:e38df0c603e3e727043aa36f198fe5c2b56c661009262d1db8656ca668659ff1`. Custody certificate: `sha256:d74fcd2a35755fe07cd20835e3be116db9f53629849b9acd27645f0fa845b294`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/P04536 (sha256:b60eba64e32f00b5bcb7ecf818b25abc53208476546555ae05349a52cbd7e7b6)

Registered target identities:

- interface-transfer-iupac-composite from IUPAC-GOLD-BOOK-INTERFACE-TRANSFER-COMPOSITE-2026

Meaning. Catalysis and interfaces preserve catalyst return, alternative pathways, reaction networks, autocatalysis, adsorption, colloids and cross-boundary transfer without fitted rates. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no catalyst dictionary, kinetic table, reaction network, surface model, colloid table, transfer equation, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no chemical carrier, phase identity, interface or reaction step may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite two-phase boundary with named donor, receiver and chemical carrier, paired crossing events, complete balance and source-bound rate conditions.

Immutable identities. Source manifest `sha256:a3ac97ebcb8ff635bf61442b1a8d0dfbbd8f52f5ac2c3a6b5276174a2e891d39`; derivation seal `sha256:4fdc50af81ae5440e6c0d7e6d9762ab95019915dc8c2dca15aa99c9ef5915cad`; engine receipt `sha256:6a6f5dde7f37947d6ea43b84dee4726844e0397ddf9f34667254e4c796bc1c4a` at `receipts/engine/model_admitted/SFT-CHEM-INTERFACE-TRANSFER-001-6a6f5dde7f37947d.json`; empirical hash `sha256:0e0e49c608e196154bfce21e5735870fd3340be1aa5c83e9ae2f0d175e723bbe`; measurement receipt `sha256:96e6ca08d7e05fe76facc2f4cc855b2564a0d7f597ccf589a7342bde233489f7`.

16. Stereochemistry Organic Polymer

Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition.

68. Chirality and non-superposable held orientation

Claim identity: `SFT-CHEM-STEREO-CHIRALITY-001`

Question and theorem. Chirality is the retained distinction between a complete oriented molecular carrier and its exact reflection when exhaustive identity-preserving proper superposition admits no equivalence.

> *complete-three-dimensional-object__exact-mirror-image-generated__composition-preserved__adjacency-preserved__no-proper-superposition*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`

- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-MOL-GEOMETRY-001`
- `SFT-CHEM-MOL-ISOMER-001`
- `SFT-CHEM-NET-REACTION-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite complete three-dimensional chemical structure with named atoms, adjacency, orientation and exhaustive identity-preserving proper maps.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-three-dimensional-object__exact-mirror-image-generated__composition-preserved__adjacency-preserved__all-proper-finite-relabelings-enumerated__no-proper-superposition__orientation-and-map-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`complete-three-dimensional-object`	`formula-only`	A formula has no spatial orientation.
`reflection`	`exact-mirror-image-generated`	`mirror-image-not-generated`	Chirality cannot be decided.
`composition`	`composition-preserved`	`mirror-composition-changed`	The comparison is not reflective identity.
`adjacency`	`adjacency-preserved`	`mirror-adjacency-changed`	The comparison changes constitution.
`superposition`	`all-proper-finite-relabelings-enumerated`	`visual-guess`	A guess cannot establish equivalence.
`result`	`no-proper-superposition`	`arbitrary-handedness-name`	A name cannot force distinction.
`record`	`orientation-and-map-trace`	`chirality-label-only`	The result cannot be reproduced.
`extra`	`no-extra-rule`	`free-chirality-rule`	An extra rule can classify anything.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:a90fecae5d272bff9642b6c1cbb2246da4a5e5ea36336d27de3e3f38c61b0842`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:2f4df44ccf7ea37f290000c67426ecfaea88c1d5dc75908b21ddeba196961327`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:e30626a49e357397a5b503d28c26a3b0fb8439bd2b6284caa64566cde654d7d2.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt sha256:2dc9b28c06452315912058f81e49812f2abb24e5d4a343fce7464529a67affe1.

Independent reconstruction. Implementation hash sha256:c58e29aeec8dd7045a2a6a0a35bac932bb4ebe42211a40aa3378176cb79256d6; independent certificate sha256:c5dba95a5d23f251698226727ab9cad978861019221b4f9eeeff049ad2237d87; external-validation hash sha256:77217828c3c566132373f558fb9b133c20e0e7bd631c4e7945069e06aef99672. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: True. All registered rows were preserved: True. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-CHIRALITY-COMPOSITE-2026

Comparison records:

- chirality-iupac-composite: predicted object-not-superposable-on-mirror-image__chirality-retained__reflection-distinction__observation-boundary-held; source-derived object-not-superposable-on-mirror-image__chirality-retained__reflection-distinction__observation-boundary-held; exact match True

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks reflection, non-superposability, retained chirality or an observation boundary, or if a tampered row is accepted.

Measurement receipt: sha256:2f858469a86812414592d37f8c551eb2cab9aeefa37411935749a5a0ce5fbec0. Isolation certificate: sha256:d81385cebada96333bdc67b2222c699f3cab100ecc5a0aad4bd27fd5223f9bf6. Custody certificate: sha256:7d44cc5ed9943c5e98870d49e914e1630ded60d1a79629b9a8e716a8558d8a15.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C01058 (sha256:a787109fa5a2c606b0628013669773016b0245479ed6387584ba027d43e5f01b)

Registered target identities:

- chirality-iupac-composite from IUPAC-GOLD-BOOK-CHIRALITY-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate

- no numerical zero, negative, irrational, imaginary or floating proof quantity

- no free, fitted, learned or target-derived parameter

- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased

- absence is an Empty structural form rather than numerical zero

- external target content remains inaccessible until the prediction is sealed

- Every finite complete three-dimensional chemical structure with named atoms, adjacency, orientation and exhaustive identity-preserving proper maps.

Immutable identities. Source manifest sha256:b81da2d3eb9742af6442f41fee14e6564d3d67853348cd4ecb1b8f2b37aec59d; derivation seal sha256:2273f6d6e92d30bb607b5e7a6e75b44c980946bc18cf3630bf0b06c70ebdf659; engine receipt `sha256:00ee8b785fd197058e6a47b394fd8fd0b9de6f6108599558989f4c190d45de07` at `receipts/engine/model_admitted/SFT-CHEM-STEREO-CHIRALITY-001-00ee8b785fd19705.json`; empirical hash `sha256:d4d540b31689d011277f61ffb31004d673d3eeea320a7e8b44eacf812429c28f`; measurement receipt `sha256:2f858469a86812414592d37f8c551eb2cab9aeefa37411935749a5a0ce5fbec0`.

69. Enantiomer pair and chiral observation boundary

Claim identity: `SFT-CHEM-STEREO-ENANTIOMER-001`

Question and theorem. Enantiomers are the paired complete stereoisomer carriers related by exact reflection and by no identity-preserving proper superposition, with opposed configurations retained.

> *paired-stereoisomers__same-composition-and-connectivity__exact-mirror-relation__non-superposable-pair__opposed-configurations-held*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`

- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-MOL-GEOMETRY-001`
- `SFT-CHEM-MOL-ISOMER-001`
- `SFT-CHEM-NET-REACTION-001`
- `SFT-CHEM-STEREO-CHIRALITY-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite pair of equal-composition, equal-connectivity oriented molecular carriers with exhaustive mirror and proper-superposition tests.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `paired-stereoisomers__same-composition-and-connectivity__exact-mirror-relation__non-superposable-pair__opposed-configurations-held__chiral-observation-boundary__paired-map-and-observation-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`members`	`paired-stereoisomers`	`one-structure-only`	No pair exists.
`constitution`	`same-composition-and-connectivity`	`different-connectivity`	They are constitutional isomers.
`mirror`	`exact-mirror-relation`	`unrelated-orientations`	The pair is not enantiomeric.
`superposition`	`non-superposable-pair`	`superposable-pair`	The reflected structures are identical.
`orientation`	`opposed-configurations-held`	`orientation-erased`	Opposed configurations merge.
`environment`	`chiral-observation-boundary`	`universal-property-difference`	Response depends on observation context.
`record`	`paired-map-and-observation-trace`	`pair-name-only`	The mirror test cannot be checked.
`extra`	`no-extra-rule`	`free-enantiomer-rule`	A discretionary label can force the pair.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:78874adb9051f2cda593f82c2378adc94592159d0fce0287299f8bc1501a94dc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:80898deff1518c63ea5d791407d62d7bbb29dc25b5cbfc6477a8240f1135088d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d892bdf9ea15e6523b592d1b7b9439d6d5022e55032d91d191b7bdc843c221d5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7f8cd4ae6999f6643eb7752e0d261eb4389e0a29ea01b45e51c0da078e636c67`.

Independent reconstruction. Implementation hash `sha256:b9e0407b93ac2315487b7c6dcda9bc34f65c4a439fa518cbdc0f817c39a0321d`; independent certificate `sha256:2c44dff49dba2fb9f7f9259ec34e3ed7e75956aab0a1073f1a5df99c26b52d23`; external-validation hash `sha256:036249ae0933ab4c85a240207a4ff934ea7dea61a49ea5ee614624e486148ad4`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ENANTIOMER-COMPOSITE-2026

Comparison records:

- enantiomer-iupac-composite: predicted molecular-entities-mirror-images__non-superposable__opposed-configurations__chiral-observation-bounded; source-derived molecular-entities-mirror-images__non-superposable__opposed-configurations__chiral-observation-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks molecular mirror relation, non-superposability, opposed configuration or chiral observation boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:dc426d9ff89781d78446229965847bd3cf8b3a077a7c62418cbe9319c8d5f5d8`. Isolation certificate: `sha256:58dbd4c37e772e1bf231b949816368e91ce331d05bbd3d9c03b2111ab4b9997c`. Custody certificate: `sha256:c44b309051d6d5348249284b7a529dbc8f0dce32e6dbd0b46c0db61ff49b6441`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/E02069 (sha256:833912292d5eba7d3528d2a31a748a95f4cd4d28b92e896f6e1b5d9ef914935d)

Registered target identities:

- enantiomer-iupac-composite from IUPAC-GOLD-BOOK-ENANTIOMER-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed

- Every finite pair of equal-composition, equal-connectivity oriented molecular carriers with exhaustive mirror and proper-superposition tests.

Immutable identities. Source manifest `sha256:f416c38a39383152bd2d37cec4f40adf10e1161504ca7b9274af54f9004eee94`; derivation seal `sha256:e6500f21f86255a34516550e01be6b61647115c7c539be982e874b43360113ba`; engine receipt `sha256:41db74ec3403458e6a269d9904e6768351dc7229b25ed7a50e0bd088e95a05f2` at `receipts/engine/model_admitted/SFT-CHEM-STEREO-ENANTIOMER-001-41db74ec3403458e.json`; empirical hash `sha256:d086045f16b868dea7b858347fd148adfe876e1e89bed2f596ab8ae6385bfe36`; measurement receipt `sha256:dc426d9ff89781d78446229965847bd3cf8b3a077a7c62418cbe9319c8d5f5d8`.

70. Diastereomer distinction

Claim identity: `SFT-CHEM-STEREO-DIASTEREOMER-001`

Question and theorem. Diastereomers are paired non-superposable stereoisomers of equal composition and connectivity that are not related as exact mirror images.

> *paired-stereoisomers__same-composition-and-connectivity__non-superposable-pair__not-mirror-related__complete-relative-configurations-held*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`

- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-MOL-GEOMETRY-001`
- `SFT-CHEM-MOL-ISOMER-001`
- `SFT-CHEM-NET-REACTION-001`
- `SFT-CHEM-STEREO-CHIRALITY-001`
- `SFT-CHEM-STEREO-ENANTIOMER-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite equal-constitution oriented molecular pair with exhaustive identity, reflection and proper-superposition classifications.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `paired-stereoisomers__same-composition-and-connectivity__non-superposable-pair__not-mirror-related__complete-relative-configurations-held__property-distinction-open__complete-pair-classification-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`members`	`paired-stereoisomers`	`one-structure-only`	No pair exists.
`constitution`	`same-composition-and-connectivity`	`different-composition-or-connectivity`	They are not stereoisomers.
`superposition`	`non-superposable-pair`	`superposable-pair`	No stereochemical distinction remains.
`mirror`	`not-mirror-related`	`exact-mirror-pair`	That pair is enantiomeric.
`configuration`	`complete-relative-configurations-held`	`all-centres-erased`	The difference is unreproducible.
`properties`	`property-distinction-open`	`properties-forced-equal`	Non-mirror structures need not share response.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`record`	`complete-pair-classification-trace`	`diastereomer-label-only`	The exclusions cannot be checked.
`extra`	`no-extra-rule`	`free-diastereomer-rule`	A taxonomy can select the answer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8682d8dcee95265f2e77c106b5f8b4cea6d08860c097fe761367e7042ae95966`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3a570f720ae4e97c27103ded8bc92baa50d7d553202e4053e17e83a4c0f7aebd`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:d351449dc0617402486df5ae3d907c79a641ac271c17fb87bcbc906f884a6a24`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ad5e7ea3c443b07d374848fb6d542ad5d4a5351ccb2ed46fb83b0be1f3bb9318`.

Independent reconstruction. Implementation hash `sha256:3936122354bc8e32351cbd45cb5e1b9835a7bb5b3a12546e2893154c9daf820c`; independent certificate `sha256:2c1b89fe37e89c64402624627281e4a28c22f9a9305246322c89e5ec3b433d5a`; external-validation hash `sha256:386d2400e7c726fa214c4d322547d02963af42c972567d28bf0bf539df4b4159`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-DIASTEREOMER-COMPOSITE-2026

Comparison records:

- diastereomer-iupac-composite: predicted stereoisomers__not-mirror-images__distinct-configurations__property-distinction-possible; source-derived stereoisomers__not-mirror-images__distinct-configurations__property-distinction-possible; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks stereoisomer identity, non-mirror relation, configurational distinction or possible property distinction, or if a tampered row is accepted.

Measurement receipt: `sha256:6b0cf84493406f61a95dad059b2dd6dd1657cfae00dfc4243550381a52452606`. Isolation certificate: `sha256:130abe45fb5e7ec3816e683b7ebc3ee72dd6d89bc2801a17aa2c5e777f342065`. Custody certificate: `sha256:45e736ef7791936d3786802d3d45a2aae0513047f1f81e90c58e8bf39818b96b`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/D01679 (sha256:4fa09c83930088c3eeea3fd428d370918e6bf98c361673287c0fc8ef9d394078)

Registered target identities:

- diastereomer-iupac-composite from IUPAC-GOLD-BOOK-DIASTEREOMER-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate

- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite equal-constitution oriented molecular pair with exhaustive identity, reflection and proper-superposition classifications.

Immutable identities. Source manifest `sha256:4c27425f692f832a454a838d286dbed27f7f84abc5d5162c23df40333f3479d8`; derivation seal `sha256:5fce9da04cde73ee2735571bad53578d233459db70b035d84bd266cdb7f8d3f2`; engine receipt `sha256:661fb2a1428aae2c60554c1101fb05f81fe985b7d8fbf5eb3877134c439aa00a` at `receipts/engine/model_admitted/SFT-CHEM-STEREO-DIASTEREOMER-001-661fb2a1428aae2c.json`; empirical hash `sha256:7ec2cac6638c1561a917b26cd83c4d5a34fe22251c807c77be745662e0ae089f`; measurement receipt `sha256:6b0cf84493406f61a95dad059b2dd6dd1657cfae00dfc4243550381a52452606`.

71. Functional-group recurrence and reaction role

Claim identity: `SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001`

Question and theorem. A functional group is a retained atom-or-group subcarrier whose complete local structure recurs across molecular embeddings and carries a characteristic generated reaction role within a declared context.

> *named-atom-or-group-support__molecular-embedding-held__recurs-across-molecules__characteristic-reaction-role__group-identity-retained*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`

- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite molecular carrier with complete named substructures, boundary bonds and generated source-bound reaction traces.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `named-atom-or-group-support__molecular-embedding-held__recurs-across-molecules__characteristic-reaction-role__group-identity-retained__molecular-context-bounded__substructure-and-reaction-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`support`	`named-atom-or-group-support`	`whole-molecule-only`	No recurrent substructure is identified.
`embedding`	`molecular-embedding-held`	`isolated-fragment`	A functional group acts in a molecule.
`recurrence`	`recurs-across-molecules`	`one-molecule-only`	No group class is established.
`reaction`	`characteristic-reaction-role`	`shape-without-transition-role`	Function requires chemical behavior.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`identity`	`group-identity-retained`	`context-erases-group`	The group cannot be followed.
`context`	`molecular-context-bounded`	`universal-context-free-response`	Neighboring structure can modify response.
`record`	`substructure-and-reaction-trace`	`group-name-only`	Recurrence cannot be reproduced.
`extra`	`no-extra-rule`	`free-functional-taxonomy`	An imported list selects the group.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:c188cf693c44cb41b1a98a4cac9a337f8c0045f92bc18d237416427b14b2f689`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3727abd174787d207fb36092b3e137e2f11ec5878725228dbc473d2ad541183b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bd73c500dd3af7832d2bc4f118d46f20ceda9568f900c6b69d93ab2c7001b32b`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:72208840975e03044131c459563f053a4d989014820fecebf1b28b8d6f418397`.

Independent reconstruction. Implementation hash `sha256:600186025be20195d13b50b988a6c1454172503b1e3c9af181c2c3579850101b`; independent certificate `sha256:93cb8ec0892c4409a11bf050cc142e7c068898f82bfeb071fcc325d3df3b5f39`; external-validation hash `sha256:26d535307714a6d4c78a06397ac3b9b746b85da168b6e540e61fea7586c9fd78`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-FUNCTIONAL-GROUP-COMPOSITE-2026

Comparison records:

- functional-group-iupac-composite: predicted atom-or-group-with-characteristic-properties__recurs-across-molecules__reaction-role-retained__molecular-context-bounded; source-derived atom-or-group-with-characteristic-properties__recurs-across-molecules__reaction-role-retained__molecular-context-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks an atom/group, characteristic behavior, recurrence, reaction role or context boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:f0b1699714ef6972d4d8895729103d4adde9a22bf84b29f24f56cfa15fbd3c94`. Isolation certificate: `sha256:465ae863c10469dbc677f8c13c18a0075fa9c5abf9b0a0b575fc77f547dfd0b3`. Custody certificate: `sha256:db6f6803ea248e418b3f3459559cdda8568041733155229399e273d96d9c1451`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/F02555 (sha256:1fc0f254f488921a762c25454615ea9ec4f40d54736be524fa9f4ff8d79869d1)

Registered target identities:

- functional-group-iupac-composite from IUPAC-GOLD-BOOK-FUNCTIONAL-GROUP-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite molecular carrier with complete named substructures, boundary bonds and generated source-bound reaction traces.

Immutable identities. Source manifest `sha256:298e6a9e7080ce74fd46a9c94cce0d11d896a24cc7edb03f29e882a412f113b0`; derivation seal `sha256:3e542a809254cdb62a1e8758b3205b2c139c674b3490f26cc9684bbbcc344dc5`; engine receipt `sha256:35355b53a7cfcfb013366ae51b0830d9ba686362b669f5feb9cb16895d9eb502` at `receipts/engine/model_admitted/SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001-35355b53a7cfcfb0.json`; empirical hash `sha256:dfa2a272245ab4d917ccf3e8cb92b0614e0da545db9afeb73661b8447229c0da`; measurement receipt `sha256:f0b1699714ef6972d4d8895729103d4adde9a22bf84b29f24f56cfa15fbd3c94`.

72. Organic reaction-family compositional law

Claim identity: `SFT-CHEM-ORGANIC-REACTION-FAMILY-001`

Question and theorem. An organic reaction family is the equivalence class of complete balanced reactant-product maps sharing one minimal bond-change pattern and role correspondence without assuming a common mechanism.

> *multiple-generated-reactions__complete-reactant-product-maps__shared-bond-change-pattern__reactant-product-role-correspondence__elemental-carriers-balanced*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`

- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-MOL-GEOMETRY-001`
- `SFT-CHEM-MOL-ISOMER-001`
- `SFT-CHEM-NET-REACTION-001`
- `SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite generated set of balanced organic reactant-product atom maps with complete bond changes, functional roles, conditions and separate mechanism records.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `multiple-generated-reactions__complete-reactant-product-maps__shared-bond-change-pattern__reactant-product-role-correspondence__elemental-carriers-balanced__mechanism-not-assumed__member-and-pattern-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`members`	`multiple-generated-reactions`	`one-reaction-only`	No family is formed.
`encoding`	`complete-reactant-product-maps`	`reaction-names-only`	Names do not establish common structure.
`pattern`	`shared-bond-change-pattern`	`unrelated-bond-changes`	No shared family relation exists.
`roles`	`reactant-product-role-correspondence`	`reactant-product-roles-erased`	Pattern orientation is ambiguous.
`composition`	`elemental-carriers-balanced`	`carrier-balance-erased`	The transformations cannot close.
`mechanism`	`mechanism-not-assumed`	`mechanism-assumed-from-pattern`	Equal net changes need not share a mechanism.
`record`	`member-and-pattern-trace`	`family-label-only`	Membership cannot be audited.
`extra`	`no-extra-rule`	`free-reaction-taxonomy`	An imported list can force membership.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2794e829a8c2cd1c9b2bf0a50c8294a70400ffe90be534b840b1363dec4b90c6`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:35977ac840a0e2d15e7593c90926ed8ba21498624ab5ca81adce3b34fbc24762`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:94e838e1e737d341840056cf2b4c397e9745eaed535e4a80dd9e9b7e65e12e93`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:db912a2311b33366ee25d520989015f8231a0f4959c831e03149d92dad4561c3`.

Independent reconstruction. Implementation hash `sha256:1af395b5f6f38579aaffb318538190387ebb8f470c3d46f7881ec9ad2c3a6cf6`; independent certificate `sha256:ff5a8998035aab50f52a103e7a6886cfcf88a42d9f1049d282fd448408ba77c7`; external-validation hash `sha256:8a804b572fd73c3596d472ae7977e57d5008e5a60a3ab8d82dfaa1878d3e5e98`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ORGANIC-REACTION-FAMILY-COMPOSITE-2026

Comparison records:

- organic-reaction-family-iupac-composite: predicted shared-bond-change-pattern__reactant-product-role-correspondence__mechanism-not-assumed__conditions-source-bounded; source-derived shared-bond-change-pattern__reactant-product-role-correspondence__mechanism-not-assumed__conditions-source-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks shared reaction pattern, reactant/product roles, separation from mechanism or condition boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:4c0191c6b4bb019cc609b2a3a976737f275d49e25f6f56a832d96644e636f42f`.
Isolation certificate: `sha256:735875b659aed84e0f9ff94dbe41f4f49063103f4a42bf6d886ba7b390a9cde2`.
Custody certificate: `sha256:27927674882d94957fa8d8a5a595f358bc93d37a0011dc7c822ab202255d5411`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/R05164 (sha256:e7253285c30faeb217a73b62d4ca4cea7636a626c1306e2eebe8b71960abf6eb)

Registered target identities:

- organic-reaction-family-iupac-composite from IUPAC-GOLD-BOOK-ORGANIC-REACTION-FAMILY-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite generated set of balanced organic reactant-product atom maps with complete bond changes, functional roles, conditions and separate mechanism records.

Immutable identities. Source manifest `sha256:7a5883890cde64ed26c6bfb0ca4038354f1366b346a20ef2b3ae9eadb667f346`; derivation seal `sha256:ae06f0a1fc25bb0951b6f2f2662ab64d4d21b5c8ca404ea8be82f4bce944d8d8`; engine receipt `sha256:73b8569eaabf2253d83ba94bf9f729a0ff89709bcb6c64500bff2856abc64672` at `receipts/engine/model_admitted/SFT-CHEM-ORGANIC-REACTION-FAMILY-001-73b8569eaabf2253.json`; empirical hash `sha256:d855f814b6d2d5332bba692d17dabf1ae9a1b6f4d69d37242571e31c74673f0f`; measurement receipt `sha256:4c0191c6b4bb019cc609b2a3a976737f275d49e25f6f56a832d96644e636f42f`.

73. Polymer chain from repeated monomer identity

Claim identity: `SFT-CHEM-POLYMER-CHAIN-001`

Question and theorem. A polymer chain is one finite connected macromolecular carrier formed by covalently ordered occurrences of a retained repeat-unit identity with exact positive extent and explicit ends.

> *connected-macromolecular-carrier__repeated-low-mass-unit-identity__covalent-sequence-connections__ordered-chain-incidence__positive-finite-repeat-count*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`

- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-ORGANIC-FUNCTIONAL-GROUP-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite connected molecular incidence path containing repeated complete low-mass unit traces, covalent connections and explicit terminal boundaries.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `connected-macromolecular-carrier__repeated-low-mass-unit-identity__covalent-sequence-connections__ordered-chai` `n-incidence__positive-finite-repeat-count__chain-end-boundary-retained__repeat-connection-end-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`connected-macromolecular-carrier`	`unconnected-monomer-mixture`	No chain exists.
`repeat`	`repeated-low-mass-unit-identity`	`anonymous-segments`	Repeat identity is lost.
`connection`	`covalent-sequence-connections`	`noncovalent-coincidence`	The sequence is not a polymer chain.
`order`	`ordered-chain-incidence`	`occurrence-order-erased`	Chain topology is lost.
`extent`	`positive-finite-repeat-count`	`fixed-universal-length`	Length is population- and synthesis-bound.
`ends`	`chain-end-boundary-retained`	`infinite-chain-assumed`	A finite carrier needs a boundary.
`record`	`repeat-connection-end-trace`	`polymer-name-only`	The chain cannot be reconstructed.
`extra`	`no-extra-rule`	`free-polymer-rule`	A threshold can arbitrarily select chains.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8726c5ae36f1d1dda15c6109697ba8f767017f4b5806a911dc62ff6d46f82db7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:4527583bc01617f404e6e917d56a214b3f33732d24bb19ab270d9e50793a8f99`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3154c241dd7a4f1d0d151c880c2e340905664003bc6411691de8ba3be063ae7d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7e6b7adcf2ff991f470a76322fad0d788027a04b2ee07e6087c6a331bf625206`.

Independent reconstruction. Implementation hash `sha256:fc9ba42e65fefb02b61cbf5a7b32554e3c38cf5d592a5371b70f07f620fc7a16`; independent certificate `sha256:8e5fd2dd33f54399c8a6a3416e84050d860244c45d6b8b202d877bd8c1de9268`; external-validation hash `sha256:fe77ad02dfd37c3137f2ca0b48e948bedee53aa0ba55f040722a061cceb4f99d`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-POLYMER-CHAIN-COMPOSITE-2026

Comparison records:

- polymer-chain-iupac-composite: predicted high-relative-molecular-mass-molecule__repeated-low-mass-units__covalent-sequence__chain-end-boundary; source-derived high-relative-molecular-mass-molecule__repeated-low-mass-units__covalent-sequence__chain-end-boundary; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks a macromolecular carrier, repeated units, covalent sequence or chain boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:223181a892b117c6316b5ea0b1fbd98c3be3cb4a3ff1516886ebf4ec3c705b5c`. Isolation certificate: `sha256:c1a94a17425a77377d1353ffc7f70eca79c658f079eaa129bc92295d9e6d57d5`. Custody certificate: `sha256:77b60028ee938c7506691af30ade2c909f7f5f7f742df34595b20b9066bf9247`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/M03667 (sha256:b38c6c106e35763169204b065edaabf18df22bf97f4ea9094db3b8b72383e85a)

Registered target identities:

- polymer-chain-iupac-composite from IUPAC-GOLD-BOOK-POLYMER-CHAIN-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite connected molecular incidence path containing repeated complete low-mass unit traces, covalent connections and explicit terminal boundaries.

Immutable identities. Source manifest `sha256:820cbc89ab68ac2cefc56ec7564a14bd8e3f0d3339799384f62d303cc0e5519d`; derivation seal `sha256:1fcd451ad68862206f24ed642c27aa2c8aa62529d27f19183d459a0413fe7a2f`; engine receipt `sha256:f84c4fa8a4329d7f4d1ef445df7cdb7988d1d976c0dd68fece9701469df20cda` at `receipts/engine/model_admitted/SFT-CHEM-POLYMER-CHAIN-001-f84c4fa8a4329d7f.json`; empirical hash `sha256:730ba9addbe68f27c6fd2c64f95e545fd456286daa8ea1d0ccc065395033b02e`; measurement receipt `sha256:223181a892b117c6316b5ea0b1fbd98c3be3cb4a3ff1516886ebf4ec3c705b5c`.

74. Polymer population and retained chain-length distribution

Claim identity: `SFT-CHEM-POLYMER-DISTRIBUTION-001`

Question and theorem. A polymer distribution is the complete source-bound census of resolved chain carriers by repeat count and positive multiplicity, with number and mass weightings retained separately.

> *all-observed-chain-carriers__exact-chain-length-counts__positive-chain-multiplicities__number-and-mass-support-separated__complete-molar-mass-distribution*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`

- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-POLYMER-CHAIN-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite observed polymer sample with complete chain identities, repeat counts, positive multiplicities, weighting rule, sample and method boundary.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `all-observed-chain-carriers__exact-chain-length-counts__positive-chain-multiplicities__number-and-mass-support-separated__complete-molar-mass-distribution__method-and-sample-bounded__chain-population-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`population`	`all-observed-chain-carriers`	`single-representative-chain`	A sample distribution is erased.
`length`	`exact-chain-length-counts`	`mean-only`	A mean loses alternatives.
`multiplicity`	`positive-chain-multiplicities`	`duplicate-chains-collapsed`	Population frequency is lost.
`measure`	`number-and-mass-support-separated`	`one-universal-average`	Different summaries retain different information.
`distribution`	`complete-molar-mass-distribution`	`answer-scalar-only`	The population cannot be reconstructed.
`method`	`method-and-sample-bounded`	`method-free-census`	Resolution changes the observed population.
`record`	`chain-population-trace`	`distribution-name-only`	The census cannot be audited.
`extra`	`no-extra-rule`	`free-distribution-fit`	A fitted shape can hide unfavorable chains.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:88065ddd66a529ae65468a34e9385d4efa760e54ccd0c87359658a17f83ea231`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:3e9fe0202d7069000f6003a37591693635f37214510a56dd32ee5d33c3c7881c`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:90d31eee582fad2b4a6b2b14cc405a233e8a7c2c0a7db750908f4137db7b3800`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:ee81365154d6fc7ee014cdac864f9d47eb8e41328b830b95829704096f87ed57`.

Independent reconstruction. Implementation hash `sha256:8628fe82a0a16f901776576b549a688e90d8c857ac9004825836c20252b39bd3`; independent certificate `sha256:ffe61525dbbc336a4802001ad25570d65a61f25408c301d1484cf53898dda99d`; external-validation hash `sha256:33f7aec26af0ea140a0bb4712ffb155c16954cdd88dde3c8306a0fe0c6f222fb`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-POLYMER-DISTRIBUTION-COMPOSITE-2026

Comparison records:

- polymer-distribution-iupac-composite: predicted polymer-sample-many-chain-lengths__molar-mass-distribution__number-and-mass-fractions-retained__method-bounded; source-derived polymer-sample-many-chain-lengths__molar-mass-distribution__number-and-mass-fractions-retained__method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks chain-length diversity, molar-mass distribution, distinct weightings or method boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:e3a7d9f2e424c9405dd02ac15d6c658092dd0c6088ae782d3e671ff1ad0b4fa5`. Isolation certificate: `sha256:83629e16de4f587b5217518bff300797dca00881c165503e791ec9574970b890`. Custody certificate: `sha256:c225e04d254605315c9d7ac6be911f896e9183f41ed8732cc895c79ee4f4457f`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/12215 (sha256:98c415c9e851b41083ffc7ee3b9e1ba2d2b9e81bc6e05bb0dce170dd9733e91a)

Registered target identities:

- polymer-distribution-iupac-composite from IUPAC-GOLD-BOOK-POLYMER-DISTRIBUTION-COMPOSITE-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite observed polymer sample with complete chain identities, repeat counts, positive multiplicities, weighting rule, sample and method boundary.

Immutable identities. Source manifest `sha256:dfb937231e96d9d92dff71bb7720da00f9ccbd2532d0cf61e14989378fa5ae2f`; derivation seal `sha256:a3fdedaaab5e27ef897debf2cb9874778e0a6843eca446bc7c148dfe75a0cf1f`; engine receipt `sha256:b7cde3ce4ce07e58a22eb8f654a9cd4074b9b41148d59e4ee9c15b73455a86a0` at `receipts/engine/model_admitted/SFT-CHEM-POLYMER-DISTRIBUTION-001-b7cde3ce4ce07e58.json`; empirical hash `sha256:565c4e20bea7f88ee39a2977f1f1d9ce31ddd683902ec45267cefbf347fb3e39`; measurement receipt `sha256:e3a7d9f2e424c9405dd02ac15d6c658092dd0c6088ae782d3e671ff1ad0b4fa5`.

75. Chemistry-to-biology molecular handoff boundary

Claim identity: `SFT-CHEM-BIOMOLECULAR-BOUNDARY-001`

Question and theorem. The chemistry-to-biology boundary passes only a complete source-bound molecular identity, structure and chemical state; biological function is a separately retained downstream relation and cannot select the chemical law.

> *chemical-molecular-identity-retained__composition-preserved__molecular-structure-preserved__chemical-state-boundary-held__biological-function-separately-held*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- SFT-FOUNDATION-FORM-ENFORCEMENT-001
- SFT-MATH-EXACT-ARITHMETIC-001
- SFT-MATH-DISCRETE-001
- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001

- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-POLYMER-CHAIN-001
- SFT-CHEM-ORGANIC-REACTION-FAMILY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete carrier preservation, exact equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite chemical molecular carrier used by a downstream biological process, with complete composition, structure, state, conditions and separately registered function relation.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `chemical-molecular-identity-retained__composition-preserved__molecular-structure-preserved__chemical-state-boundary-held__biological-function-separately-held__chemistry-to-biology-handoff__composition-structure-state-function-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`chemical-molecular-identity-retained`	`biological-function-without-molecule`	Chemistry requires a molecular carrier.
`composition`	`composition-preserved`	`composition-erased-at-handoff`	The object cannot be traced chemically.
`structure`	`molecular-structure-preserved`	`structure-erased-at-handoff`	Biological recognition loses its substrate.
`state`	`chemical-state-boundary-held`	`timeless-molecule`	Function acts on physical chemical states.
`function`	`biological-function-separately-held`	`function-as-chemical-identity`	Different functions need not change chemical identity.
`boundary`	`chemistry-to-biology-handoff`	`biology-imported-into-chemistry`	A downstream law cannot select chemistry.
`record`	`composition-structure-state-function-trace`	`biomolecule-name-only`	The handoff cannot be audited.
`extra`	`no-extra-rule`	`free-life-criterion`	A life label can select any molecule.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:d24fc84b3d2eb24db795ab0e5e24264436d5b446a3d0db66271770e515baf85f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:d9532c7da21be65210d77703ceafb00a1d0ed36e06cb60b83d3ed2582d4e9a9f`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:e3b5635d8c33cd0e99f9d01f3d9368a5206a4d053087c72ecc3fc2a813a78e46`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7c47d008b17ce95a41abeb1071ec1c3281cfac6d126816e7529602799b36c302`.

Independent reconstruction. Implementation hash `sha256:d831eace1b996519fb40bc5b13e8897cdf6c007f87d9159e66bf77edeef4130a`; independent certificate `sha256:f4d1b90f1ef528e8bb267bdc296905e85dcbd67d0ea2eee3e84ff09c71e7561e`; external-validation hash `sha256:ce2e18c75f02c7a9ac082c3edb4160da04c0050bc6994594fdf9901c79a2f37f`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026

Comparison records:

- biomolecular-boundary-authority-composite: predicted chemical-molecular-identity-retained__biological-function-not-chemical-identity__structure-composition-handoff__biology-boundary-explicit; source-derived chemical-molecular-identity-retained__biological-function-not-chemical-identity__structure-composition-handoff__biology-boundary-explicit; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence cannot separate chemical molecular identity from biological function at an explicit handoff boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:1fd01394c060361f41594cb2e5faeafacf6c2f2f4c6f9a57a1d93548da7df75a`. Isolation certificate: `sha256:819826e2239d7b8e241efa0e71ca7e1ca417dd34a948063166a5fd66af8a9f79`. Custody certificate: `sha256:3434cfb36ec513b6089ed50cea8cb2fb844fca1392239b585b2c82af1f3db026`.

Registered source descriptions:

- EMBL-European Bioinformatics Institute: https://www.ebi.ac.uk/chebi/ (sha256:d862deab28de555e80751d9fce3fecf4ac4de970d17d9a4c1481dc6976731f24)

Registered target identities:

- biomolecular-boundary-authority-composite from EMBL-EBI-CHEBI-BIOMOLECULAR-BOUNDARY-2026

Meaning. Stereochemical and polymer distinctions arise from complete finite structure equivalence, held orientation, recurrence and connected composition. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no stereochemical dictionary, organic reaction taxonomy, polymer table, biomolecular database, measured target or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no atom, bond, orientation, repeat unit, chain or observation distinction may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical molecular carrier used by a downstream biological process, with complete composition, structure, state, conditions and separately registered function relation.

Immutable identities. Source manifest `sha256:ac63afa7c56f2f87a630caef127128b2787a10a4b96693836c0bb0f47d76f537`; derivation seal `sha256:cc93dcd2e0b68c5cd2d497b751efc1292cb3a845fb29e86a07b8b7d9cf4c5a14`; engine receipt `sha256:610945f40717478ae45485c17632b717085f1801b29cf794ae1dd1b7c008193e` at `receipts/engine/model_admitted/SFT-CHEM-BIOMOLECULAR-BOUNDARY-001-610945f40717478a.json`; empirical hash `sha256:3070285b3783ed838c144def1aca376ee3b6a5facdaa5215b5c971ebb5ebc572`; measurement receipt `sha256:1fd01394c060361f41594cb2e5faeafacf6c2f2f4c6f9a57a1d93548da7df75a`.

17. Analytical Spectroscopic

Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law.

76. Analytical sample, analyte and matrix distinction

Claim identity: `SFT-CHEM-ANALYTICAL-SAMPLE-001`

Question and theorem. An analytical sample is a declared material portion whose complete resolved component support is partitioned by held roles into the analyte and retained matrix, together with its source-to-sample map.

> *declared-material-portion__complete-resolved-component-support__analyte-role-held__matrix-support-held__source-to-sample-map-held*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`

- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite declared material portion with complete resolved components, one or more analyte roles and retained non-analyte support.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `declared-material-portion__complete-resolved-component-support__analyte-role-held__matrix-support-held__source-to-sample-map-held__component-response-map-held__sample-analyte-matrix-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
portion	declared-material-portion	unbounded-material-name	No sampled carrier is identified.
components	complete-resolved-component-support	component-support-erased	The portion cannot be compositionally audited.
analyte	analyte-role-held	analyte-not-designated	No measurand-bearing component is selected.
matrix	matrix-support-held	non-analyte-components-discarded	Interference support is hidden.
sampling	source-to-sample-map-held	sample-equals-whole-source	A portion need not reproduce the source whole.
observation	component-response-map-held	response-without-component-map	A response cannot be assigned.
record	sample-analyte-matrix-trace	result-only	The sample distinction cannot be reproduced.
extra	no-extra-rule	free-sample-rule	A discretionary rule can select any portion.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3efc490e1bb313eb3a34c6f86d105d7b0feb050e3dc07907cb22f9e557e3db2a`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:f6628399c538353c7fb79523e499076b0985aa2d7845782e3bab7e269ba2402e`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:54733cd21625ad9d2061bffcbfc18170a824966c03c94ce4abd76a18e66cdfcb`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7c171463044bd6ddcb0a3699fa40084a8b5f356e72f0015643dd009118f5dab8`.

Independent reconstruction. Implementation hash `sha256:2e248b730cee3eeb2aeb40d4018c0beafc12e8ac5692a7b8c6ed0b72657168bc`; independent certificate `sha256:29057279273fb54b1e5c5c6f56b99df6749458dbbb89aec2895105ee80e84ac8`; external-validation hash `sha256:2f3f30c535e2eddb27a8d50f12b69be618efdb99175f206525cdf726e1b86970`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ANALYTICAL-SAMPLE-COMPOSITE-2026

Comparison records:

- analytical-sample-iupac-composite: predicted sample-is-declared-material-portion__analyte-is-component-under-determination__matrix-is-other-sample-components__sampling-boundary-held; source-derived sample-is-declared-material-portion__analyte-is-component-under-determination__matrix-is-other-sample-components__sampling-boundary-held; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks sample portion, analyte, matrix or sampling boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:9dd5852a454c591de6380c54af7fd342025d09cd158b7f4af562639bef615080`. Isolation certificate: `sha256:2b6ebe84cd1474932eb55716fa6f13b30fe3b952ecbf9af3224490cf5259286f`. Custody certificate: `sha256:03c9a215a94e78ae118236a2ad453098c1dcac55244e58a3e3525578612faa7f`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/S05451 (sha256:f2d630c6a9a8ec0ab1304a6e9e3acf1bc495b9b3f2d5f3958375aa0c3fafa184)

Registered target identities:

- analytical-sample-iupac-composite from IUPAC-GOLD-BOOK-ANALYTICAL-SAMPLE-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite declared material portion with complete resolved components, one or more analyte roles and retained non-analyte support.

Immutable identities. Source manifest `sha256:59738f8ee3fa617ea9ee488d7328b171fff70fcf05b7a0adf148ab96a42e1f91`; derivation seal `sha256:3247a2f5a28abfd5d60dab628748e12a960f6d7b406fd52e7824b34df1bd3803`; engine receipt `sha256:1880410ee99b23f159fdc8775f6c1186f2094319030db175661c992b48590551` at `receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-SAMPLE-001-1880410ee99b23f1.json`; empirical hash `sha256:96d710f063d8db488f420012076fd14288519c901b11c1d733ae55cf4c5197f2`; measurement receipt `sha256:9dd5852a454c591de6380c54af7fd342025d09cd158b7f4af562639bef615080`.

77. Analytical calibration and traceable comparison

Claim identity: `SFT-CHEM-ANALYTICAL-CALIBRATION-001`

Question and theorem. Calibration is the complete traceable comparison of held reference input/response pairs with a sample response inside an explicit finite method domain, retaining uncertainty and every row.

> *traceable-reference-input__reference-response-pair-held__exact-generated-comparison-order__sample-compared-with-reference-map__declared-calibration-domain*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`

- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-ANALYTICAL-SAMPLE-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite registered reference-response set and sample response sharing one declared observation organization.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `traceable-reference-input__reference-response-pair-held__exact-generated-comparison-order__sample-compared-with-reference-map__declared-calibration-domain__reference-and-sample-uncertainty-held__reference-map-and-result-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`reference`	`traceable-reference-input`	`unknown-reference-input`	Comparison has no fixed support.
`response`	`reference-response-pair-held`	`reference-response-erased`	The comparison cannot be reconstructed.
`ordering`	`exact-generated-comparison-order`	`unordered-fit`	A fit can conceal incompatible rows.
`sample`	`sample-compared-with-reference-map`	`sample-outside-reference-map`	No calibrated comparison is possible.
`range`	`declared-calibration-domain`	`universal-unbounded-map`	Calibration support is finite and method-bound.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`uncertainty`	`reference-and-sample-uncertainty-held`	`uncertainty-erased`	Alternatives are falsely collapsed.
`record`	`reference-map-and-result-trace`	`calibrated-answer-only`	Traceability is lost.
`extra`	`no-extra-rule`	`free-calibration-parameter`	A fitted value can select the answer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:032e873ba65355fe014ee4ea11bc71860ad6fe7164823becca09a813425962ad`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ec401bf322205d24affee40106306cd23ff06e1726aaa862af3db593374ff9bb`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:454c698f8da30d9662a41b9526c30f56193e6a98cee4296f04011056542b30dc`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:f2d3b9679de44e3fec73bc19dec2b45f5bf4a9a76a71918982afc544d84b3980`.

Independent reconstruction. Implementation hash `sha256:1dbf78592e06bd73224c7c6afeee79718cf93ce2936d633abbc26654f817b49b`; independent certificate `sha256:94a382fa81641e7ff83e6cef4cb4683a3db0e00ef1acce42986f34d6b71c5a72`; external-validation hash `sha256:fbcd85191232ce6b12444ae069c1d3f22e0da138e5eab25fd8eee4051877efa0`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ANALYTICAL-CALIBRATION-COMPOSITE-2026

Comparison records:

- analytical-calibration-iupac-composite: predicted calibration-relates-response-to-reference-values__traceability-chain-retained__calibration-domain-explicit__uncertainty-method-bounded; source-derived calibration-relates-response-to-reference-values__traceability-chain-retained__calibration-domain-explicit__uncertainty-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks response/reference relation, traceability, domain or uncertainty/method boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:3ca1469a4ebf546a35e8913fb787a87358e5be94e75d96e21717da05b0f72d72`. Isolation certificate: `sha256:1a8973b8fb35f45380983bcc6b57642e32a3a2231060c37d09bb1800aefcb708`. Custody certificate: `sha256:b26d8132d69aeb2dd353a3738f78de034ea5318793651caa7d3e49a55c2889db`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/C00775 (sha256:9a50ffb4e909ae78ac6ccf2cfa69db372a7ff659db4fb88b48e89e79322b69c0)

Registered target identities:

- analytical-calibration-iupac-composite from IUPAC-GOLD-BOOK-ANALYTICAL-CALIBRATION-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite registered reference-response set and sample response sharing one declared observation organization.

Immutable identities. Source manifest `sha256:6fc05197ce55d2639e3bb3f6cfeeda3802c9003ac56c3621df0ef186ce5790da`; derivation seal `sha256:6c4f143e466e9cafc4b5205774ab8c543b3a201d57cc9b87c1aa38770a0797b1`; engine receipt `sha256:4ac2379caade53b802030b6fe2ee43633bb794d2a50b9f7cf4f3fd2982bf5cf1` at `receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-CALIBRATION-001-4ac2379caade53b8.json`; empirical hash `sha256:4b5eea2cc1d8a15f1e589ab0511927364ab680f5f8d7ed9fa3fee62332031bb7`; measurement receipt `sha256:3ca1469a4ebf546a35e8913fb787a87358e5be94e75d96e21717da05b0f72d72`.

78. Analytical selectivity and interference boundary

Claim identity: `SFT-CHEM-ANALYTICAL-SELECTIVITY-001`

Question and theorem. Analytical selectivity is exact distinguishability of the analyte response from every declared interferent response within a retained method-and-condition boundary; merged responses are recorded as interference.

> *analyte-response-held__declared-interferent-support__all-analyte-interferent-responses-compared__distinguishable-analyte-response__interference-boundary-retained*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`

- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-MOL-GEOMETRY-001`
- `SFT-CHEM-MOL-ISOMER-001`
- `SFT-CHEM-NET-REACTION-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-TRACEABILITY-001`
- `SFT-CHEM-BIOMOLECULAR-BOUNDARY-001`
- `SFT-CHEM-ANALYTICAL-SAMPLE-001`
- `SFT-CHEM-ANALYTICAL-CALIBRATION-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite analyte/interferent response support under one registered method and condition class.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `analyte-response-held__declared-interferent-support__all-analyte-interferent-responses-compared__distinguishable-analyte-response__interference-boundary-retained__method-and-condition-bounded__complete-response-comparison-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
target	analyte-response-held	response-without-analyte	Selectivity has no target.
alternatives	declared-interferent-support	other-components-erased	Competing support is hidden.
comparison	all-analyte-interferent-responses-compared	single-response-only	No discrimination is tested.
decision	distinguishable-analyte-response	name-based-assignment	A name cannot prove attribution.
interference	interference-boundary-retained	failed-alternatives-deleted	Unfavorable rows disappear.
conditions	method-and-condition-bounded	universal-selectivity	Response depends on method and conditions.
record	complete-response-comparison-trace	selectivity-scalar-only	The tested alternatives cannot be audited.
extra	no-extra-rule	free-selectivity-threshold	An arbitrary threshold can force success.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:3b6ea28b86431cb84b37ffd90b03311ea422cc2dc805d76224971189594641e8`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a7d04dc9d2f7bc12b43d5e5e8df3b7a4bcf26911119584e555d5d74f01160b64`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:18a0bf0c39f89ee23bcf5b4c7a0092ac625f88fea4c3d15c8e860f04698bc715`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:98ad52c643c40c247122849ca5db493c0f3ab8e4cb17c141826be16d1d16b702`.

Independent reconstruction. Implementation hash `sha256:62b133e342f7dd1974b887e976c0df6a472e824750afb8ec0872087d4d5427bf`; independent certificate `sha256:fe120fe14ad5e2b636c45d8014360b4858a8c3d87b8d79139fc67719289f4487`; external-validation hash `sha256:208b713a88a9bfa39042681fadab45152cc4ddfcaac7f90ea134e10ec17a3d43`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ANALYTICAL-SELECTIVITY-COMPOSITE-2026

Comparison records:

- analytical-selectivity-iupac-composite: predicted analyte-response-distinguished-amid-components__interference-retained__selectivity-not-universal__method-condition-boundary-held; source-derived analyte-response-distinguished-amid-components__interference-retained__selectivity-not-universal__method-condition-boundary-held; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks analyte discrimination, interference, non-universality or method/condition boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:da6f121d768e5e3cd227d4fa02a27a1ae951bed717f206f0509500450a09524c`.
Isolation certificate: `sha256:e400a8788587c5690c8d366b7d8e8b335e69773736267c741044f3d1df90f51f`.
Custody certificate: `sha256:c5b528577459443b0a49f2c21284d46b9012f08eb4f2c09ec59529ea2582d97f`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/S05564 (sha256:24cfb5654b5ba14eb99cc4dfe0c01efa7a251be91937a33000f362ba8a9730f6)

Registered target identities:

- analytical-selectivity-iupac-composite from IUPAC-GOLD-BOOK-ANALYTICAL-SELECTIVITY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite analyte/interferent response support under one registered method and condition class.

Immutable identities. Source manifest `sha256:ff8fa9f556403b99cb8df4a5b8f602389cf76e2eec62851e961b0e9df189525f`; derivation seal `sha256:e3044df347a42d228e3b0e52ba0dfe680bc14e2a71dbe97beb680c3d7560040d`; engine receipt `sha256:b00b15ae23003f159f654f5f6bad65e0aae8268fe5769873ad0b3ab0e1859e8b` at `receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-SELECTIVITY-001-b00b15ae23003f15.json`; empirical hash `sha256:59a8da91f9ca9b1d0ef6fcd6aafb566445163284aa455228f415f976afa39a22`; measurement receipt `sha256:da6f121d768e5e3cd227d4fa02a27a1ae951bed717f206f0509500450a09524c`.

79. Mass-spectral composition correspondence

Claim identity: `SFT-CHEM-SPEC-MASS-001`

Question and theorem. A mass-spectral record is the complete map from generated ionized carriers to retained mass/charge response classes and counted abundances, including precursor-fragment relations and observation conditions.

> *generated-ionized-carriers__composition-and-isotope-support-held__mass-charge-classes-held__class-position-and-abundance-held__precursor-fragment-map-held*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`

- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-MOL-GEOMETRY-001`
- `SFT-CHEM-MOL-ISOMER-001`
- `SFT-CHEM-NET-REACTION-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-TRACEABILITY-001`
- `SFT-CHEM-BIOMOLECULAR-BOUNDARY-001`
- `SFT-CHEM-ANALYTICAL-SAMPLE-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite generated ionized-carrier support with complete composition, isotope, mass/charge, response and precursor-fragment records.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `generated-ionized-carriers__composition-and-isotope-support-held__mass-charge-classes-held__class-position-and-abundance-held__precursor-fragment-map-held__ionization-and-instrument-boundary__complete-carrier-response-trace__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`generated-ionized-carriers`	`neutral-sample-label-only`	No detected carrier class is generated.
`identity`	`composition-and-isotope-support-held`	`carrier-composition-erased`	Composition correspondence is impossible.
`partition`	`mass-charge-classes-held`	`undifferentiated-response`	Distinct carrier classes merge.
`response`	`class-position-and-abundance-held`	`peak-position-only`	Multiplicity support is lost.
`fragments`	`precursor-fragment-map-held`	`fragment-origin-erased`	Products cannot be related to precursors.
`conditions`	`ionization-and-instrument-boundary`	`universal-spectrum`	Carrier production is method-bound.
`record`	`complete-carrier-response-trace`	`spectrum-image-only`	The composition map cannot be checked.
`extra`	`no-extra-rule`	`free-peak-assignment`	A library guess can select composition.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b30aae3c4fac29492c023e1ddecc2f8fcb59015da79eeaadf2fb7ea23baaca22`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:7023aee7d974957bfd7bd46f6c83b997fa078f56c68a69d46fb051980153b357`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:5fbc7a983050ed25c72382178f46875fdf3fbb11d6ec7c91dc77044761a41e49`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e39d6c8b12dfe47a2149f642070ebdeba1fad62702440d164fd805569d2c2f98`.

Independent reconstruction. Implementation hash `sha256:63f8223999cb6f43867dfa34e17c98dc20e6a5a1c49bc7286e9867776ab6c544`; independent certificate `sha256:68951ba69b1e384489a5e141bf0855dc916c1454d8548dcaab442b39f328392d`; external-validation hash `sha256:63571f9167c8b27006e3ed0cb71992f6cd7d04030ec1c7337a208e19ffea1aba`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-MASS-SPECTROSCOPY-COMPOSITE-2026

Comparison records:

- mass-spectroscopy-iupac-composite: predicted ions-separated-by-mass-to-charge__mass-spectrum-retains-relative-abundance__composition-fragment-correspondence__ionization-method-bounded; source-derived ions-separated-by-mass-to-charge__mass-s pectrum-retains-relative-abundance__composition-fragment-correspondence__ionization-method-bounded; exact match True

- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks ion mass/charge separation, abundance, composition/fragment correspondence or method boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:77f7def06c3bc419f995ccf48f17f8d46ed68d2f835561979150017924e59336`. Isolation certificate: `sha256:2b81b6c4e5626f4d9e1e9ce7b5152e70c1ef70b8967af710a3ce5f0ea2a7b1e4`. Custody certificate: `sha256:2a1d5bc32a4354ccae5b3e58f09b16852cb112ec7ad240c15aee2376e5b7b20d`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/M03749 (sha256:bb7b4631fab65d61fd3c0f8b6257ae91bc87da91ceea23a961f4c824f20b4e56)

Registered target identities:

- mass-spectroscopy-iupac-composite from IUPAC-GOLD-BOOK-MASS-SPECTROSCOPY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite generated ionized-carrier support with complete composition, isotope, mass/charge, response and precursor-fragment records.

Immutable identities. Source manifest `sha256:4b19515587686f660712aa91b2eccaef9918c1714c30bd5f711de9a2dedbc933`; derivation seal `sha256:5800598040da59b271b7dffad5de0528b95c6782508fc03815e14bc1cbae40f9`; engine receipt `sha256:e155dbf2c8cd4be28f11d225025424f0d4005a458783853b4588977745edc60e` at `receipts/engine/model_admitted/SFT-CHEM-SPEC-MASS-001-e155dbf2c8cd4be2.json`; empirical hash `sha256:21d06af9ba02bdae4c8382a1f7f65aeeacaa23b77e410ca9cc5401a32e644740`; measurement receipt `sha256:77f7def06c3bc419f995ccf48f17f8d46ed68d2f835561979150017924e59336`.

80. Infrared recurrence and molecular distinction

Claim identity: `SFT-CHEM-SPEC-INFRARED-001`

Question and theorem. Infrared organization is the complete condition-bound pattern of incident recurrence classes coupled to retained molecular vibrational state transitions, with whole-pattern rather than isolated-line identity.

> *incident-infrared-support-held__complete-molecular-carrier-held__vibrational-state-transition-held__dipole-response-change-required__complete-frequency-response-pattern*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`

- SFT-MATH-COMBINATORICS-001
- SFT-MATH-GRAPH-NETWORK-001
- SFT-MATH-ORDER-LATTICE-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001
- SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001
- SFT-PHYS-QUANTUM-EXCLUSION-001
- SFT-PHYS-MATTER-CONSERVED-LABELS-001
- SFT-PHYS-THERMO-EQUILIBRIUM-001
- SFT-CHEM-ELEM-ION-001
- SFT-CHEM-STOICH-CONSERVATION-001
- SFT-CHEM-BOND-COVALENT-001
- SFT-CHEM-BOND-IONIC-001
- SFT-CHEM-MOL-MOLECULE-001
- SFT-CHEM-MOL-INTERMOLECULAR-001
- SFT-CHEM-AB-ACID-BASE-001
- SFT-CHEM-AB-PROTON-TRANSFER-001
- SFT-CHEM-AB-LEWIS-001
- SFT-CHEM-AB-AMPHOTERIC-001
- SFT-CHEM-AB-BUFFER-001
- SFT-CHEM-ELECTRONEGATIVITY-001
- SFT-CHEM-BOND-POLARITY-001
- SFT-CHEM-REDOX-OXIDATION-STATE-001
- SFT-CHEM-REDOX-COUPLING-001
- SFT-CHEM-ELECTROCHEM-CELL-001
- SFT-CHEM-RXN-IDENTITY-001
- SFT-CHEM-RXN-MECHANISM-001
- SFT-CHEM-KIN-ACTIVATION-001
- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001

- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-TRACEABILITY-001`
- `SFT-CHEM-BIOMOLECULAR-BOUNDARY-001`
- `SFT-CHEM-PHOTOCHEM-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite molecular carrier with generated internal recurrence states, paired transitions, interaction-change tests and a declared observation domain.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `incident-infrared-support-held__complete-molecular-carrier-held__vibrational-state-transition-held__dipole-response-change-required__complete-frequency-response-pattern__whole-pattern-molecular-distinction__state-method-condition-boundary__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`input`	`incident-infrared-support-held`	`unregistered-incident-recurrence`	No excitation comparison is defined.
`molecule`	`complete-molecular-carrier-held`	`molecular-identity-erased`	Response cannot distinguish carriers.
`transition`	`vibrational-state-transition-held`	`static-geometry-only`	A spectrum requires a state change.
`coupling`	`dipole-response-change-required`	`all-transitions-assumed-visible`	Observation requires a changed interaction trace.
`spectrum`	`complete-frequency-response-pattern`	`one-line-answer`	Alternative recurrences are erased.
`identity`	`whole-pattern-molecular-distinction`	`universal-one-line-identifier`	Different carriers may share a line.
`conditions`	`state-method-condition-boundary`	`condition-free-spectrum`	State and method affect observation.
`extra`	`no-extra-rule`	`free-infrared-line-rule`	A lookup can force assignment.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:b2ffe399a008258b480ff5f4c6f502fe06f1554a8544521f5d1ece040955890a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:a603e622aeaed9f12c66289143a2bd1b22cec3d5a1b7d996404b77fa9d144334`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:1d6b9d317916812bb704757723d4399c03e03f6d00a9e5e985f0bbc82060bbad`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:3d6cea107788fcc04f575ed6126f2ad493ed4dc3187751c7aa2e56e002b9db24`.

Independent reconstruction. Implementation hash `sha256:9623a7a8ad4f940f91f869bb1b7224df95a3399ea5ee013763d34c92a621724b`; independent certificate `sha256:81480db63e0f925d2062ec153fa02b0bb1df1d38b15c2471b5794949c032bd86`; external-validation hash `sha256:9529ae6b8296de786c171d9400d2fcf33d01478c255fb759e569522a707360a1`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-INFRARED-SPECTROSCOPY-COMPOSITE-2026

Comparison records:

- infrared-spectroscopy-iupac-composite: predicted infrared-absorption-corresponds-to-molecular-vibration__frequency-pattern-retained__molecular-distinction-uses-whole-spectrum__state-method-bounded; source-derived infrared-absorption-corresponds-to-molecular-vibration__frequency-pattern-retained__molecular-distinction-uses-whole-spectrum__state-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks infrared/vibrational correspondence, a retained frequency pattern, whole-spectrum molecular distinction or state/method boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:66439cdd064a13c442bd3feb7372e4f0a016e24e4eae0ee368dcbacc91b7f8f9`. Isolation certificate: `sha256:e4c85f682b1d10051b9f4667e9da502cd896e4b47d0219201be45dddb5396e19`. Custody certificate: `sha256:f775a666c4853e98b0560c25dcbe0e626f85f0f743e1a5945b5ebc6f19336ff2`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/08618 (sha256:0be3cc15940bfcf345e61af54ef378022de8f2c32b7e566b7d4ed25f69c553ab)

Registered target identities:

- infrared-spectroscopy-iupac-composite from IUPAC-GOLD-BOOK-INFRARED-SPECTROSCOPY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite molecular carrier with generated internal recurrence states, paired transitions, interaction-change tests and a declared observation domain.

Immutable identities. Source manifest `sha256:c07ca280f1bf8c27df6ee5f4c840dd086848066a03f5e4bb4b10cd3f5cca37df`; derivation seal `sha256:c1815e5aaa2678f960ef5ea2fab48ad01421689c29e088440df015d9a874a0d6`; engine receipt `sha256:b1d149a0252e24a9e4ae697f92c986f6ee3fd0bc34698d43d39b19f5cab0721b` at `receipts/engine/model_admitted/SFT-CHEM-SPEC-INFRARED-001-b1d149a0252e24a9.json`; empirical hash `sha256:30acd4a106af6be00731ab9c87d59bbb211d389aefda98155f76a354a97b0c43`; measurement receipt `sha256:66439cdd064a13c442bd3feb7372e4f0a016e24e4eae0ee368dcbacc91b7f8f9`.

81. Electronic absorption and molecular-state distinction

Claim identity: `SFT-CHEM-SPEC-UVVIS-001`

Question and theorem. Ultraviolet-visible organization is the complete condition-bound response pattern of molecular or ionic carriers under retained transitions between ordered electronic states.

> *ultraviolet-visible-support-held__molecular-or-ionic-carrier-held__paired-electronic-states__higher-state-transition-held__wavelength-response-pattern-held*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`

- SFT-CHEM-KIN-RATE-001
- SFT-CHEM-EQ-CHEMICAL-001
- SFT-CHEM-THERMO-REACTION-001
- SFT-CHEM-THERMO-DIRECTION-001
- SFT-CHEM-PHASE-CHEMICAL-001
- SFT-CHEM-MOL-GEOMETRY-001
- SFT-CHEM-MOL-ISOMER-001
- SFT-CHEM-NET-REACTION-001
- SFT-CHEM-MEAS-CHEMICAL-ENTITY-001
- SFT-CHEM-MEAS-UNCERTAINTY-001
- SFT-CHEM-MEAS-TRACEABILITY-001
- SFT-CHEM-BIOMOLECULAR-BOUNDARY-001
- SFT-CHEM-PHOTOCHEM-001

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite chemical carrier with paired generated electronic state classes, ordered transitions and a declared ultraviolet-visible observation domain.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `ultraviolet-visible-support-held__molecular-or-ionic-carrier-held__paired-electronic-states__higher-state-transition-held__wavelength-response-pattern-held__carrier-state-response-map__state-method-condition-boundary__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`input`	`ultraviolet-visible-support-held`	`unregistered-radiation`	No excitation support is identified.
`carrier`	`molecular-or-ionic-carrier-held`	`chemical-carrier-erased`	Absorption cannot be assigned.
`state`	`paired-electronic-states`	`single-state-only`	No transition occurs.
`transition`	`higher-state-transition-held`	`energy-order-erased`	Absorption direction is ambiguous.
`response`	`wavelength-response-pattern-held`	`absorption-number-only`	Spectral support is lost.
`composition`	`carrier-state-response-map`	`response-detached-from-carrier`	Molecular distinction cannot be tested.
`conditions`	`state-method-condition-boundary`	`universal-response`	Environment and method affect observation.
`extra`	`no-extra-rule`	`free-electronic-band-rule`	A fitted band can select the answer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:8f109438a2184ff5bd94ed5c3baa2779939b48fa5908c03a9494ead18ff4c43a`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:bdaad010665344c3bc6d4315e843eea1f36ae20f9d6016475c3f06bac89b5953`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:256566afdc578e393902758a27a005c03759954a45509d5445c7b8427a425f5a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:4cf9ae33fa257656b34c7fe81af112869f07b904db81a4937acccd502270d63a`.

Independent reconstruction. Implementation hash `sha256:09021bd00f42035e6774890cefcb66fd971761de04775ef636494b024c7a45c8`; independent certificate `sha256:cdb28a031fb92bd003449669ff7410e83e34c5809bdcd5cc065f98c9471e5c7c`; external-validation hash `sha256:3b6f0f0ce8d093a1c6ece6539a8f88d62e09cea6999635a2c02be6f99743d299`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-UVVISIBLE-SPECTROSCOPY-COMPOSITE-2026

Comparison records:

- uv-visible-spectroscopy-iupac-composite: predicted ultraviolet-visible-absorption-involves-electronic-transition__carrier-and-state-retained__response-pattern-recorded__state-method-bounded; source-derived ultraviolet-visible-absorption-involves-electronic-transition__carrier-and-state-retained__response-pattern-recorded__state-method-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks UV-visible absorption, electronic transition, retained carrier/state response or method boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:ab68ae086496094b34addbd6f2eb057ece27c88e0bee1c99e8dc2b6579499f9e`. Isolation certificate: `sha256:bf41ea75746962dc9f01f22df3c1a0a081ba9342674ae8ce36bdf302a91d1895`. Custody certificate: `sha256:69eab65a16c11a8b416bb56c52622248ca35f957a0c69f2335f2bf94b2fd5803`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/S05696 (sha256:bcf4022ed1743c6445d1c6a2dc88d616adad7f833c299734f1b3fc3b45652a7d)

Registered target identities:

- uv-visible-spectroscopy-iupac-composite from IUPAC-GOLD-BOOK-UVVISIBLE-SPECTROSCOPY-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite chemical carrier with paired generated electronic state classes, ordered transitions and a declared ultraviolet-visible observation domain.

Immutable identities. Source manifest `sha256:3365544b98ddc2a8728c20b4b5ebaae3a1c83aadee3e8ee956b61be2574d6788`; derivation seal `sha256:e01d957ce5faf81059e38d012818e21411f565c169add8875dc9a60c50836ff6`; engine receipt `sha256:6286103d4fa407f19df94df7529d42c98998bce4f1873bfa1f71bad69af17e37` at `receipts/engine/model_admitted/SFT-CHEM-SPEC-UVVIS-001-6286103d4fa407f1.json`; empirical hash `sha256:c226e877cef95636b84f27d10a5efee1a48cb1ae5dad9de2cf5b81b80b33dfe9`; measurement receipt `sha256:ab68ae086496094b34addbd6f2eb057ece27c88e0bee1c99e8dc2b6579499f9e`.

82. Rotational-vibrational molecular spectra

Claim identity: `SFT-CHEM-SPEC-ROT-VIB-001`

Question and theorem. A rotational-vibrational spectrum is the complete condition-bound transition organization of retained molecular orientation-recurrence and internal relative-displacement recurrence classes.

> *complete-molecular-geometry-held__rotational-state-classes-held__vibrational-state-classes-held__rotational-vibrational-composition__allowed-state-transition-trace*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`
- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`

- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-MOL-GEOMETRY-001`
- `SFT-CHEM-MOL-ISOMER-001`
- `SFT-CHEM-NET-REACTION-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-TRACEABILITY-001`
- `SFT-CHEM-BIOMOLECULAR-BOUNDARY-001`
- `SFT-CHEM-SPEC-INFRARED-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite molecular geometry with generated rotational and vibrational recurrence classes, paired transitions and declared observation support.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-molecular-geometry-held__rotational-state-classes-held__vibrational-state-classes-held__rotational-vibrational-composition__allowed-state-transition-trace__complete-transition-frequency-pattern__state-method-condition-boundary__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`complete-molecular-geometry-held`	`point-label-only`	Internal molecular motion is absent.
`rotation`	`rotational-state-classes-held`	`orientation-history-erased`	Rotational recurrence cannot be distinguished.
`vibration`	`vibrational-state-classes-held`	`relative-displacement-erased`	Vibrational recurrence cannot be distinguished.
`composition`	`rotational-vibrational-composition`	`motions-collapsed-together`	Joint spectral structure is lost.
`transition`	`allowed-state-transition-trace`	`states-without-transition-map`	No spectral event is generated.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`spectrum`	`complete-transition-frequency-pattern`	`one-frequency-only`	Resolved organization is erased.
`conditions`	`state-method-condition-boundary`	`universal-spectrum`	Population and resolution are bounded.
`extra`	`no-extra-rule`	`free-rotor-oscillator-equation`	An imported model can select the spectrum.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:4323d8285eafe3506586879204c1788f573d36d9a713b75d9b52ab8745e41bfa`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:ca67c32795fdd77c1b6ba5a17ba0a5984f0c5433217db50572ebcc84a840f588`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:bef4f75c40ebc0d0e5933659a7ea33f77a3bad65f0ed79a5877261fbbf03c980`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:179ca61d9ff6baa1547972cfa7013cebb029156f005a5193a807e86ff46661d3`.

Independent reconstruction. Implementation hash `sha256:833c5b45b93d721aa1ab38cc1261be7e6e312fd3be9721a6ba087236b26d7eb1`; independent certificate `sha256:416fbcd13691a5a9887624f22611dcd6b60a31f2f6e113167ace8efeb96a0183`; external-validation hash `sha256:d184133bed088a2864cb839289022239ee561c40235a83dd2e4d4d026f95c7e5`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026

Comparison records:

- rotation-vibration-iupac-composite: predicted molecular-spectra-retain-rotational-and-vibrational-transitions__state-composition-held__transition-pattern-complete__population-resolution-bounded; source-derived molecular-spectra-retain-rotational-and-vibrational-transitions__state-composition-held__transition-pattern-complete__population-resolution-bounded; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks rotational/vibrational transitions, state composition, transition pattern or population/resolution boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:9e6705743272cc7a980dbff088a552cc9fa032666221bce0201399ba9c83dc25`. Isolation certificate: `sha256:d6e4b8971fd420ccafcaae0d29e24ca2a51f85f504f5ac52e284dea925f10e45`. Custody certificate: `sha256:d6097c94596e1d228f13e4b8cc8fd50852a2dc8dcab8791cb81630c7fa40dd89`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://goldbook.iupac.org/terms/view/08676 (sha256:fa1bdd34ab2e93cce09dd22311bba6d9c1e3ee4221a285510eca2b7ce8e3e647)

Registered target identities:

- rotation-vibration-iupac-composite from IUPAC-GOLD-BOOK-ROTATION-VIBRATION-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite molecular geometry with generated rotational and vibrational recurrence classes, paired transitions and declared observation support.

Immutable identities. Source manifest `sha256:89d52a84711f33006402cfa91a8cd63bd29135b9e350468641221d220f5cd859`; derivation seal `sha256:214611a96dc7ce7ed35126a179a111b50b66db259a3ac7be747818b07e920aef`; engine receipt `sha256:c3ab121f56b37e99216f98fd7435428eee357b453bfda395ce375032e8de808f` at `receipts/engine/model_admitted/SFT-CHEM-SPEC-ROT-VIB-001-c3ab121f56b37e99.json`; empirical hash `sha256:6e5b5994425799fb11bed90270db138170775d0e10412ae47cd19fdfb829b2ae`; measurement receipt `sha256:9e6705743272cc7a980dbff088a552cc9fa032666221bce0201399ba9c83dc25`.

83. Complete analytical result and falsification record

Claim identity: `SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001`

Question and theorem. A complete analytical result retains the measurand, sample/analyte/matrix, method, conditions, reference chain, every raw and derived observation, uncertainty, failed and tampered controls, and an explicit falsification condition.

> *measurand-and-purpose-held__sample-analyte-matrix-record__method-and-condition-record__calibration-and-reference-chain__all-raw-and-derived-rows-held__uncertainty-and-resolution-held__failed-tampered-and-falsification-record*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-FOUNDATION-FORM-ENFORCEMENT-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-DISCRETE-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-GRAPH-NETWORK-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-MATH-LOGIC-PROOF-001`
- `SFT-INFO-SYMBOL-DISTINCTION-001`
- `SFT-INFO-CONSERVATION-LOSS-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-FIELD-ELECTRIC-POTENTIAL-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-PHYS-THERMO-EQUILIBRIUM-001`
- `SFT-CHEM-ELEM-ION-001`
- `SFT-CHEM-STOICH-CONSERVATION-001`

- `SFT-CHEM-BOND-COVALENT-001`
- `SFT-CHEM-BOND-IONIC-001`
- `SFT-CHEM-MOL-MOLECULE-001`
- `SFT-CHEM-MOL-INTERMOLECULAR-001`
- `SFT-CHEM-AB-ACID-BASE-001`
- `SFT-CHEM-AB-PROTON-TRANSFER-001`
- `SFT-CHEM-AB-LEWIS-001`
- `SFT-CHEM-AB-AMPHOTERIC-001`
- `SFT-CHEM-AB-BUFFER-001`
- `SFT-CHEM-ELECTRONEGATIVITY-001`
- `SFT-CHEM-BOND-POLARITY-001`
- `SFT-CHEM-REDOX-OXIDATION-STATE-001`
- `SFT-CHEM-REDOX-COUPLING-001`
- `SFT-CHEM-ELECTROCHEM-CELL-001`
- `SFT-CHEM-RXN-IDENTITY-001`
- `SFT-CHEM-RXN-MECHANISM-001`
- `SFT-CHEM-KIN-ACTIVATION-001`
- `SFT-CHEM-KIN-RATE-001`
- `SFT-CHEM-EQ-CHEMICAL-001`
- `SFT-CHEM-THERMO-REACTION-001`
- `SFT-CHEM-THERMO-DIRECTION-001`
- `SFT-CHEM-PHASE-CHEMICAL-001`
- `SFT-CHEM-MOL-GEOMETRY-001`
- `SFT-CHEM-MOL-ISOMER-001`
- `SFT-CHEM-NET-REACTION-001`
- `SFT-CHEM-MEAS-CHEMICAL-ENTITY-001`
- `SFT-CHEM-MEAS-UNCERTAINTY-001`
- `SFT-CHEM-MEAS-TRACEABILITY-001`
- `SFT-CHEM-BIOMOLECULAR-BOUNDARY-001`
- `SFT-CHEM-ANALYTICAL-SAMPLE-001`
- `SFT-CHEM-ANALYTICAL-CALIBRATION-001`
- `SFT-CHEM-ANALYTICAL-SELECTIVITY-001`

Generated grammar. Generate the literal product of the eight registered binary coordinates; decide every form by complete distinction preservation, exact recurrence/equivalence enumeration, minimality and absence of an extra rule.

Boundary: Every finite analytical execution with registered question, material, method, reference, observations, transformations, uncertainty and controls.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `measurand-and-purpose-held__sample-analyte-matrix-record__method-and-condition-record__calibration-and-reference-chain__all-raw-and-derived-rows-held__uncertainty-and-resolution-held__failed-tampered-and-falsification-record__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`question`	`measurand-and-purpose-held`	`unstated-measurand`	The result has no declared target.
`sample`	`sample-analyte-matrix-record`	`sample-identity-erased`	The result cannot be located.
`method`	`method-and-condition-record`	`method-erased`	The transformation cannot be reproduced.
`reference`	`calibration-and-reference-chain`	`reference-erased`	Traceability is lost.
`observation`	`all-raw-and-derived-rows-held`	`accepted-values-only`	Unfavorable evidence disappears.
`uncertainty`	`uncertainty-and-resolution-held`	`single-value-certainty`	Admissible alternatives are hidden.
`falsification`	`failed-tampered-and-falsification-record`	`success-only-report`	The claim cannot fail publicly.
`extra`	`no-extra-rule`	`free-reporting-omission`	Discretion can hide a contradiction.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:0cab94f55d8f3ec6788fc21b068becad07822404dccecdfaa4aaef265212cad1`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:1ee4fcf2da1f0174c3b1e3c980363e6bca1731c16e17bea678247da58b52565d`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:93a5466a246fc746f2db197b298846a920a9544ce5df6ee0b6ebd4bbb02b6726`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:739798d50cb8adcec6ce6d95e38de2516a30dd5354ebca6d8c46646e40ec048e`.

Independent reconstruction. Implementation hash `sha256:2ca15f1a1c1c4a464dbbe10d8860ab9a34743a8ac02dfd0ab04a032a5c999881`; independent certificate `sha256:a82ede737db1b63ea05e93848b8c61587572baf23629fcc9fd3127be01fc3249`; external-validation hash `sha256:7809c436f7c32024339ec7030d9ecbb6b3618c0c77f20180bbb081a5fe7c65e9`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-NIST-COMPLETE-ANALYTICAL-RECORD-COMPOSITE-2026

Comparison records:

- complete-analytical-record-authority-composite: predicted analytical-result-identifies-measurand-sample-and-method__traceability-and-uncertainty-retained__all-results-and-controls-preserved__falsification-condition-explicit; source-derived analytical-result-identifies-measurand-sample-and-method__traceability-and-uncertainty-retained__all-results-and-controls-preserved__falsification-condition-explicit; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The claim fails if authority evidence lacks result identity, traceability/uncertainty, retained controls or explicit falsification boundary, or if a tampered row is accepted.

Measurement receipt: `sha256:c7dfa8ab47c6d7b6328e0ce79bb0f55bb8c85329ff82744e552936a6ac22449a`. Isolation certificate: `sha256:3c73e9d2545d47c2d1839231134fcbffa1ae7fc9808941257c095e078217d549`. Custody certificate: `sha256:4423955a12bb76535a4949f952ddcbeae75aa34aeb77da18350c29ab5ecd4f69`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry and National Institute of Standards and Technology: https://goldbook.iupac.org/terms/view/M03796 (sha256:f27104514cc2af6ad31064e74244121a076fc3c795f3f636e4f7f8face60e9a1)

Registered target identities:

- complete-analytical-record-authority-composite from IUPAC-NIST-COMPLETE-ANALYTICAL-RECORD-COMPOSITE-2026

Meaning. Analytical chemistry keeps sample, calibration, selectivity, spectra, uncertainty, adverse controls and falsification records separate from derivational law. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no analytical glossary, calibration curve, spectral database, measured line, fitted profile, target source or V2 answer may select a candidate
- no numerical zero, negative, irrational, imaginary or floating proof quantity
- no free, fitted, learned or target-derived parameter
- no sample, component, response, transition, reference or unfavorable observation may be created, copied or silently erased
- absence is an Empty structural form rather than numerical zero
- external target content remains inaccessible until the prediction is sealed
- Every finite analytical execution with registered question, material, method, reference, observations, transformations, uncertainty and controls.

Immutable identities. Source manifest `sha256:9fd499aac36241625375469b74d0a4a57b88545b50a126df9bf9d2b1cf53c0f4`; derivation seal `sha256:19bd6f6461040e3eea0a099949c134066d4d2121ce5bc762ecd595c32e89e0b0`; engine receipt `sha256:e851e1cf0fa3a72278c1782e7858a862bbd9621d130034b3fdfeb33bf149ce50` at `receipts/engine/model_admitted/SFT-CHEM-ANALYTICAL-COMPLETE-RECORD-001-e851e1cf0fa3a722.json`; empirical hash `sha256:b6ace663ae95961dfe00c270b9ba6162f5bd4f03abe159ca77baf69c5ba5d0d6`; measurement receipt `sha256:c7dfa8ab47c6d7b6328e0ce79bb0f55bb8c85329ff82744e552936a6ac22449a`.

18. Gblock Smithium

The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction.

84. Generated g-block structural prediction

Claim identity: `SFT-CHEM-PRED-G-BLOCK-001`

Question and theorem. The admitted Fold orbit capacities ordered by increasing joint principal/orbit cover, with principal rank breaking a cover tie, generate noble closures 2, 10, 18, 36, 54, 86 and 118; fill 8s at 119-120; and open the 5g block at 121. The known prefix is externally checked and the g-block placement remains a sealed unobserved prediction.

> *The exact walk has capacities 2,6,10,14,18; noble closures 2,10,18,36,54,86,118; 8s occupations at 119-120; and first 5g occupation at 121.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001`
- `SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001`
- `SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001`
- `SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001`

Generated grammar. Generate the complete eight-axis product of positive orbit carrier, capacity, joint-cover order, predicted coordinate, provenance, target custody, evidence status and extra-rule forms.

Boundary: All finite subshell walks built from the admitted positive-rank capacities, complete valid principal/orbit cells, increasing joint cover and source-rank tie order before the atomic endpoint.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `positive-principal-and-orbit-ranks__admitted-Fold-orbit-capacity__increasing-joint-cover-then-principal-rank__first-5g-occupation-at-121__complete-fill-and-dependency-trace__official-table-opened-after-seal__known-prefix-validated-future-coordinate-unobserved__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`positive-principal-and-orbit-ranks`	`borrowed-orbital-notation`	A borrowed notation has no Fold support trace.
`capacity`	`admitted-Fold-orbit-capacity`	`imported-subshell-widths`	Imported widths would supply the answer.
`order`	`increasing-joint-cover-then-principal-rank`	`memorized-aufbau-list`	A memorized order is a target-bearing table.
`coordinate`	`first-5g-occupation-at-121`	`selected-future-coordinate`	A selected coordinate is not forced.
`provenance`	`complete-fill-and-dependency-trace`	`answer-only-prediction`	An answer-only prediction cannot be reproduced.
`target`	`official-table-opened-after-seal`	`official-table-readable-before-seal`	The known table could select the walk.
`status`	`known-prefix-validated-future-coordinate-unobserved`	`future-prediction-called-measured`	That conflates unobserved consequence with empirical confirmation.
`extension`	`no-extra-rule`	`free-exception-or-fit`	An exception can be tuned to a desired element.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:7029631ba2389d152c39777bc1b5fdc51d436ec265ac0641b9f3cef50e51fc70`.

- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:250a87885493ce97efb4e3db74f1eb38976bc61c1cb2b6271fce279423030e19`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8e07b41044d8c88aea1e8e8bc250609590e03205172694eac518f05cc0012c3a`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:7f0ea0abcc06647ebf2729218c6ad037a15810eced930bfe1b51e9914c4447ab`.

Independent reconstruction. Implementation hash `sha256:e323670a7c5c5d5a5c69b3138b71f1339902d495c29f0e7a9569905a7ad707d3`; independent certificate `sha256:eec03859c29d9208b967fb97d8f354e997dcea1b0f8026466fb738a677ff1bff`; external-validation hash `sha256:5e283a27959797e090c59a2f7f65d56cb204acc528b24f5f34d39b516cd13880`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- IUPAC-G-BLOCK-KNOWN-PREFIX: predicted known-noble-closures-2-10-18-36-54-86-118__g-block-121-standing-unobserved-prediction; source-derived known-noble-closures-2-10-18-36-54-86-118__g-block-121-standing-unobserved-prediction; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The generated known closure prefix differs from the official source, the 5g opening is not 121, a target enters before sealing, a future prediction is called observed, or a tampered row is accepted.

Measurement receipt: `sha256:f90a1aabda3ab41d3955dcf4b1ed2e8f96540de855f7ca982d0eaa623a49e97e`. Isolation certificate: `sha256:d11aee024e8a048521aa12fc28e2a03eca797223f52d918cd36ae5f7721eb501`. Custody certificate: `sha256:f38e70328919f4e9f6ac89c2ce639f974e1f8bd41c9d40d67c849e49c9de9b91`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- IUPAC-G-BLOCK-KNOWN-PREFIX from IUPAC-PERIODIC-TABLE-2022

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary. The official record checks only the known prefix; every coordinate beyond element 118 remains a standing unobserved prediction.

Explicit exclusions.

- no imported aufbau list or observed shell width
- no V1/V2 answer as a derivational input
- no semantic numerical zero, negative or nonexact proof value
- no claim that element 121 has been observed

Immutable identities. Source manifest `sha256:b40e7b8ec262ceec1fe073069a7392a7f4bd2d01a0d6c5a89e5335954d33cd18`; derivation seal `sha256:289d7818e0571477f6a481a2145684d578166225407f8e84db7d7f6306042113`; engine receipt `sha256:3d646a5267b619daa6cd184814fbcbb36a85718702f9c5a338c275d1ad488bb6` at `receipts/engine/model_admitted/SFT-CHEM-PRED-G-BLOCK-001-3d646a5267b619da.json`; empirical hash `sha256:72c25069eea465471b7de6aebaa729e84513f44faf0563dc492261ae794a39cc`; measurement receipt `sha256:f90a1aabda3ab41d3955dcf4b1ed2e8f96540de855f7ca982d0eaa623a49e97e`.

85. Smithium element-126 standing prediction

Claim identity: `SFT-CHEM-PRED-SMITHIUM-001`

Question and theorem. The independently sealed nuclear recurrence places the next proton/neutron double closure at 126/184. The generated electronic walk gives element 126 the active configuration 8s2 5g6, mass coordinate 310, eight valence carriers and structurally admissible positive oxidation counts +2 through +8.

> *Smithium is the standing prediction Z=126, N=184, A=310 with active configuration 8s2 5g6, valence count 8 and predicted positive oxidation counts +2 through +8.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001`
- `SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001`
- `SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001`
- `SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001`
- `SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001`
- `SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001`
- `SFT-CHEM-PRED-G-BLOCK-001`

Generated grammar. Generate the complete eight-axis product of positive orbit carrier, capacity, joint-cover order, nuclear/electronic coordinate, provenance, target custody, evidence status and extra-rule forms.

Boundary: All double-closure/electronic-fill consequences at the next generated proton closure after 82 using the admitted nuclear recurrence and complete subshell walk.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `positive-principal-and-orbit-ranks__admitted-Fold-orbit-capacity__sealed-joint-cover-fill-walk__next-double-closure-126-184-with-8s2-5g6__complete-fill-and-dependency-trace__official-table-opened-after-seal__known-prefix-validated-future-coordinate-unobserved__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`positive-principal-and-orbit-ranks`	`borrowed-orbital-notation`	A borrowed notation has no Fold support trace.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`capacity`	`admitted-Fold-orbit-capacity`	`imported-subshell-widths`	Imported widths would supply the answer.
`order`	`sealed-joint-cover-fill-walk`	`memorized-aufbau-list`	A memorized order is a target-bearing table.
`coordinate`	`next-double-closure-126-184-with-8s2-5g6`	`selected-future-coordinate`	A selected coordinate is not forced.
`provenance`	`complete-fill-and-dependency-trace`	`answer-only-prediction`	An answer-only prediction cannot be reproduced.
`target`	`official-table-opened-after-seal`	`official-table-readable-before-seal`	The known table could select the walk.
`status`	`known-prefix-validated-future-coordinate-unobserved`	`future-prediction-called-measured`	That conflates unobserved consequence with empirical confirmation.
`extension`	`no-extra-rule`	`free-exception-or-fit`	An exception can be tuned to a desired element.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2b9281d1a762cc7bdb1579504ba24551a5b444ea492bab0951a4287e69276fbc`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:fef001517e2f8109bea8ec2718b66087f655d84ce9032d5dcf4bc6f32e033f52`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3f0509e56d6e645f0f27d9d64972dc7be8ce260b8fc027edfd1ccecf7e3b2e6d`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:e3c97ef11cec09172621574fe62e1ceeea7995cc9ff6eaac97785d3456443933`.

Independent reconstruction. Implementation hash `sha256:dbf2dee9cba1c6c7ccb65a677bb3214e5f4aeefb986c74c37f57d5305bf0a956`; independent certificate `sha256:c46787ad2d1ee4ec941ade01601086040d8389a934fb9f2de55f817cf3467695`; external-validation hash `sha256:5e3d12781d9854b133abb666e2e465d4e7e5a92db4c46b727a63c168c3208db1`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- IUPAC-SMITHIUM-OBSERVATION-BOUNDARY: predicted official-table-through-118__smithium-126-standing-unobserved-prediction; source-derived official-table-through-118__smithium-126-standing-unobserved-prediction; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: A future verified element 126 lacks the sealed nuclear/electronic coordinates, the official known boundary differs, target data enter before seal, or a tampered row is accepted.

Measurement receipt: `sha256:1f0a538e836b85d88fc7001c9d0c7c42a932f8913d40f946b9f518fe5e4833c4`. Isolation certificate: `sha256:9cea68a9339c7df811842de45a9ed772999b45cd89c7a81d512aaa8152cd5d6c`. Custody certificate: `sha256:dcbdf5cb5cfe46f2ad04414d23477b0cc211f2b88b5d7281aa3a6b5466b77a3a`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- IUPAC-SMITHIUM-OBSERVATION-BOUNDARY from IUPAC-PERIODIC-TABLE-2022

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary. The official record checks only the known prefix; every coordinate beyond element 118 remains a standing unobserved prediction.

Explicit exclusions.

- no observed Smithium claim
- no imported island-of-stability coordinate
- no fitted nuclear coupling or chemical exception
- no semantic numerical zero, negative oxidation magnitude or nonexact proof value

Immutable identities. Source manifest `sha256:749eca244a42d91c3d13f9ba0c36d281c17744006c6493988dda57a677e160a3`; derivation seal `sha256:4139210c4b7c81bc194e47786feef97691d365eb343dbc3a1d76a28e78038c6a`; engine receipt `sha256:2f05e2fe825d74621cbd063b397f6f5b9132a37ebc38cf2772c55942111c37c7` at `receipts/engine/model_admitted/SFT-CHEM-PRED-SMITHIUM-001-2f05e2fe825d7462.json`; empirical hash `sha256:f3b32f3a2fd94d31e17ba472e47f7a769d3c79245b30d6394042692ec6247136`; measurement receipt `sha256:1f0a538e836b85d88fc7001c9d0c7c42a932f8913d40f946b9f518fe5e4833c4`.

86. Generated periodic endpoint standing prediction

Claim identity: `SFT-CHEM-PRED-PERIODIC-ENDPOINT-001`

Question and theorem. The independently sealed exact inverse fine-structure ratio and One binding ceiling admit positive whole atomic coordinate 137 and reject its successor 138. Chemistry therefore records 137 as the model's structural endpoint while the official observed table remains source-bounded through 118.

> *The model's exact structural periodic endpoint is positive atomic number 137; the IUPAC 2022 observed boundary remains 118, so 119-137 including the endpoint are unobserved standing predictions.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001`
- `SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001`
- `SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-ORDER-LATTICE-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-CHEM-ELEM-PERIODIC-RECURRENCE-001`
- `SFT-CHEM-ELEM-PERIODIC-BOUNDARY-001`

- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-CHEM-PRED-G-BLOCK-001`
- `SFT-CHEM-PRED-SMITHIUM-001`

Generated grammar. Generate the complete eight-axis product of positive atomic carrier, exact binding boundary, greatest-whole construction, predicted coordinate, provenance, target custody, evidence status and extra-rule forms.

Boundary: All Chemistry endpoint consequences of the admitted exact atomic-existence boundary and positive atomic-number order, separated from the source-dated observed table boundary.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `positive-principal-and-orbit-ranks__admitted-Fold-orbit-capacity__sealed-exact-binding-boundary__structural-endpoint-137-successor-138-excluded__complete-fill-and-dependency-trace__official-table-opened-after-seal__known-prefix-validated-future-coordinate-unobserved__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`positive-principal-and-orbit-ranks`	`borrowed-orbital-notation`	A borrowed notation has no Fold support trace.
`capacity`	`admitted-Fold-orbit-capacity`	`imported-subshell-widths`	Imported widths would supply the answer.
`order`	`sealed-exact-binding-boundary`	`memorized-aufbau-list`	A memorized order is a target-bearing table.
`coordinate`	`structural-endpoint-137-successor-138-excluded`	`selected-future-coordinate`	A selected coordinate is not forced.
`provenance`	`complete-fill-and-dependency-trace`	`answer-only-prediction`	An answer-only prediction cannot be reproduced.
`target`	`official-table-opened-after-seal`	`official-table-readable-before-seal`	The known table could select the walk.
`status`	`known-prefix-validated-future-coordinate-unobserved`	`future-prediction-called-measured`	That conflates unobserved consequence with empirical confirmation.
`extension`	`no-extra-rule`	`free-exception-or-fit`	An exception can be tuned to a desired element.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:9d93b8a29fb07ab3a7f8f5d4efe373c45864448cb4bc6b54ac7656b4f2ae0a7a`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:314897440a8fb21e21bbe6d1be41620f1cc2b1435261325748c8d401dbc6ab28`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4c7ebcab86120fc5c11fb4089e51b6de3225b9f60c844de96937fe2e3d2370a8`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:a9260d97a3d1af2d03cce0d6cf6037fcfb8039e34a7f91e66000ce1a4e28ad78`.

Independent reconstruction. Implementation hash `sha256:419962bc01b6a1a8c4399dddc62b15cb7b621808d204ef66530146b89b8d85a6`; independent certificate `sha256:ee90ca1e5a00142feb28aacce9965f6f8aab8f8d9cdcec7a1ec2673ebc898507`; external-validation hash `sha256:69a4057c8a7c7554c76a3aa71de7b535033a2fab3f9a498f9be0418d08b7908b`. The independent

program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IUPAC-PERIODIC-TABLE-2022

Comparison records:

- IUPAC-ENDPOINT-OBSERVATION-BOUNDARY: predicted official-table-through-118__endpoint-137-standing-unobserved-prediction; source-derived official-table-through-118__endpoint-137-standing-unobserved-prediction; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: A verified neutral element beyond 137 satisfies the registered One-ceiling boundary, the exact prerequisite fails replay, the observed/source boundary is conflated with the structural law, or a tampered row is accepted.

Measurement receipt: `sha256:2b27c0299574bb1deea7d18b7dc0cb4d1562a24275c206e6b9664ee650c73e36`. Isolation certificate: `sha256:b4318304b667dfd2262d19cebdb73044d925778b93851de42eec87fd5b5c398c`. Custody certificate: `sha256:51eea197c861b56a2f7d339a3e2e19d0df5c3d5a24d69df7f6248a5e0613358a`.

Registered source descriptions:

- International Union of Pure and Applied Chemistry: https://iupac.org/wp-content/uploads/2022/07/IUPAC_Periodic_Table-04May22_CRA.pdf (sha256:ef6ca2f6d46554f96e30ad3a60693d6630fe45ad81ce83cb14e508c6cbb7d3b3)

Registered target identities:

- IUPAC-ENDPOINT-OBSERVATION-BOUNDARY from IUPAC-PERIODIC-TABLE-2022

Meaning. The final subbranch applies separately admitted exact physical prerequisites to a complete positive-rank fill walk and distinguishes known-domain validation from unobserved prediction. This particular result is closed only at its exact registered grammar and evidence boundary. The official record checks only the known prefix; every coordinate beyond element 118 remains a standing unobserved prediction.

Explicit exclusions.

- no observed absence promoted to impossibility
- no bounded endpoint scan
- no imported critical-charge value
- no claim that element 137 has been observed

Immutable identities. Source manifest `sha256:79311bd68d7c9ee7bc096c1c47696c737df00176d6cabc10b49b88f050d9ea66`; derivation seal `sha256:cce4325ca809e604d40b0f29f51767e1607e0a4568b84ec8809f8c32579ff3e5`; engine receipt `sha256:ab9af61c4eccd228e24a01a2fe8b5967116818bb285f7ddfc168e37e4db4b267` at `receipts/engine/model_admitted/SFT-CHEM-PRED-PERIODIC-ENDPOINT-001-ab9af61c4eccd228.json`; empirical hash `sha256:9ab75a13a17bf7d8b1f1e7b29c6b21840fea3a048a8c73247f3f135d72a4c8c4`; measurement receipt `sha256:2b27c0299574bb1deea7d18b7dc0cb4d1562a24275c206e6b9664ee650c73e36`.

19. Exact upstream prerequisite and validation appendix

These seven claims are not hidden assumptions. They are separately engine-admitted V3 laws with their own complete evidence.

87. Terminal exact inverse fine-structure Fold ratio

Claim identity: `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`

Question and theorem. The binary predecessor and generator-three recurrence force cover depths five and seven. Their typed tower, boundary, complete three-direction cover, returning One and finite one-direction-at-a-time promotion ladder force the exact terminal ratio 503846395469/3676744786 before measurement.

> *The terminal inverse fine-structure Fold ratio is exactly 503846395469/3676744786; its leading rung is exactly 34259/250.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-FOUNDATION-COUNT-001`
- `SFT-FOUNDATION-PART-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar. Generate the complete typed product of cover, depth, tower, boundary, return, join, promotion, termination, target custody and extension forms.

Boundary: All type-correct assemblies that use each forced binary/generator cover block in its generated role, admit only the One as recurrence correction, and promote each of the three interchangeable cover directions at most once.

The engine regenerated `1024` candidates and `1024` one-for-one decisions. Exactly one survived: `least-complete-binary-covers__cover-generator-cross-lock__binary-at-up-depth__generator-at-boundary-rank__one-return-over-complete-cover__tower-plus-dilated-boundary__one-remaining-direction-per-rung__all-three-directions-promoted__measurement-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`cover`	`least-complete-binary-covers`	`asserted-depths`	Asserted depths have no support census.
`depth`	`cover-generator-cross-lock`	`single-route-depths`	One route cannot expose a changed carrier.
`tower`	`binary-at-up-depth`	`free-leading-whole`	A free leading whole is a parameter.
`boundary`	`generator-at-boundary-rank`	`free-boundary-factor`	A free coefficient is not structural.
`return`	`one-return-over-complete-cover`	`fitted-small-correction`	A fitted correction reads the target.
`join`	`tower-plus-dilated-boundary`	`untyped-arithmetic-permutation`	Permuting typed carriers erases their roles.
`promotion`	`one-remaining-direction-per-rung`	`selected-refinement-series`	A selected series adds unfixed terms.
`termination`	`all-three-directions-promoted`	`infinite-or-truncated-by-fit`	A fit-selected truncation is not closure.
`target`	`measurement-inaccessible-until-seal`	`measured-alpha-visible`	That makes measurement an answer-bearing premise.
`extension`	`no-extra-rule`	`extra-scale-rule`	An extra scale is a free parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:ca5b4e49ab3bedece92f0ec11667e4305baafe8624283410439868e1b79e151c`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:0d516338a2c561300eae04d43e460a74cae268ab52d41ce1ace635214b048ab5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:937d252d67fbb9ad1f3a8bcbd5000226d4d7bde3b511429cbe0ee917e4986843`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:debfd802d48d0d9c9b5e1df0a6bdbf950f19bd012c0bdf52cfeec2505494619b`.

Independent reconstruction. Implementation hash `sha256:97d0cdb7077804e63fb1bab56244287f15fbc1ee6fdeece167b59a309604117f`; independent certificate `sha256:ec1a7c48b4269d74564b029d286350d600f1feca69383bf655ad7415714f71f4`; external-validation hash `sha256:d2a29ed3e7a828cd1adc21abfff803d43cce478e12935dc83818e0a79af26a67`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest `sha256:624239a4a1a47a06acec655b1a55db66675f5d32fe9f67c6a4305d52eb56b99a`; derivation seal `sha256:9dfbee307fc2a9b96180abd41ec62bdf2a131a6e7aa702951e7acbf0ad0459b1`; engine receipt `sha256:4999a85b5862afebcf24604c2191c0ce8b96aa57f988b3fab83b469c3506a7c9` at `receipts/engine/model_admitted/SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001-4999a85b5862afeb.json`; empirical hash `None`; measurement receipt `None`.

88. Atomic orbit-cell capacity from three-space boundary orientation

Claim identity: `SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001`

Question and theorem. At positive orbit rank r, three-space supplies one central orientation and the rank-two boundary supplies one held pair for every predecessor step. Two exclusion-distinct Fold labels occupy each orientation, forcing capacity 2(1+2(r-1)) and the sequence 2, 6, 10, 14, 18 without observation.

> *For every positive orbit rank r, capacity is exactly 2(1+2(r-1)); ranks one through five have capacities 2, 6, 10, 14 and 18.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-SPACE-BOUNDARY-RANK-TWO-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-COMBINATORICS-001`

Generated grammar. Generate the complete product of carrier, central class, successor extension, boundary multiplicity, Fold label, exclusion, capacity, generality, target custody and extension forms.

Boundary: All orbit-cell capacity laws generated from one central orientation, positive orbit succession, the complete rank-two boundary, two held Fold labels and exclusion.

The engine regenerated `1024` candidates and `1024` one-for-one decisions. Exactly one survived: `held-orientation-cell__one-central-orientation__one-boundary-orientation-pair__forced-rank-two-pair__both-Fold-labels__one-of-each-label-per-cell__labels-times-complete-orientations__constant-boundary-pair-successor__widths-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`held-orientation-cell`	`unlabelled-cell-count`	An unlabelled count loses orientation identity.
`central`	`one-central-orientation`	`empty-numerical-origin`	Numerical zero is not an SFT carrier.
`successor`	`one-boundary-orientation-pair`	`free-rank-increment`	A free increment is a parameter.
`boundary`	`forced-rank-two-pair`	`single-or-three-direction-addition`	One omits a boundary side and three re-adds the held normal.
`label`	`both-Fold-labels`	`label-erased-or-selected`	Erasure or selection violates complete Fold support.
`exclusion`	`one-of-each-label-per-cell`	`duplicate-cell-label`	Duplication destroys distinguishability.
`capacity`	`labels-times-complete-orientations`	`linear-or-doubling-without-boundary`	Those former survivors ignore the newly admitted boundary discriminator.
`generality`	`constant-boundary-pair-successor`	`finite-width-list`	A list has no next-rank proof.
`target`	`widths-inaccessible-until-seal`	`width-list-visible`	That reverses derivation and test.
`extension`	`no-extra-rule`	`extra-degeneracy-rule`	An added degeneracy is a parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:1ed87436e21f59d043967ca432026eaebd71e5e1f8be1907692eab24d946c6d5`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:625315de8974967651ba198f65bcbd903e36bfac7a0cc349d449aed6298445b7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:4bf7aa1fcc6d37bda661b6b898af0d9b3a48ba75d0f12afa04f8db6b363f4d61`.

- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:1c45ec73b2de2f2c722af6fcddecc09f844cab496f9bb77bdbdda1f0c85f80ab`.

Independent reconstruction. Implementation hash `sha256:86c0b909d7fb32a35993571e1c3646d129604a21264f1da55aabad2b205c428e`; independent certificate `sha256:dd12a5fc73eacfd47d2eeaaf16cc2925d756b0b88926041d958ecc2867474ec0`; external-validation hash `sha256:4ac2066f98eb5a6803d8f83eb315180dfaccda40a4fdf6575c2208668792f67a`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest `sha256:7b624a62e7d39d55c7a235163b8d0f93f6ffeebf172a7b94bf8fd4790be796f8`; derivation seal `sha256:6cd7fec86b24f0f4624a70894b507f836d4358eb2229515b61498407d958f993`; engine receipt `sha256:2b8928b0ddf5b87a7bf56813a5f430ba4a99f715b1cfc1dd1db98106216e3d70` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001-2b8928b0ddf5b87a.json`; empirical hash `None`; measurement receipt `None`.

89. Exact colour-sector Fold coupling

Claim identity: `SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001`

Question and theorem. The complete two-label Fold distinction transported over the generator-three carrier forces the exact positive coupling ratio two-thirds, with no measured or fitted interaction strength.

> *The exact generator-three colour-sector Fold coupling is 2/3.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-STRUCT-GENERATOR-THREE-001`
- `SFT-FOUNDATION-FOLD-001`
- `SFT-FOUNDATION-FOLD-ASSEMBLY-001`
- `SFT-PHYS-FIELD-INTERACTION-CLASSES-001`
- `SFT-PHYS-MATTER-COMPOSITE-HADRONS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar. Generate every product of carrier, numerator, denominator, orientation, normalization, target custody and extension forms.

Boundary: All exact normalized ratios formed by transporting the complete Fold-label support across one generator-three interaction carrier.

The engine regenerated `128` candidates and `128` one-for-one decisions. Exactly one survived: `generator-three-carrier__both-Fold-labels__complete-generator-support__labels-held-through-carrier__two-over-three__closures-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`generator-three-carrier`	`free-colour-count`	A free carrier count is a parameter.
`numerator`	`both-Fold-labels`	`selected-single-label`	One label omits half the Fold distinction.
`denominator`	`complete-generator-support`	`external-normalization`	External normalization is fitted.
`orientation`	`labels-held-through-carrier`	`labels-collapsed`	Collapse loses the interaction distinction.
`normalization`	`two-over-three`	`reciprocal-or-unscaled-count`	Those forms reverse roles or omit normalization.
`target`	`closures-inaccessible-until-seal`	`magic-list-visible`	A closure list would fit the coupling.
`extension`	`no-extra-rule`	`extra-coupling-rule`	An added strength is a free parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:e4e7687a9a1d832e556174d015e0aa64f11f64f92e06734a6051d3b987b676db`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:e2edad0e6d45799bbf3bb96bb8c4e89f8be5a715fc0d921e462d57b1d7d2e8f0`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:12e6496b82b65e70e5458cdc2c1f49daecc9625414c9a58da9cda056fea5e5c6`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:5947e11fa098dddaf9eb6ce47ae72cd6b0492d99a75eeb6452299129b69e0996`.

Independent reconstruction. Implementation hash `sha256:babedcbfa7852074ccccf0885740154b359515c4b90ca34e584e8544bc59f31b`; independent certificate `sha256:8a8b86ffa63d4555d21aaab69918f4d60af798ad0b31d7e9c2014eb38779646e`; external-validation hash `sha256:c9a0a0d4ee3a15a784312da4b2c7bff456b6a17ddc1b6073c1dc58244065e846`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest `sha256:55e3c26bda6cce369b8f47cf4b459b445cabed93342517e8e9ee93424b0ca57f`; derivation seal `sha256:4893061724f8e857a5efe97692521d1df83c013f9723fe61d4804c3bae4baa77`; engine receipt `sha256:d38dc32f0926ae5f06e432f6bd7546d8680a8ea648c211dfbe965fc44e8179e5` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001-d38dc32f0926ae5f.json`; empirical hash `None`; measurement receipt `None`.

90. Complete Fold nuclear-closure sequence

Claim identity: `SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001`

Question and theorem. Complete weak compositions across three spatial directions with two exclusion labels force the base filled-shell counts. The independently sealed two-thirds colour coupling reaches its first whole spin-orbit gap at angular count three and forces the reordered closure sequence 2, 8, 20, 28, 50, 82, 126, 184 with a depth-independent successor law.

> *The depth-independent nuclear closure law gives the exact prefix 2, 8, 20, 28, 50, 82, 126, 184; its next proton closure after 82 is 126 and the next closure after 126 is 184.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-ATOMIC-CELL-ORBIT-CAPACITY-001`
- `SFT-PHYS-NUCLEAR-COLOUR-COUPLING-001`
- `SFT-PHYS-SPACE-DIMENSION-THREE-001`
- `SFT-PHYS-QUANTUM-SPIN-001`
- `SFT-PHYS-QUANTUM-EXCLUSION-001`
- `SFT-PHYS-NUCLEAR-BINDING-001`
- `SFT-PHYS-NUCLEAR-LEVELS-001`
- `SFT-PHYS-NUCLEAR-REACTIONS-001`
- `SFT-MATH-COMBINATORICS-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar. Generate the complete product of spatial composition, label multiplicity, base closure, coupling source, shift, threshold, reordering, recurrence, target custody and extension forms.

Boundary: All filled-shell closure schedules built from complete weak compositions over forced three-space, both exclusion labels and the first whole-gap action of the independently forced colour coupling.

The engine regenerated `1024` candidates and `1024` one-for-one decisions. Exactly one survived: `complete-three-direction-weak-compositions__two-exclusion-distinct-labels__positive-rank-composition-form__sealed-two-thirds-colour-coupling__top-angular-half-label-pair__least-positive-whole-gap__complete-top-orbit-capacity__piecewise-positive-successor-law__sequence-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`composition`	`complete-three-direction-weak-compositions`	`selected-shell-degeneracy`	A selected degeneracy imports a model.
`labels`	`two-exclusion-distinct-labels`	`single-or-duplicated-label`	Omission or duplication violates Fold support and exclusion.
`base`	`positive-rank-composition-form`	`closure-list`	A list has no generative proof.
`coupling`	`sealed-two-thirds-colour-coupling`	`fitted-spin-orbit-strength`	A fitted strength reads the closures.
`shift`	`top-angular-half-label-pair`	`free-shell-shift`	A free shift is a parameter.
`threshold`	`least-positive-whole-gap`	`selected-rank-threshold`	A selected threshold fits the sequence.
`reorder`	`complete-top-orbit-capacity`	`arbitrary-intruder-count`	An arbitrary intruder count is fitted.
`recurrence`	`piecewise-positive-successor-law`	`finite-eight-value-table`	A table does not close later ranks.
`target`	`sequence-inaccessible-until-seal`	`magic-numbers-visible`	That is target leakage.
`extension`	`no-extra-rule`	`extra-nuclear-rule`	An added selector is a parameter.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:5c771cf5592a53cc672843d7cdadcfbe7e340db992858decb1ab49eb0d695e8b`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:6e2dc9948609f6a9bcffed8b9e06971fa9e84b851c60221af6c0726a7a4bf601`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:8069e2016e09a7983c9c51f36313fbec85fe529516e27d9f31877bb68d776e78`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:9df3facdacb8b6591dcebf69efd2b75b8b300056734782cf444b7e08cdb38fc6`.

Independent reconstruction. Implementation hash `sha256:0c39e11b369cf6d388ac93d2942d49a037aa20f3cb2550601df04805eee0dcaa`; independent certificate `sha256:988e6fd342d2c09cabea9a075e383ac3d673f1e74279b820e4d79c2bbb3a37ea`; external-validation hash `sha256:4a0b588d1c3b8b4ea798c27237d953cdee1dcb0918ef3b436678c0aa798fccfa`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest `sha256:239be83e36611d89d86421f4dc8e517f20288187660251a1313146246bfe5133`; derivation seal `sha256:f063dc762cb725c66340c946953cd0086648754569448 11ad902315d6b79b69d`; engine receipt `sha256:35f7fe50e217d2a61383f4169c314f1d4d5b4671a1def5baa011d26a1d89af9f` at `receipts/engine/model_admitted/SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001-35f7fe50e217d2a6.json`; empirical hash `None`; measurement receipt `None`.

91. Exact atomic existence boundary

Claim identity: `SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001`

Question and theorem. A neutral atomic coordinate remains structurally admissible only while its positive whole charge does not exceed the exact inverse fine-structure Fold ratio. The unique greatest whole below that sealed ratio is 137, with 137 below and its successor 138 above the boundary.

> *The unique greatest positive whole atomic coordinate under the exact Fold binding ceiling is 137; 137 <= 503846395469/3676744786 < 138.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-PHYS-QUANTUM-DISCRETE-SPECTRA-001`
- `SFT-PHYS-FIELD-ELECTRIC-DISTINCTION-001`
- `SFT-PHYS-MATTER-CONSERVED-LABELS-001`
- `SFT-CHEM-ELEM-ELEMENT-001`
- `SFT-CHEM-ELEM-ATOMIC-NUMBER-001`
- `SFT-CHEM-ELEM-PERIODIC-ORDER-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`

Generated grammar. Generate the complete product of charge carrier, coupling, ceiling, endpoint construction, lower witness, upper witness, generality, target custody and extension forms.

Boundary: All greatest-positive-whole atomic-coordinate conclusions under the One binding ceiling and the independently sealed exact inverse fine-structure ratio.

The engine regenerated `512` candidates and `512` one-for-one decisions. Exactly one survived: `positive-held-charge-count__charge-over-sealed-inverse-ratio__complete-One-ceiling__greatest-whole-division-certificate__exact-lower-inequality__successor-above-plus-order__depth-independent-greatest-whole-law__observations-inaccessible-until-seal__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`charge`	`positive-held-charge-count`	`unbound-number`	An unbound number has no element identity.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
coupling	charge-over-sealed-inverse-ratio	fitted-binding-equation	A fitted equation imports the endpoint.
ceiling	complete-One-ceiling	external-critical-charge	An external critical charge is target input.
endpoint	greatest-whole-division-certificate	bounded-neighborhood-walk	A chosen search interval does not close all successors.
lower	exact-lower-inequality	endpoint-label-only	A label has no inequality witness.
upper	successor-above-plus-order	single-next-value-check-only	One check without order closure does not exclude later values.
generality	depth-independent-greatest-whole-law	finite-coordinate-census	A finite census leaves later coordinates open.
target	observations-inaccessible-until-seal	observed-elements-visible	That would turn an observation boundary into a law.
extension	no-extra-rule	extra-size-parameter	A size correction is a different, parameterized model.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the form missing the first structural preservation; observed reject the form missing the first structural preservation; receipt `sha256:4160fd712b17226fbf3a8cc047e5446f5375e7bd3fa14fd1c64641c517a5e38e`.
- `tampered_source`: passed; expected reject a changed source identity; observed reject a changed source identity; receipt `sha256:eba05c35e0e176c852c05a3f09315b28f7bd0cbe56b2511e8b5f44780bd2472a`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:fe957e7c888b14fab4f760796244349be17a05c08bb8a440bb09417b28d8a928`.
- `boundary`: passed; expected reject forbidden values, target feedback and extra laws; observed reject forbidden values, target feedback and extra laws; receipt `sha256:a37c7574414b2ddd6b071742183623ac6758ef060b905ba5bbf38a6617730b62`.

Independent reconstruction. Implementation hash `sha256:55d5c09b7d7696f1d73843966a64fb278a666537289fd9472b7c3e557a83e55c`; independent certificate `sha256:d4b358d1dc307a24df8800cec04b9db1f575e9e52f920356fabfbf7739b7e2d2`; external-validation hash `sha256:eb370cb694ac8bb024e4020fbed03b0cfbec1a28e9e0af2d2abf7141d17c999c`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. This is a sealed formal prerequisite. Its empirical status is recorded only by a separate downstream validation claim.

Registered source descriptions:

- None.

Registered target identities:

- None.

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no V1/V2 proof artifact, measured constant, observed shell list or intended Chemistry answer as a premise
- no semantic numerical zero, negative, irrational, imaginary or floating proof value
- no fitted coupling, selected candidate neighborhood or target-selected form
- no unrecorded external rule or measurement-to-derivation flow

Immutable identities. Source manifest `sha256:97c4ce7770ec307c488fa3e908f713ef94eb5057771ebba5848675592d99d879`; derivation seal `sha256:6baa991e8391efbd50b0d7235617136cbf6b14f2df03464d9e05d075d2bd3476`; engine receipt `sha256:5cedeb1cb99ec83a668db0c80839b248c93e83bdeeafbd067fa379fb8e45eb17` at `receipts/engine/model_admitted/SFT-PHYS-ATOMIC-EXISTENCE-BOUNDARY-001-5cedeb1cb99ec83a.json`; empirical hash `None`; measurement receipt `None`.

92. Post-seal CODATA check of the terminal inverse fine-structure ratio

Claim identity: `SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001`

Question and theorem. After V3 independently seals the exact terminal inverse fine-structure ratio, the complete NIST CODATA 2022 central value and stated standard uncertainty are converted to exact rational interval endpoints. The sealed ratio is checked for interval membership without fitting or rewriting it.

> *The exact terminal ratio 503846395469/3676744786 lies inside the complete 2022 CODATA interval 137.035999177 +/- 0.000000021.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-CONSTANT-INVERSE-FINE-STRUCTURE-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-PHYS-MEAS-UNCERTAINTY-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`
- `SFT-MATH-ORDER-LATTICE-001`

Generated grammar. Generate the complete eight-axis post-seal relation, provenance, custody, row, exact-interval and no-extra-rule product.

Boundary: All exact post-seal comparisons between the immutable terminal ratio and the one complete registered CODATA inverse-alpha row including its uncertainty.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-fold-carrier__sealed-terminal-ratio-versus-complete-codata-interval__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-terminal-ratio-versus-complete-codata-interval`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:2460ed9e9112ca2db7f5fec2ba9676993debc0b85cb8d1040342b8a51dd34e68`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:20a184a1749db08e208fa09d5609987741171219db6cfdb7e3c1db52cbf51806`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:3921cc46ead2ad22ee5ae1bb3ad7f8b9184ac08981590c2f580ab371eda892a3`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:9d42267b960bf9dfdd82343e855943f7fd5e75c3f418041c4cee03165d0ad02f`.

Independent reconstruction. Implementation hash `sha256:31aaffa05397abd35a390640284e0b7e2c2bd7c1793d599a60aadc3916ff1695`; independent certificate `sha256:4b54876e16a0163c485b0fe3c7b79c5c12fd3c53c49dd90bdd7fa11ed7248116`; external-validation hash `sha256:bfb80b39991e704b6d7b7fc4a0a419ef35d89db78ec3db6f95bef05e54606135`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- NIST-CODATA-2022-ALL-CONSTANTS

Comparison records:

- NIST-CODATA-2022-INVERSE-ALPHA: predicted sealed-terminal-inverse-alpha-inside-complete-codata-2022-interval; observed sealed-terminal-inverse-alpha-inside-complete-codata-2022-interval; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: The sealed ratio lies outside the complete reported interval, the source row/hash changes, the uncertainty is omitted, or a tampered comparison is accepted.

Measurement receipt: `sha256:060cad30845d985ab813c4e4114eabee8d690c4f62d18860ab215d35061db1d7`. Isolation certificate: `sha256:b3e8f58f7e96427308176390df6c109d246ad97efe3734bcb6250ad59ec0cdb3`. Custody certificate: `sha256:b5389bb0d662254fceddf1684eadabe1f77b1f3354e7f690b23622eaa5d61d54`.

Registered source descriptions:

- National Institute of Standards and Technology / CODATA Task Group on Fundamental Constants: https://physics.nist.gov/cuu/Constants/Table/allascii.txt (sha256:77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67)

Registered target identities:

- NIST-CODATA-2022-INVERSE-ALPHA from NIST-CODATA-2022-ALL-CONSTANTS

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no CODATA value accessible to the formal derivation
- no fitted digit or best-match selection
- no floating-point comparison
- no omitted uncertainty or unfavorable control

Immutable identities. Source manifest `sha256:fb0058c462a90eec0073158ae31c88e7bad16311f630152e83ed5ae277d5282a`; derivation seal `sha256:5f16c61d3619dddb76f43b0199161bf2920453b1ecbfa49e2e24a858fcf70d95`; engine receipt `sha256:cfe0c0b068864707942c03909ef4dd16660476c808fe5e9567970c60d8640bbd` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-INVERSE-FINE-STRUCTURE-001-cfe0c0b068864707.json`; empirical hash `sha256:5eed8e7391a59d9ffdbf11d158d63dc889357300a19a27203f3719606f5e6877`; measurement receipt `sha256:060cad30845d985ab813c4e4114eabee8d690c4f62d18860ab215d35061db1d7`.

93. Post-seal IAEA check of the nuclear-closure sequence

Claim identity: `SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001`

Question and theorem. After V3 independently seals the positive-rank nuclear recurrence, the complete first eight outputs are compared in order with the registered IAEA Nuclear Data Section magic-number record.

> *The sealed V3 closure prefix (2, 8, 20, 28, 50, 82, 126, 184) exactly matches the complete registered IAEA sequence.*

Rooted dependency chain. The registration names the single premise-free root theorem and accepts only these already admitted dependencies:

- `SFT-PHYS-NUCLEAR-CLOSURE-SEQUENCE-001`
- `SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001`
- `SFT-PHYS-MEAS-CAPABILITY-PREDICTION-001`
- `SFT-PHYS-MEAS-TARGET-CUSTODY-001`
- `SFT-PHYS-MEAS-HOSTILE-PACKAGE-001`
- `SFT-PHYS-MEAS-VALUE-RECORD-001`
- `SFT-MATH-EXACT-ARITHMETIC-001`

Generated grammar. Generate the complete eight-axis post-seal ordered-sequence, provenance, custody, row-retention and no-extra-rule product.

Boundary: All exact order-preserving post-seal comparisons between the immutable first-eight closure output and the complete registered IAEA positive magic-number sequence.

The engine regenerated `256` candidates and `256` one-for-one decisions. Exactly one survived: `complete-fold-carrier__sealed-closure-prefix-versus-complete-iaea-sequence__source-bound-proof-trace__capability-closed-prediction__separate-measurement-record__complete-registered-rows__one-successor-closure__no-extra-rule`. Closure is `depth_independent`; minimality and named-shape uniqueness are recorded in the closure certificate.

Coordinate-by-coordinate uniqueness. Each row compares the survivor with every otherwise-surviving alternative on that coordinate.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`carrier`	`complete-fold-carrier`	`answer-only-scalar`	An answer-only scalar erases the generated physical carrier.
`relation`	`sealed-closure-prefix-versus-complete-iaea-sequence`	`imported-or-fitted-relation`	An imported or fitted relation lets consensus or target data select the law.

Axis	Forced form	Eliminated alternatives	Exact rejection basis
`provenance`	`source-bound-proof-trace`	`unbound-provenance`	Unbound provenance cannot demonstrate which premises produced the result.
`prediction`	`capability-closed-prediction`	`target-readable-prediction`	A target-readable predictor can copy or optimize against the measurement.
`record`	`separate-measurement-record`	`proof-measurement-conflation`	Treating an external decimal record as a proof scalar violates the measured-value boundary.
`rows`	`complete-registered-rows`	`selected-favourable-rows`	Selecting favorable rows cannot falsify the relation.
`generality`	`one-successor-closure`	`finite-answer-lookup`	An answer lookup has no successor certificate.
`extension`	`no-extra-rule`	`free-extra-rule`	A free constant, scale or exception can select the measured answer.

Adverse controls. All mandatory controls passed:

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt `sha256:6f99243f846465531b8e8d04eafd231fa57515365c2ce894180820a930c4f6be`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt `sha256:856dc18ebff37c0d3ec1fe6a114c55e16f6bd261778ca0abd9f5830bf7c82bb8`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt `sha256:77a9bb7213b8ff916417351bf0dada8cf20622b8fef94e29c8a977e5c2c2eb9e`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt `sha256:745f64af58a7835963b0d86ea1dd07931f1d68b4035bc3d091798fd4b90ac198`.

Independent reconstruction. Implementation hash `sha256:305b02c334ed9f2efcdc389e5a3601348dabb2f28d1b4b9e9a0c91f36ba7716b`; independent certificate `sha256:0c4548de901d5c3e1ab8a7be0eb1ce51e3801104365a796d733b907dd9ca9642`; external-validation hash `sha256:7098173bec13b8fbba7b194c44cf71c486b50c1411683a505d48c61764a57da5`. The independent program regenerates the declared product and sole survivor without importing the derivation implementation.

Post-seal external check. Target content was opened only after the matching prediction seal: `True`. All registered rows were preserved: `True`. The capability-isolation and custody certificates passed. External source identities:

- IAEA-INDC-NDS-0452-MAGIC-NUMBERS

Comparison records:

- IAEA-INDC-NDS-0452-MAGIC-SEQUENCE: predicted sealed-eight-closure-prefix-exactly-matches-iaea-reported-sequence; observed sealed-eight-closure-prefix-exactly-matches-iaea-reported-sequence; exact match True
- deliberately tampered unfavorable control rejected

Falsification boundary: Any sealed/source coordinate differs, either ordered sequence is incomplete, the source identity/hash changes, or a tampered comparison is accepted.

Measurement receipt: `sha256:aae2b19de5952d1488ac8ec859b09819c13784069993670087O9aa7b423a7f08`. Isolation certificate: `sha256:264c93c4ab7661fdc40750e550143a79427eeca64dc3cf4508db4d52fba2e913`. Custody certificate: `sha256:5ca4ac8aaff5471643b0ccb92f484218242a81626c97d4d9590494215809e94d`.

Registered source descriptions:

- International Atomic Energy Agency Nuclear Data Section: https://www-nds.iaea.org/publications/indc/indc-nds-0452-part1.pdf (sha256:ce642d62698b97fa509b8101a5661e578d6524f8705c19e38f7426efe7a0f6a6)

Registered target identities:

- IAEA-INDC-NDS-0452-MAGIC-SEQUENCE from IAEA-INDC-NDS-0452-MAGIC-NUMBERS

Meaning. This upstream claim supplies an exact prerequisite used by the final Chemistry predictions. This particular result is closed only at its exact registered grammar and evidence boundary.

Explicit exclusions.

- no magic-number value accessible to the formal derivation
- no selected favorable subsequence
- no fitted coupling or threshold
- no omitted source or tampered control

Immutable identities. Source manifest `sha256:1d815786712147e935c6d17ddb736acd9863124d56a42adab83e860b22b0d131`; derivation seal `sha256:56679e7ef35dd6a054b48cb85d9a365e04483d1876703a9d2e8ef6e52858f811`; engine receipt `sha256:e55e9e67e3e4b389dd46c7a5cd6260674d11a9c832d55835d92100936f365bbd` at `receipts/engine/model_admitted/SFT-PHYS-VALIDATION-NUCLEAR-CLOSURES-001-e55e9e67e3e4b389.json`; empirical hash `sha256:bfb9984c3d44e47579ac8b6b710cc22f04875b3c99fbfc07a179ccb8905a036a`; measurement receipt `sha256:aae2b19de5952d1488ac8ec859b09819c1378406999367008709aa7b423a7f08`.

20. Integrated result and scientific meaning

The branch unifies chemical identity, exact composition, bonding, reactions, thermochemistry, interfaces, stereochemistry, polymers, analysis and periodic organization under one provenance-preserving state-transition framework. Chemical laws do not begin with a periodic-table diagram or a fitted Hamiltonian. They begin with generated distinguishability, held labels, complete joint support, exclusion, recurrence, conservation and observation classes. Conventional names appear only at the correspondence boundary.

The terminal sequence has two independent locks. First, three-space and its rank-two boundary force orbit orientation growth; the two Fold labels and exclusion force capacities 2, 6, 10, 14 and 18. Second, complete three-direction compositions and the separately sealed 2/3 colour coupling force the nuclear closures. The joint-cover fill walk then reproduces the known noble closures and extends to the 5g prediction. The terminal inverse-alpha object independently supplies the atomic existence ceiling. Agreement of the upstream known support does not constitute observation of the future coordinates.

21. Reproducibility route

The one-command repository validator rechecks schemas, source identities, engine unit/E2E coverage and every admitted receipt in order. The Chemistry publication verifier additionally requires 86/86 claim coverage, all evidence files, one survivor per claim, every control, every empirical custody record, paper inclusion of every receipt, exact evidence-map identities and a visually rendered PDF. This paper build is a read-only projection and never changes a derivation.

22. Limitations and falsifiers

- IUPAC correspondence through 118 does not confirm element 121, Smithium or element 137.
- The structural endpoint is the registered point-carrier One-ceiling result; other physical models cannot be silently substituted into its claim boundary.
- The predicted Smithium oxidation counts are structural model consequences awaiting direct chemical observation.
- Any official target mismatch, missing row, source-hash change, replay mismatch or accepted tampered control fails the corresponding claim.
- Open V1/V2 lineage groups outside Chemistry remain mandatory V3 work; this paper does not paper them over.

23. Conclusion

At its frozen boundary, Chemistry is fully admitted in V3: 86 required claims, every candidate census, one survivor each, complete controls, independent reconstruction and post-seal evidence where empirical. The paper preserves the difference between exact derivation, known-domain correspondence and unobserved prediction. Its authority is the open evidence chain from the premise-free root theorem to immutable receipts, not credential, consensus or opaque prediction.

References and authoritative records

- International Union of Pure and Applied Chemistry, *Compendium of Chemical Terminology (Gold Book)*, https://goldbook.iupac.org/.
- International Union of Pure and Applied Chemistry, *Periodic Table of Elements*, release dated 4 May 2022, https://iupac.org/what-we-do/periodic-table-of-elements/.
- National Institute of Standards and Technology / CODATA, *2022 Recommended Values of the Fundamental Physical Constants*, https://physics.nist.gov/constants.
- International Atomic Energy Agency Nuclear Data Section, INDC(NDS)-0452 Part 1, nuclear shell-structure record.
- Smith, Maria, *There Is No Nothing*, Ernos Labs methods and foundation paper.
- Smith, Maria, *From One to Fold*, Foundation branch paper.
- Smith, Maria, *From Fold to Mathematics*, Mathematics branch paper.
- Smith, Maria, *From Distinction to Information*, Information Science branch paper.
- Smith, Maria, *After Turing: The Fold Machine*, Classical Computation branch paper.
- Smith, Maria, *The Quantum Fold Machine*, Reversible and Quantum Computation branch paper.
- Smith, Maria, *From Fold to Physics*, Physics branch paper.